The working (6,7): a verified graphical identity

A forward derivation with connected braid strips, every sphere move identified, and an independent boundary-action ending.

BRIEF SUMMARY: (6,7) -> fifty -> an actual standard twenty plus thirty -> two verified sphere identities.

The new 154-letter representative is our working (6,7). Join it to the saved calculation chart, follow the checked named substitutions to fifty positive units, and isolate a standard twenty in seven Hurwitz moves and one sphere rechart. Rewrite the remaining thirty on a five-point disk; its exact based action is the double outer-boundary twist. Cap that double boundary and close the twenty independently.

Proof scope. The ending is an exact factor-level identity certificate using a faithful disk-action calculation. It is not a purely elementary Hurwitz normal form W0 W1. The isolated thirty has permutation orbit sizes 3,2,1, so it cannot alone be Hurwitz-equivalent to W1. The classification theorem is not used as an identity proof.

How to read the figures. Each support is drawn with the shared Arc/Loop library and paired with the braid that represents that pictured support. Blue dots separate factors, and braid strand colors continue downward. A yellow SPHERE SUPPORT CHART label means the display is an explicitly certified sphere representative. Actual sphere derivation moves are labeled SPHERE MOVE. Raw factors, disk discrepancies, global frames and lift checks are retained in the online certificates.

Navigation. This detailed book retains every old atomic substitution instead of hiding it in a group. Its front section is a brief explanation; the subsequent pages give the complete diagrams. The online view offers chapter/group navigation and side-scrolling columns. The final disk cut-system comparison is jointly certified; the other drawings are individual factor supports.

Contents

Section pages are listed below. PDF bookmarks provide direct section navigation.

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II. The named-substitution route to fifty 12

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V. The decisive joint cut-system certificate 211

VI. Sources, limitations and downloadable certificates 212

Every atomic move is retained. Sphere moves are explicitly labeled; certificates are linked in Section VI.

What the steps establish

1. Start from the working (6,7). Three inverse adjacent Hurwitz moves, an explicit global change of chart, and exact disk support presentations join the working seed to the saved source. This does not use a sphere identity.

2. Reach the actual positive fifty. The preserved 145 atomic moves use transported PC lantern/half-lantern substitutions, disk Hurwitz moves and explicitly recorded sphere changes. The previous inverse pair is replaced using a squared sphere relation, preserving the genus-two lift.

3. Extract the standard twenty. Move F3 upward past the transported {5,6} pair. Rechart that pair as round T56 on the sphere, then move it unchanged past six neighbors. The resulting Q A Q A block is a disk conjugate of W0, where A=43211234 and Q=55. Its closure is checked independently.

4. Close the thirty in a five-point chart. Put point 6 at infinity. One pair loop becomes its complementary four-point loop; its expanded disk-core length increases by ten. The exact ordered five-point model has forty positive core letters and the same action on all five based meridians as the double outer-boundary twist. That is a faithful disk braid equality.

5. Cap the double boundary. The double cap is a separate sphere move with trivial genus-two lift discrepancy. A single five-point boundary twist has lift -I and would be wrong. No closure hypothesis for the starting word is used in the final disk-action calculation.

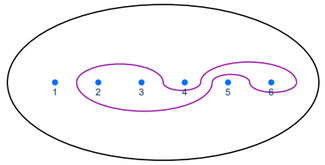
I. The working (6,7) and the calculation chart

The working 154-letter factorization joins the literal saved source by checked disk operations and one explicit global conjugation. No sphere substitution occurs in this joining phase.

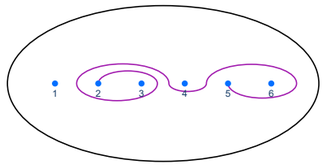
The working (6,7): complete factorization

FULL ORDERED FACTORIZATION. Read down the left column, then down the right. Braids continue with strand colors retained.

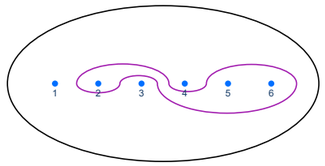
1. F1 | T3^2



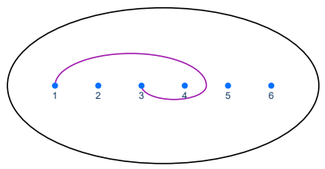
2. F2 | H



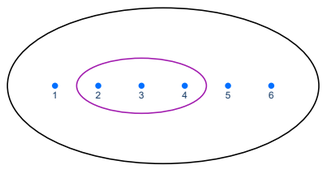
3. F4 | T3^2



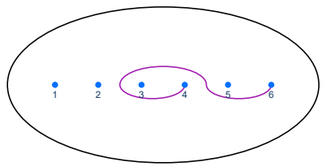
4. F3 | H

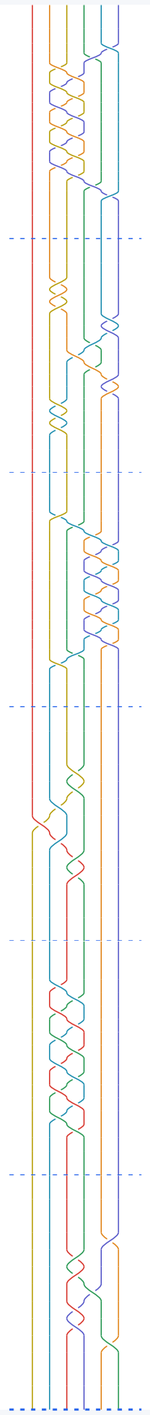


5. F5 | T3^2

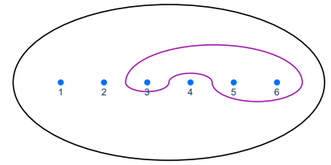


6. F6 | H

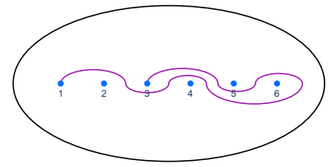




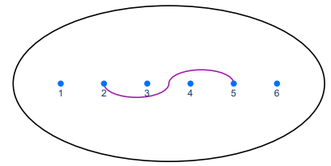
7. F8 | T3^2



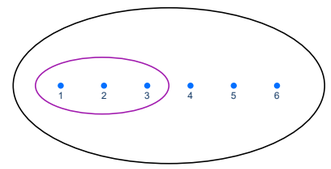
8. F9 | H



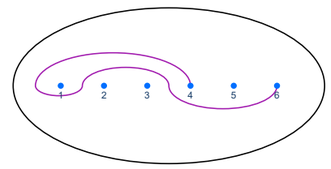
9. F7 | H



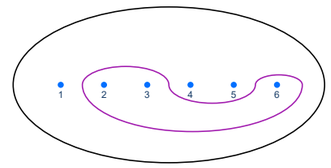
10. F10 | T3^2 | SPHERE SUPPORT CHART

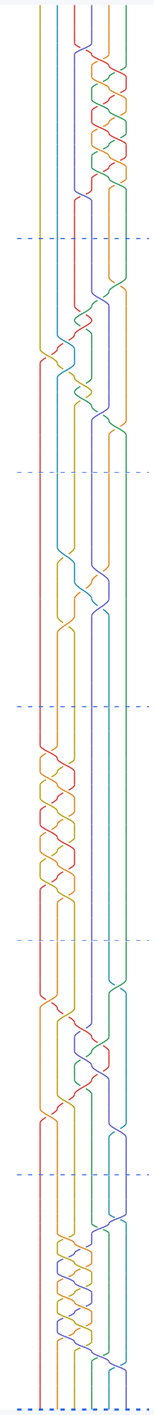


11. F11 | H



12. F12 | T3^2 | SPHERE SUPPORT CHART

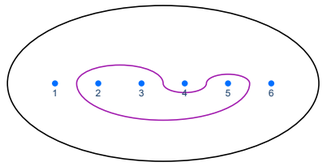


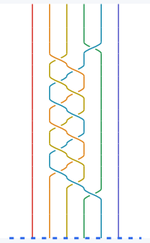


The working (6,7): complete factorization - continued

FULL ORDERED FACTORIZATION. Read down the left column, then down the right. Braids continue with strand colors retained.

13. F13 | T3^2





I - step 001

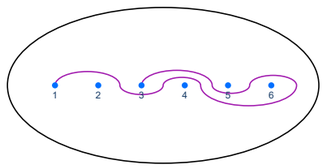
Change the working ordering by an inverse Hurwitz move

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

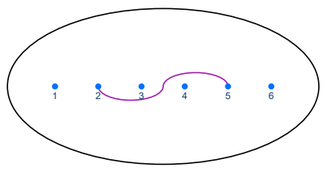
7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

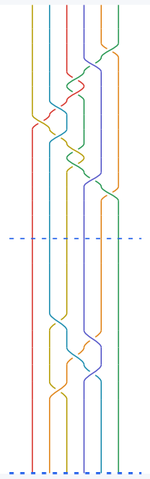
Before

8. F9 | H



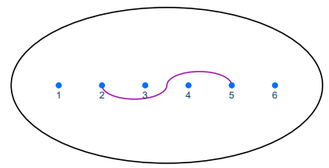
9. F7 | H



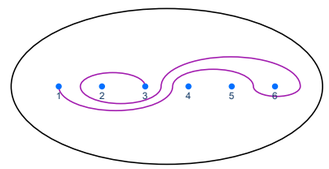


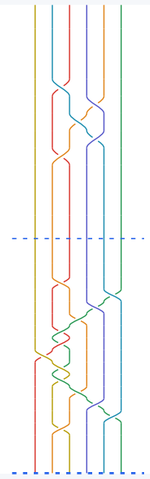
After

8. F7 | H



9. F9 | H





I - step 002

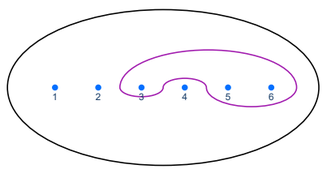
Change the working ordering by an inverse Hurwitz move

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

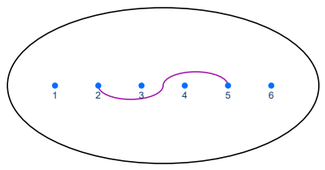
6 unchanged displayed factors above; 5 below. Only the changed interval is shown.

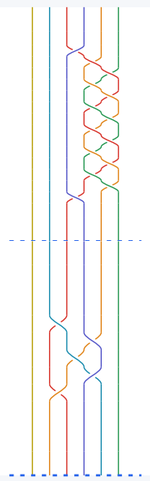
Before

7. F8 | T3^2



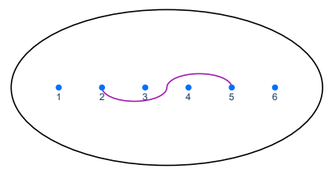
8. F7 | H



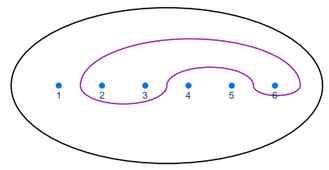


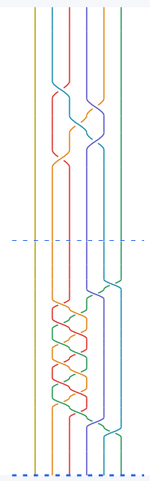
After

7. F7 | H



8. F8 | T3^2





I - step 003

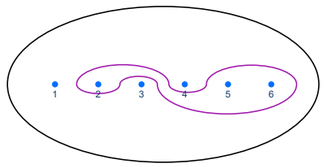
Change the working ordering by an inverse Hurwitz move

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

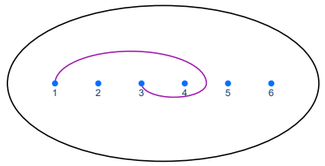
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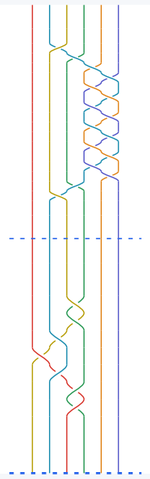
Before

3. F4 | T3^2



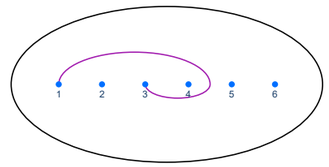
4. F3 | H



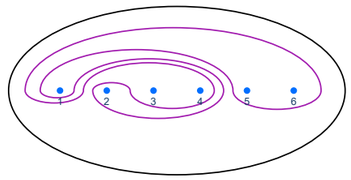


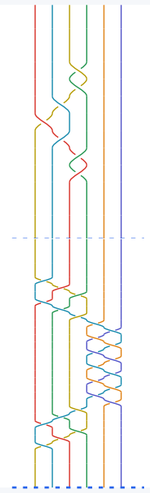
After

3. F3 | H



4. F4 | T3^2





I - step 004

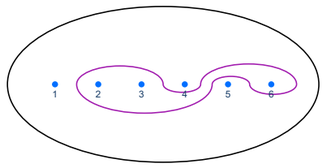
Global conjugation by 3 -5 -2 -3 -2 into the saved calculation chart

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

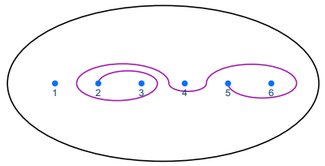
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

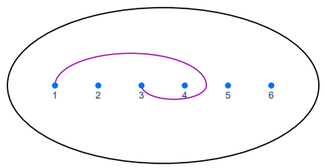
1. F1 | T3^2



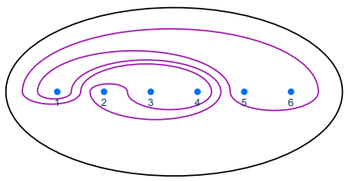
2. F2 | H



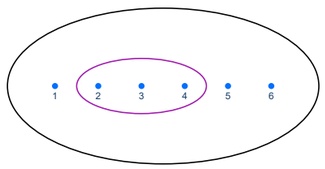
3. F3 | H

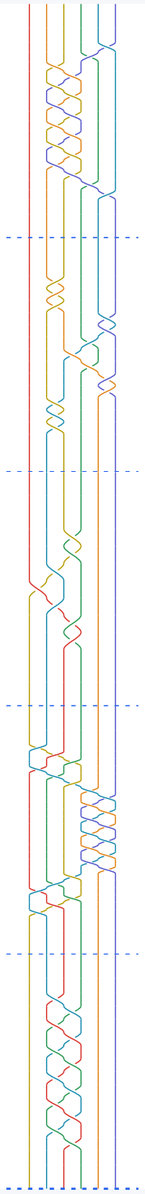


4. F4 | T3^2



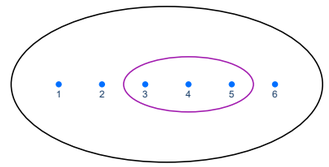
5. F5 | T3^2



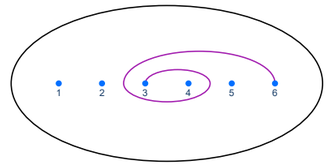


After

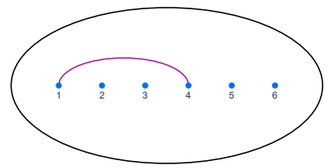
1. F1 | T3^2



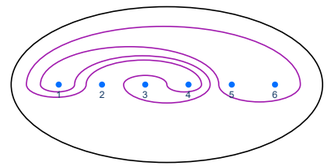
2. F2 | H



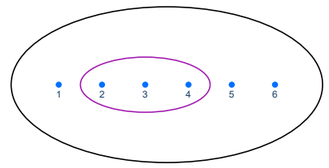
3. F3 | H

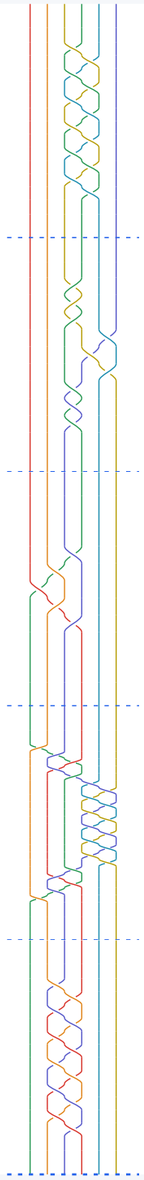


4. F4 | T3^2



5. F5 | T3^2





I - step 004 - continued

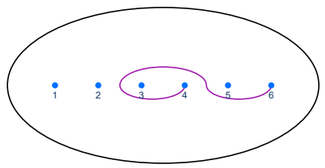
Global conjugation by 3 -5 -2 -3 -2 into the saved calculation chart

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

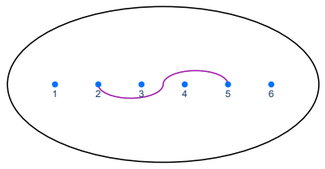
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

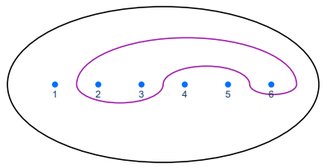
6. F6 | H



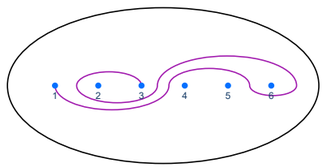
7. F7 | H



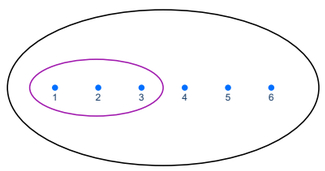
8. F8 | T3^2

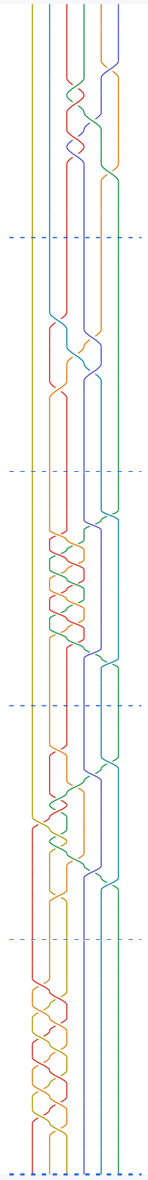


9. F9 | H



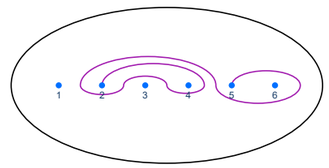
10. F10 | T3^2 | SPHERE SUPPORT CHART



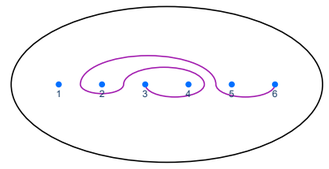


After

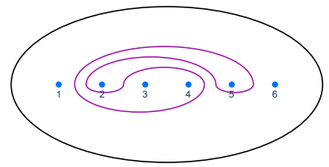
6. F6 | H



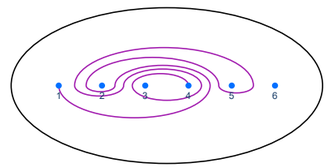
7. F7 | H



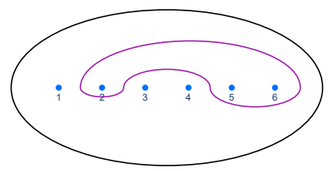
8. F8 | T3^2

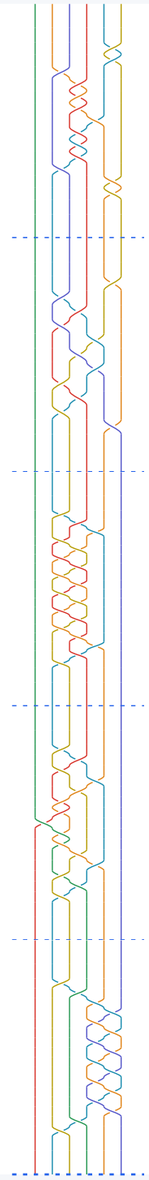


9. F9 | H



10. F10 | T3^2





I - step 004 - continued

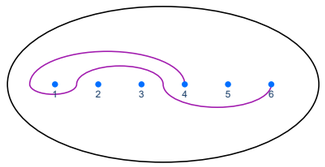
Global conjugation by 3 -5 -2 -3 -2 into the saved calculation chart

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

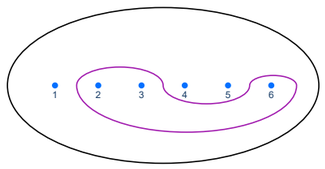
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

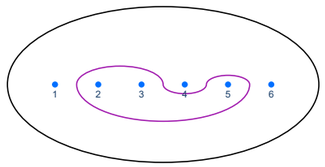
11. F11 | H

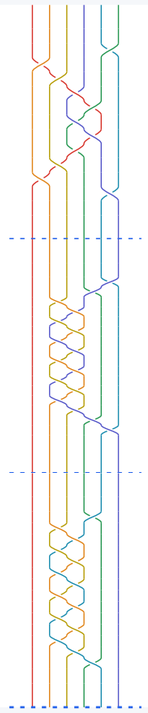


12. F12 | T3^2 | SPHERE SUPPORT CHART



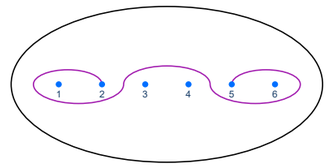
13. F13 | T3^2



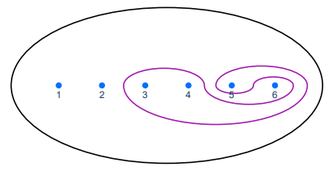


After

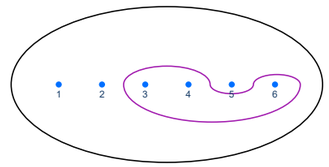
11. F11 | H

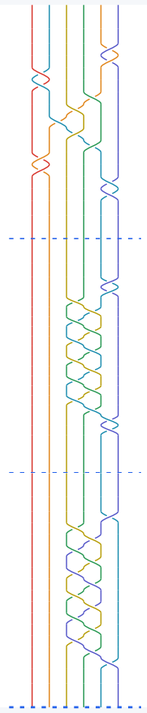


12. F12 | T3^2 | SPHERE SUPPORT CHART



13. F13 | T3^2





I - step 005

Use the literal source presentations of the saved route

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

II. The named-substitution route to fifty

All 145 saved atomic moves are retained in their forward order. Column chart frames and display-only sphere replacements are retained separately from the actual derivation moves.

II - step 001

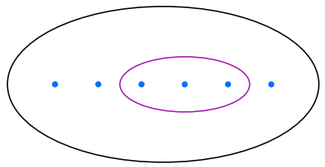
Split F1 into ordinary roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

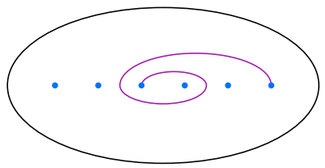
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=identity; after g=2 -4 5.

Before

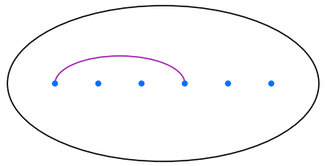
1. F1 | T3^2



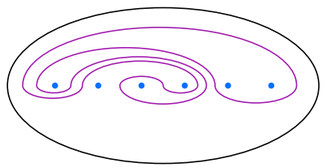
2. F2 | H



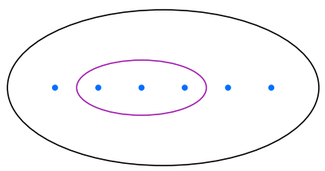
3. F3 | H

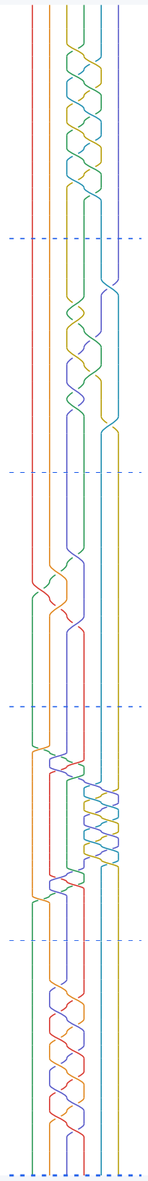


4. F4 | T3^2



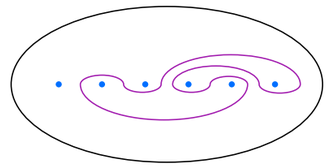
5. F5 | T3^2



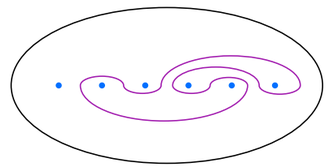


After

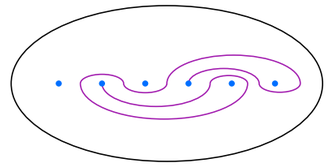
1. F1.1 | T3



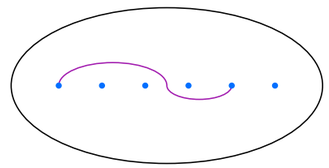
2. F1.2 | T3



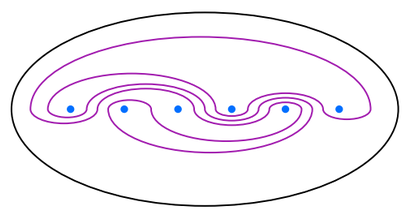
3. F2 | H

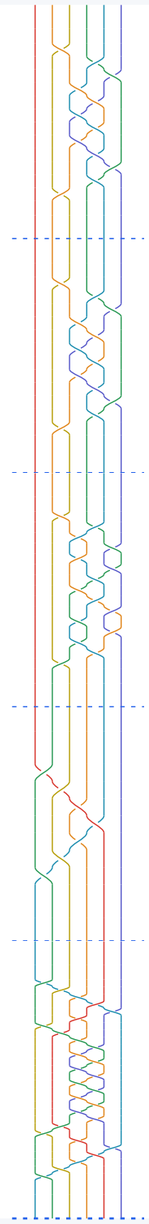


4. F3 | H



5. F4 | T3^2





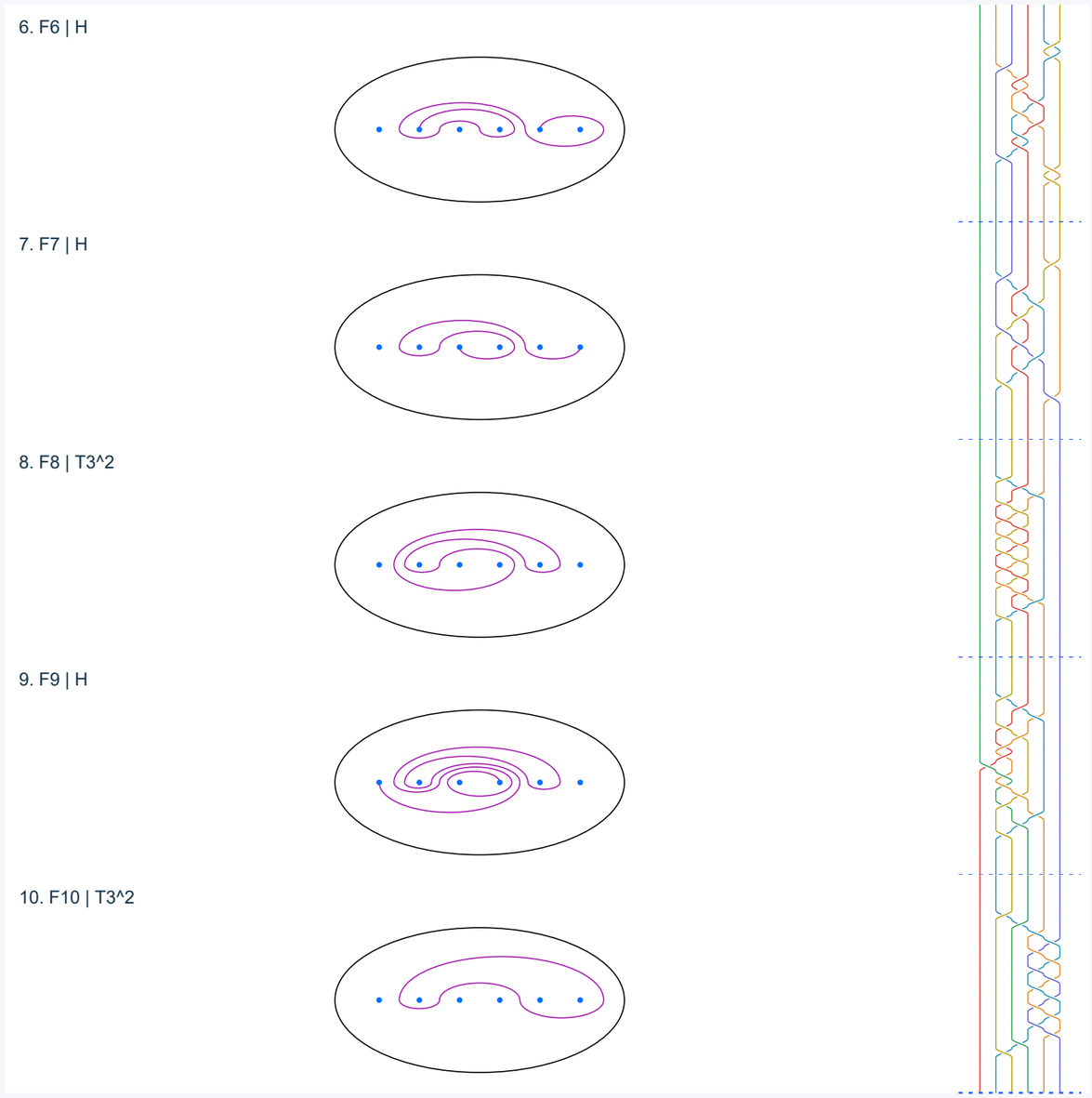
II - step 001 - continued

Split F1 into ordinary roots

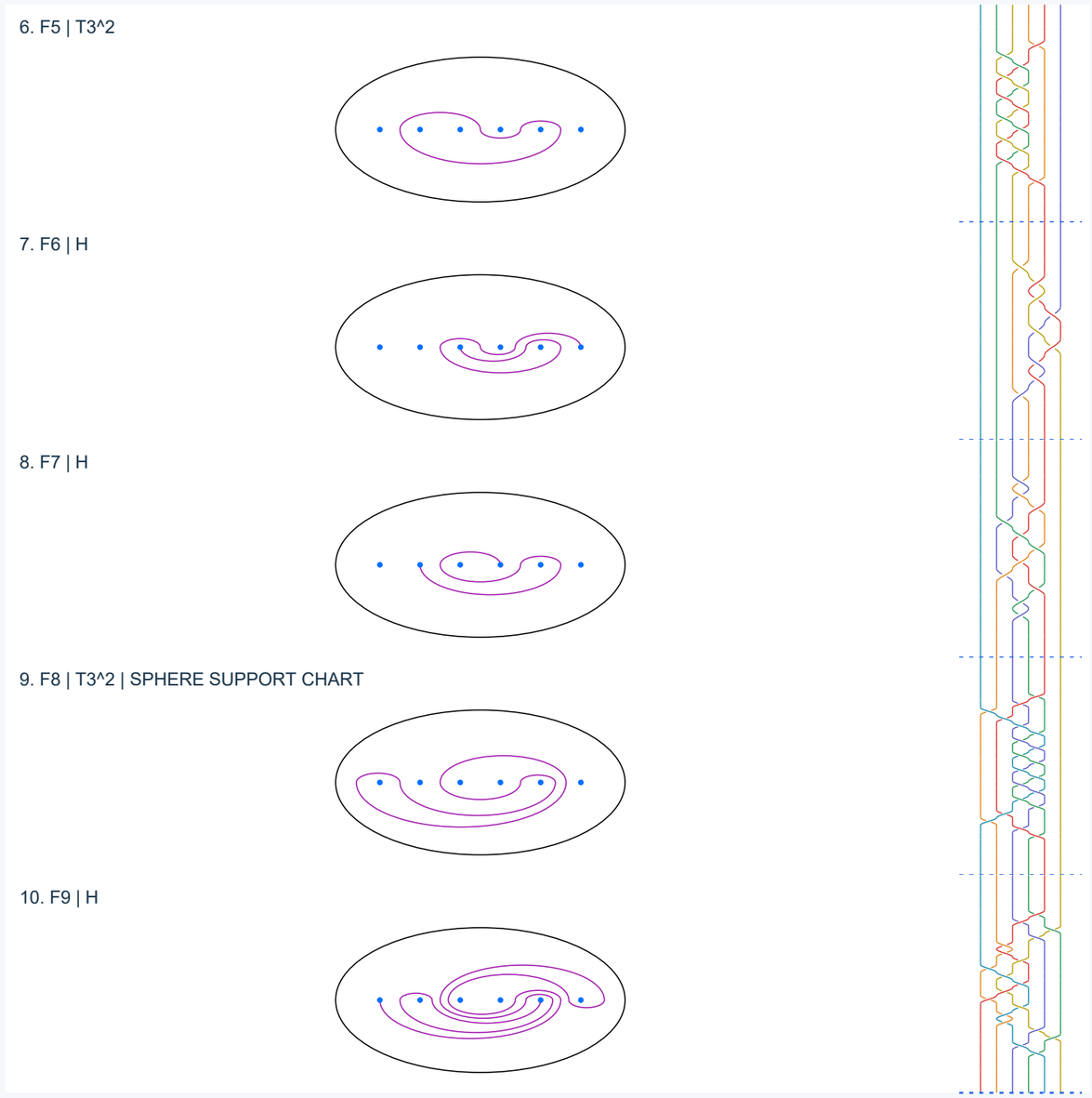
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=identity; after g=2 -4 5.

Before



After



II - step 001 - continued

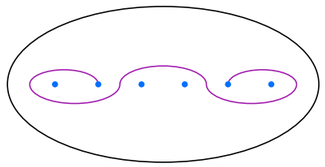
Split F1 into ordinary roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

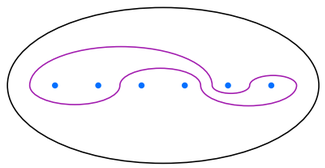
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=identity; after g=2 -4 5.

Before

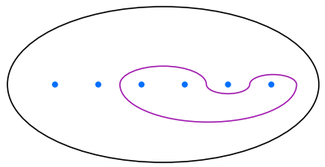
11. F11 | H

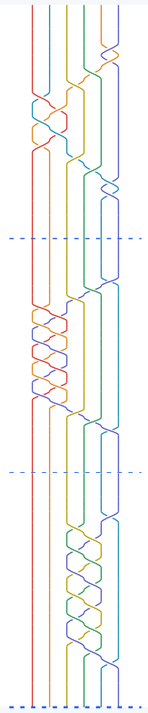


12. F12 | T3^2



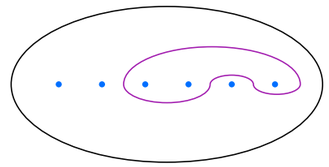
13. F13 | T3^2



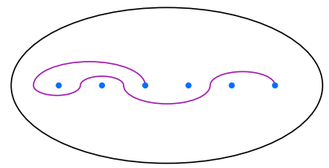


After

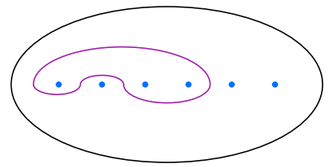
11. F10 | T3^2



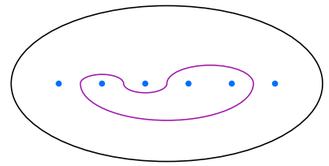
12. F11 | H

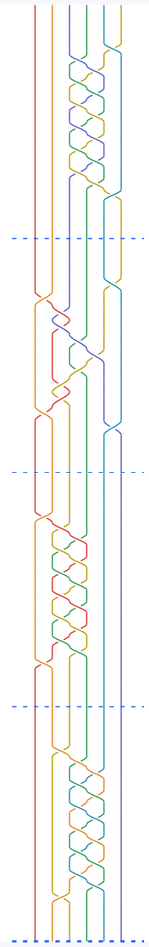


13. F12 | T3^2



14. F13 | T3^2





II - step 002

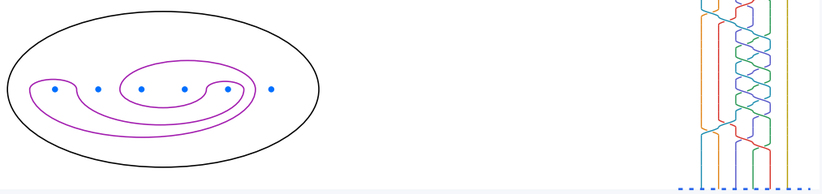
Split F8 into ordinary roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

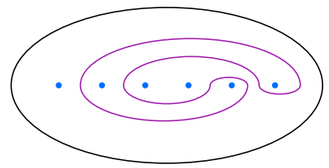
Before

9. F8 | T3^2 | SPHERE SUPPORT CHART

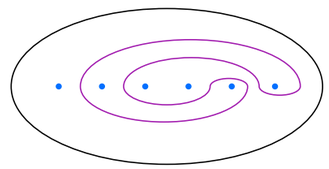


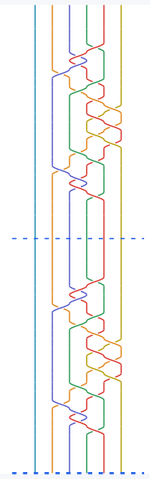
After

9. F8.1 | T3



10. F8.2 | T3





II - step 003

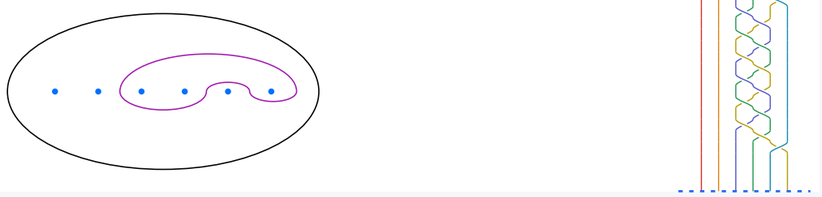
Split F10 into ordinary roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

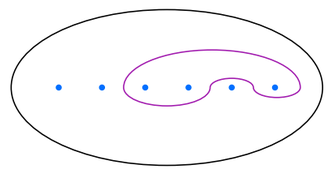
Before

12. F10 | T3^2

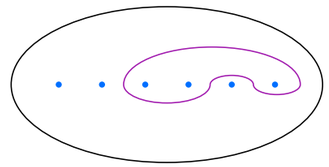


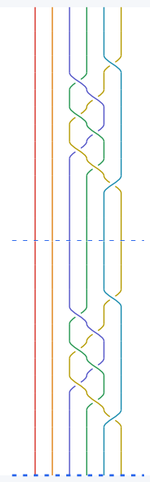
After

12. F10.1 | T3



13. F10.2 | T3





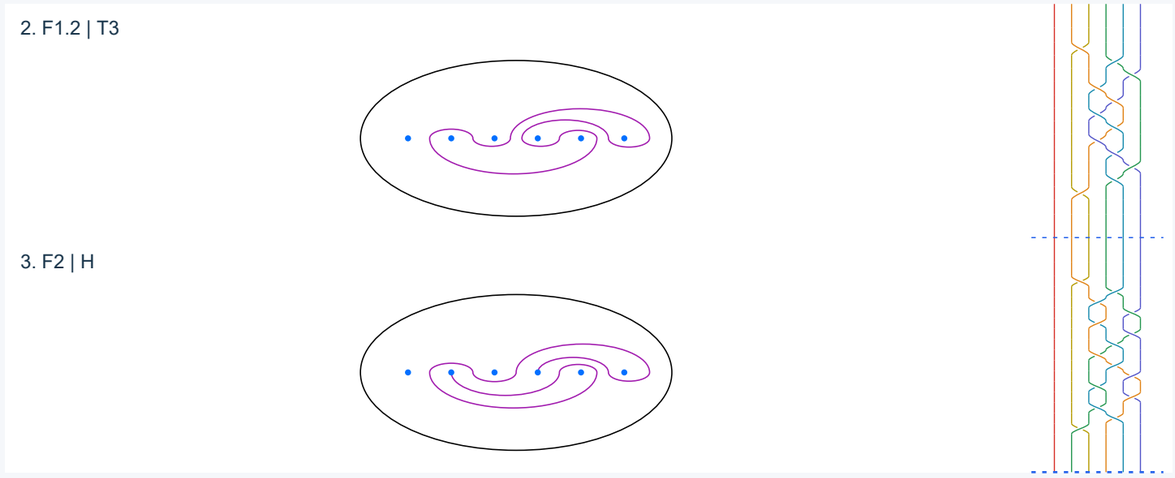
II - step 004

F1.2 down past F2

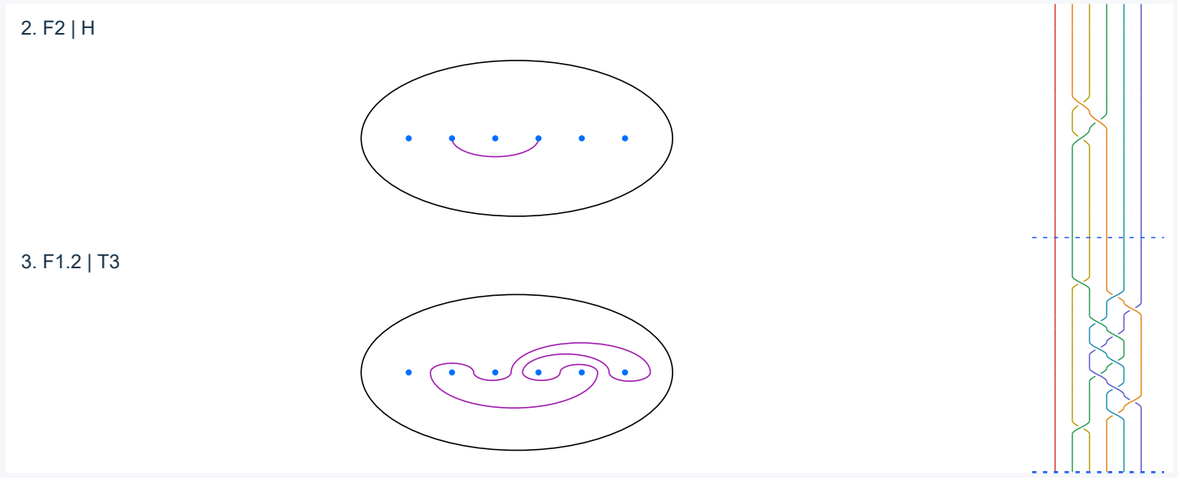
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

Before



After



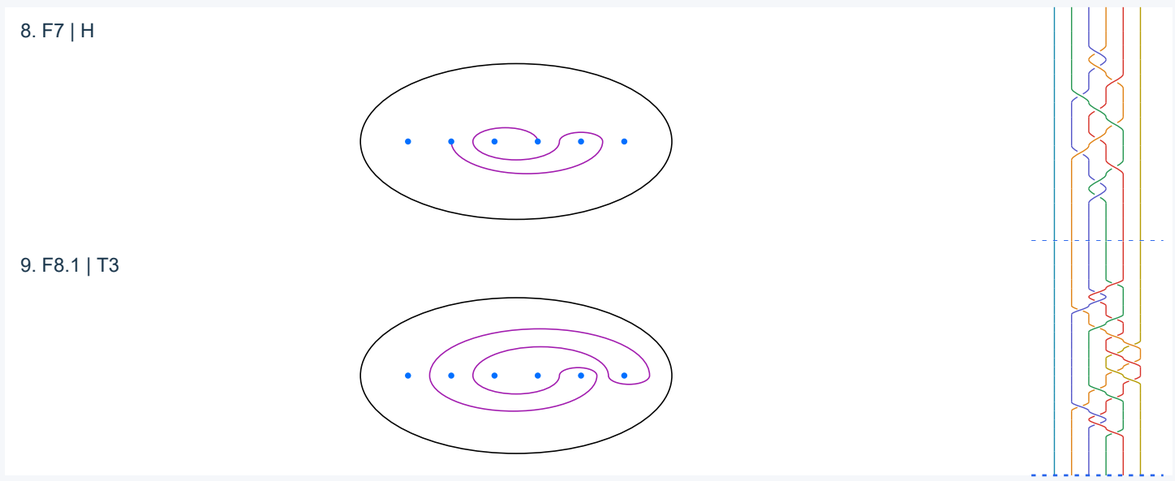
II - step 005

F8.1 up past F7

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

Before



After



II - step 006

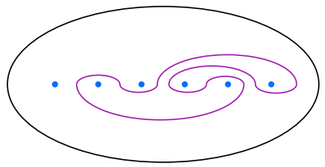
F10.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

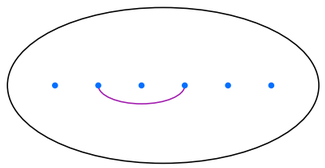
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=1 2 -4 5.

Before

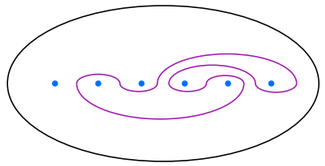
1. F1.1 | T3



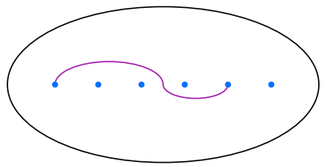
2. F2 | H



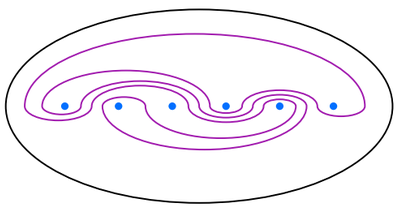
3. F1.2 | T3

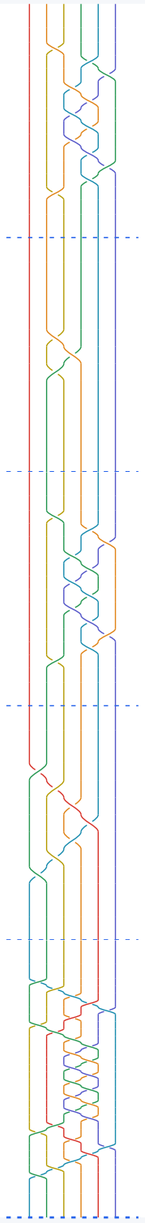


4. F3 | H



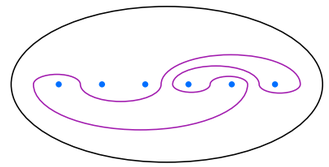
5. F4 | T3^2



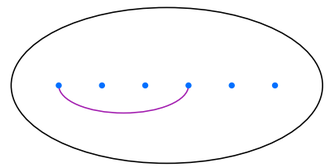


After

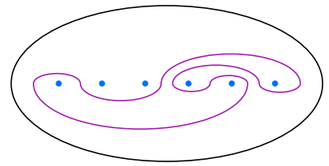
1. F1.1 | T3



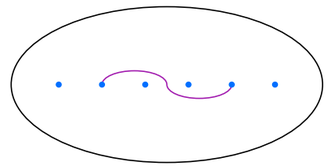
2. F2 | H



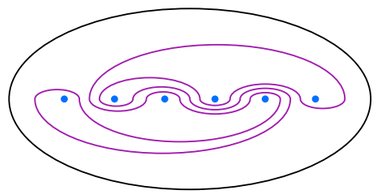
3. F1.2 | T3

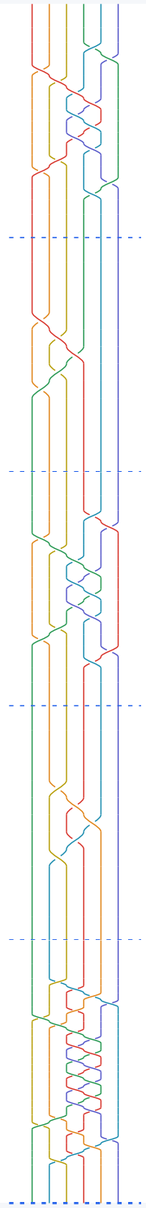


4. F3 | H



5. F4 | T3^2





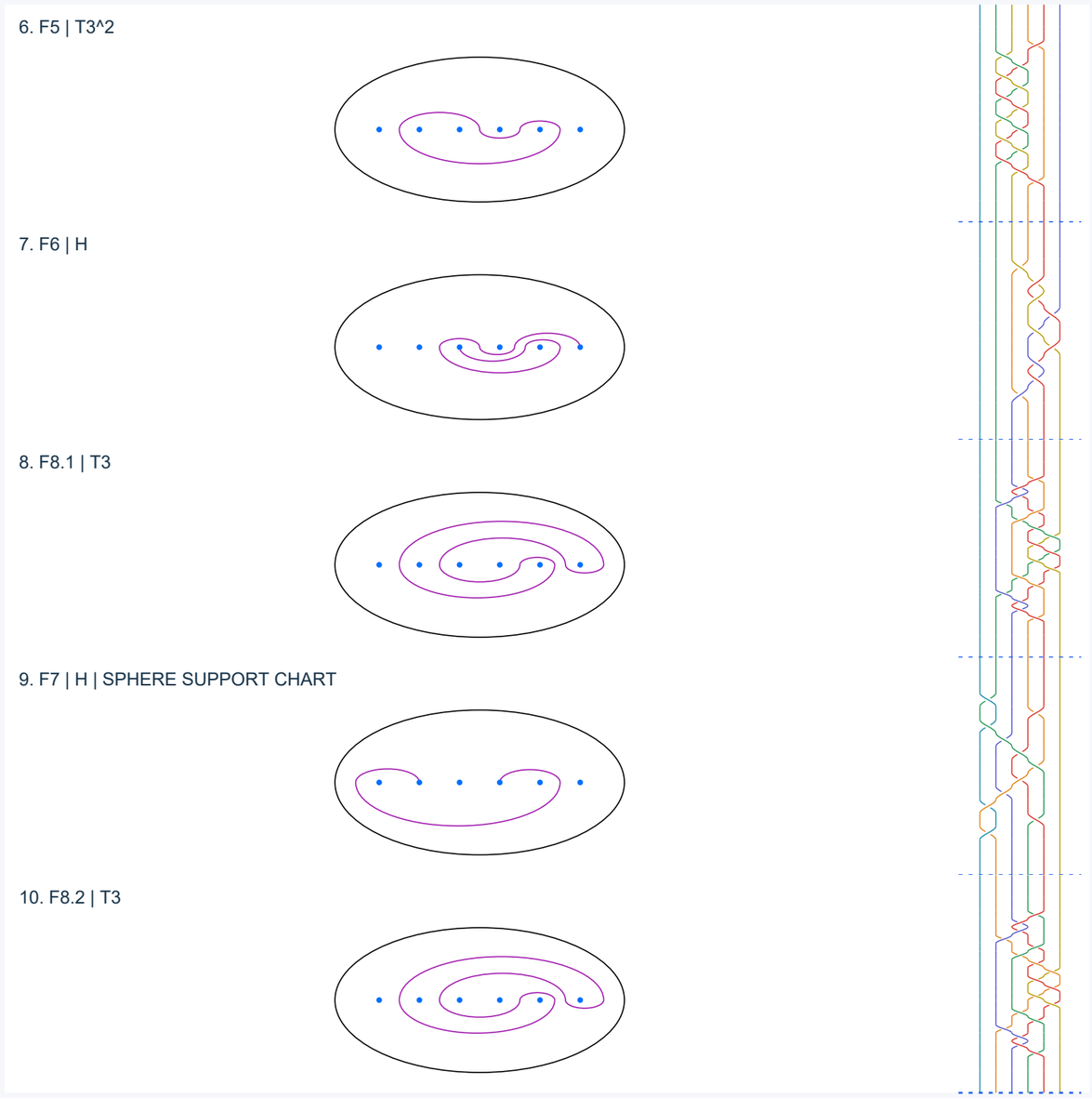
II - step 006 - continued

F10.1 up past F9

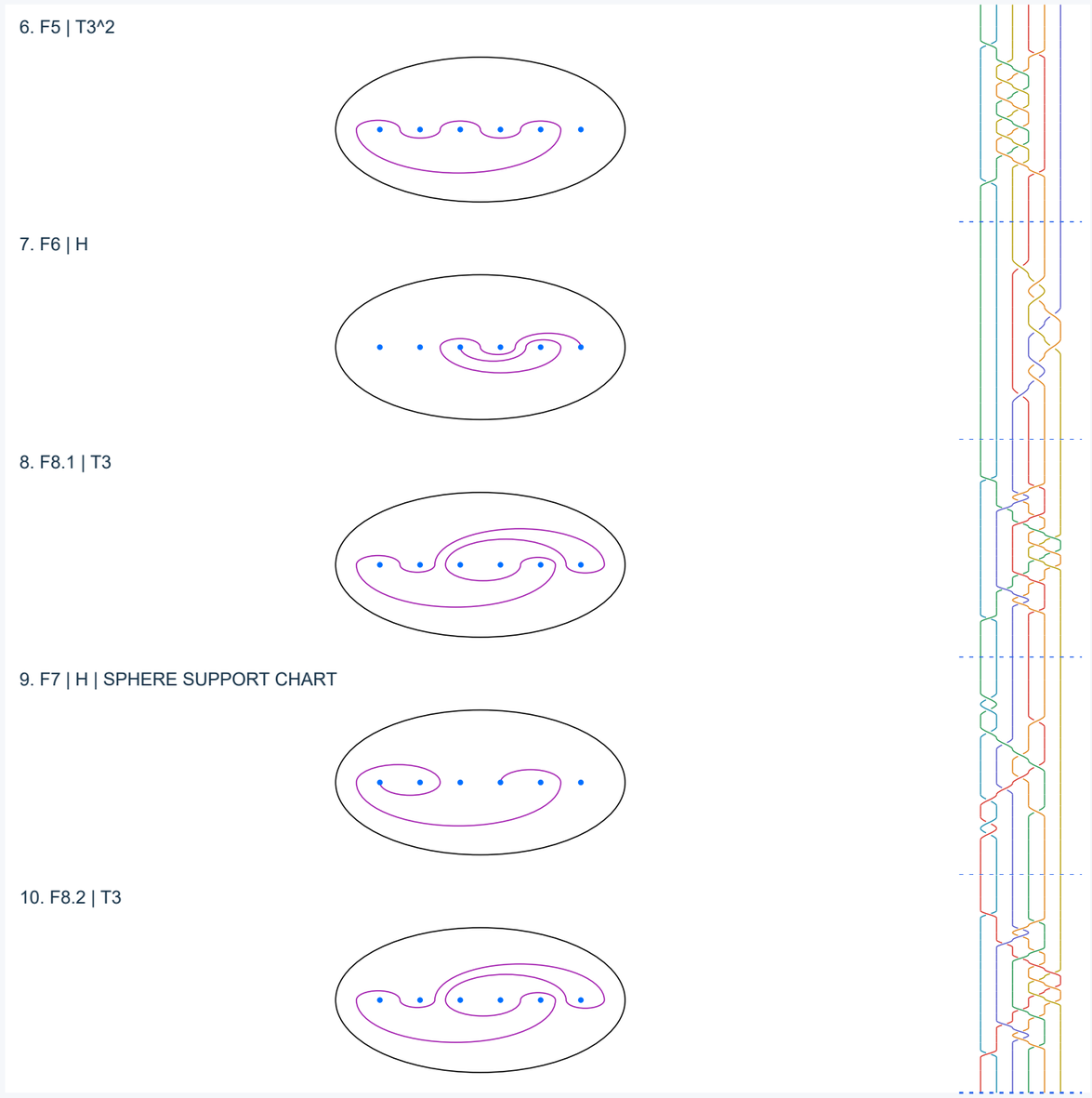
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=1 2 -4 5.

Before



After



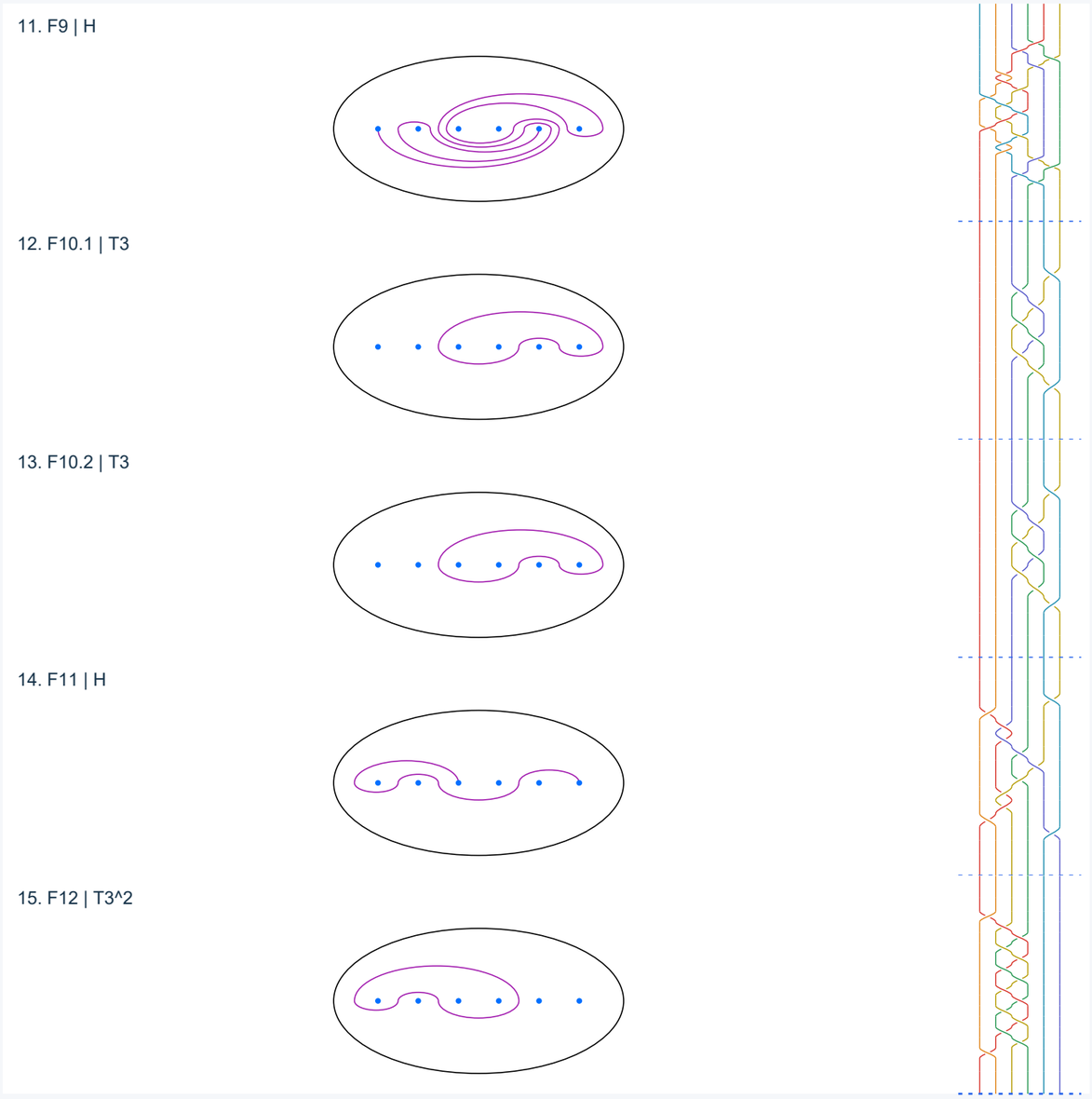
II - step 006 - continued

F10.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=1 2 -4 5.

Before



After



II - step 006 - continued

F10.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=1 2 -4 5.

Before

16. F13 | T3^2



After

16. F13 | T3^2



II - step 007

F10.1 up past F8.2

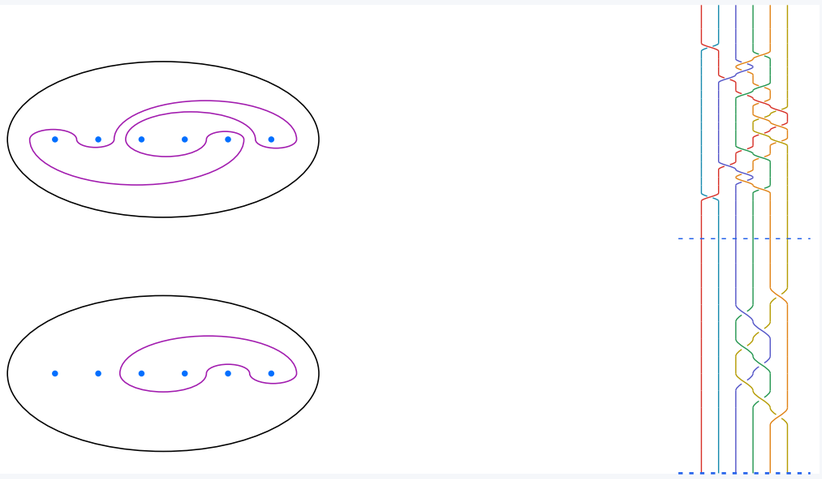
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before

10. F8.2 | T3

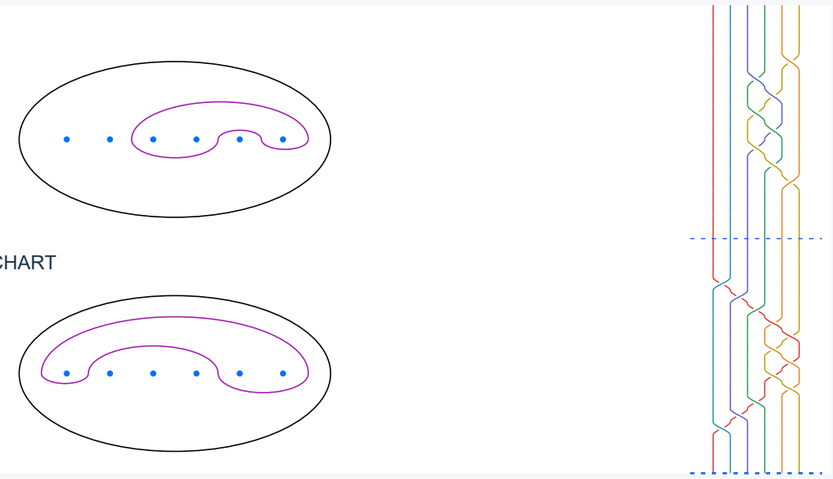
11. F10.1 | T3



After

10. F10.1 | T3

11. F8.2 | T3 | SPHERE SUPPORT CHART



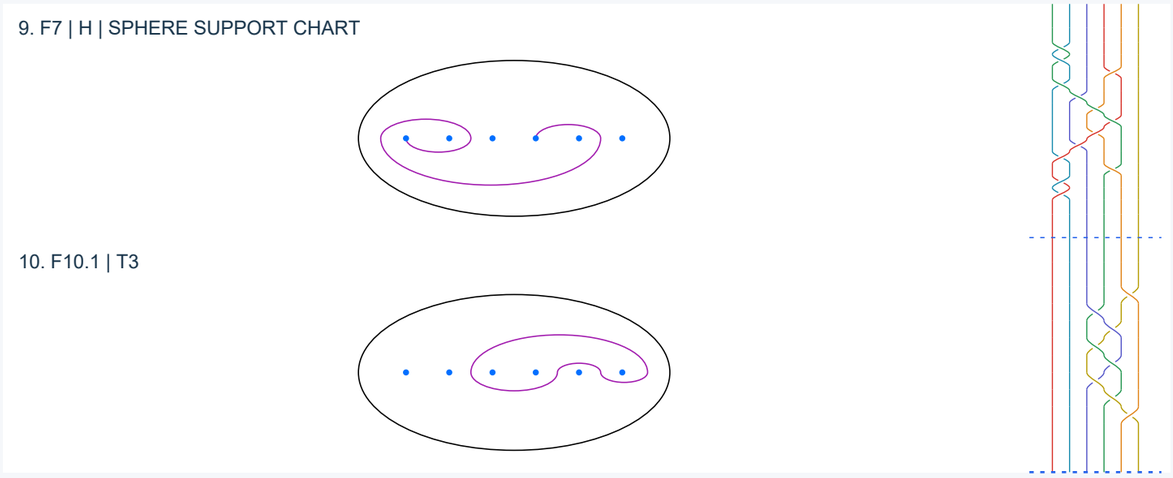
II - step 008

F10.1 up past F7

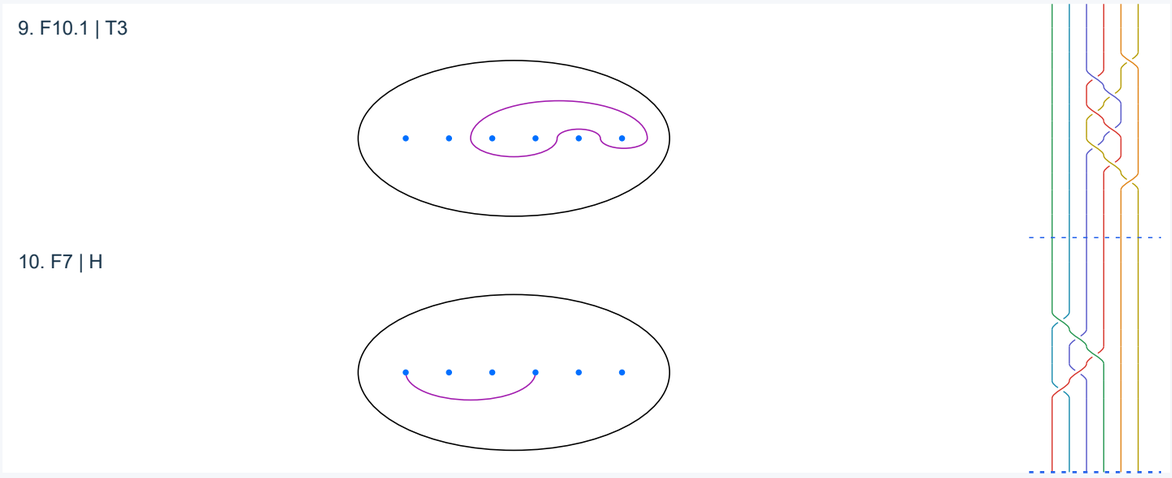
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



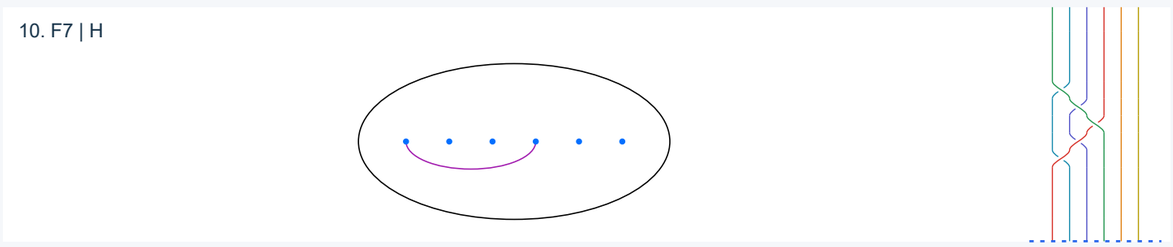
II - step 009

Use the exact disk equality of the two simple arcs

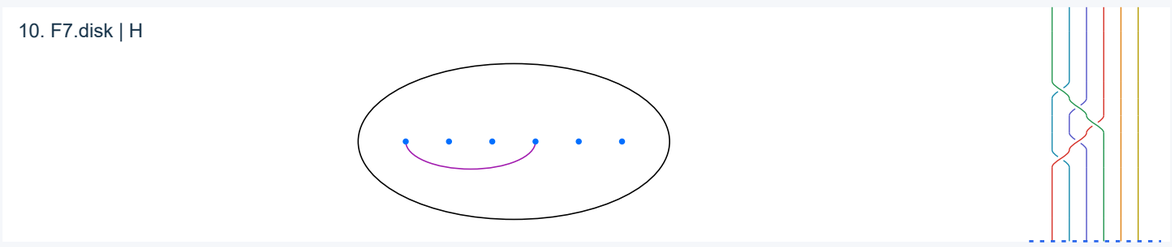
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



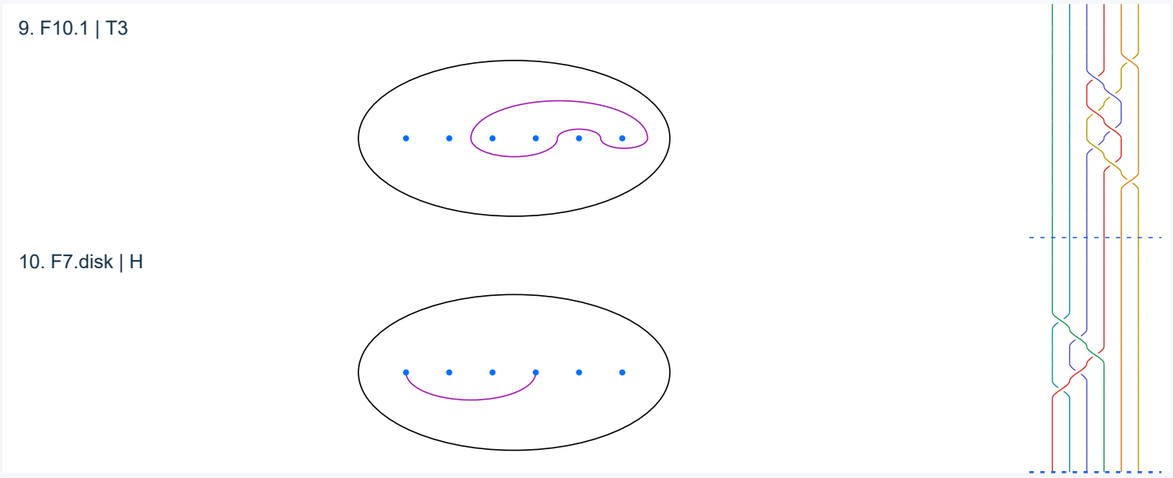
II - step 010

F7.disk up past F10.1

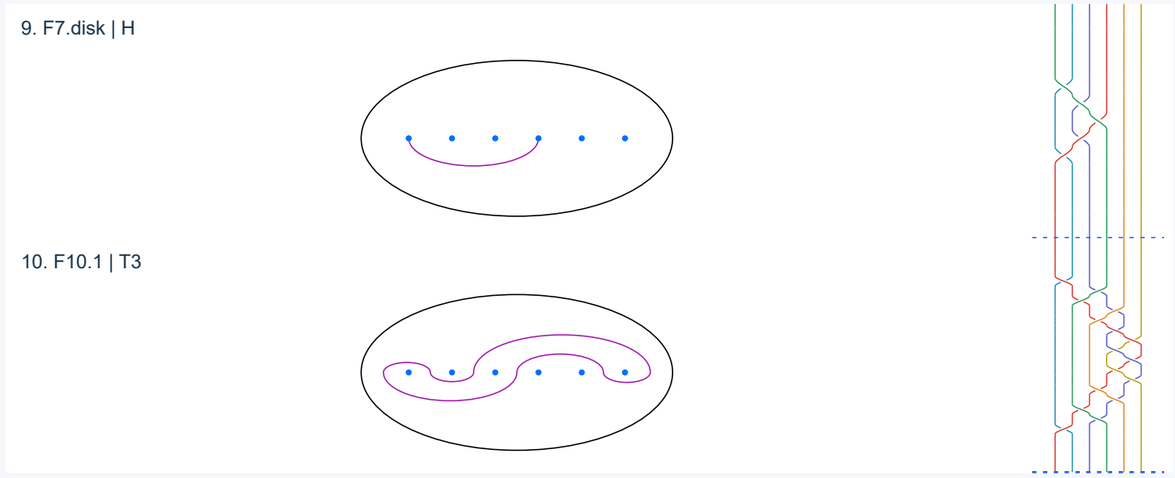
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



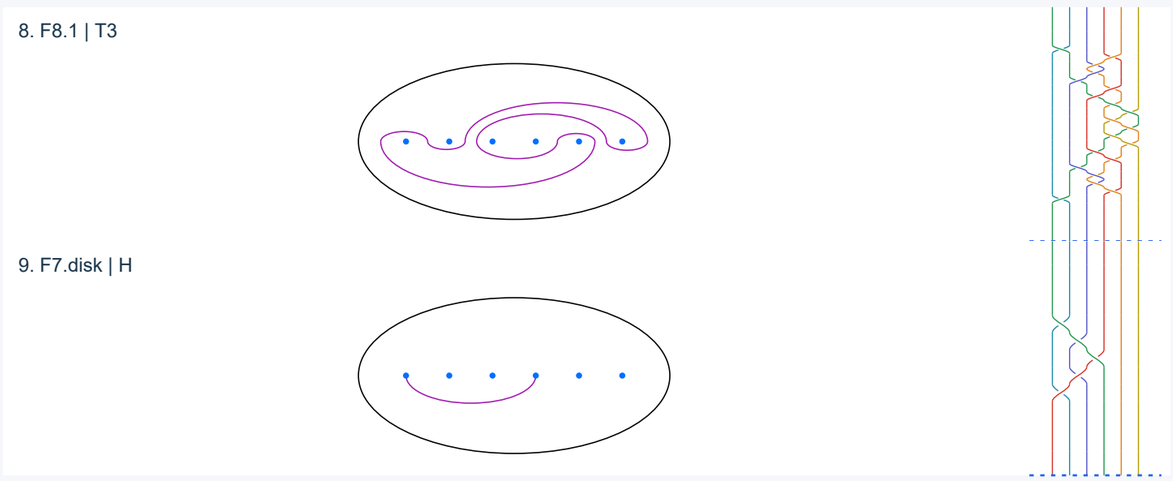
II - step 011

F7.disk up past F8.1

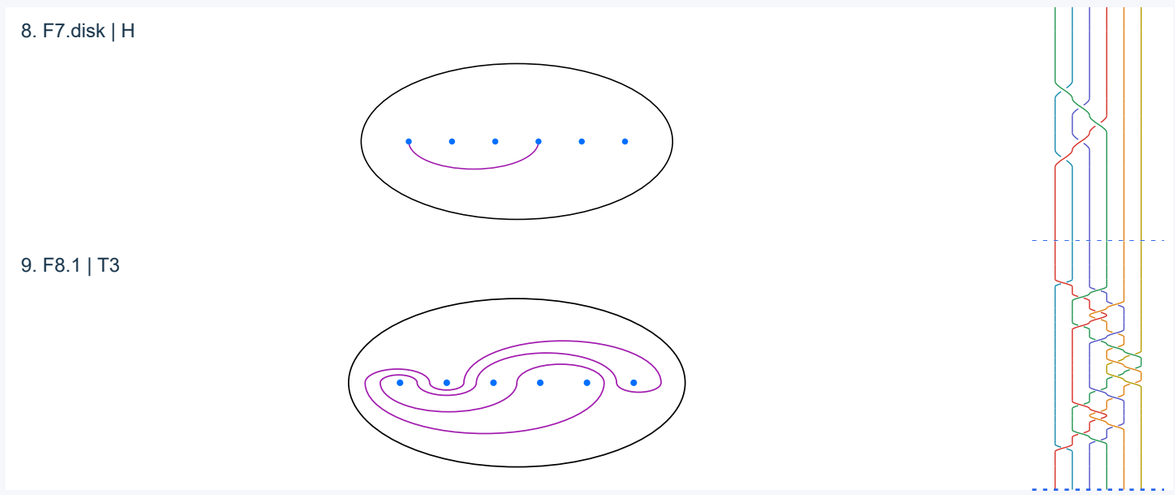
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



II - step 012

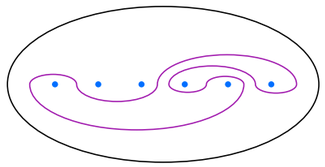
F7.disk up past F6

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

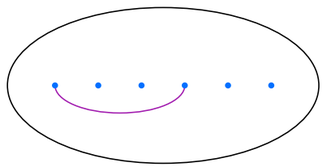
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=-4 5.

Before

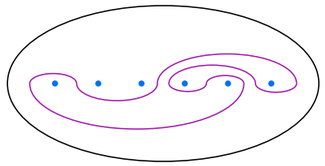
1. F1.1 | T3



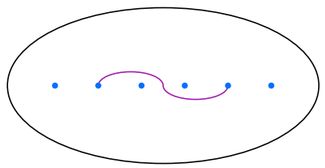
2. F2 | H



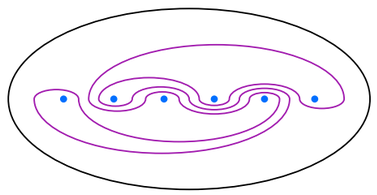
3. F1.2 | T3

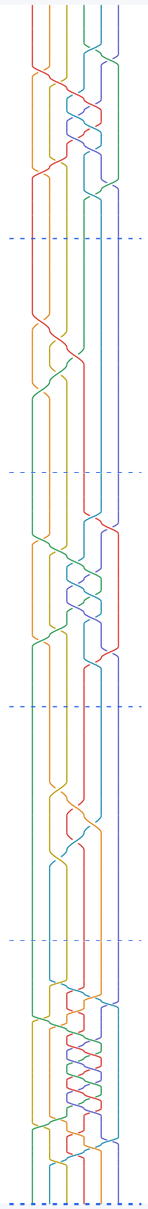


4. F3 | H



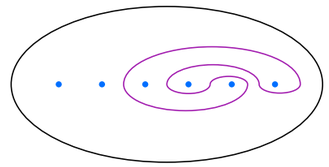
5. F4 | T3^2



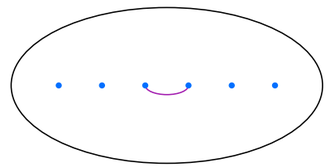


After

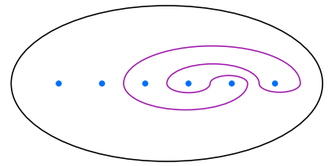
1. F1.1 | T3



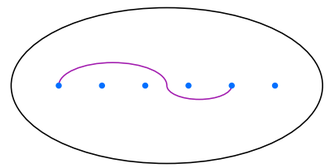
2. F2 | H



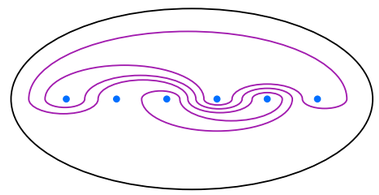
3. F1.2 | T3

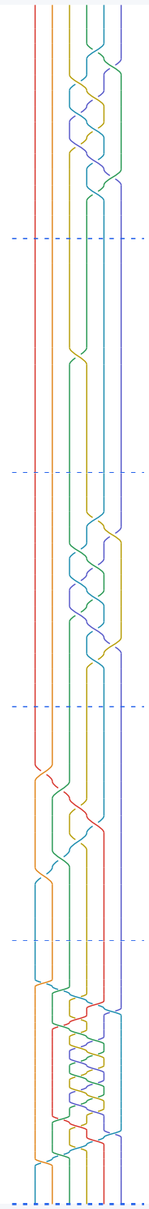


4. F3 | H



5. F4 | T3^2





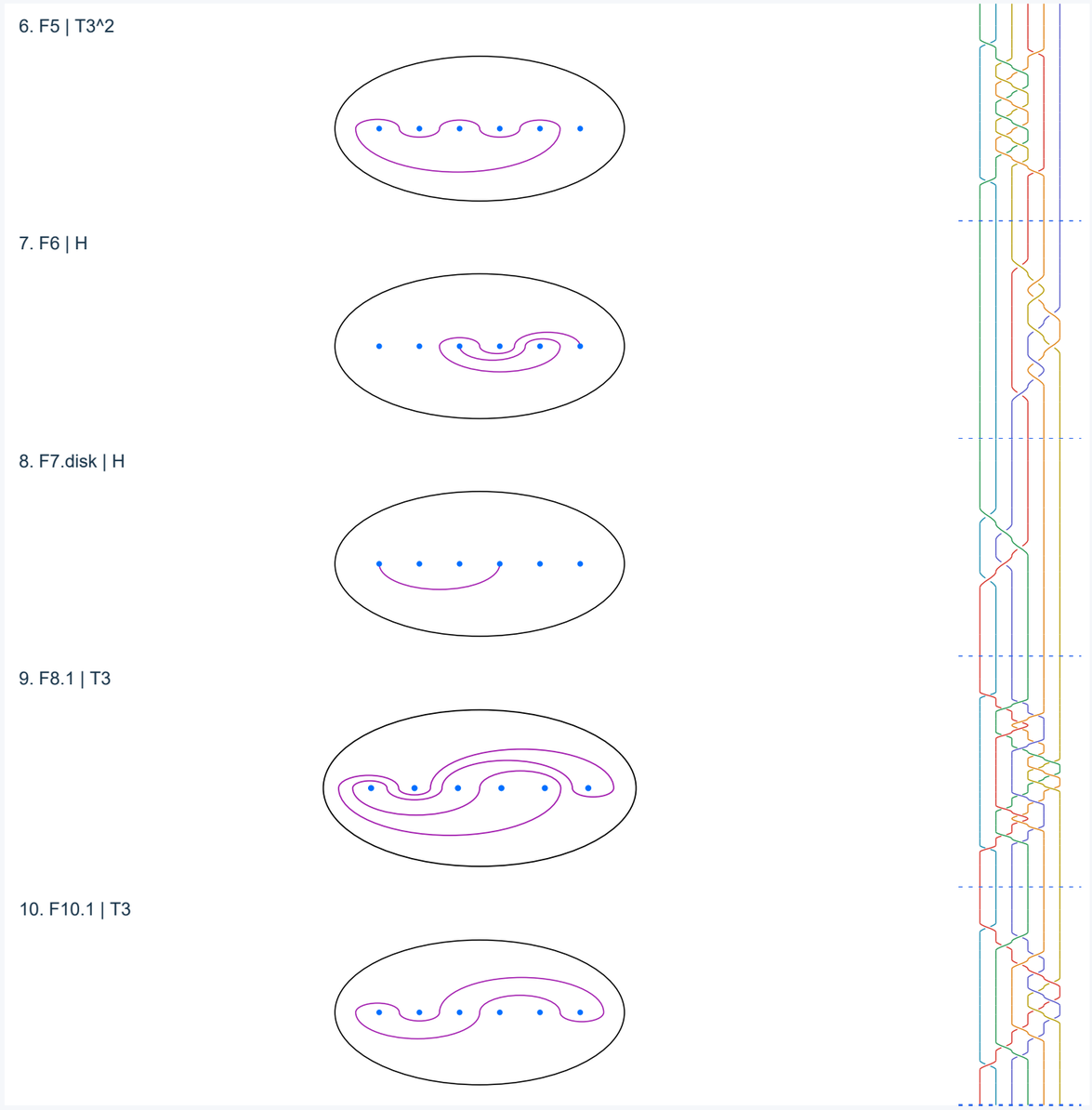
II - step 012 - continued

F7.disk up past F6

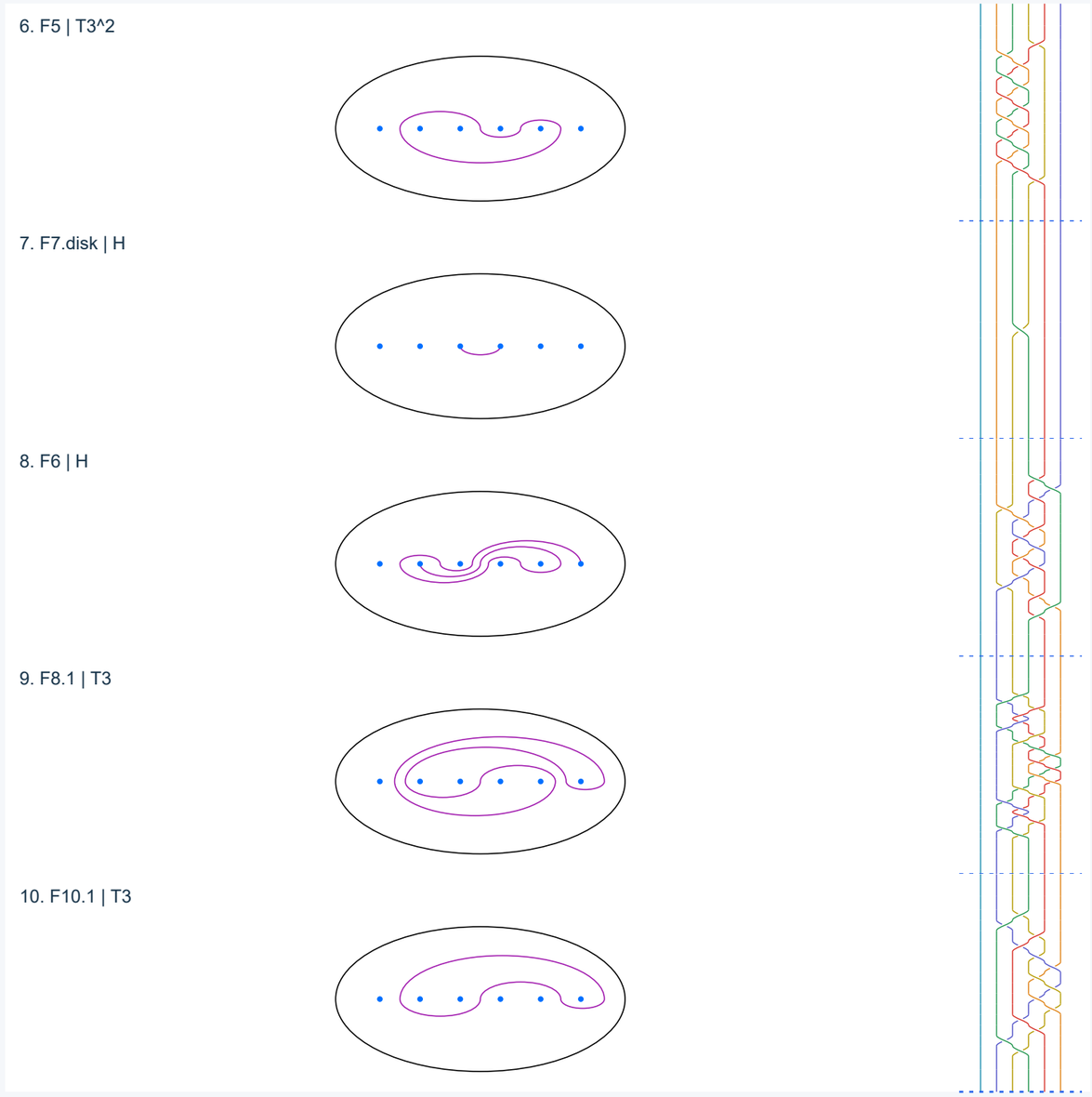
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=-4 5.

Before



After



II - step 012 - continued

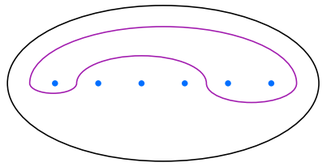
F7.disk up past F6

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

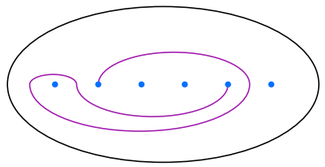
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=-4 5.

Before

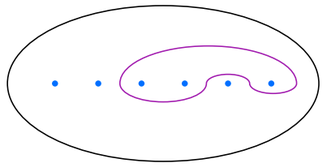
11. F8.2 | T3 | SPHERE SUPPORT CHART



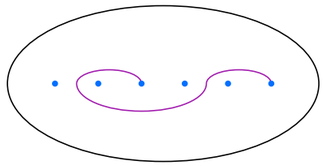
12. F9 | H | SPHERE SUPPORT CHART



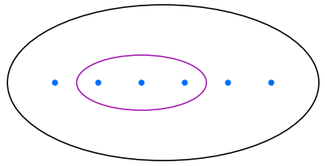
13. F10.2 | T3

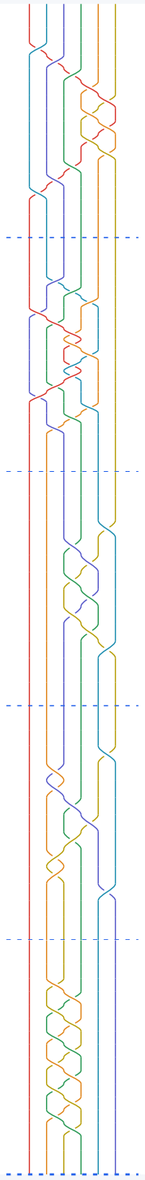


14. F11 | H



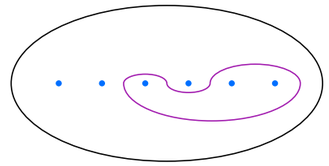
15. F12 | T3^2



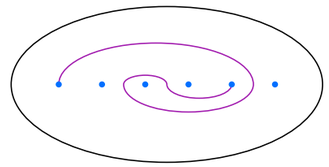


After

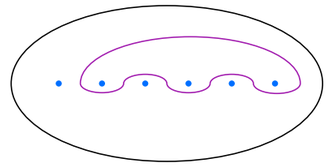
11. F8.2 | T3



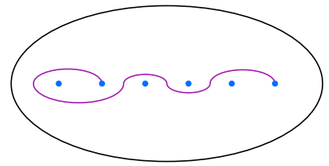
12. F9 | H | SPHERE SUPPORT CHART



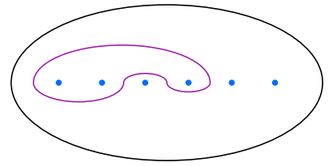
13. F10.2 | T3

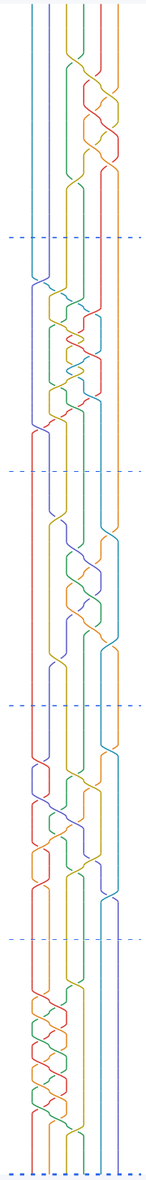


14. F11 | H



15. F12 | T3^2





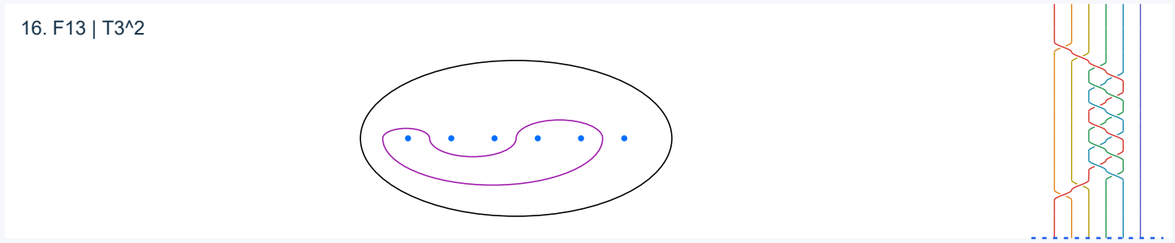
II - step 012 - continued

F7.disk up past F6

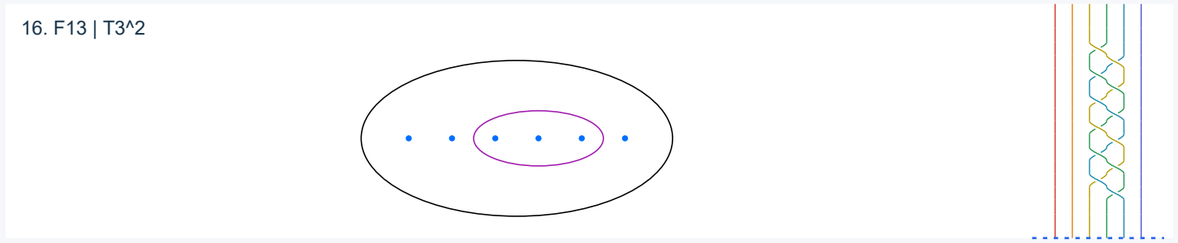
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=-4 5.

Before



After



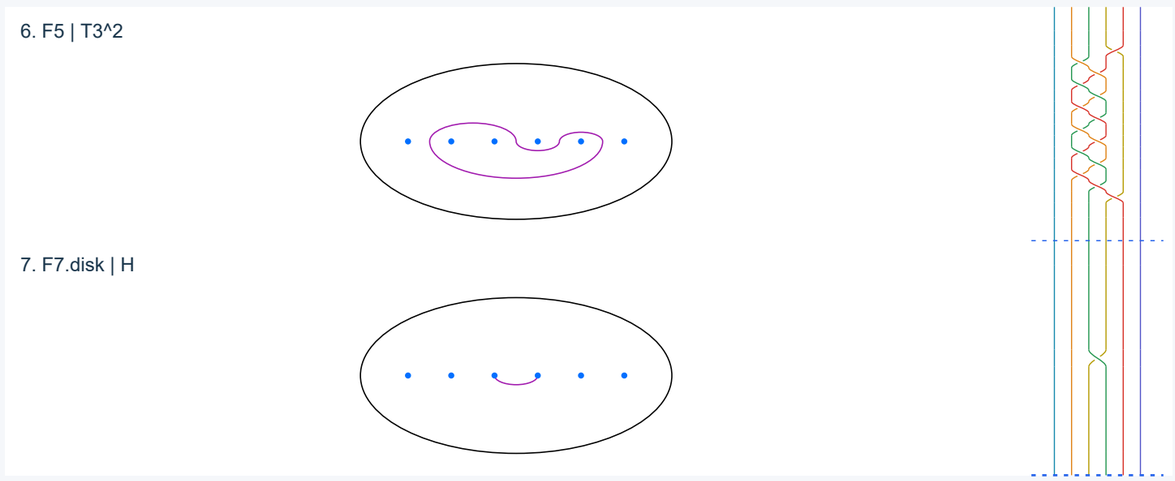
II - step 013

F7.disk up past F5

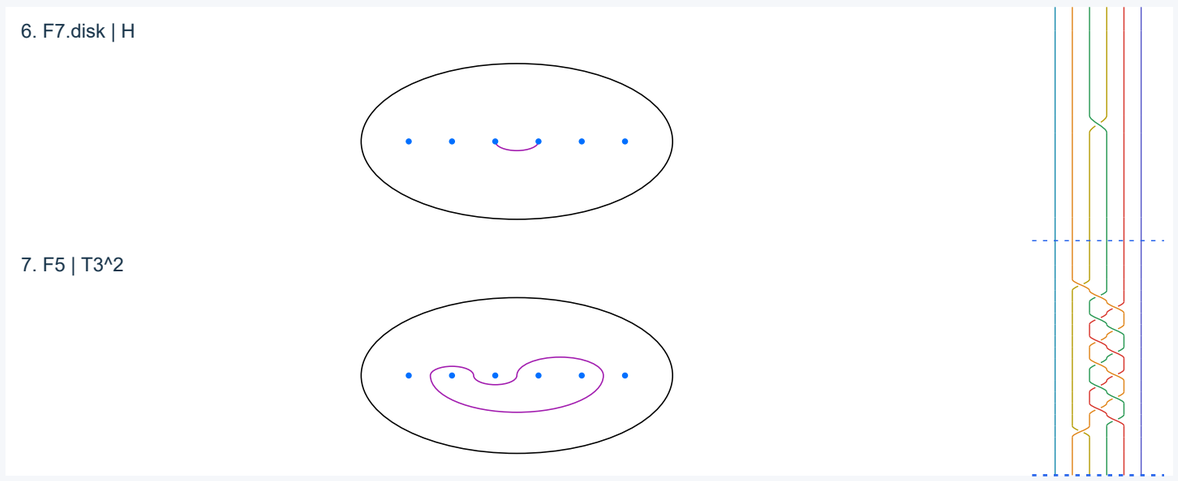
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



II - step 014

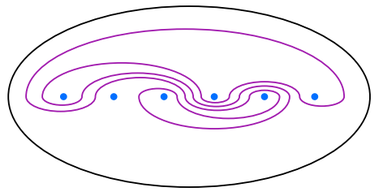
F7.disk up past F4

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

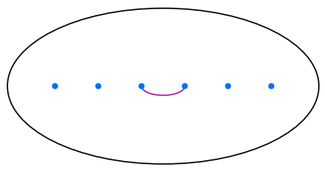
4 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

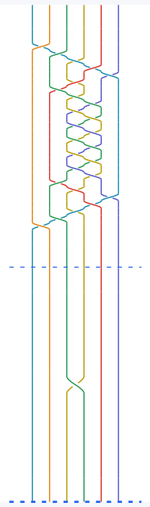
Before

5. F4 | T3^2



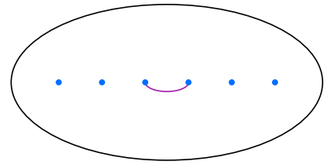
6. F7.disk | H



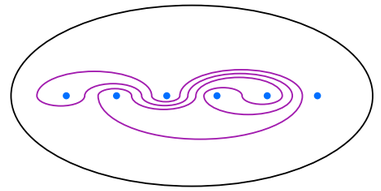


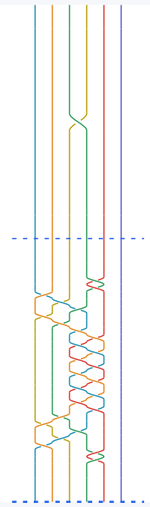
After

5. F7.disk | H



6. F4 | T3^2 | SPHERE SUPPORT CHART





II - step 015

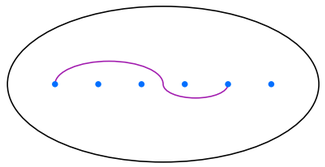
F7.disk up past F3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

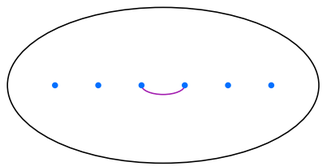
3 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

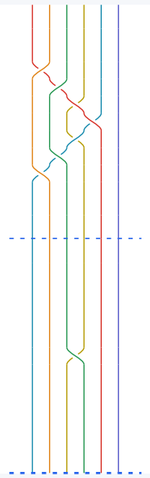
Before

4. F3 | H



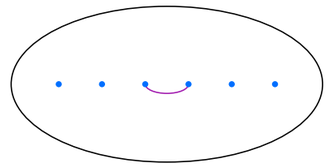
5. F7.disk | H



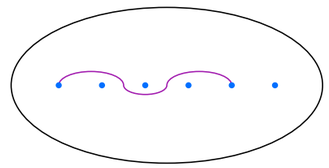


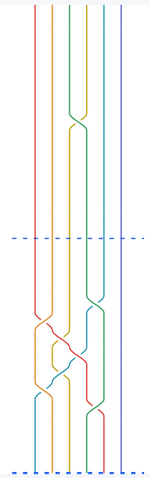
After

4. F7.disk | H



5. F3 | H





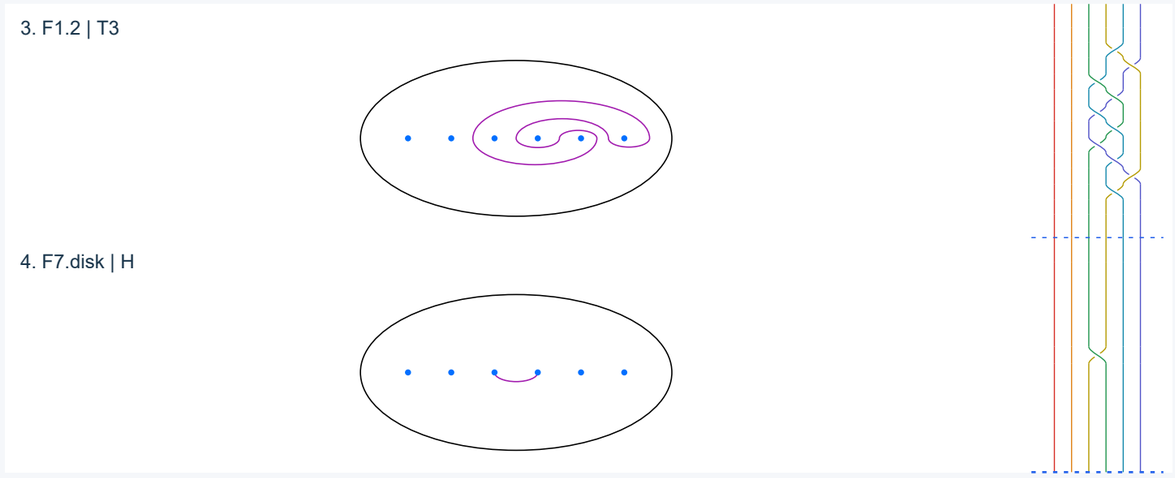
II - step 016

F7.disk up past F1.2

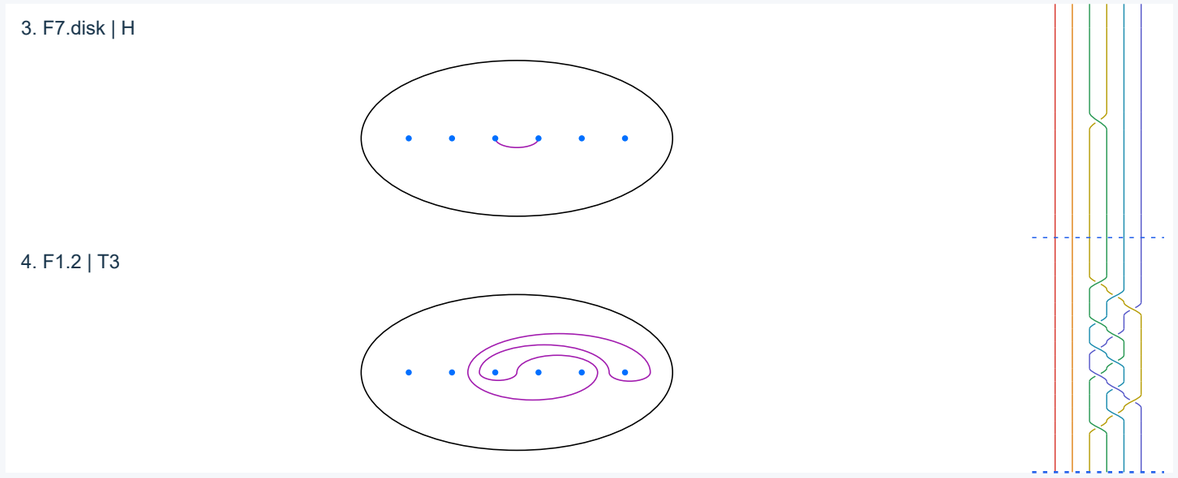
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



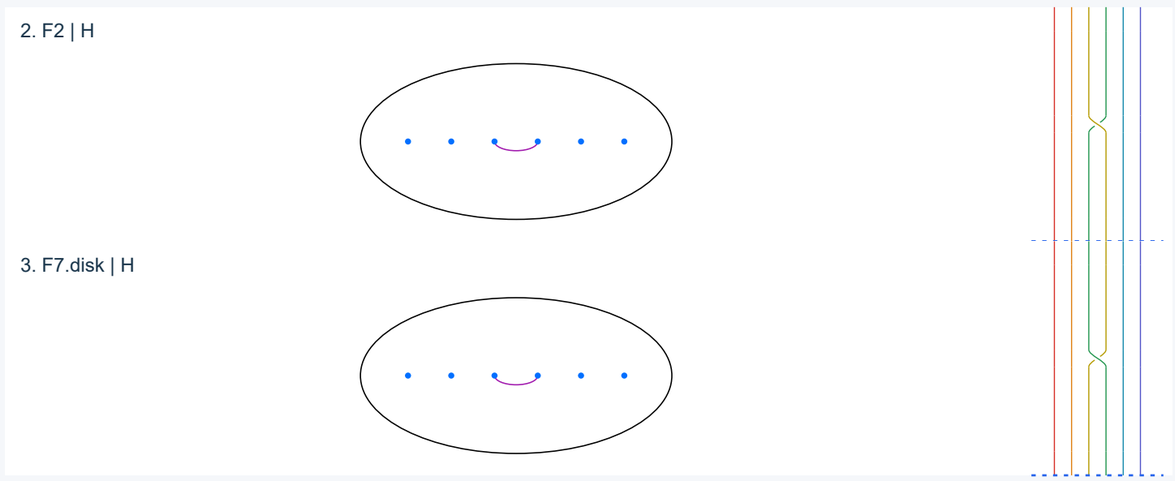
II - step 017

Combine transported F2 and F7 into a pair twist

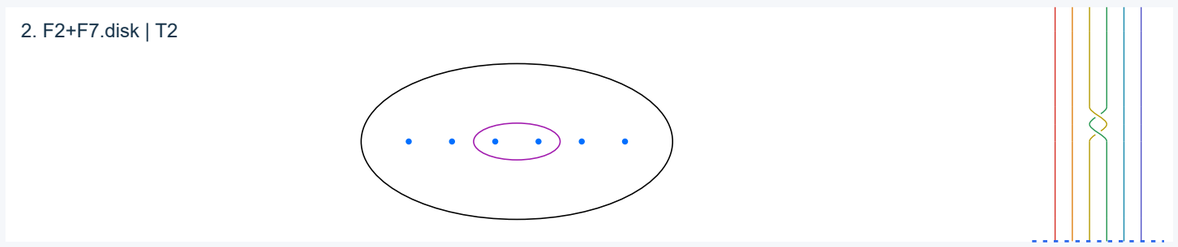
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



II - step 018

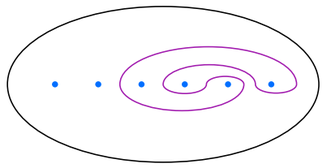
Replace that prefix by its two lantern boundary twists

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

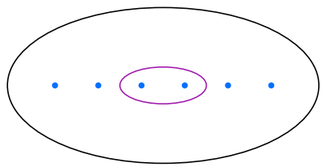
0 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before

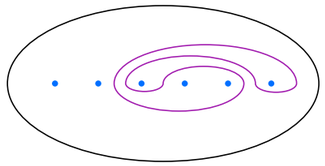
1. F1.1 | T3

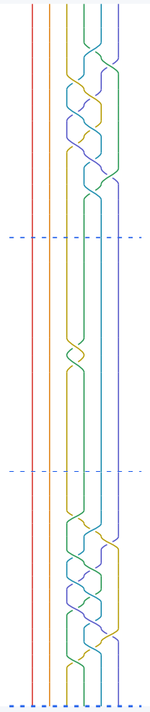


2. F2+F7.disk | T2



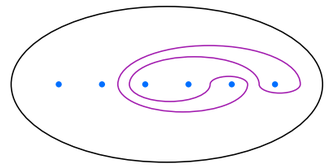
3. F1.2 | T3



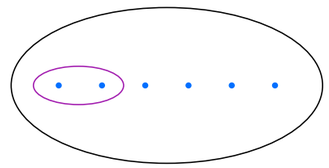


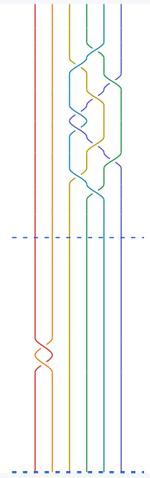
After

1. prefix.P45 | T2



2. prefix.P12 | T2





II - step 019

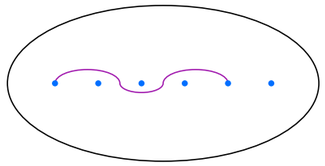
Apply the {34}, {26}, {5} cluster lantern

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

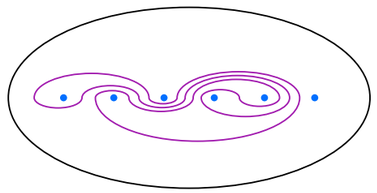
2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=1 -4 5.

Before

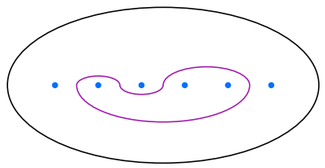
3. F3 | H



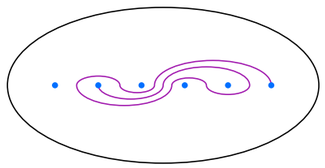
4. F4 | T3^2 | SPHERE SUPPORT CHART



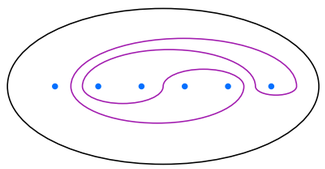
5. F5 | T3^2

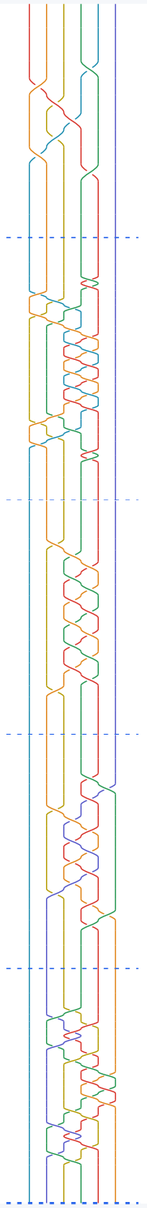


6. F6 | H



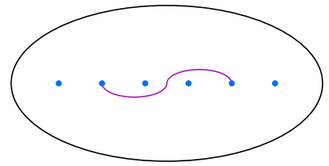
7. F8.1 | T3



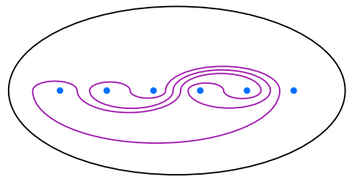


After

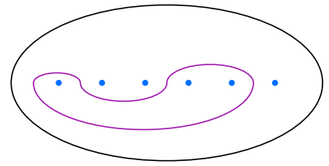
3. F3 | H



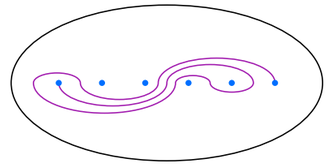
4. F4 | T3^2 | SPHERE SUPPORT CHART



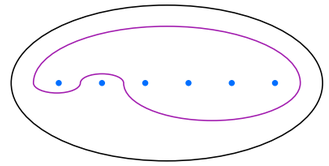
5. F5 | T3^2

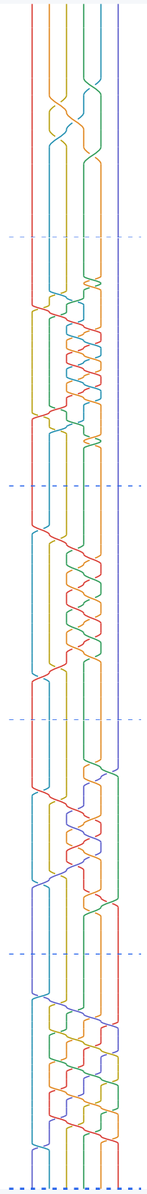


6. F6 | H



7. new.O23456 | T5 | SPHERE SUPPORT CHART





II - step 019 - continued

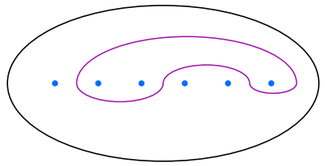
Apply the {34}, {26}, {5} cluster lantern

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

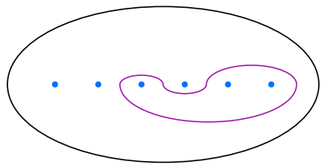
2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=1 -4 5.

Before

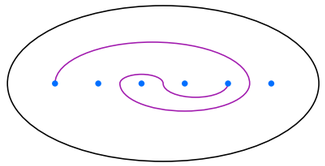
8. F10.1 | T3



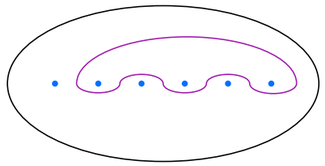
9. F8.2 | T3



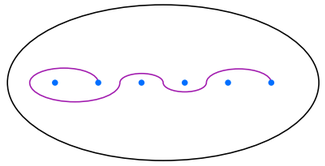
10. F9 | H | SPHERE SUPPORT CHART

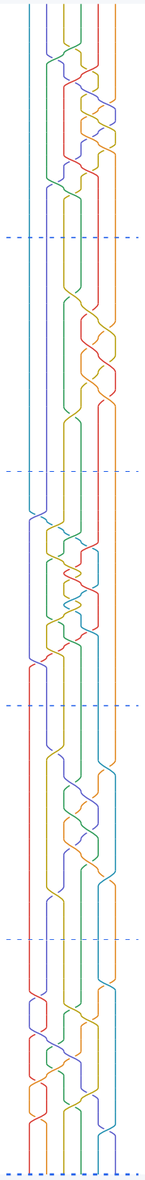


11. F10.2 | T3



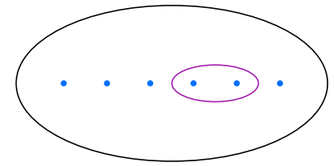
12. F11 | H



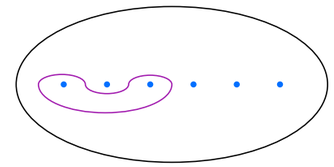


After

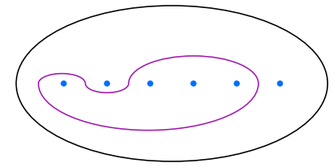
8. new.P34 | T2



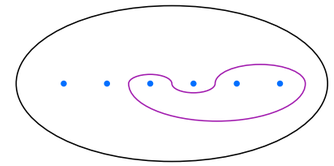
9. new.P26 | T2



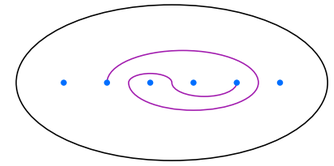
10. new.Q2346 | T4^-1

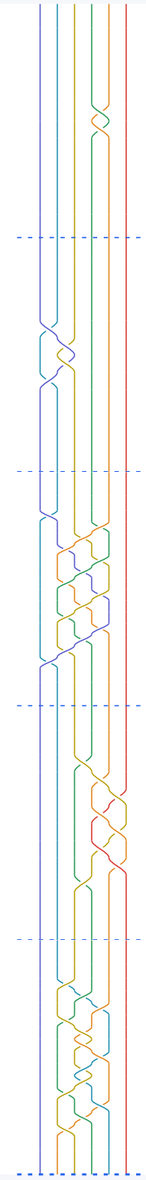


11. F8.2 | T3



12. F9 | H | SPHERE SUPPORT CHART





II - step 019 - continued

Apply the {34}, {26}, {5} cluster lantern

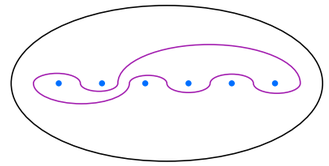
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=1 -4 5.

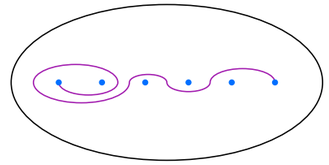
Before

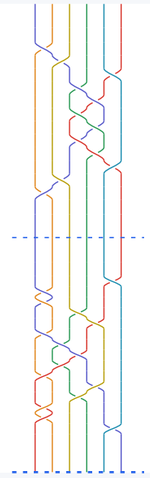
After

13. F10.2 | T3



14. F11 | H





II - step 020

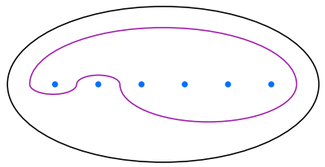
Pair the peripheral omission with the complementary-root rechart

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

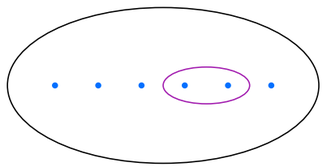
6 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=1 -4 5; after g=1 -4 5.

Before

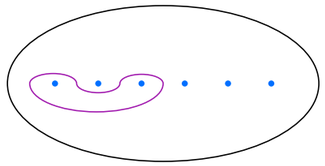
7. new.O23456 | T5 | SPHERE SUPPORT CHART



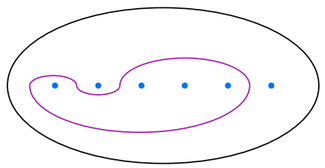
8. new.P34 | T2



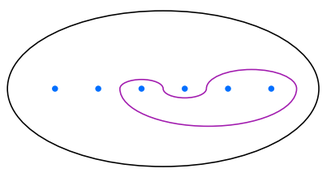
9. new.P26 | T2

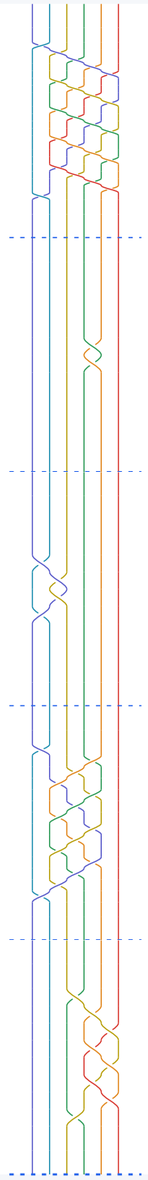


10. new.Q2346 | T4^-1



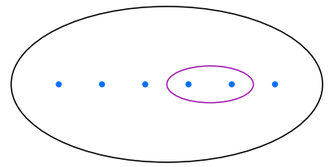
11. F8.2 | T3



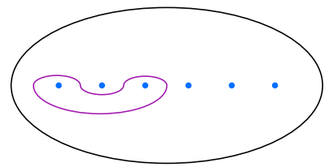


After

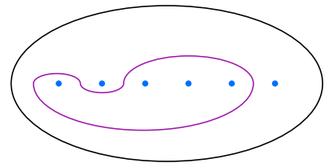
7. new.P34 | T2



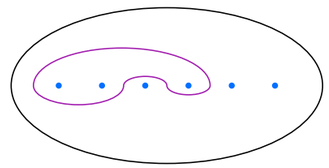
8. new.P26 | T2

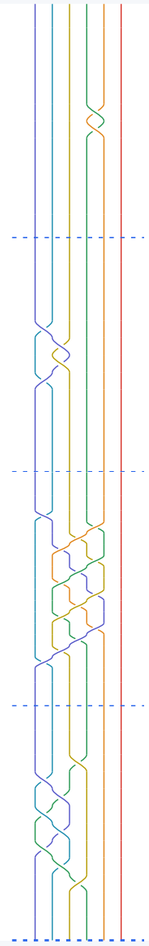


9. new.Q2346 | T4^-1



10. F12.root | T3





II - step 021

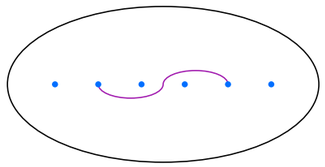
F12.root down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

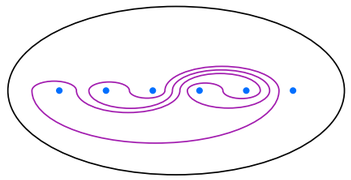
2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=1 -4 5; after g=-4 5.

Before

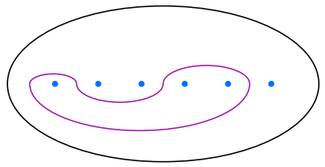
3. F3 | H



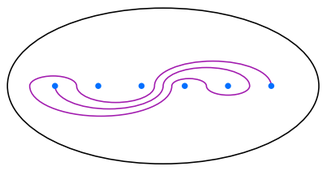
4. F4 | T3^2 | SPHERE SUPPORT CHART



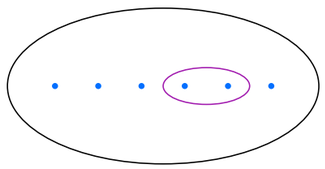
5. F5 | T3^2

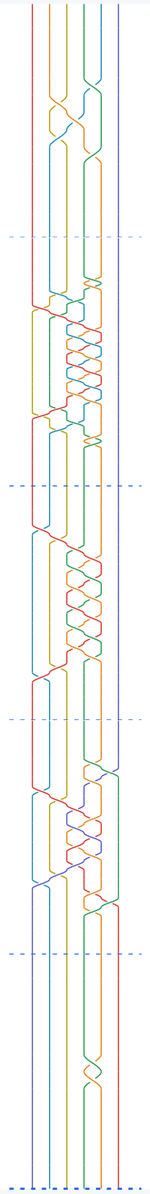


6. F6 | H



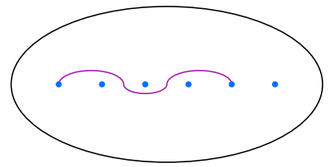
7. new.P34 | T2



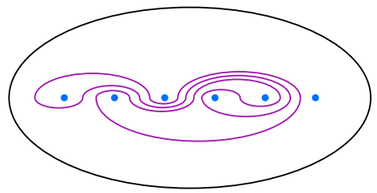


After

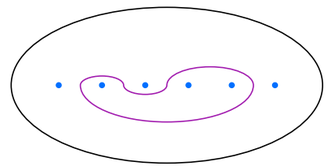
3. F3 | H



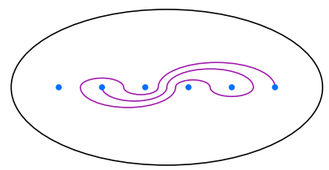
4. F4 | T3^2 | SPHERE SUPPORT CHART



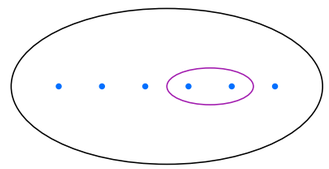
5. F5 | T3^2

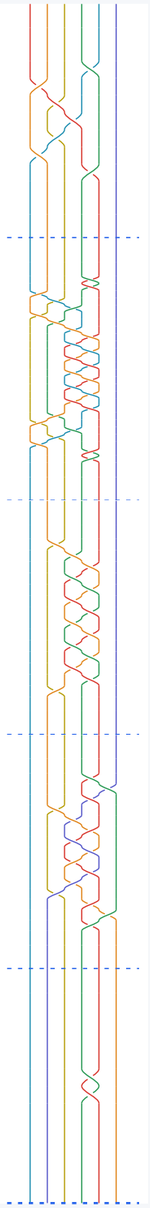


6. F6 | H



7. new.P34 | T2





II - step 021 - continued

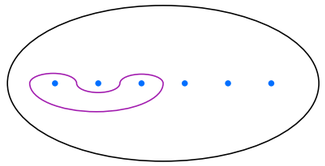
F12.root down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

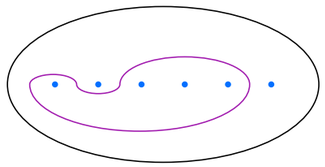
2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=1 -4 5; after g=-4 5.

Before

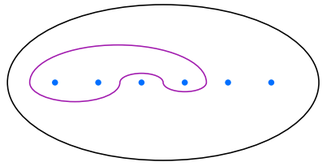
8. new.P26 | T2



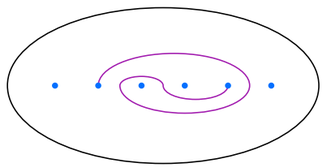
9. new.Q2346 | T4^-1



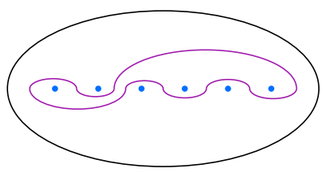
10. F12.root | T3

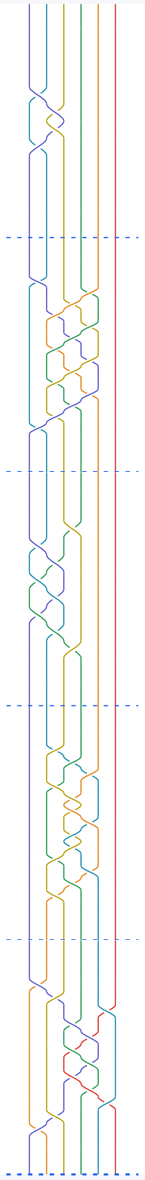


11. F9 | H | SPHERE SUPPORT CHART



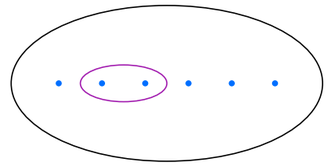
12. F10.2 | T3



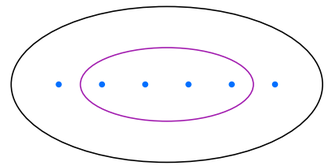


After

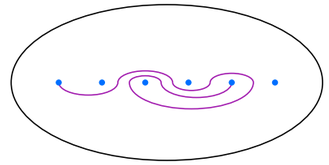
8. new.P26 | T2



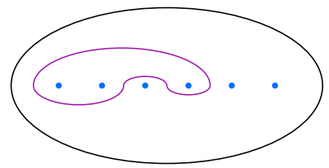
9. new.Q2346 | T4^-1



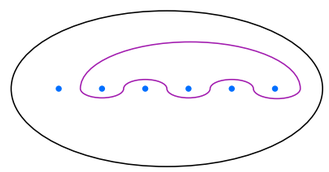
10. F9 | H | SPHERE SUPPORT CHART

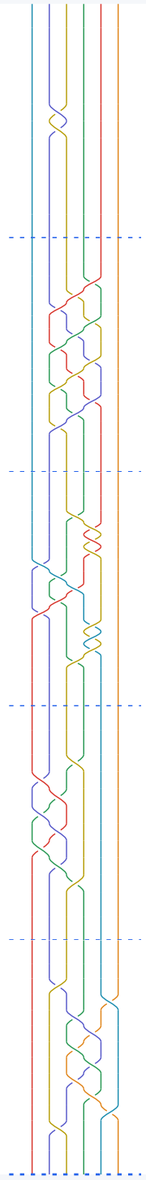


11. F12.root | T3



12. F10.2 | T3





II - step 021 - continued

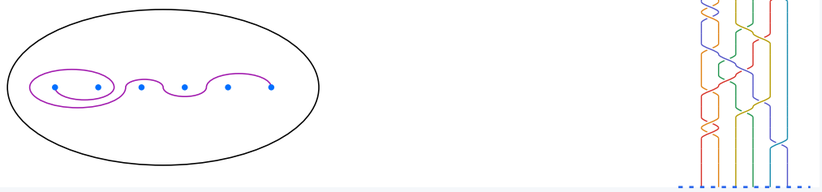
F12.root down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=1 -4 5; after g=-4 5.

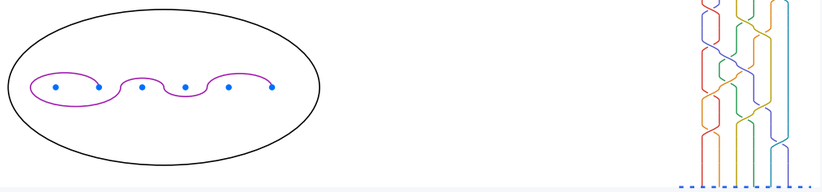
Before

13. F11 | H



After

13. F11 | H



II - step 022

F12.root down past F10.2

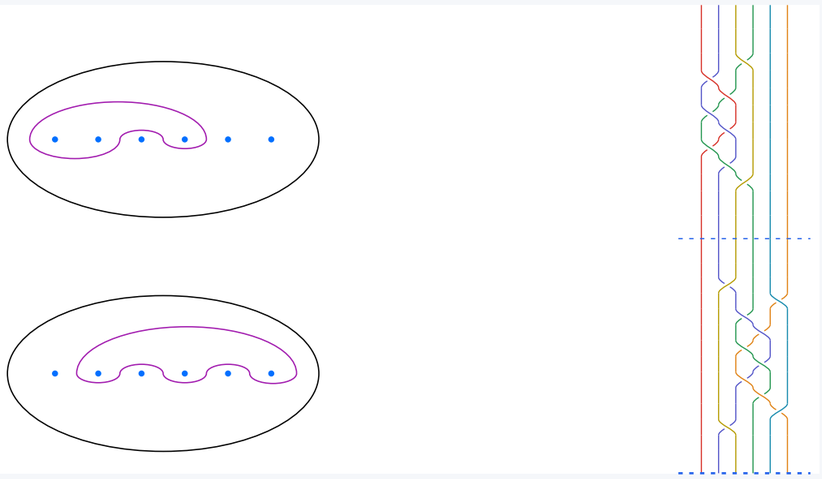
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before

11. F12.root | T3

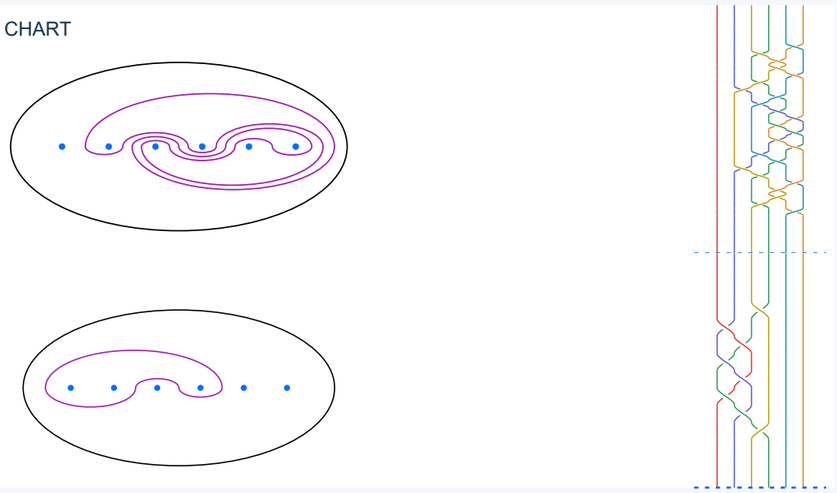
12. F10.2 | T3



After

11. F10.2 | T3 | SPHERE SUPPORT CHART

12. F12.root | T3



II - step 023

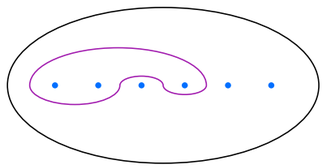
F12.root down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

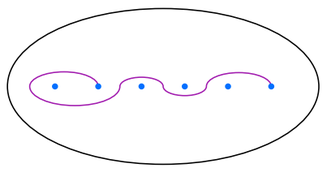
11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

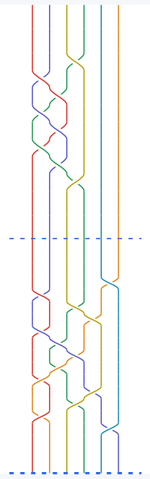
Before

12. F12.root | T3



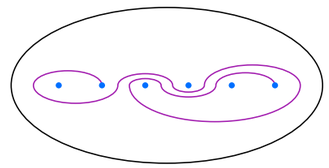
13. F11 | H



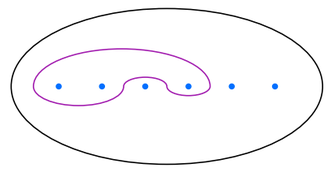


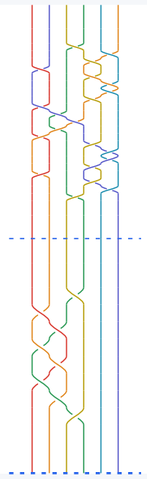
After

12. F11 | H | SPHERE SUPPORT CHART



13. F12.root | T3





II - step 024

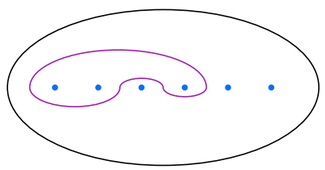
Combine the complementary root with F12 into a cube

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

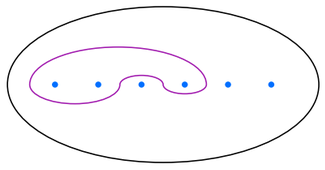
12 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

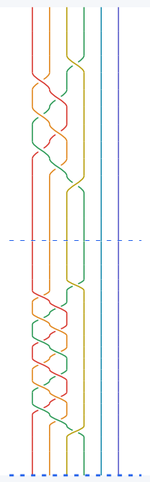
Before

13. F12.root | T3



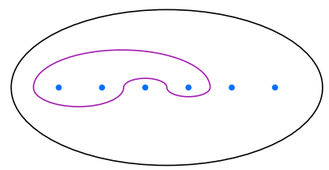
14. F12 | T3^2

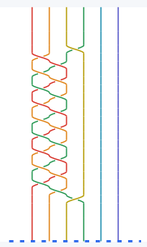




After

13. F12.root+F12 | T3^3





II - step 025

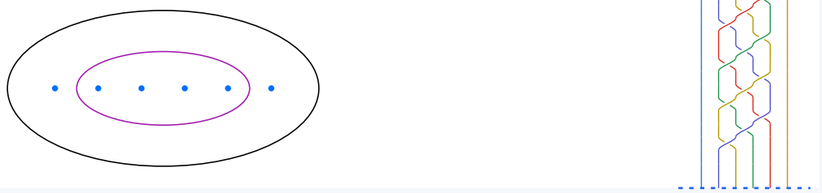
Draw the inverse Q along its complementary {1,5} pair

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

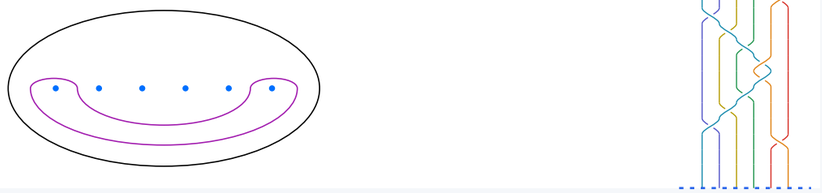
Before

9. new.Q2346 | T4^-1



After

9. new.Q.complement | T2^-1



II - step 026

Rechart the whole C13 square along its complementary triple

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before

14. F13 | T3^2



After

14. C13.complement.square | T3^2 | SPHERE SUPPORT CHART



II - step 027

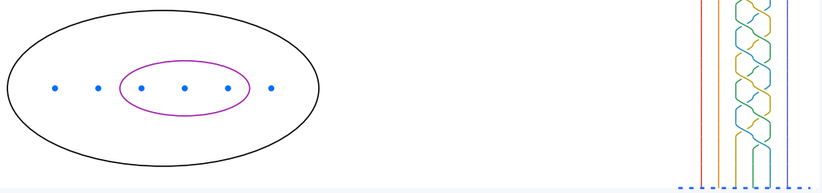
Apply the chosen PC half lantern to each C13 unit

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before

14. C13.complement.square | T3^2 | SPHERE SUPPORT CHART



After

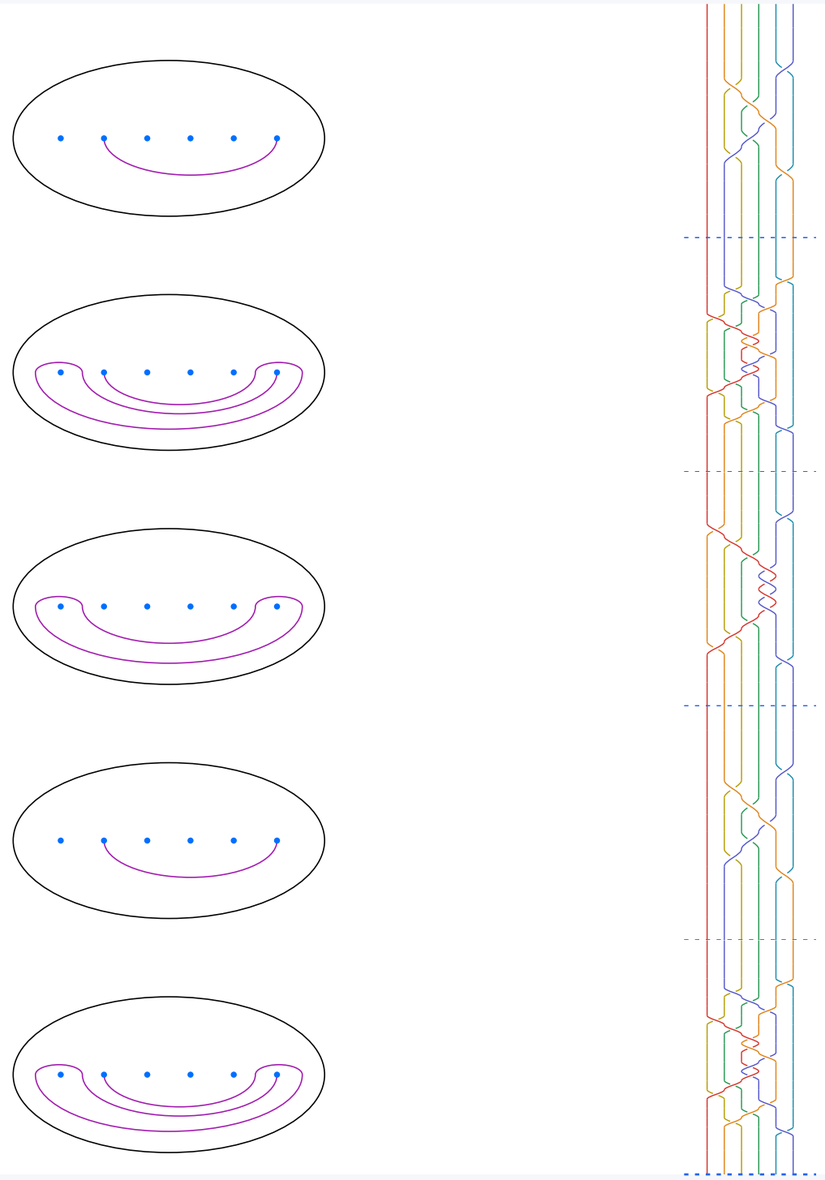
14. C13.HL1.1 | H

15. C13.HL1.2 | H

16. C13.HL1.3 | T2^2

17. C13.HL2.1 | H

18. C13.HL2.2 | H



II - step 027 - continued

Apply the chosen PC half lantern to each C13 unit

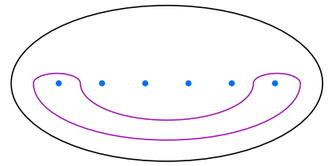
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

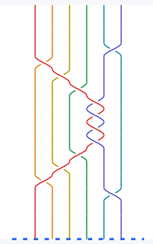
13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before

After

19. C13.HL2.3 | T2^2





II - step 028

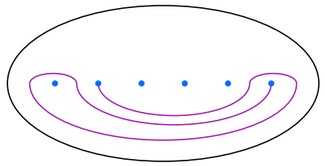
C13.HL2.3 up past C13.HL2.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

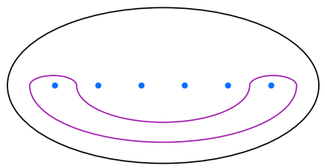
17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

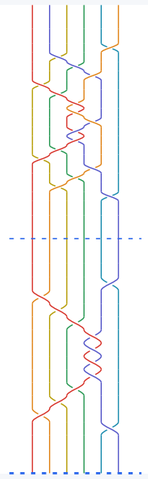
Before

18. C13.HL2.2 | H



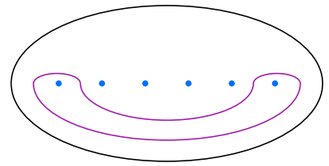
19. C13.HL2.3 | T2^2



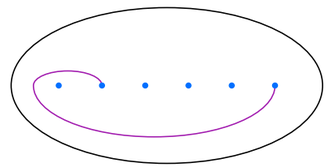


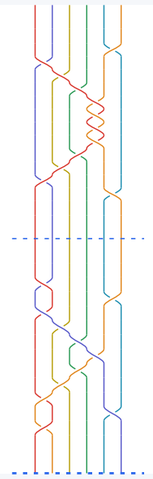
After

18. C13.HL2.3 | T2^2



19. C13.HL2.2 | H





II - step 029

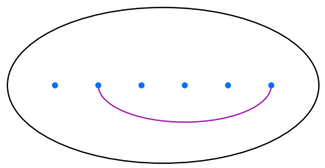
C13.HL2.3 up past C13.HL2.1

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

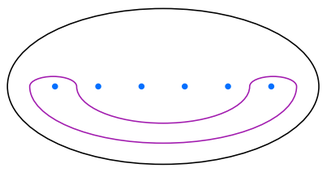
16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

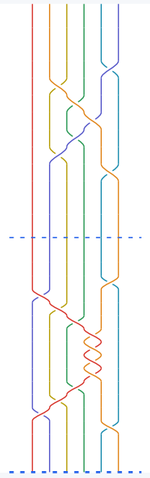
Before

17. C13.HL2.1 | H



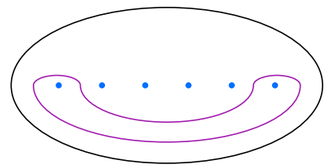
18. C13.HL2.3 | T2^2



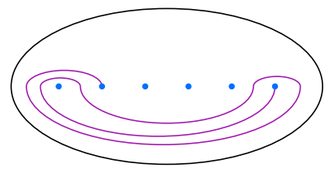


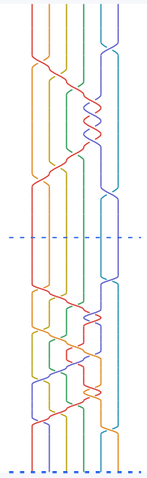
After

17. C13.HL2.3 | T2^2



18. C13.HL2.1 | H





II - step 030

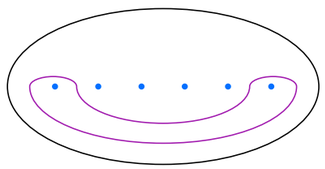
C13.HL2.3 up past C13.HL1.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

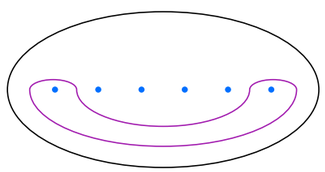
15 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

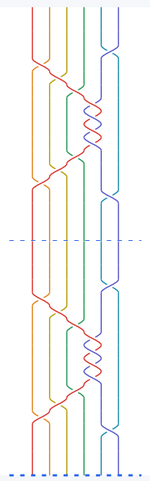
Before

16. C13.HL1.3 | T2^2



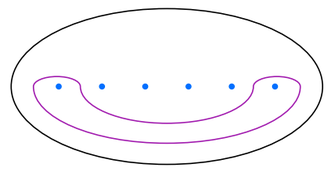
17. C13.HL2.3 | T2^2



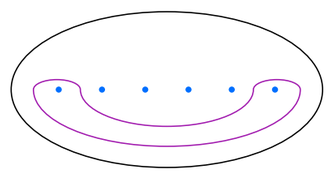


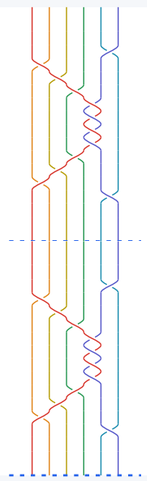
After

16. C13.HL2.3 | T2^2



17. C13.HL1.3 | T2^2





II - step 031

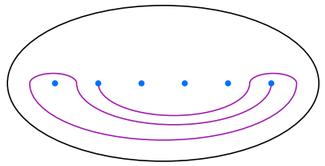
C13.HL2.3 up past C13.HL1.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

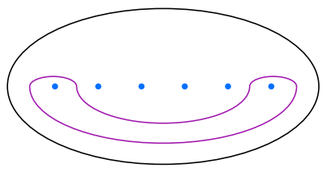
14 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

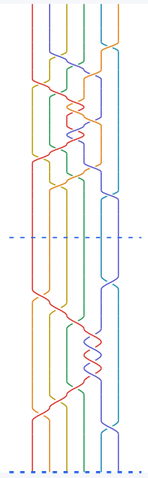
Before

15. C13.HL1.2 | H



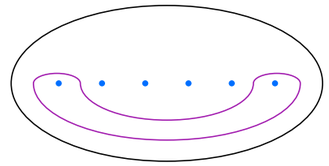
16. C13.HL2.3 | T2^2



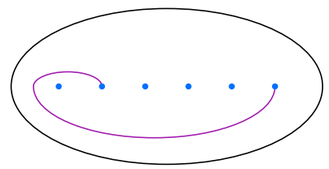


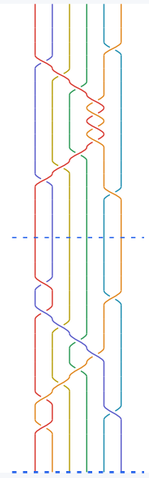
After

15. C13.HL2.3 | T2^2



16. C13.HL1.2 | H





II - step 032

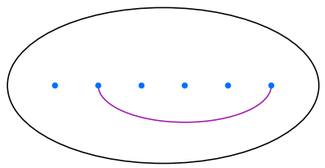
C13.HL2.3 up past C13.HL1.1

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

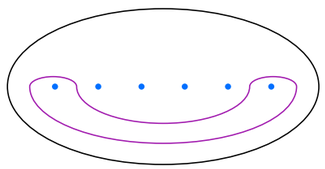
13 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

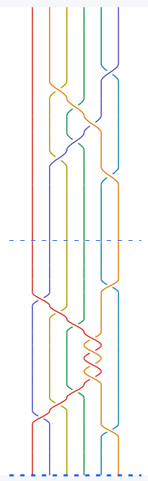
Before

14. C13.HL1.1 | H



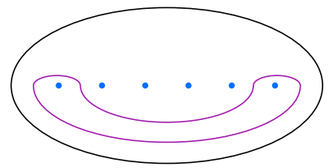
15. C13.HL2.3 | T2^2



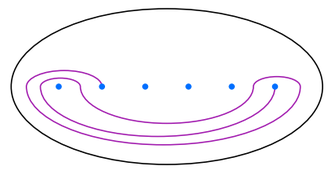


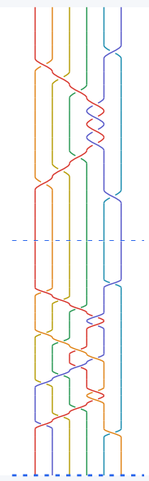
After

14. C13.HL2.3 | T2^2



15. C13.HL1.1 | H





II - step 033

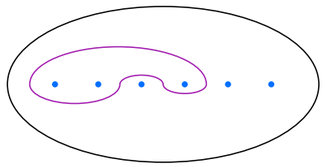
C13.HL2.3 up past F12.root+F12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

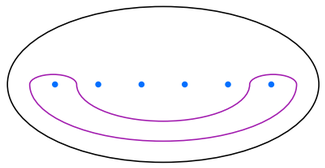
12 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

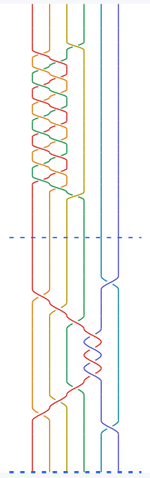
Before

13. F12.root+F12 | T3^3



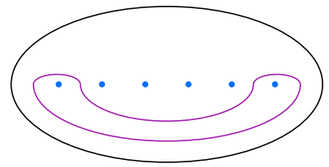
14. C13.HL2.3 | T2^2



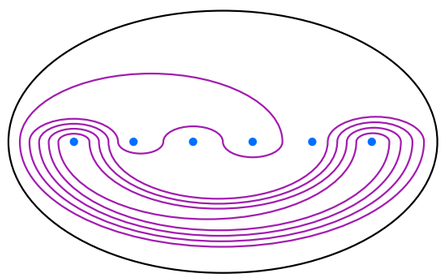


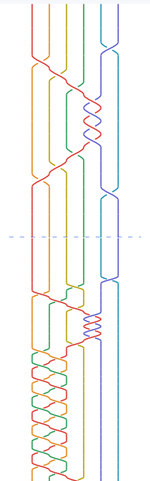
After

13. C13.HL2.3 | T2^2



14. F12.root+F12 | T3^3





II - step 034

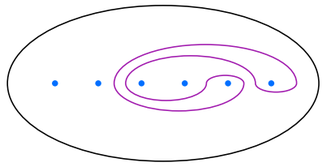
C13.HL2.3 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

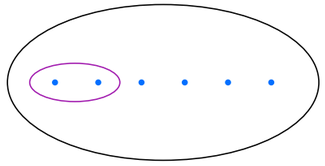
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

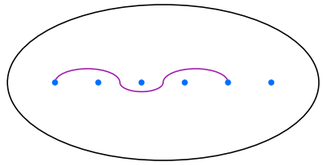
1. prefix.P45 | T2



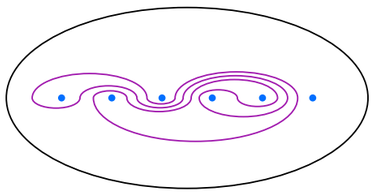
2. prefix.P12 | T2



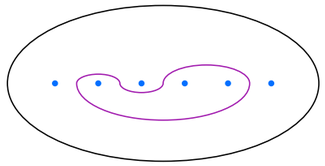
3. F3 | H

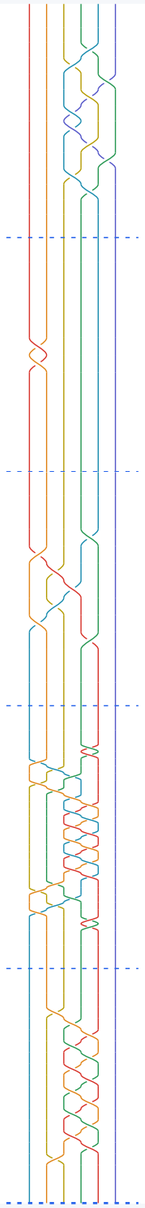


4. F4 | T3^2 | SPHERE SUPPORT CHART



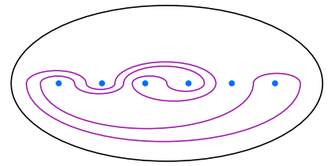
5. F5 | T3^2



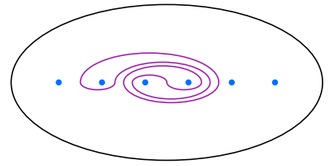


After

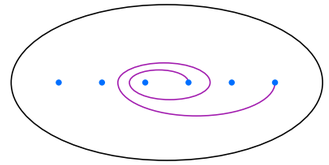
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



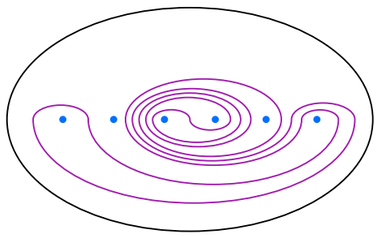
2. prefix.P12 | T2 | SPHERE SUPPORT CHART



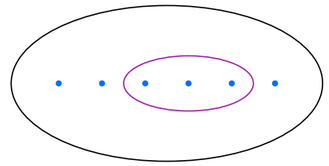
3. F3 | H

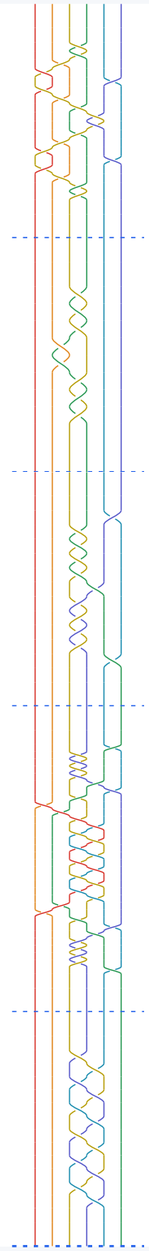


4. F4 | T3^2 | SPHERE SUPPORT CHART



5. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 034 - continued

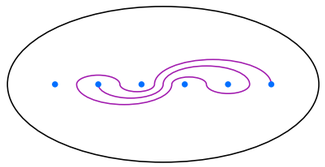
C13.HL2.3 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

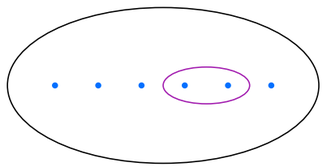
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

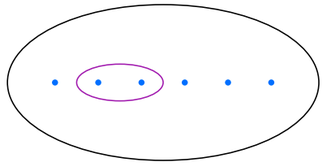
6. F6 | H



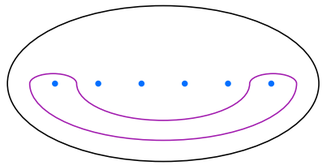
7. new.P34 | T2



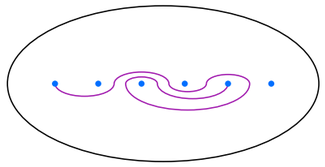
8. new.P26 | T2

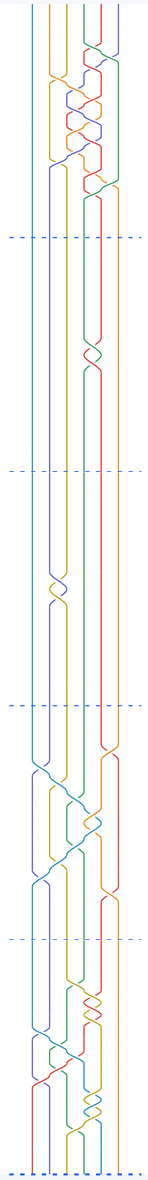


9. new.Q.complement | T2^-1



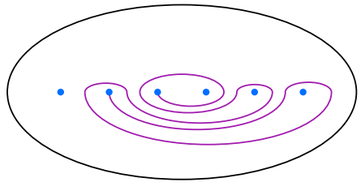
10. F9 | H | SPHERE SUPPORT CHART



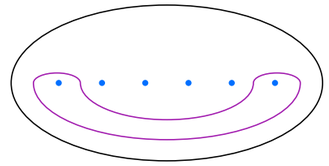


After

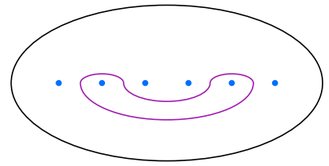
6. F6 | H | SPHERE SUPPORT CHART



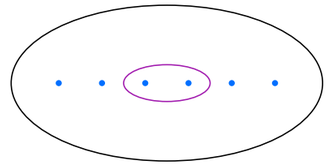
7. new.P34 | T2 | SPHERE SUPPORT CHART



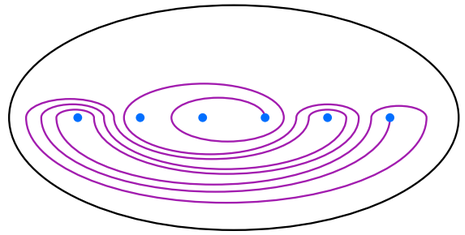
8. new.P26 | T2 | SPHERE SUPPORT CHART

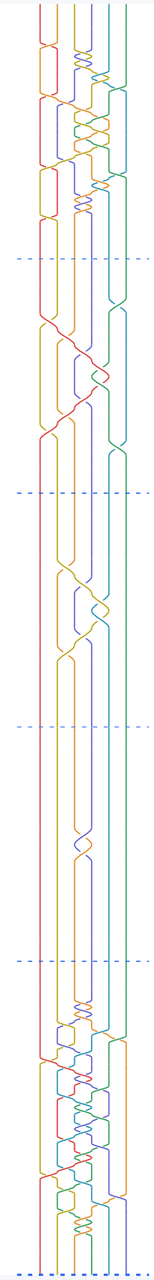


9. new.Q.complement | T2^-1 | SPHERE SUPPORT CHART



10. F9 | H | SPHERE SUPPORT CHART





II - step 034 - continued

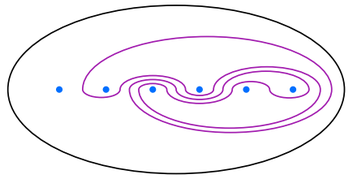
C13.HL2.3 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

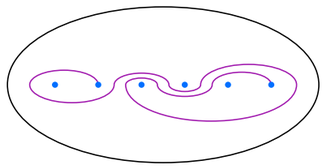
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

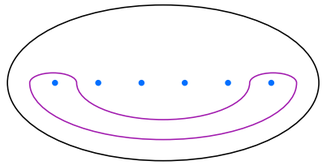
11. F10.2 | T3 | SPHERE SUPPORT CHART



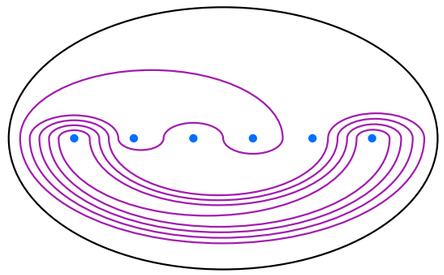
12. F11 | H | SPHERE SUPPORT CHART

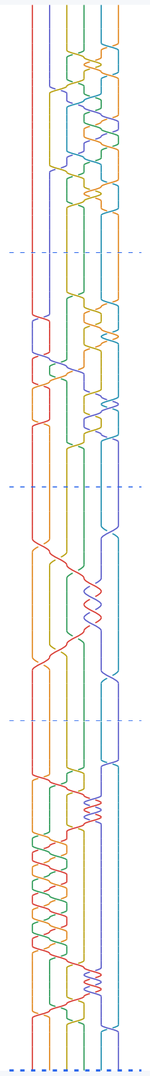


13. C13.HL2.3 | T2^2



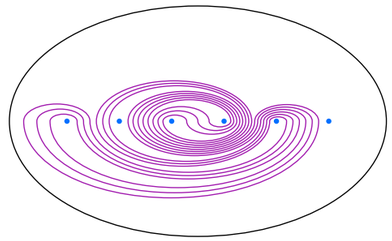
14. F12.root+F12 | T3^3



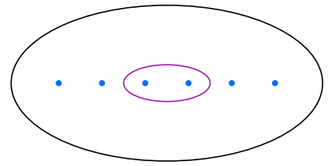


After

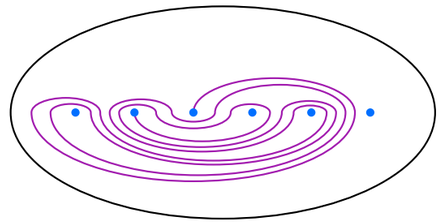
11. F10.2 | T3 | SPHERE SUPPORT CHART



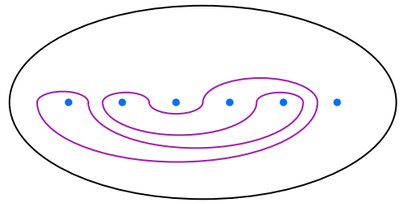
12. C13.HL2.3 | T2^2 | SPHERE SUPPORT CHART

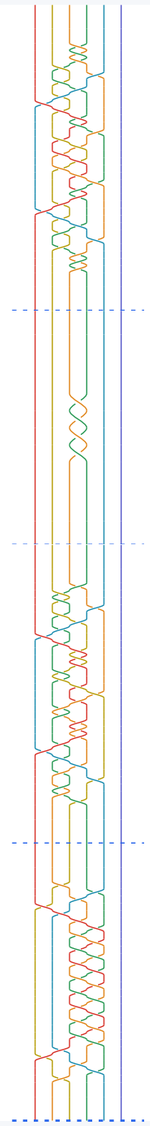


13. F11 | H | SPHERE SUPPORT CHART



14. F12.root+F12 | T3^3 | SPHERE SUPPORT CHART





II - step 034 - continued

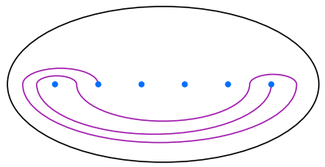
C13.HL2.3 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

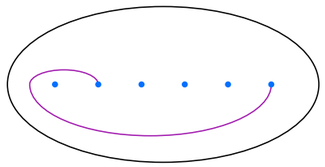
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

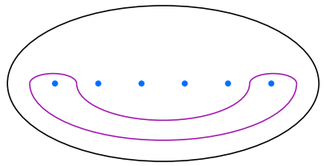
15. C13.HL1.1 | H



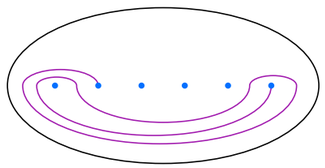
16. C13.HL1.2 | H



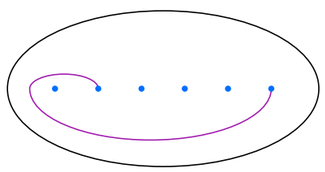
17. C13.HL1.3 | T2^2

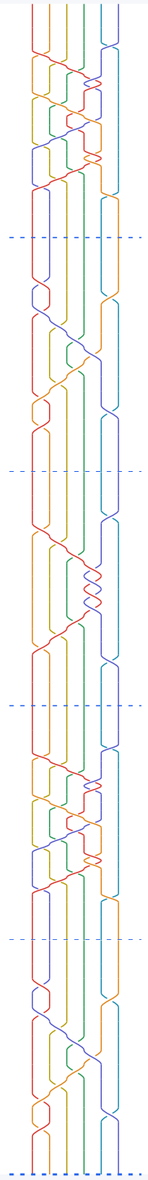


18. C13.HL2.1 | H



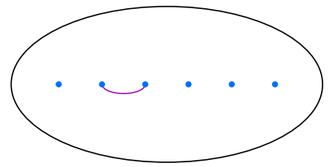
19. C13.HL2.2 | H



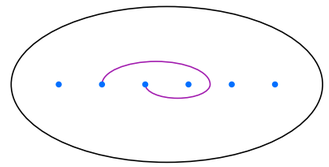


After

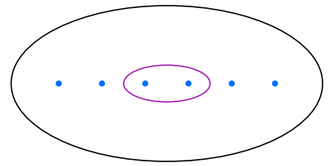
15. C13.HL1.1 | H | SPHERE SUPPORT CHART



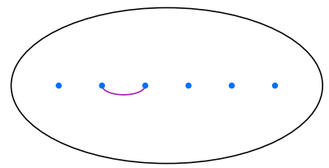
16. C13.HL1.2 | H | SPHERE SUPPORT CHART



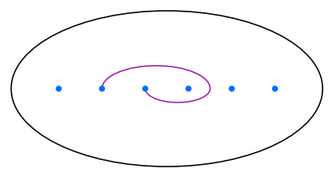
17. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART

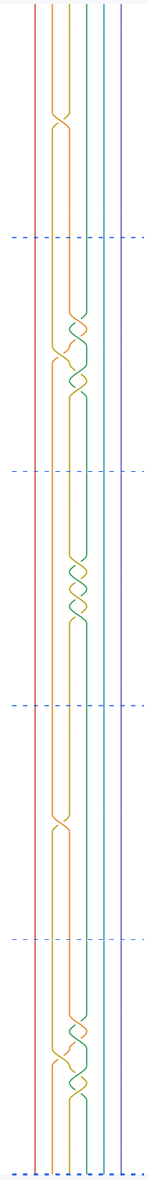


18. C13.HL2.1 | H | SPHERE SUPPORT CHART



19. C13.HL2.2 | H | SPHERE SUPPORT CHART





II - step 035

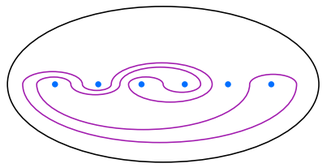
C13.HL2.3 up past F10.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

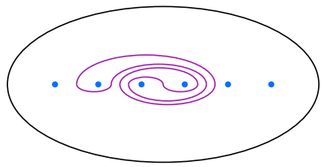
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

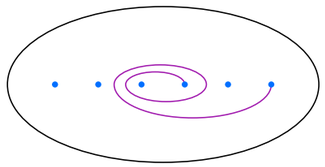
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



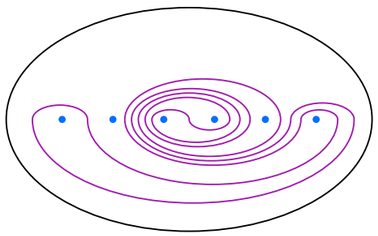
2. prefix.P12 | T2 | SPHERE SUPPORT CHART



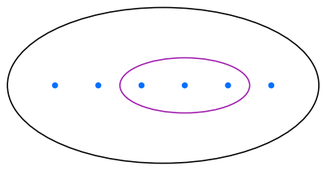
3. F3 | H

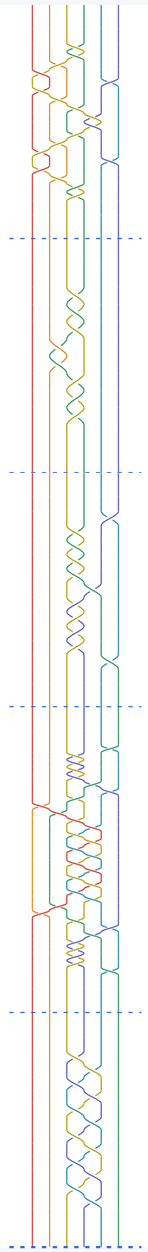


4. F4 | T3^2 | SPHERE SUPPORT CHART



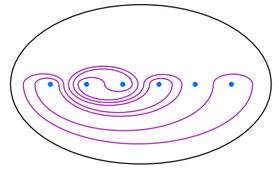
5. F5 | T3^2 | SPHERE SUPPORT CHART



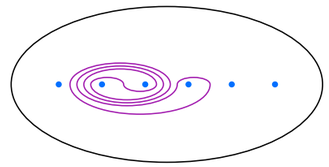


After

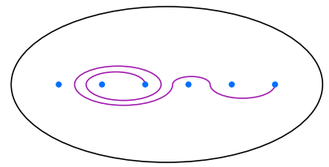
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



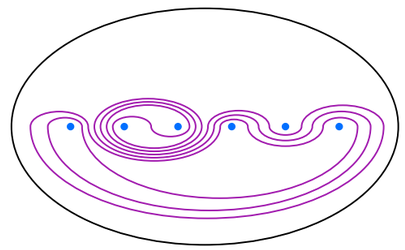
2. prefix.P12 | T2 | SPHERE SUPPORT CHART

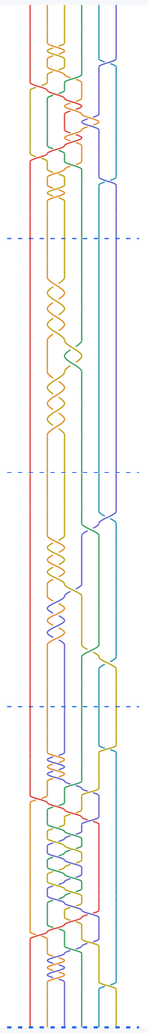


3. F3 | H



4. F4 | T3^2 | SPHERE SUPPORT CHART





II - step 035 - continued

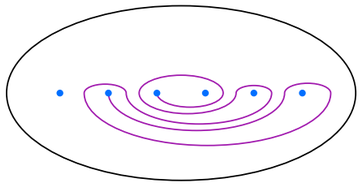
C13.HL2.3 up past F10.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

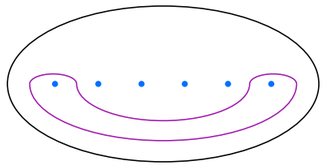
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

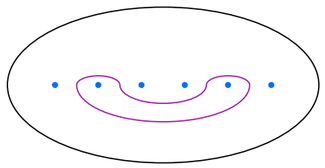
6. F6 | H | SPHERE SUPPORT CHART



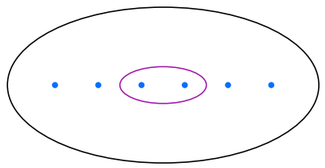
7. new.P34 | T2 | SPHERE SUPPORT CHART

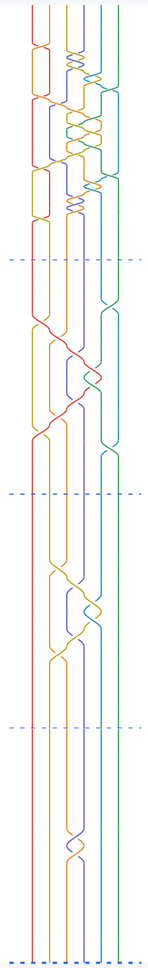


8. new.P26 | T2 | SPHERE SUPPORT CHART



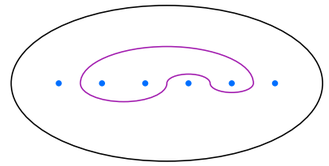
9. new.Q.complement | T2^-1 | SPHERE SUPPORT CHART



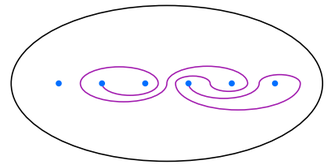


After

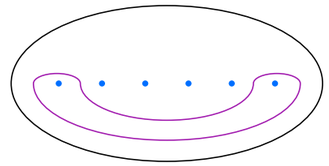
5. F5 | T3^2 | SPHERE SUPPORT CHART



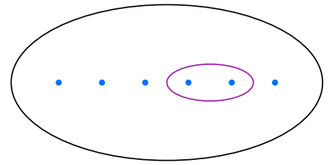
6. F6 | H | SPHERE SUPPORT CHART



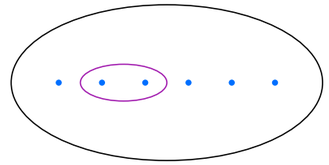
7. new.P34 | T2 | SPHERE SUPPORT CHART

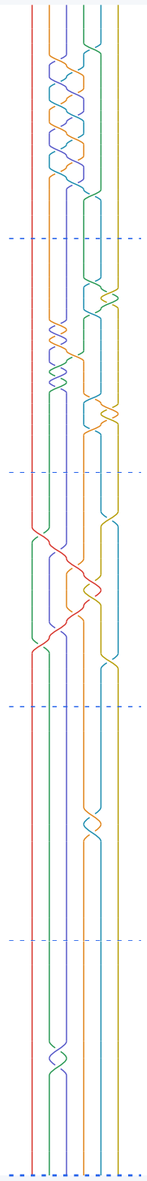


8. new.P26 | T2 | SPHERE SUPPORT CHART



9. new.Q.complement | T2^-1 | SPHERE SUPPORT CHART





II - step 035 - continued

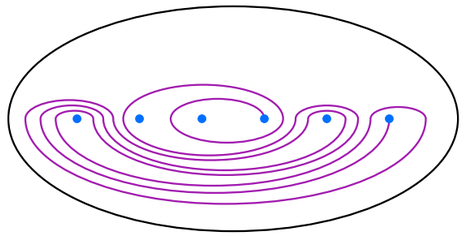
C13.HL2.3 up past F10.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

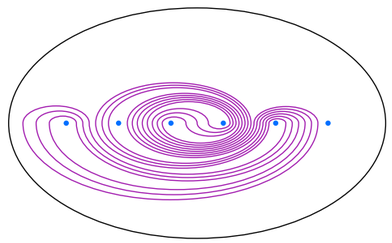
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

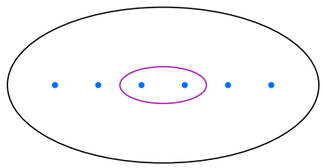
10. F9 | H | SPHERE SUPPORT CHART



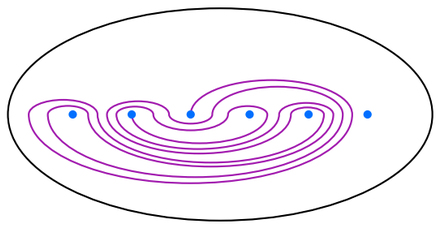
11. F10.2 | T3 | SPHERE SUPPORT CHART

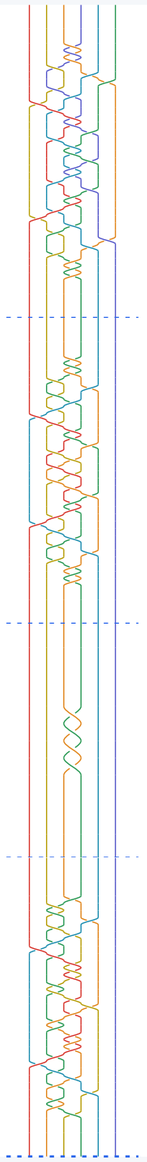


12. C13.HL2.3 | T2^2 | SPHERE SUPPORT CHART



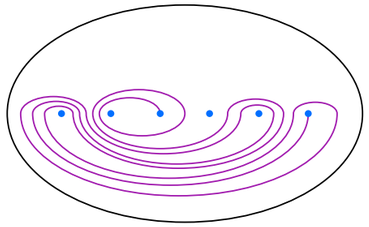
13. F11 | H | SPHERE SUPPORT CHART



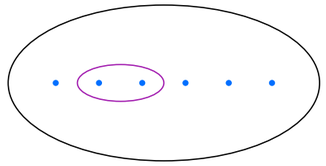


After

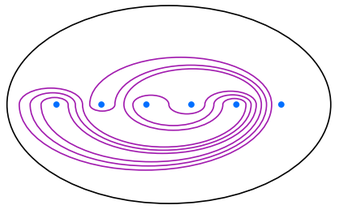
10. F9 | H | SPHERE SUPPORT CHART



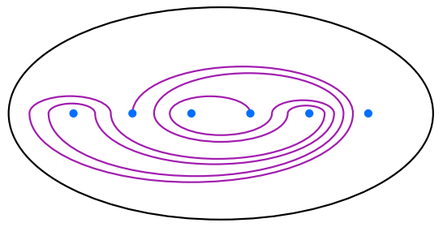
11. C13.HL2.3 | T2^2 | SPHERE SUPPORT CHART

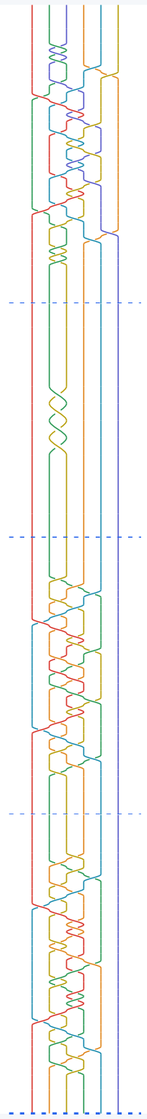


12. F10.2 | T3 | SPHERE SUPPORT CHART



13. F11 | H | SPHERE SUPPORT CHART





II - step 035 - continued

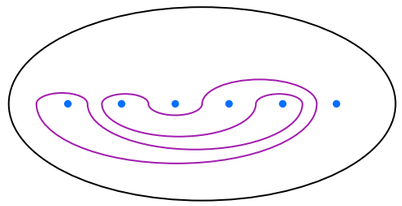
C13.HL2.3 up past F10.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

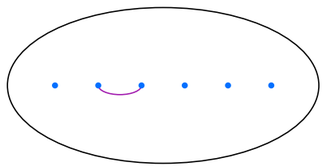
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

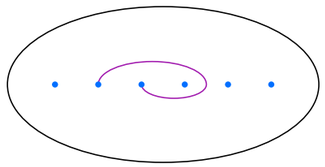
14. F12.root+F12 | T3^3 | SPHERE SUPPORT CHART



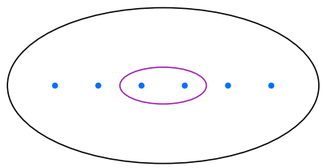
15. C13.HL1.1 | H | SPHERE SUPPORT CHART



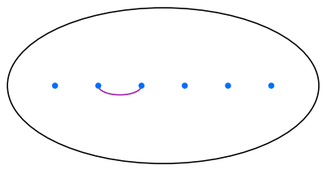
16. C13.HL1.2 | H | SPHERE SUPPORT CHART

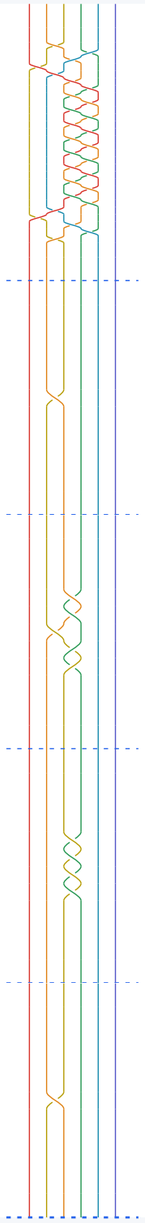


17. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART



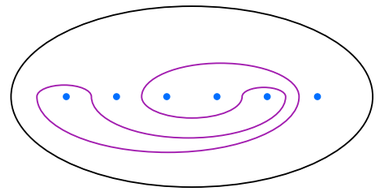
18. C13.HL2.1 | H | SPHERE SUPPORT CHART



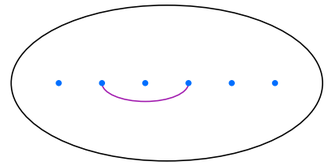


After

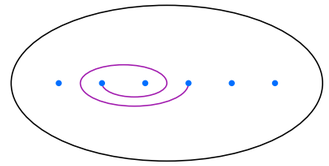
14. F12.root+F12 | T3^3 | SPHERE SUPPORT CHART



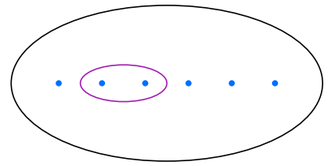
15. C13.HL1.1 | H | SPHERE SUPPORT CHART



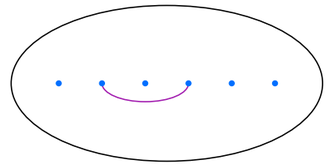
16. C13.HL1.2 | H | SPHERE SUPPORT CHART

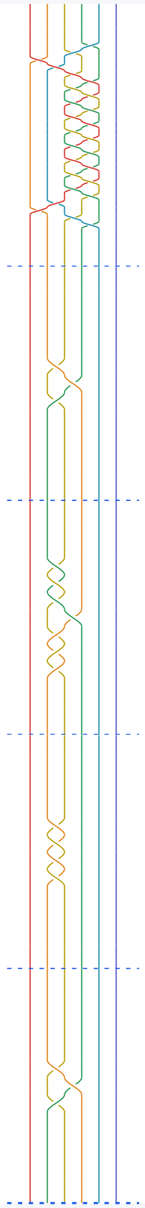


17. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART



18. C13.HL2.1 | H | SPHERE SUPPORT CHART





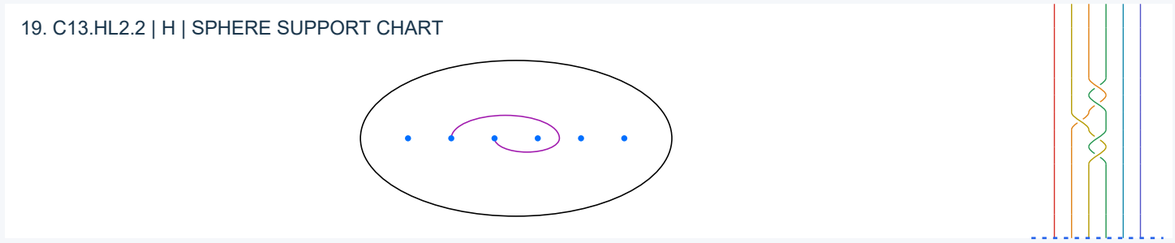
II - step 035 - continued

C13.HL2.3 up past F10.2

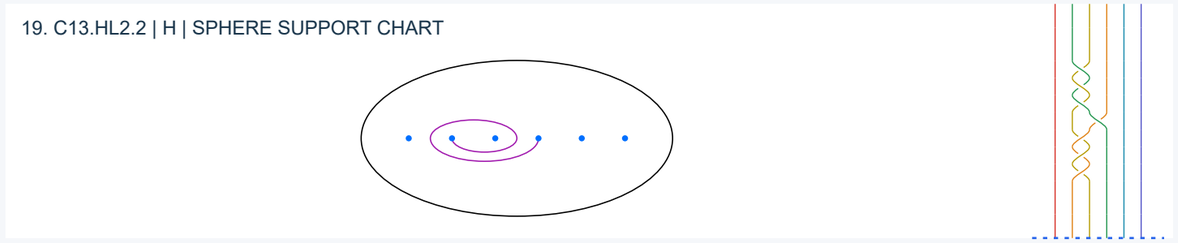
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 036

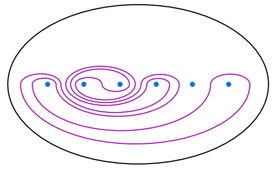
C13.HL2.3 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

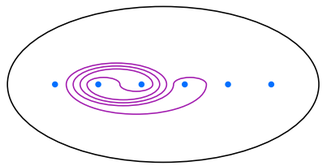
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

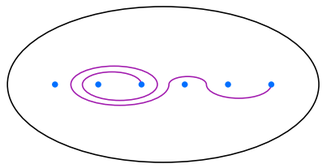
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



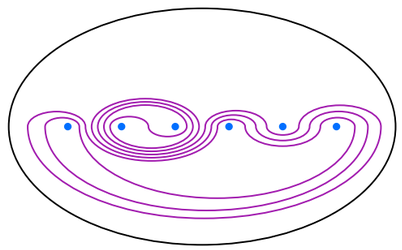
2. prefix.P12 | T2 | SPHERE SUPPORT CHART

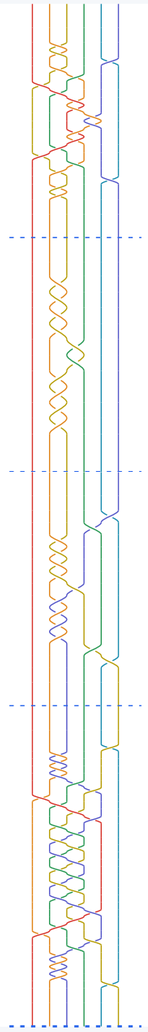


3. F3 | H



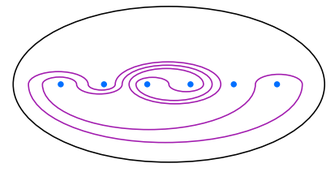
4. F4 | T3^2 | SPHERE SUPPORT CHART



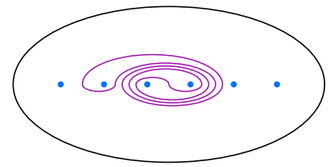


After

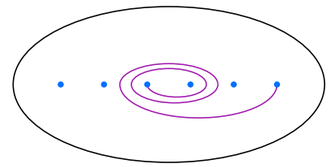
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



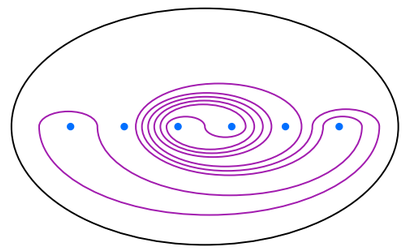
2. prefix.P12 | T2 | SPHERE SUPPORT CHART

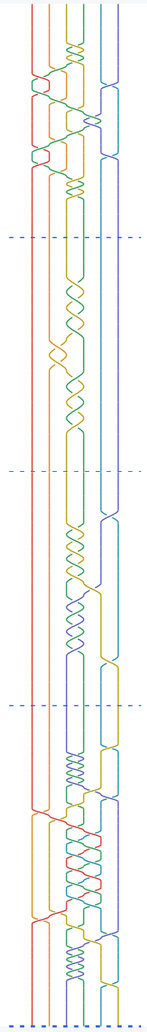


3. F3 | H



4. F4 | T3^2 | SPHERE SUPPORT CHART





II - step 036 - continued

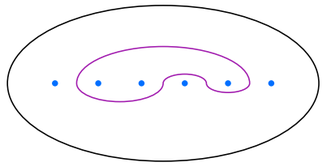
C13.HL2.3 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

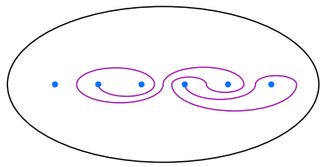
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

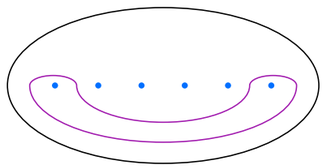
5. F5 | T3^2 | SPHERE SUPPORT CHART



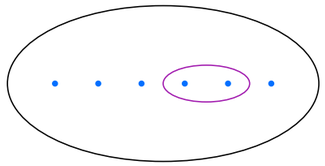
6. F6 | H | SPHERE SUPPORT CHART



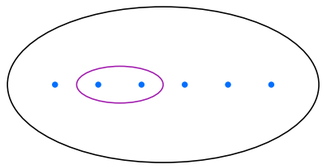
7. new.P34 | T2 | SPHERE SUPPORT CHART

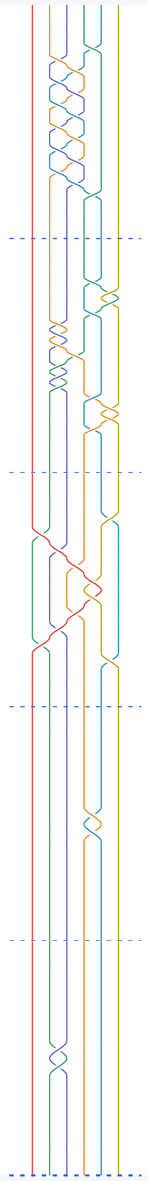


8. new.P26 | T2 | SPHERE SUPPORT CHART



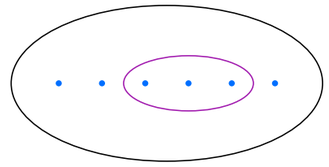
9. new.Q.complement | T2^-1 | SPHERE SUPPORT CHART



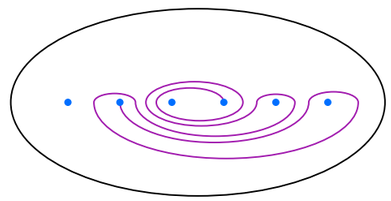


After

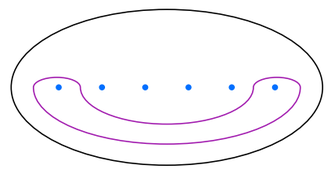
5. F5 | T3^2 | SPHERE SUPPORT CHART



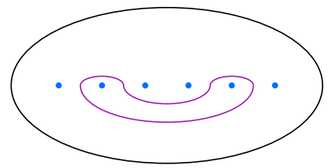
6. F6 | H | SPHERE SUPPORT CHART



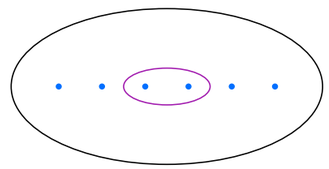
7. new.P34 | T2 | SPHERE SUPPORT CHART

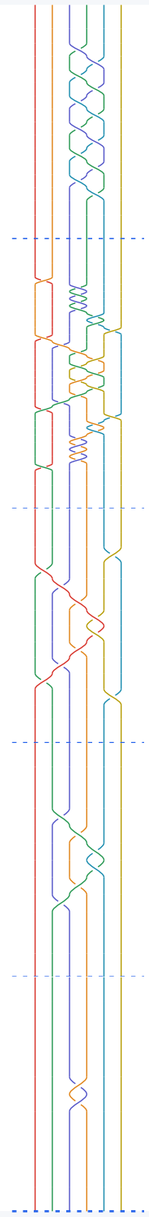


8. new.P26 | T2 | SPHERE SUPPORT CHART



9. new.Q.complement | T2^-1 | SPHERE SUPPORT CHART





II - step 036 - continued

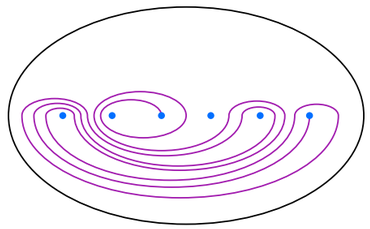
C13.HL2.3 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

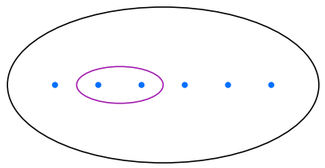
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

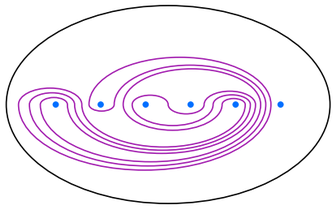
10. F9 | H | SPHERE SUPPORT CHART



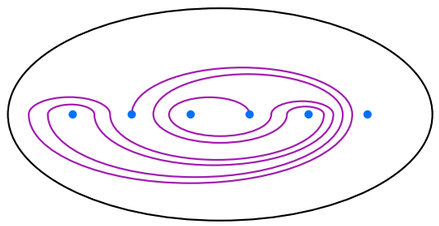
11. C13.HL2.3 | T2^2 | SPHERE SUPPORT CHART

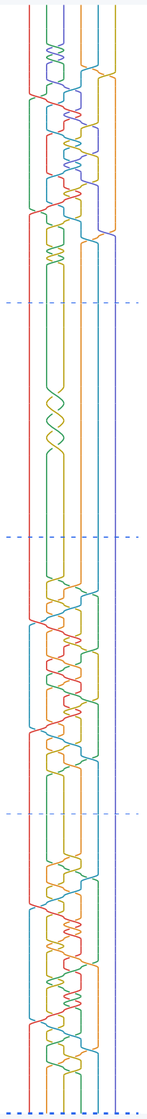


12. F10.2 | T3 | SPHERE SUPPORT CHART



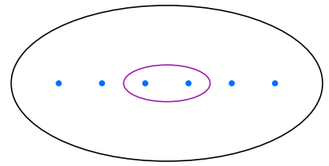
13. F11 | H | SPHERE SUPPORT CHART



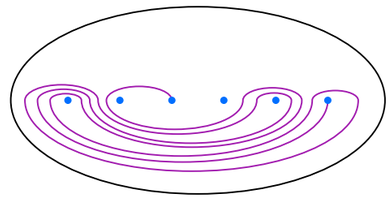


After

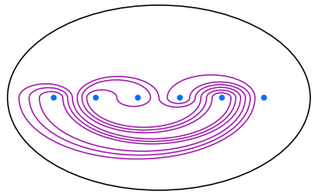
10. C13.HL2.3 | T2^2 | SPHERE SUPPORT CHART



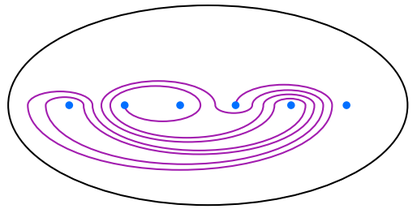
11. F9 | H | SPHERE SUPPORT CHART

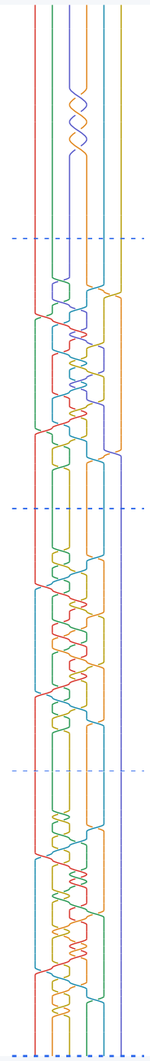


12. F10.2 | T3 | SPHERE SUPPORT CHART



13. F11 | H | SPHERE SUPPORT CHART





II - step 036 - continued

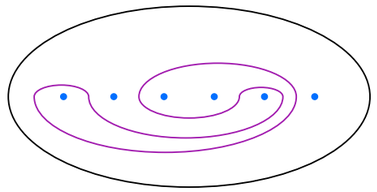
C13.HL2.3 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

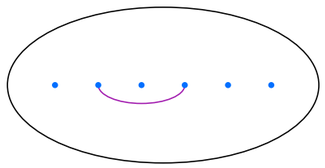
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

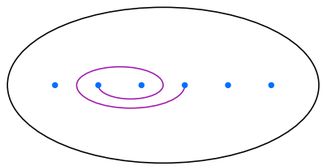
14. F12.root+F12 | T3^3 | SPHERE SUPPORT CHART



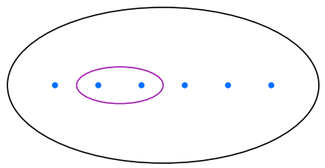
15. C13.HL1.1 | H | SPHERE SUPPORT CHART



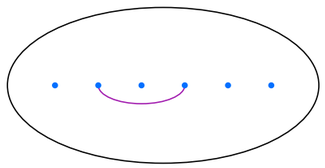
16. C13.HL1.2 | H | SPHERE SUPPORT CHART

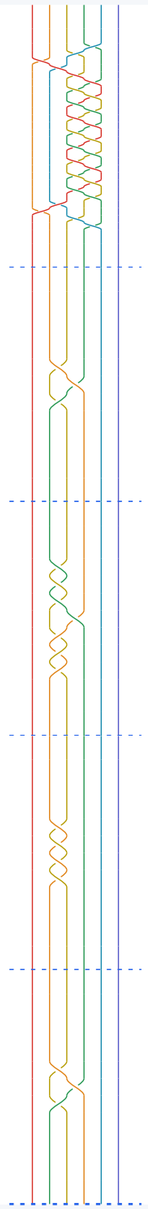


17. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART



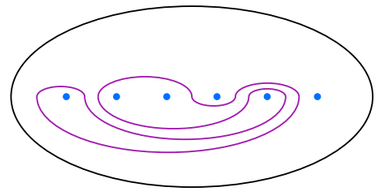
18. C13.HL2.1 | H | SPHERE SUPPORT CHART



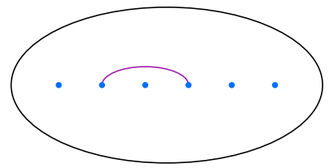


After

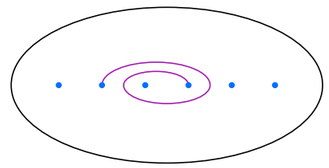
14. F12.root+F12 | T3^3 | SPHERE SUPPORT CHART



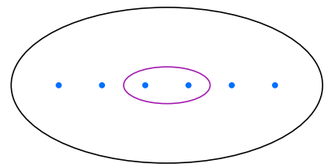
15. C13.HL1.1 | H | SPHERE SUPPORT CHART



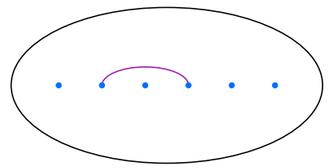
16. C13.HL1.2 | H | SPHERE SUPPORT CHART

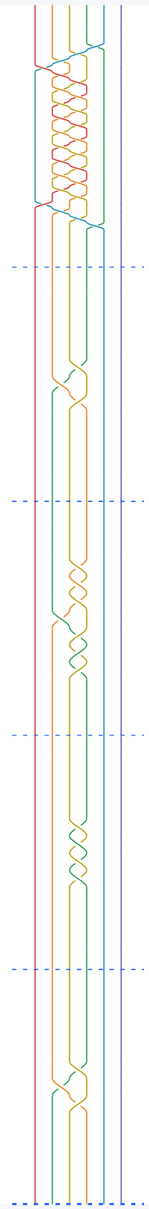


17. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART



18. C13.HL2.1 | H | SPHERE SUPPORT CHART





II - step 036 - continued

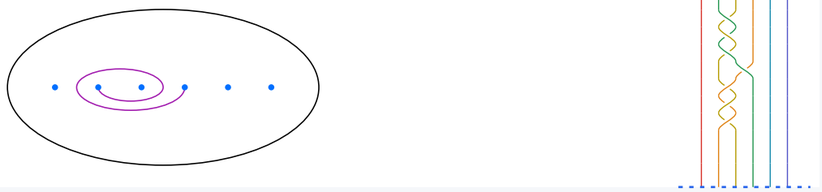
C13.HL2.3 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

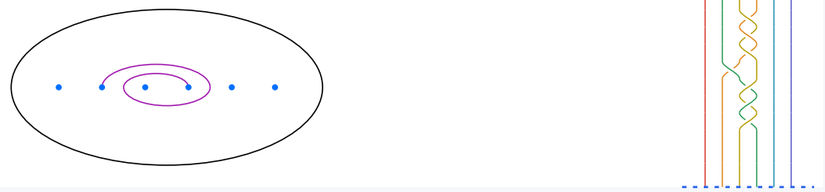
Before

19. C13.HL2.2 | H | SPHERE SUPPORT CHART



After

19. C13.HL2.2 | H | SPHERE SUPPORT CHART



II - step 037

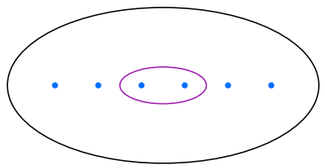
Cancel the inverse pair against one unit of the squared pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

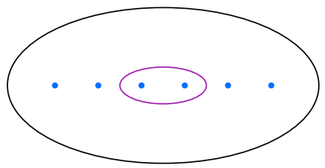
8 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

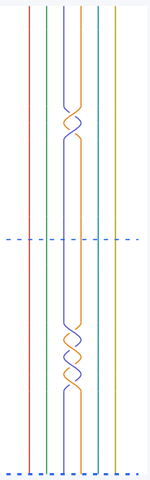
Before

9. new.Q.complement | T2^-1 | SPHERE SUPPORT CHART



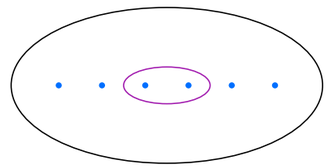
10. C13.HL2.3 | T2^2 | SPHERE SUPPORT CHART

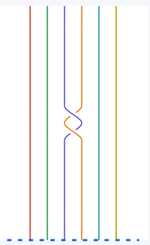




After

9. P15.after-cancellation | T2 | SPHERE SUPPORT CHART





II - step 038

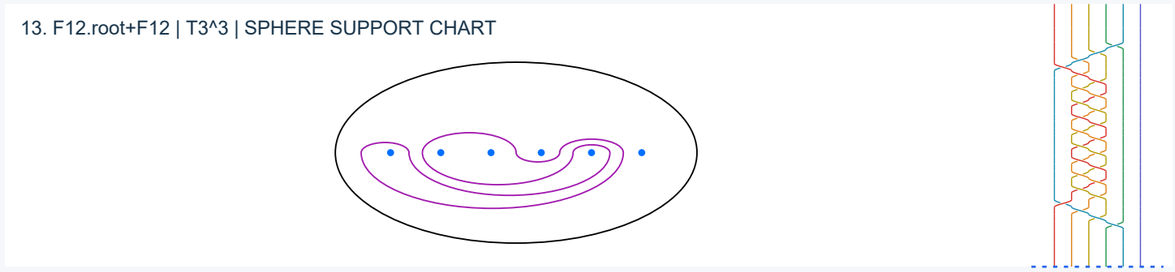
Split the C12 cube into three roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

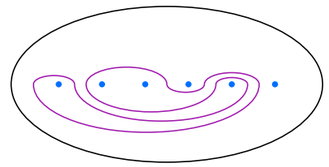
Before

13. F12.root+F12 | T3^3 | SPHERE SUPPORT CHART

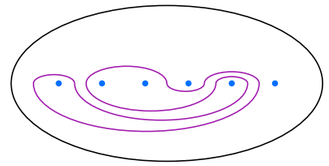


After

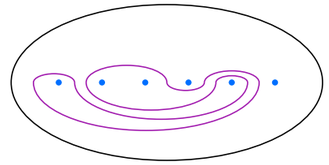
13. F12.root+F12.1 | T3 | SPHERE SUPPORT CHART

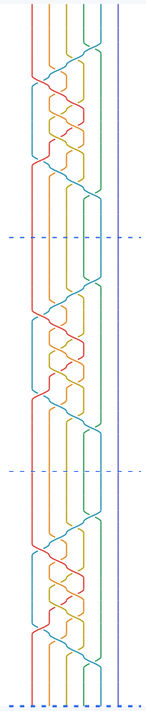


14. F12.root+F12.2 | T3 | SPHERE SUPPORT CHART



15. F12.root+F12.3 | T3 | SPHERE SUPPORT CHART





II - step 039

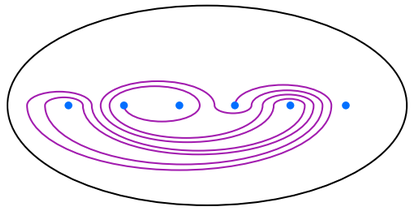
F12.root+F12.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

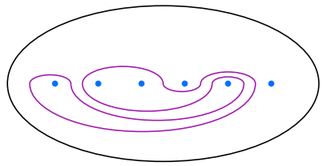
11 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

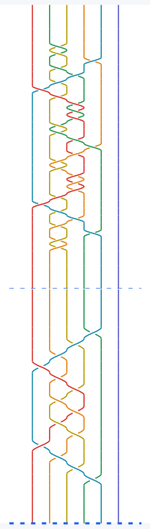
Before

12. F11 | H | SPHERE SUPPORT CHART



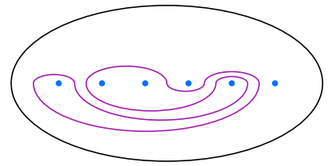
13. F12.root+F12.1 | T3 | SPHERE SUPPORT CHART



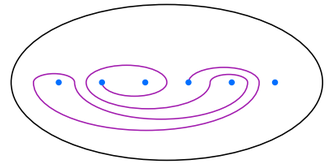


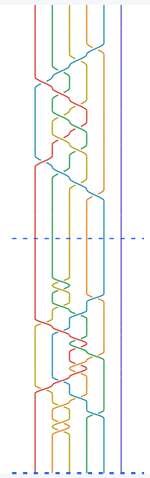
After

12. F12.root+F12.1 | T3 | SPHERE SUPPORT CHART



13. F11 | H | SPHERE SUPPORT CHART





II - step 040

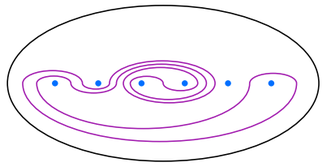
Use the residual lantern in the chart S C12, with the roots in their actual order

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

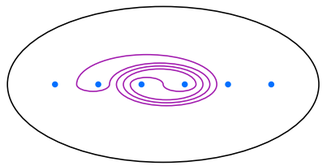
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

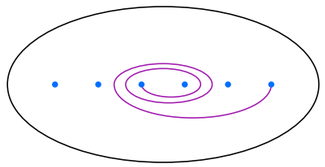
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



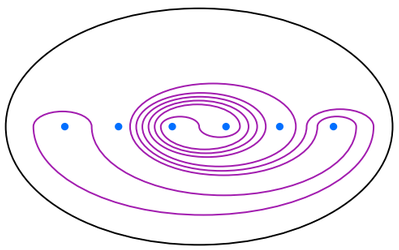
2. prefix.P12 | T2 | SPHERE SUPPORT CHART

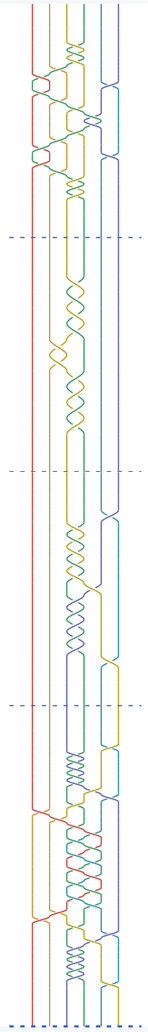


3. F3 | H



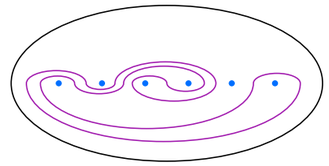
4. F4 | T3^2 | SPHERE SUPPORT CHART



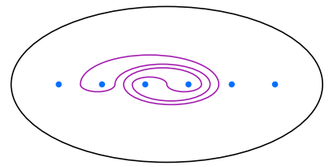


After

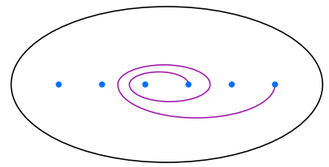
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



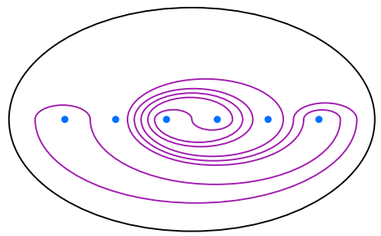
2. prefix.P12 | T2 | SPHERE SUPPORT CHART



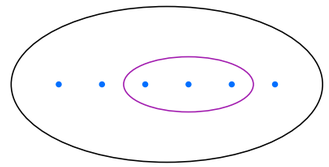
3. F3 | H

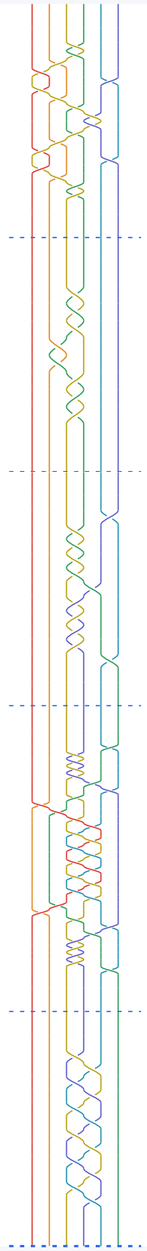


4. F4 | T3^2 | SPHERE SUPPORT CHART



5. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 040 - continued

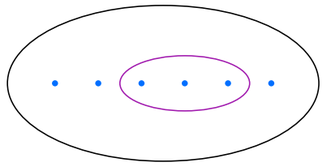
Use the residual lantern in the chart S C12, with the roots in their actual order

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

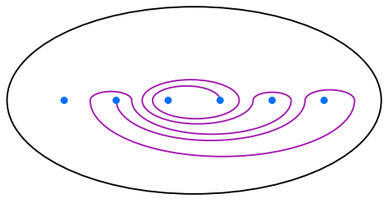
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

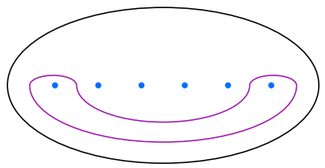
5. F5 | T3^2 | SPHERE SUPPORT CHART



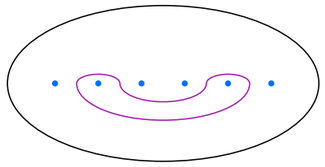
6. F6 | H | SPHERE SUPPORT CHART



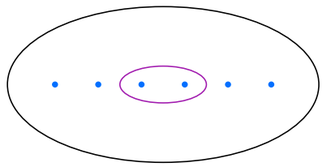
7. new.P34 | T2 | SPHERE SUPPORT CHART

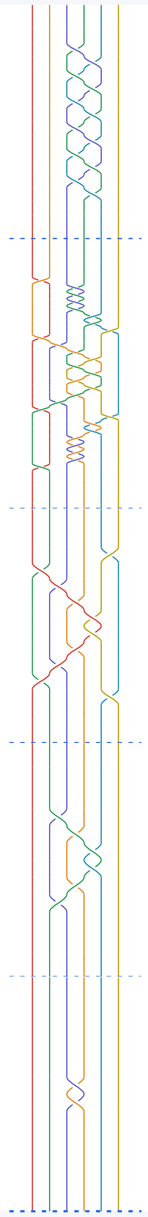


8. new.P26 | T2 | SPHERE SUPPORT CHART



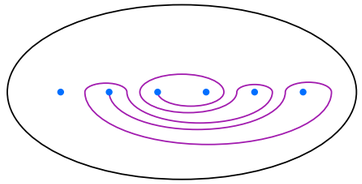
9. P15.after-cancellation | T2 | SPHERE SUPPORT CHART



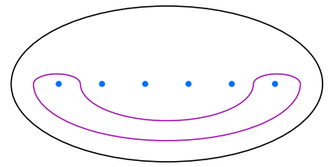


After

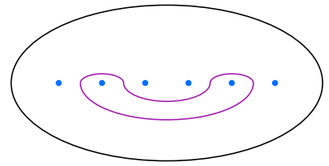
6. F6 | H | SPHERE SUPPORT CHART



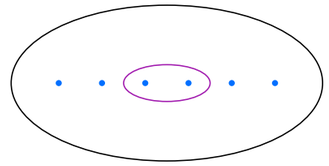
7. new.P34 | T2 | SPHERE SUPPORT CHART



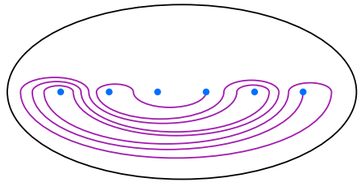
8. new.P26 | T2 | SPHERE SUPPORT CHART

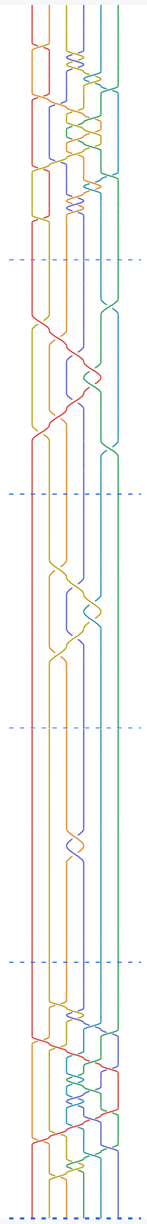


9. P15.after-cancellation | T2 | SPHERE SUPPORT CHART



10. F9 | H | SPHERE SUPPORT CHART





II - step 040 - continued

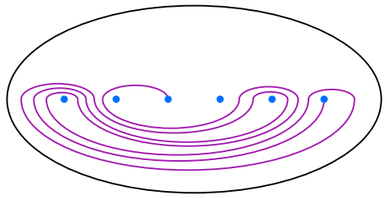
Use the residual lantern in the chart S C12, with the roots in their actual order

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

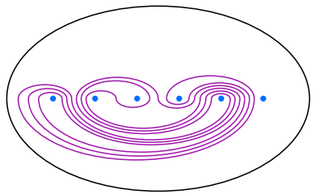
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

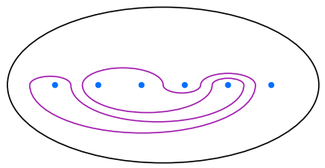
10. F9 | H | SPHERE SUPPORT CHART



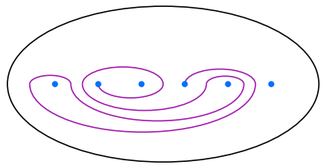
11. F10.2 | T3 | SPHERE SUPPORT CHART



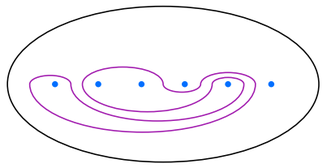
12. F12.root+F12.1 | T3 | SPHERE SUPPORT CHART

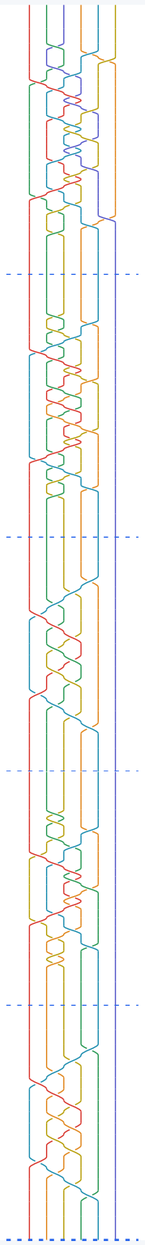


13. F11 | H | SPHERE SUPPORT CHART



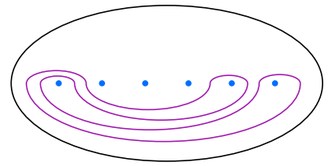
14. F12.root+F12.2 | T3 | SPHERE SUPPORT CHART



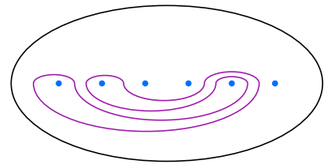


After

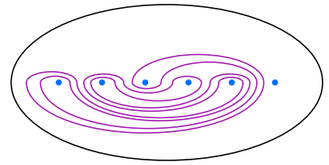
11. residual.P34 | T2 | SPHERE SUPPORT CHART



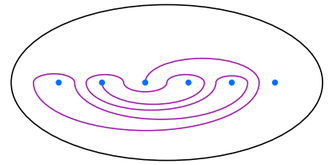
12. residual.P26 | T2 | SPHERE SUPPORT CHART



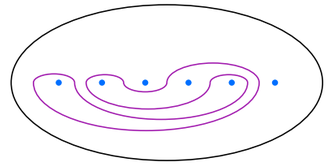
13. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART

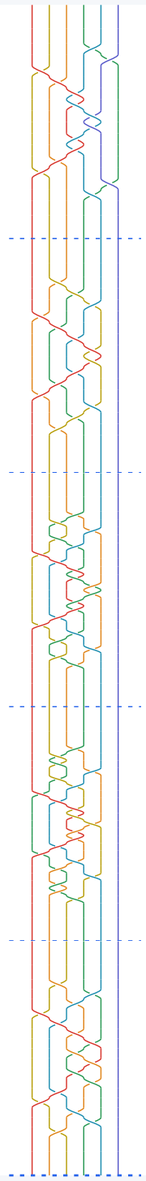


14. F11 | H | SPHERE SUPPORT CHART



15. F12.root+F12.2 | T3 | SPHERE SUPPORT CHART





II - step 040 - continued

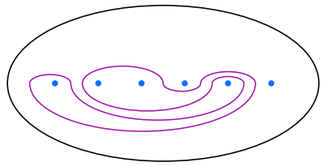
Use the residual lantern in the chart S C12, with the roots in their actual order

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

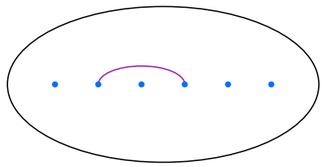
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

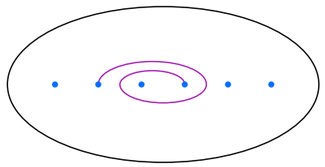
15. F12.root+F12.3 | T3 | SPHERE SUPPORT CHART



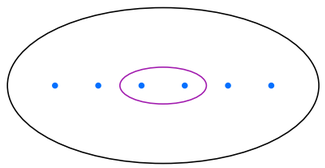
16. C13.HL1.1 | H | SPHERE SUPPORT CHART



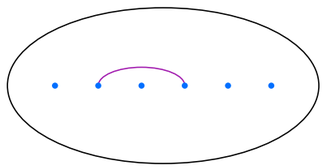
17. C13.HL1.2 | H | SPHERE SUPPORT CHART

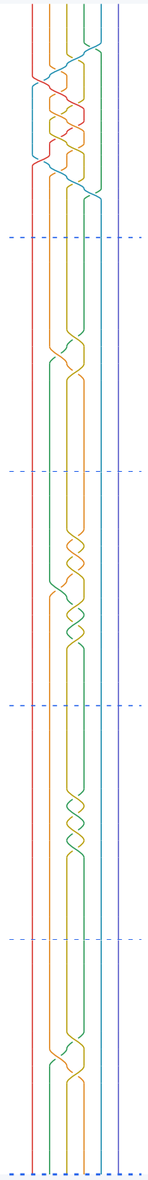


18. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART



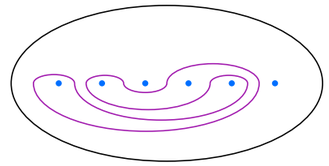
19. C13.HL2.1 | H | SPHERE SUPPORT CHART



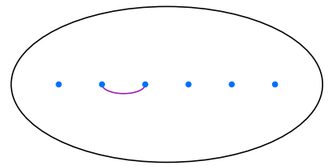


After

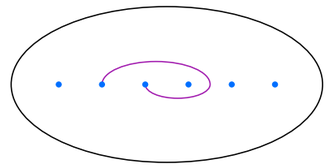
16. F12.root+F12.3 | T3 | SPHERE SUPPORT CHART



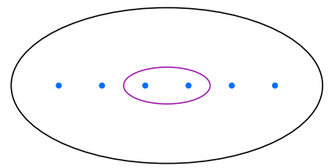
17. C13.HL1.1 | H | SPHERE SUPPORT CHART



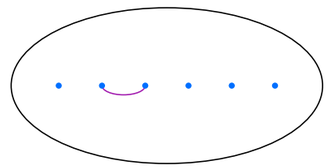
18. C13.HL1.2 | H | SPHERE SUPPORT CHART

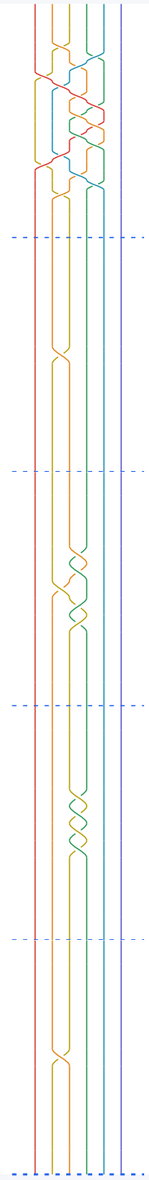


19. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART



20. C13.HL2.1 | H | SPHERE SUPPORT CHART





II - step 040 - continued

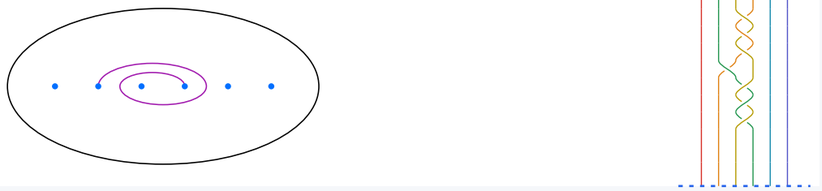
Use the residual lantern in the chart S C12, with the roots in their actual order

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

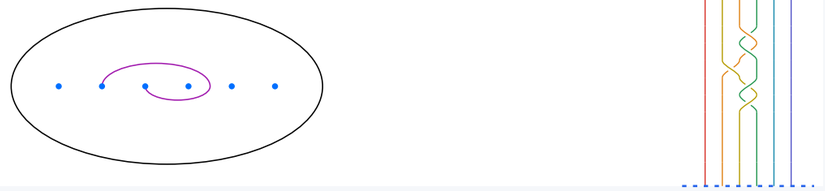
Before

20. C13.HL2.2 | H | SPHERE SUPPORT CHART



After

21. C13.HL2.2 | H | SPHERE SUPPORT CHART



II - step 041

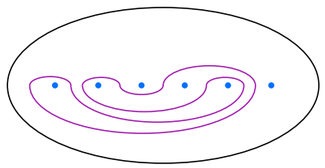
Combine the two unused C12 roots into a square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

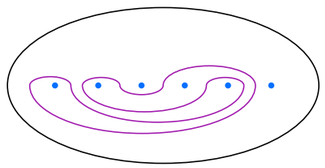
14 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

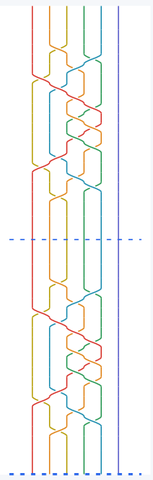
Before

15. F12.root+F12.2 | T3 | SPHERE SUPPORT CHART



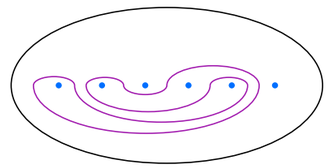
16. F12.root+F12.3 | T3 | SPHERE SUPPORT CHART

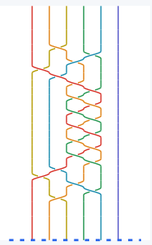




After

15. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART





II - step 042

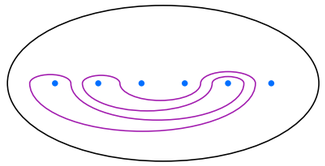
residual.P26 down past residual.Q.complement

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

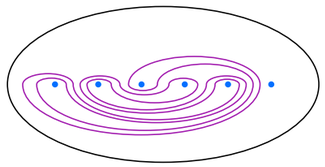
11 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

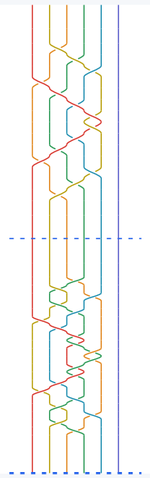
Before

12. residual.P26 | T2 | SPHERE SUPPORT CHART



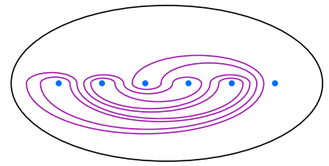
13. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



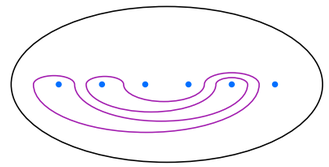


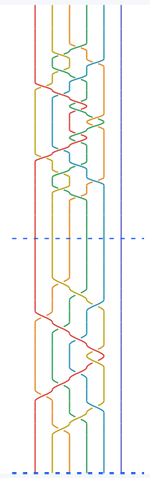
After

12. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



13. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 043

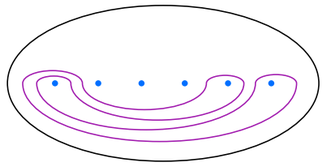
residual.P34 down past residual.Q.complement

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

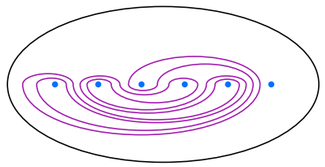
10 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

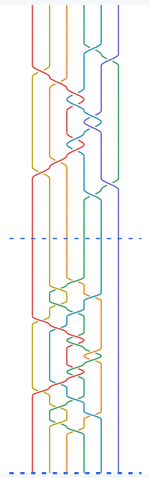
Before

11. residual.P34 | T2 | SPHERE SUPPORT CHART



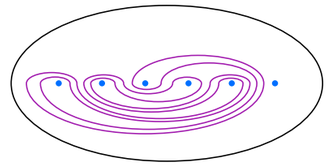
12. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



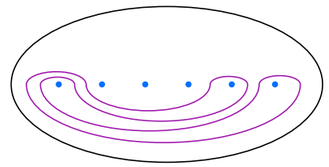


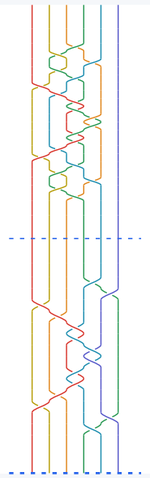
After

11. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



12. residual.P34 | T2 | SPHERE SUPPORT CHART





II - step 044

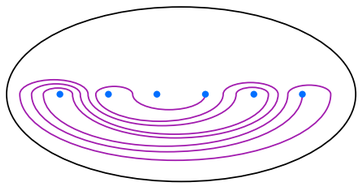
F9 down past residual.Q.complement

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

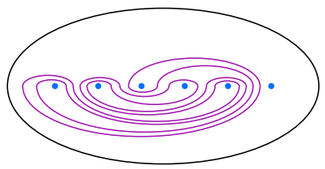
9 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

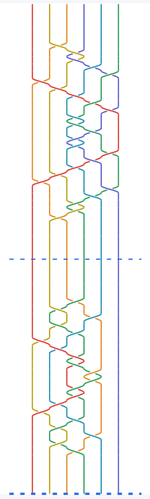
Before

10. F9 | H | SPHERE SUPPORT CHART



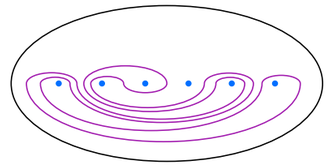
11. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



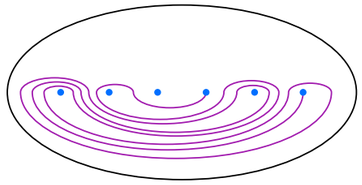


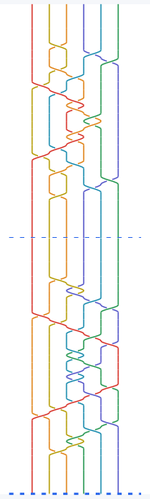
After

10. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



11. F9 | H | SPHERE SUPPORT CHART





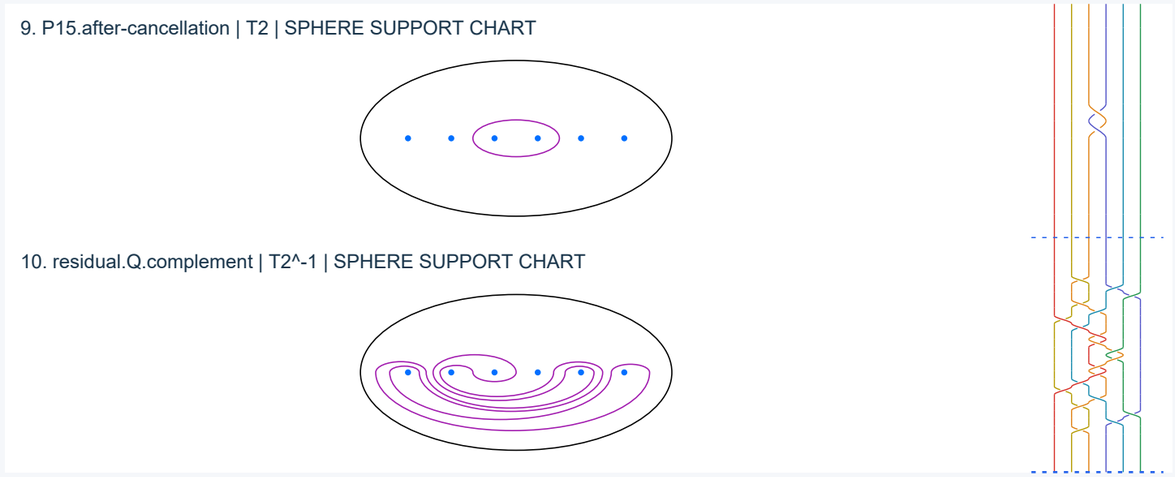
II - step 045

P15.after-cancellation down past residual.Q.complement

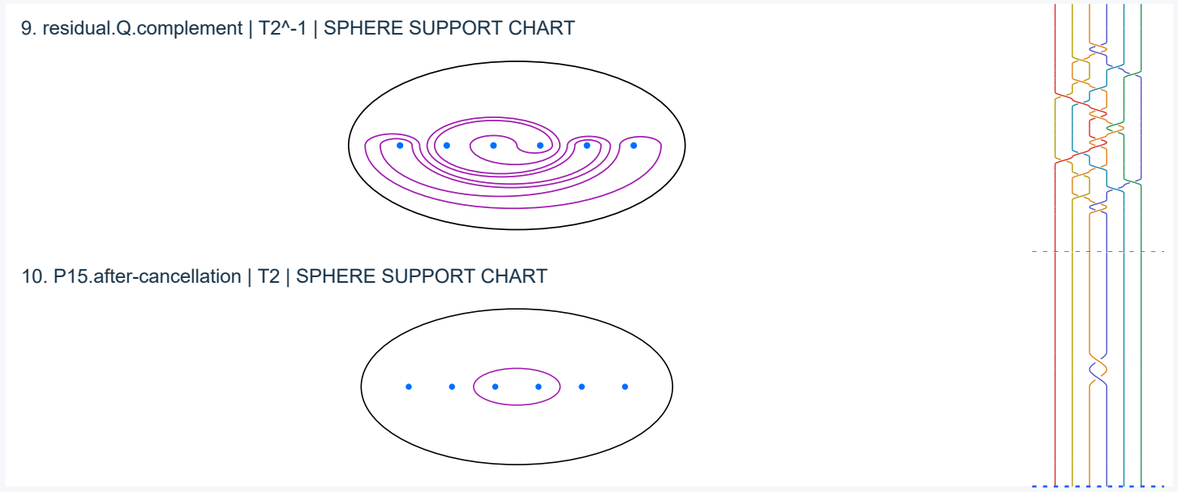
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



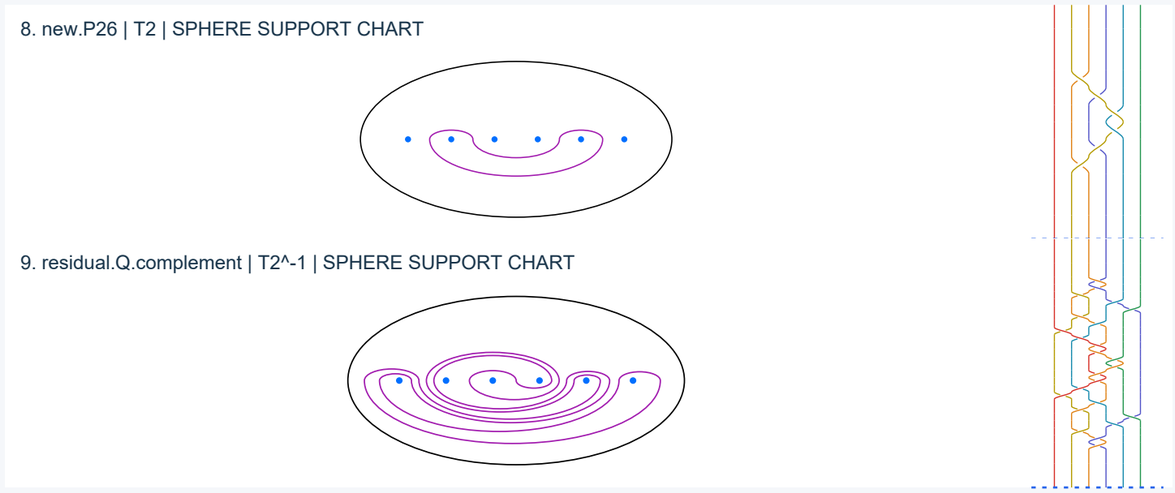
II - step 046

new.P26 down past residual.Q.complement

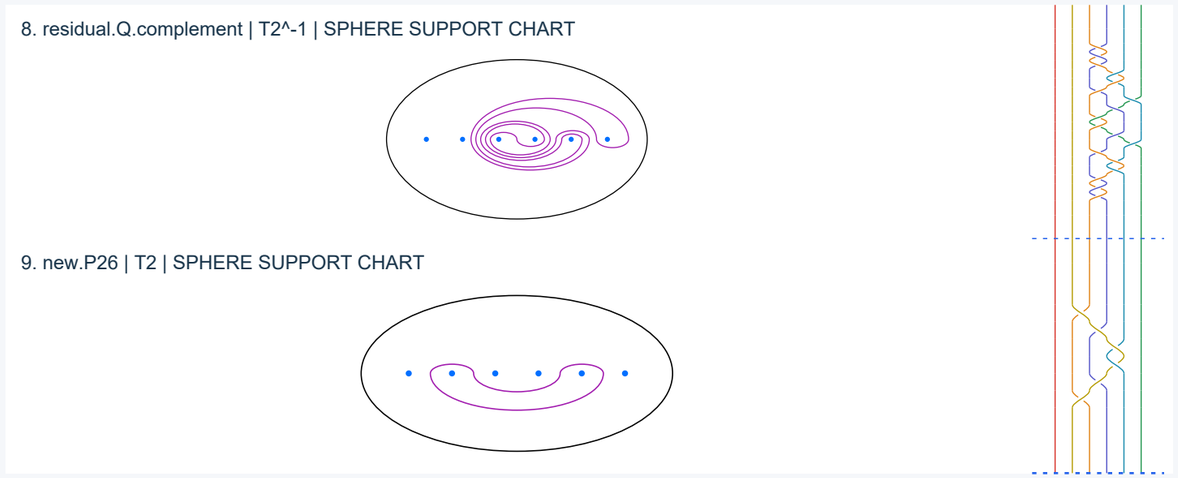
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



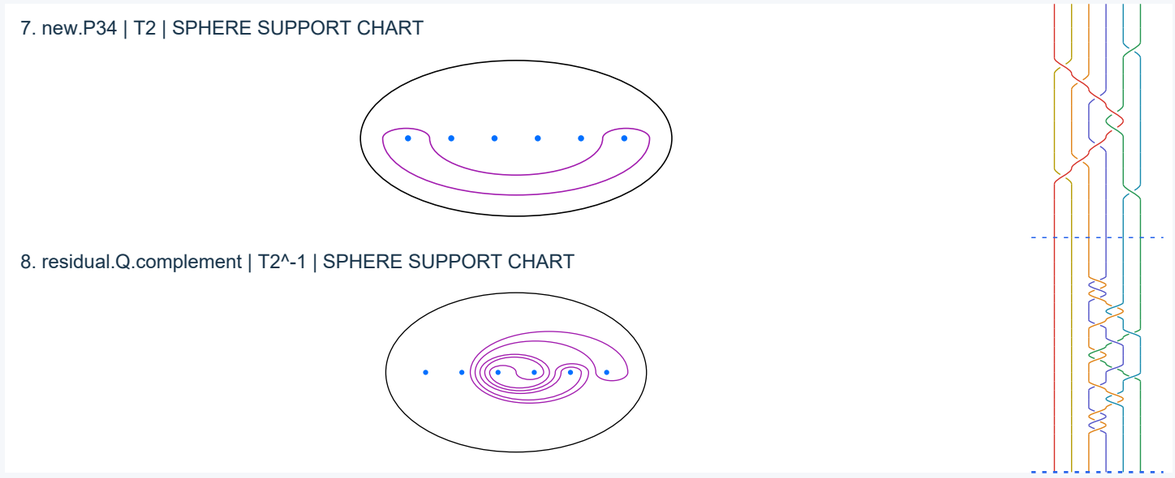
II - step 047

new.P34 down past residual.Q.complement

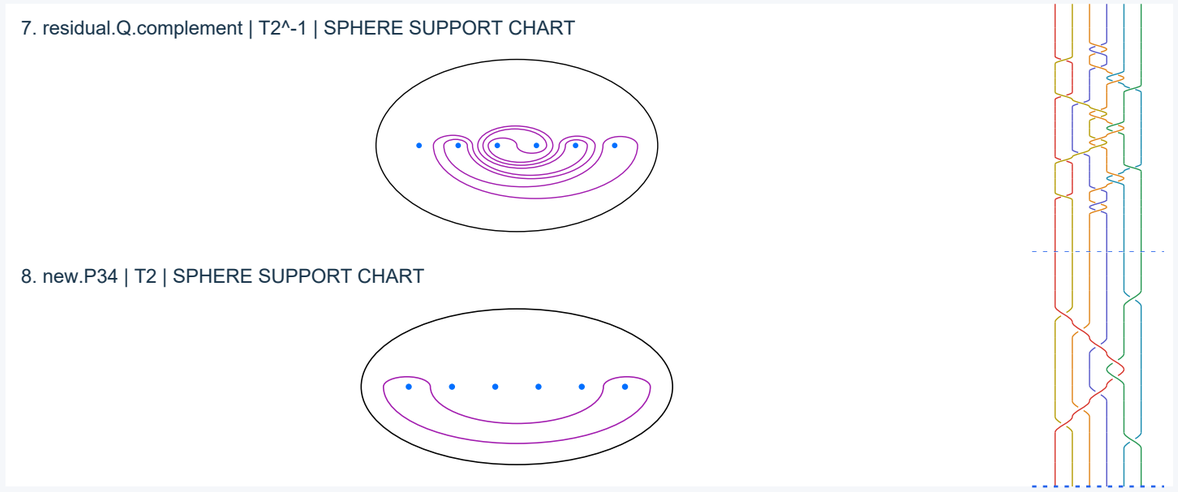
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 048

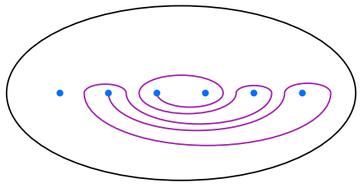
F6 down past residual.Q.complement

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

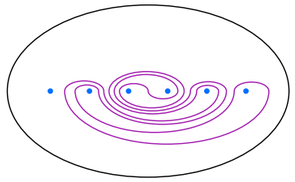
5 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

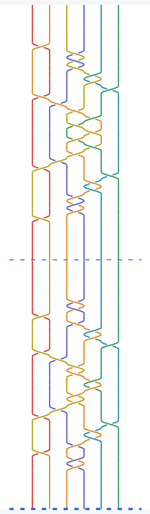
Before

6. F6 | H | SPHERE SUPPORT CHART



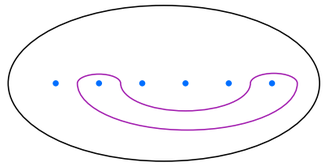
7. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



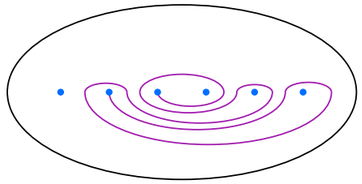


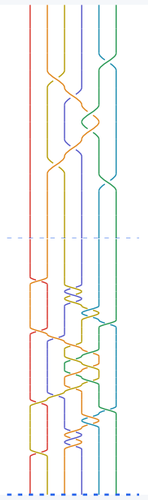
After

6. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



7. F6 | H | SPHERE SUPPORT CHART





II - step 049

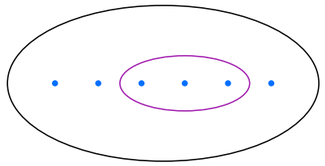
F5 down past residual.Q.complement

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

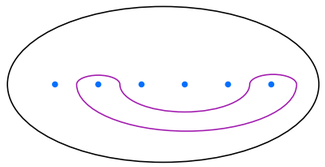
4 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

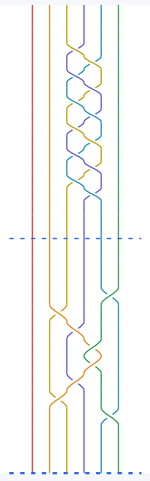
Before

5. F5 | T3^2 | SPHERE SUPPORT CHART



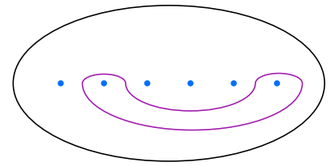
6. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



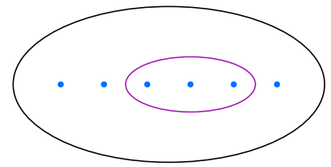


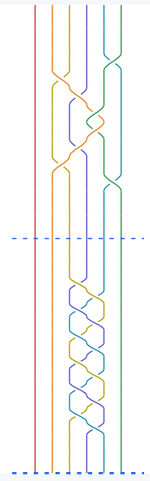
After

5. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



6. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 050

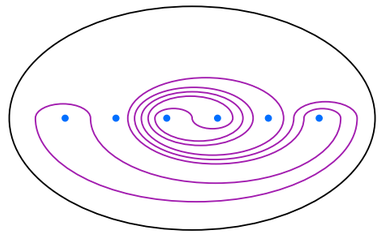
F4 down past residual.Q.complement

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

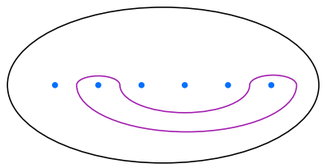
3 unchanged displayed factors above; 15 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

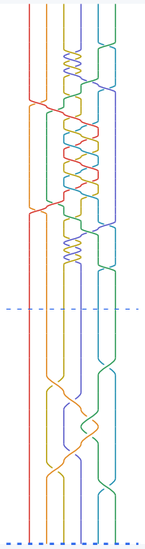
Before

4. F4 | T3^2 | SPHERE SUPPORT CHART



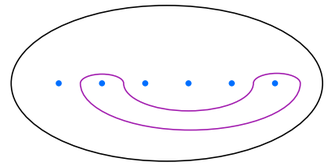
5. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



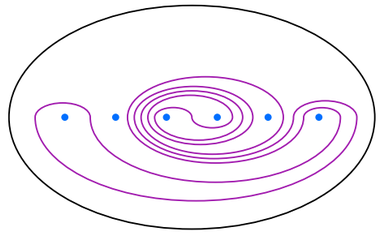


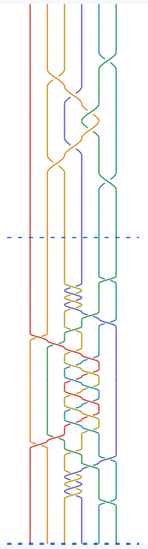
After

4. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



5. F4 | T3^2 | SPHERE SUPPORT CHART





II - step 051

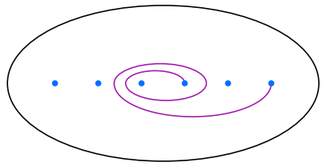
F3 down past residual.Q.complement

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

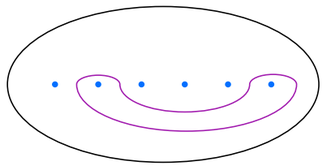
2 unchanged displayed factors above; 16 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

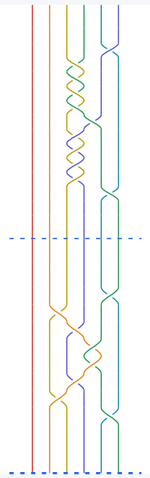
Before

3. F3 | H



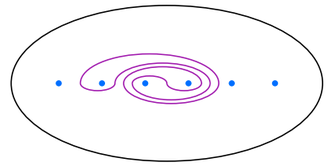
4. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



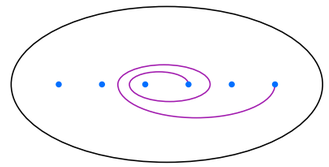


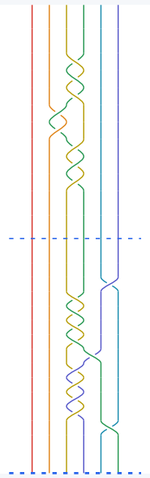
After

3. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART



4. F3 | H





II - step 052

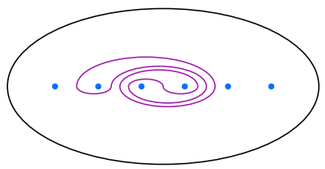
Rechart the transported inverse pair along the prefix pair on the sphere

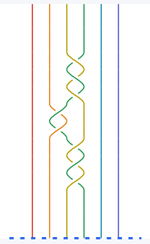
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

2 unchanged displayed factors above; 17 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

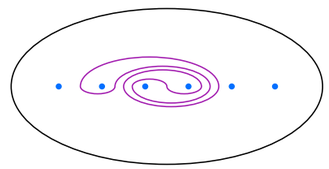
3. residual.Q.complement | T2^-1 | SPHERE SUPPORT CHART

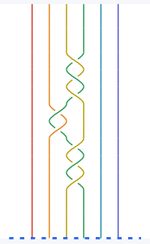




After

3. residual.inverse.prefix.P12 | T2^-1 | SPHERE SUPPORT CHART





II - step 053

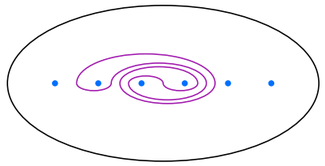
Cancel the equal positive and negative pair twists

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

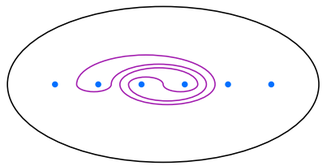
1 unchanged displayed factors above; 17 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

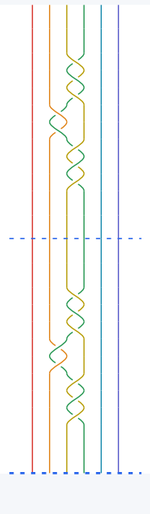
Before

2. prefix.P12 | T2 | SPHERE SUPPORT CHART



3. residual.inverse.prefix.P12 | T2^-1 | SPHERE SUPPORT CHART





After

Identity / no factors in this interval.

II - step 054

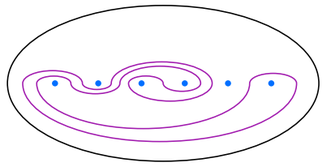
P15.after-cancellation down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

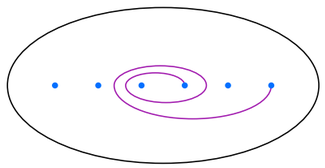
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

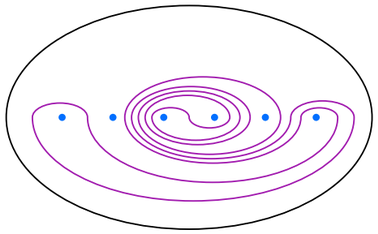
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



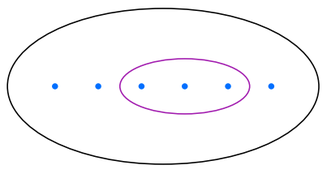
2. F3 | H



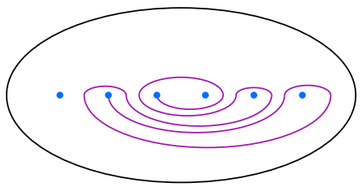
3. F4 | T3^2 | SPHERE SUPPORT CHART

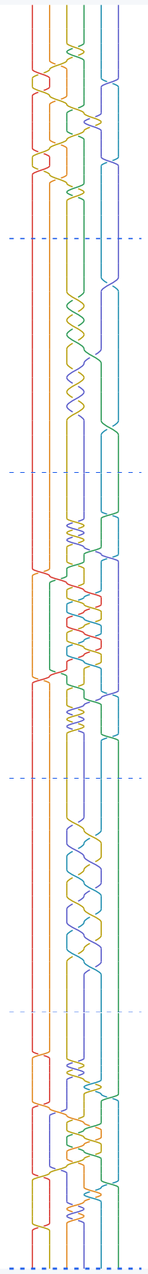


4. F5 | T3^2 | SPHERE SUPPORT CHART



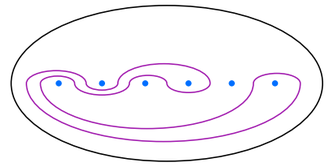
5. F6 | H | SPHERE SUPPORT CHART



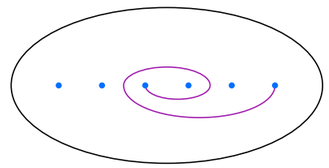


After

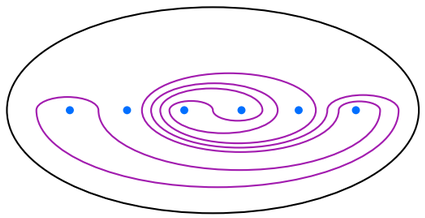
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



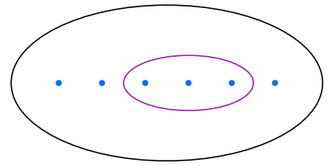
2. F3 | H



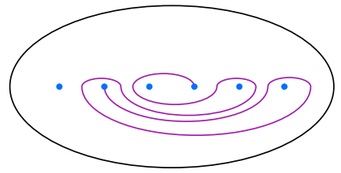
3. F4 | T3^2 | SPHERE SUPPORT CHART

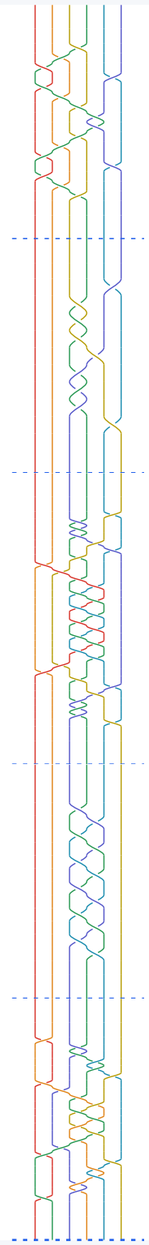


4. F5 | T3^2 | SPHERE SUPPORT CHART



5. F6 | H | SPHERE SUPPORT CHART





II - step 054 - continued

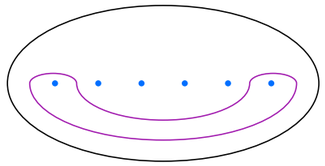
P15.after-cancellation down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

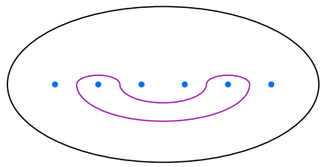
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

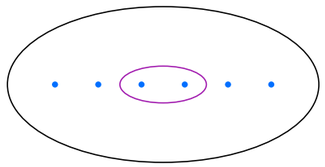
6. new.P34 | T2 | SPHERE SUPPORT CHART



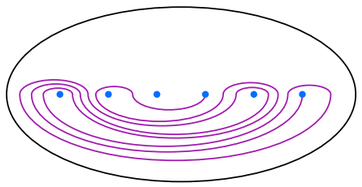
7. new.P26 | T2 | SPHERE SUPPORT CHART



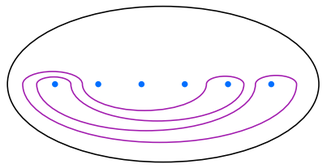
8. P15.after-cancellation | T2 | SPHERE SUPPORT CHART

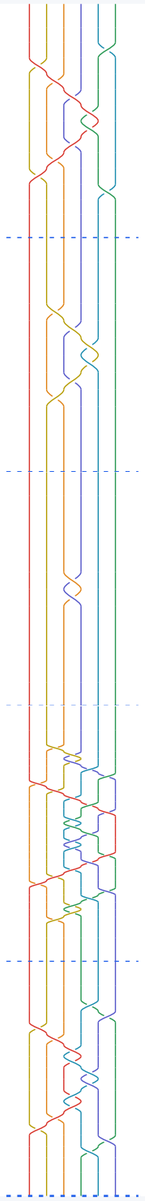


9. F9 | H | SPHERE SUPPORT CHART



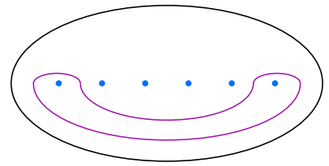
10. residual.P34 | T2 | SPHERE SUPPORT CHART



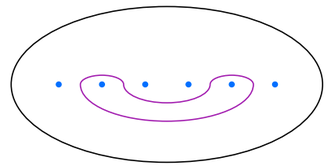


After

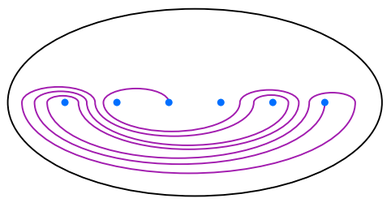
6. new.P34 | T2 | SPHERE SUPPORT CHART



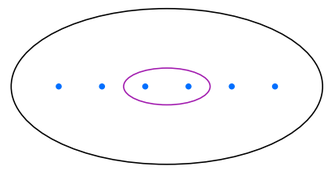
7. new.P26 | T2 | SPHERE SUPPORT CHART



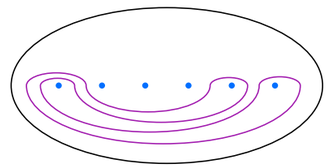
8. F9 | H | SPHERE SUPPORT CHART

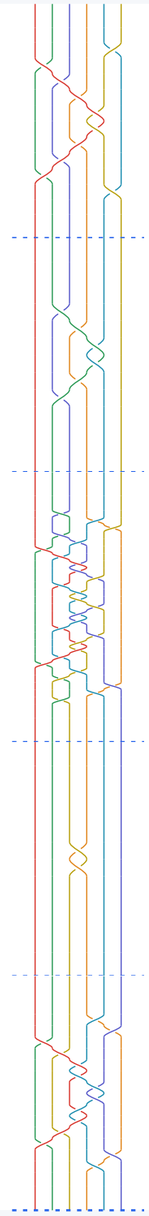


9. P15.after-cancellation | T2 | SPHERE SUPPORT CHART



10. residual.P34 | T2 | SPHERE SUPPORT CHART





II - step 054 - continued

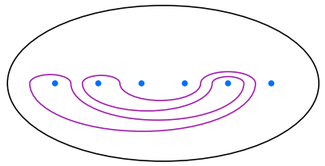
P15.after-cancellation down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

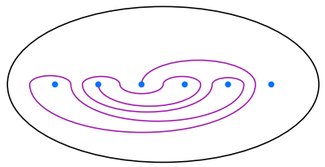
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

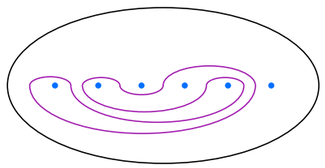
11. residual.P26 | T2 | SPHERE SUPPORT CHART



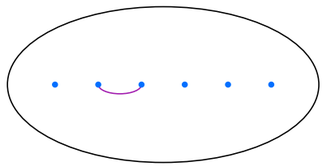
12. F11 | H | SPHERE SUPPORT CHART



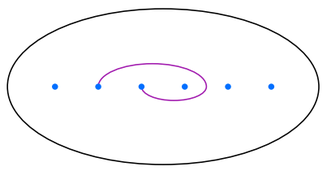
13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

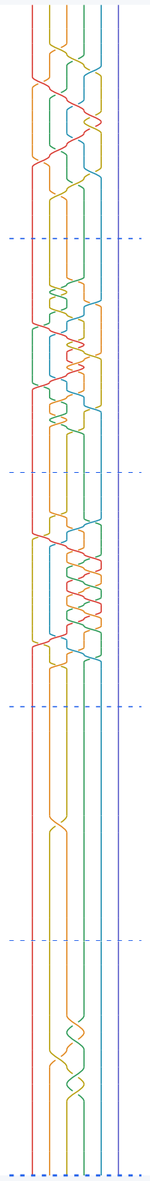


14. C13.HL1.1 | H | SPHERE SUPPORT CHART



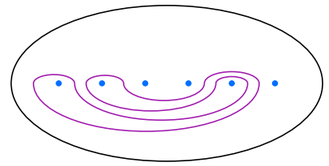
15. C13.HL1.2 | H | SPHERE SUPPORT CHART



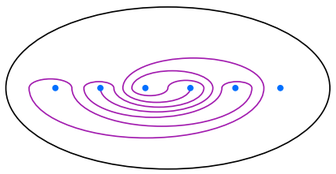


After

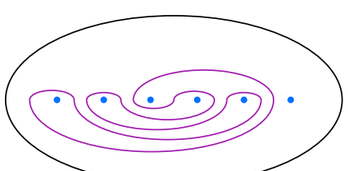
11. residual.P26 | T2 | SPHERE SUPPORT CHART



12. F11 | H | SPHERE SUPPORT CHART



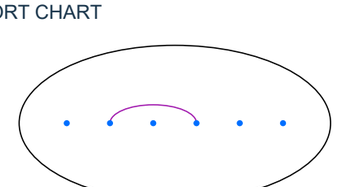
13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

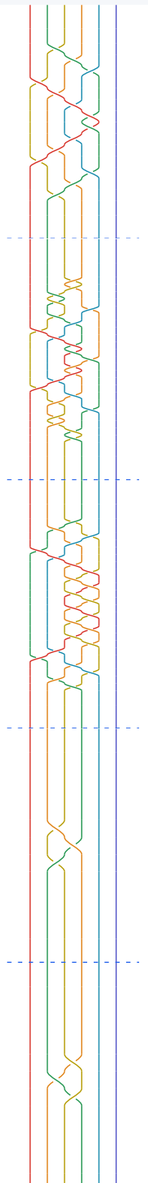


14. C13.HL1.1 | H | SPHERE SUPPORT CHART



15. C13.HL1.2 | H | SPHERE SUPPORT CHART





II - step 054 - continued

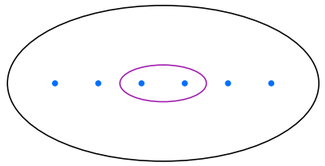
P15.after-cancellation down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

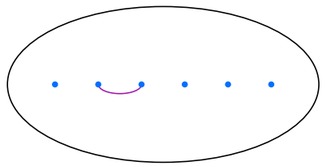
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

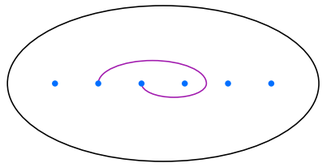
16. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART

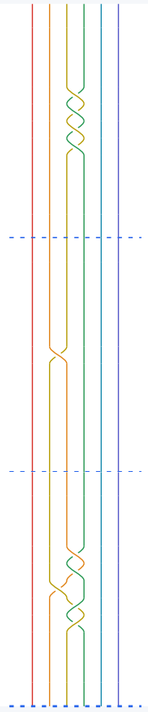


17. C13.HL2.1 | H | SPHERE SUPPORT CHART



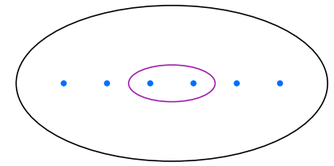
18. C13.HL2.2 | H | SPHERE SUPPORT CHART



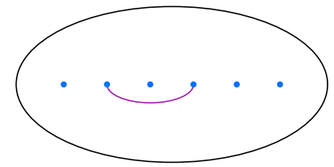


After

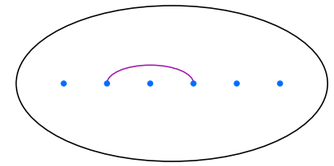
16. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART

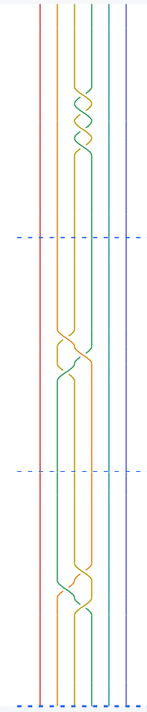


17. C13.HL2.1 | H | SPHERE SUPPORT CHART



18. C13.HL2.2 | H | SPHERE SUPPORT CHART





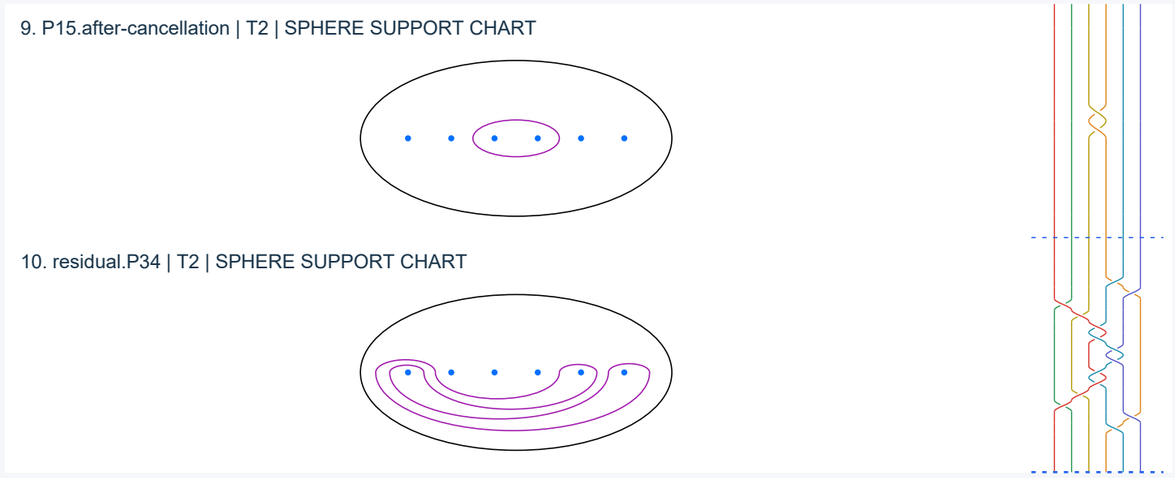
II - step 055

P15.after-cancellation down past residual.P34

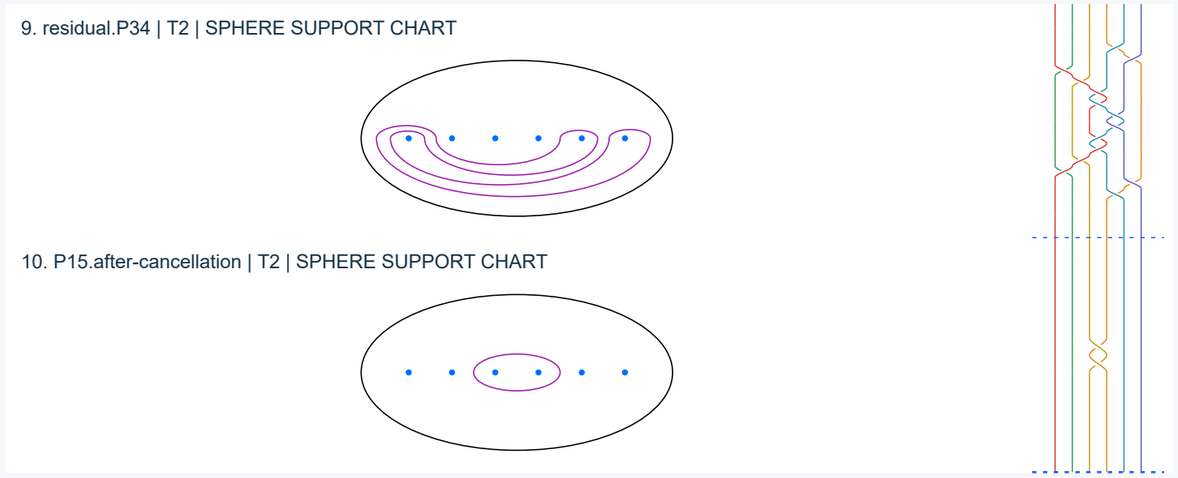
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



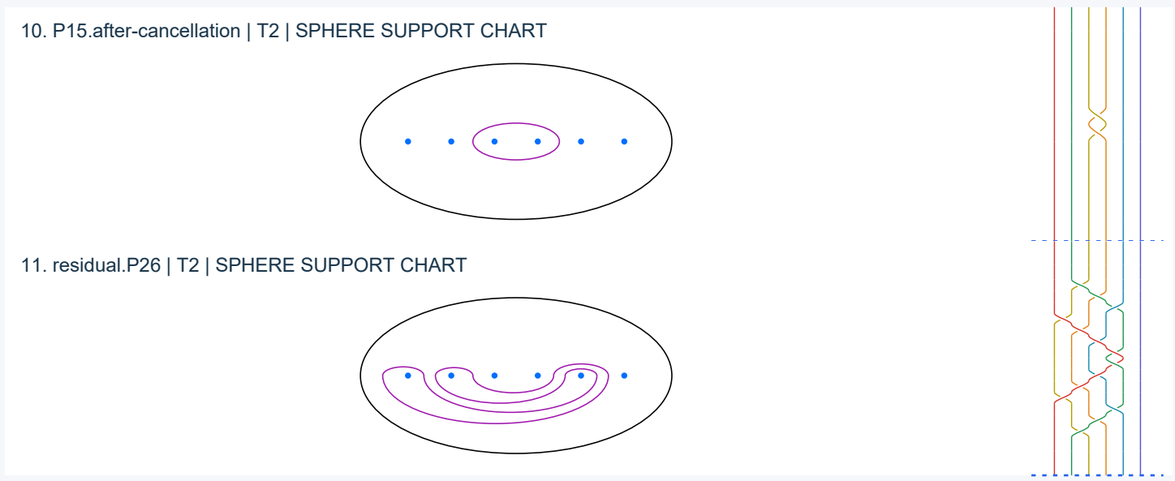
II - step 056

P15.after-cancellation down past residual.P26

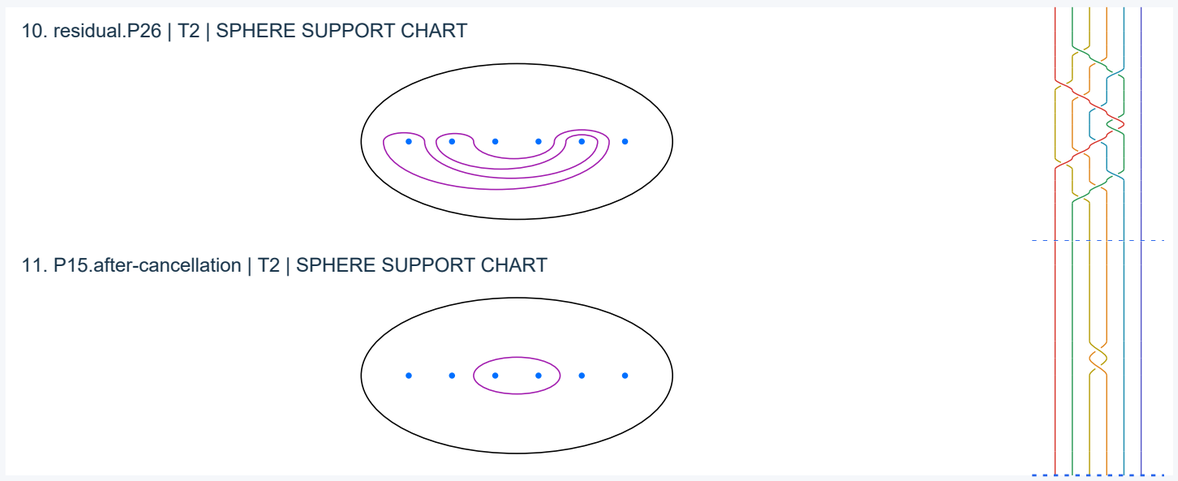
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 057

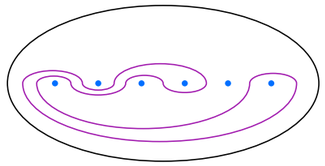
P15.after-cancellation down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

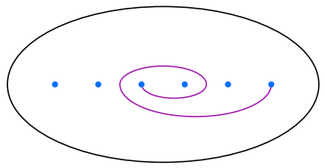
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

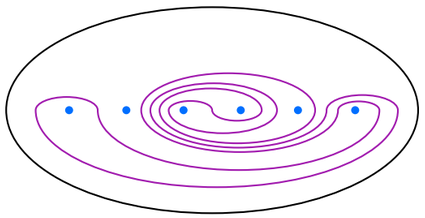
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



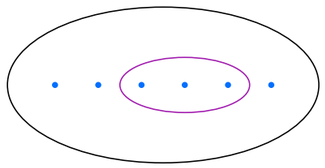
2. F3 | H



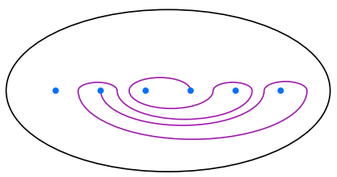
3. F4 | T3^2 | SPHERE SUPPORT CHART

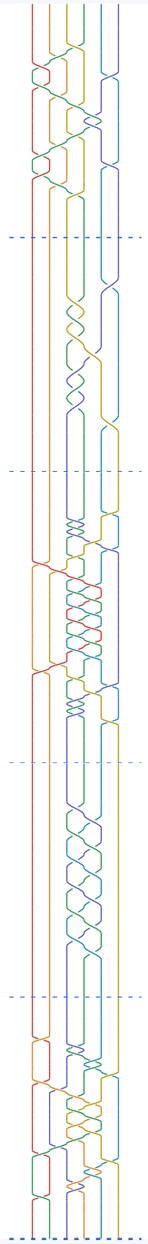


4. F5 | T3^2 | SPHERE SUPPORT CHART



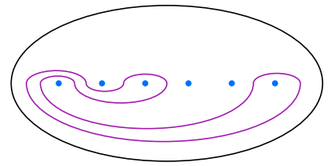
5. F6 | H | SPHERE SUPPORT CHART



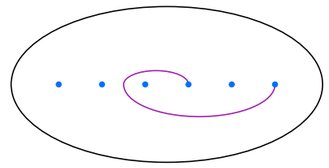


After

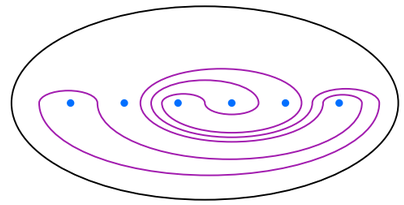
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



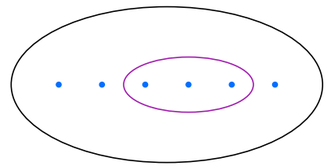
2. F3 | H



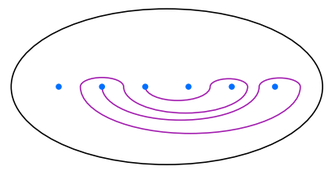
3. F4 | T3^2 | SPHERE SUPPORT CHART

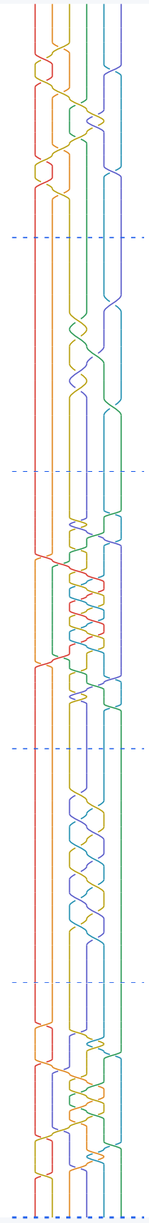


4. F5 | T3^2 | SPHERE SUPPORT CHART



5. F6 | H | SPHERE SUPPORT CHART





II - step 057 - continued

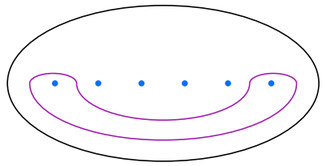
P15.after-cancellation down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

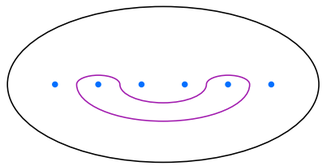
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

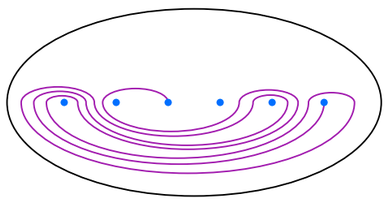
6. new.P34 | T2 | SPHERE SUPPORT CHART



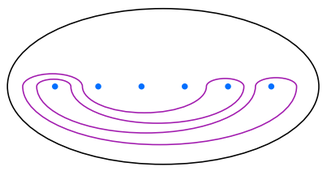
7. new.P26 | T2 | SPHERE SUPPORT CHART



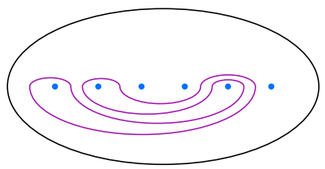
8. F9 | H | SPHERE SUPPORT CHART

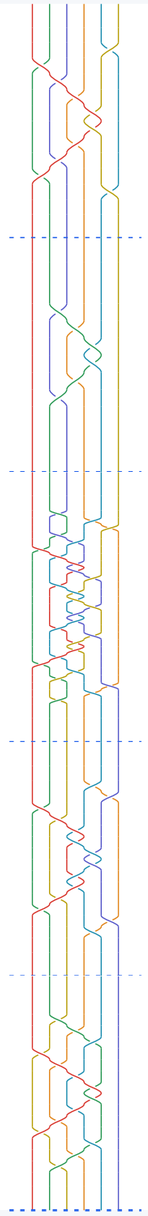


9. residual.P34 | T2 | SPHERE SUPPORT CHART



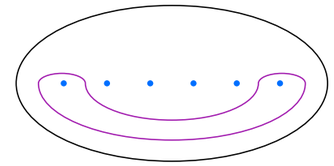
10. residual.P26 | T2 | SPHERE SUPPORT CHART



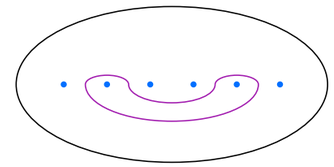


After

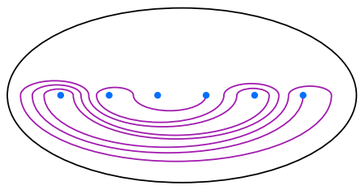
6. new.P34 | T2 | SPHERE SUPPORT CHART



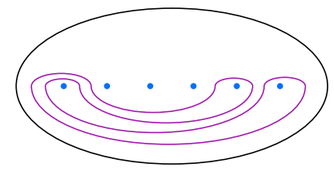
7. new.P26 | T2 | SPHERE SUPPORT CHART



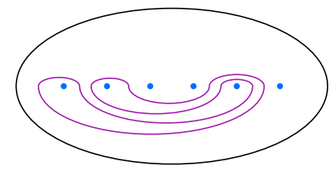
8. F9 | H | SPHERE SUPPORT CHART

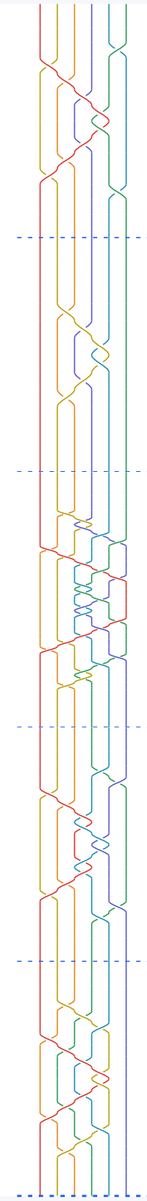


9. residual.P34 | T2 | SPHERE SUPPORT CHART



10. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 057 - continued

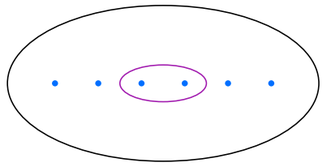
P15.after-cancellation down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

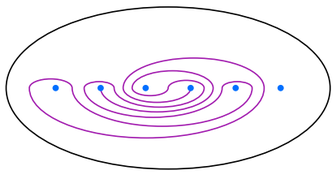
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

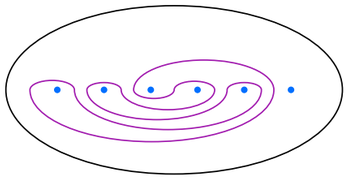
11. P15.after-cancellation | T2 | SPHERE SUPPORT CHART



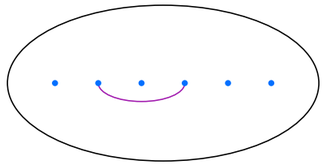
12. F11 | H | SPHERE SUPPORT CHART



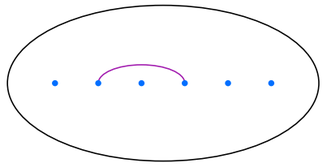
13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

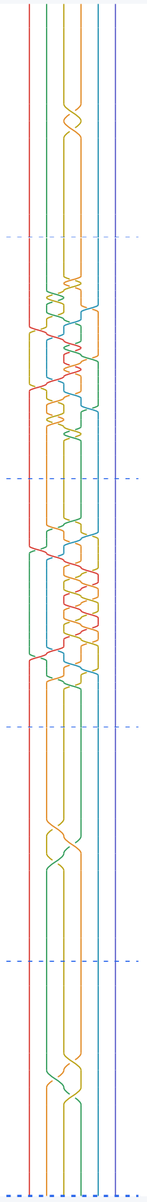


14. C13.HL1.1 | H | SPHERE SUPPORT CHART



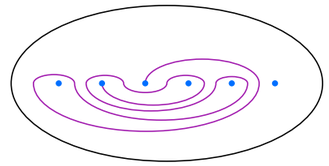
15. C13.HL1.2 | H | SPHERE SUPPORT CHART



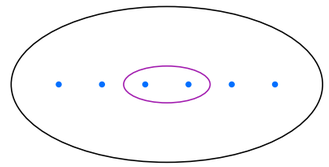


After

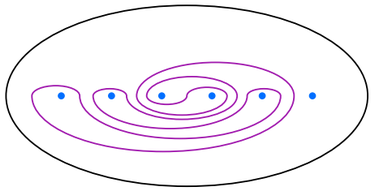
11. F11 | H | SPHERE SUPPORT CHART



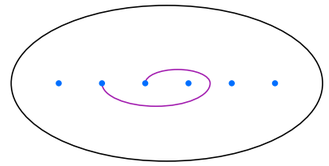
12. P15.after-cancellation | T2 | SPHERE SUPPORT CHART



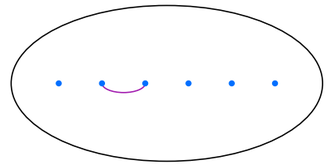
13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

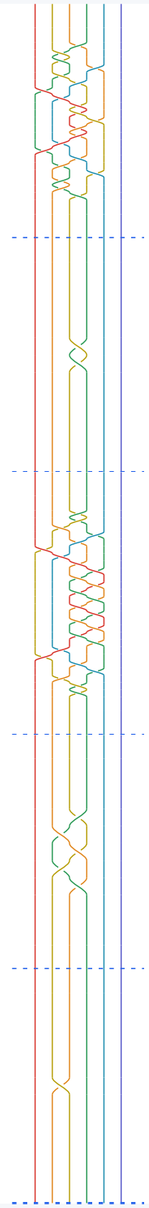


14. C13.HL1.1 | H | SPHERE SUPPORT CHART



15. C13.HL1.2 | H | SPHERE SUPPORT CHART





II - step 057 - continued

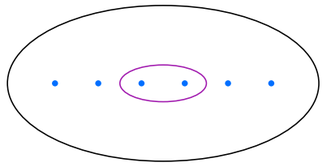
P15.after-cancellation down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

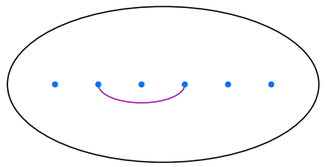
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

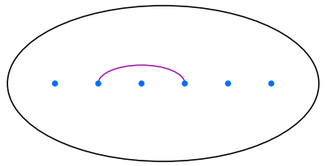
16. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART

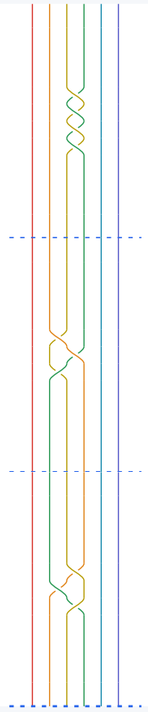


17. C13.HL2.1 | H | SPHERE SUPPORT CHART



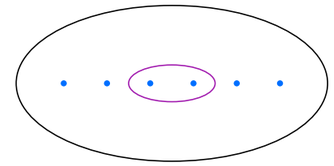
18. C13.HL2.2 | H | SPHERE SUPPORT CHART



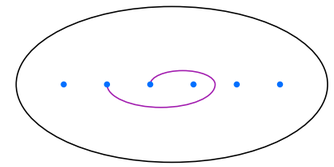


After

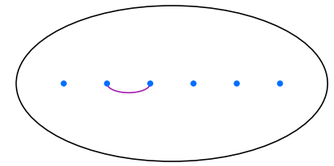
16. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART

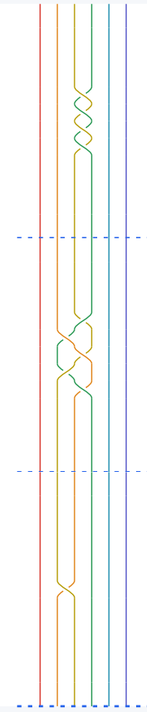


17. C13.HL2.1 | H | SPHERE SUPPORT CHART



18. C13.HL2.2 | H | SPHERE SUPPORT CHART





II - step 058

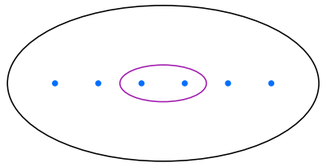
P15.after-cancellation down past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

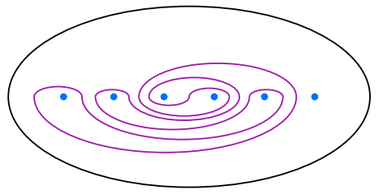
11 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

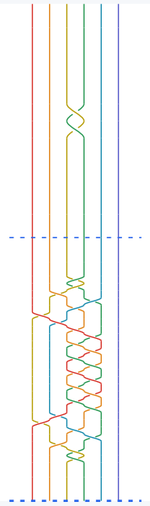
Before

12. P15.after-cancellation | T2 | SPHERE SUPPORT CHART



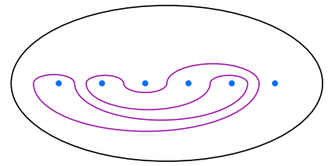
13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



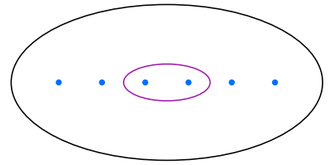


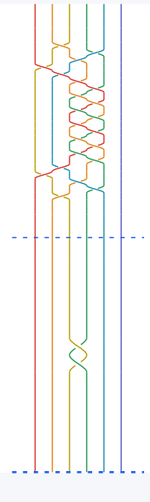
After

12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



13. P15.after-cancellation | T2 | SPHERE SUPPORT CHART





II - step 059

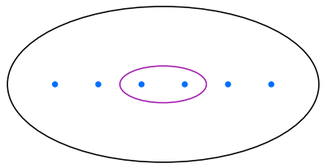
Reverse the two PC half lanterns and retain the inverse pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

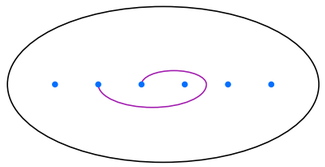
12 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

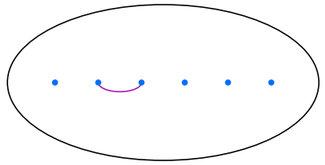
13. P15.after-cancellation | T2 | SPHERE SUPPORT CHART



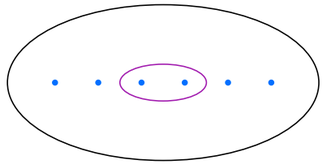
14. C13.HL1.1 | H | SPHERE SUPPORT CHART



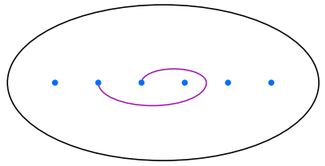
15. C13.HL1.2 | H | SPHERE SUPPORT CHART

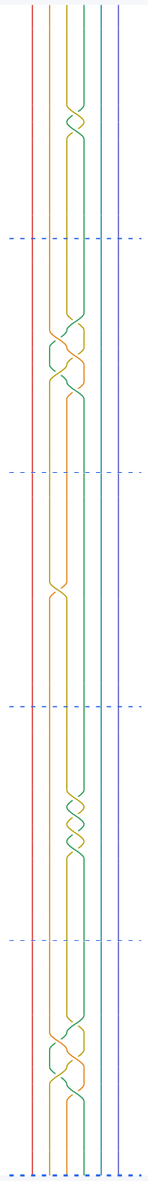


16. C13.HL1.3 | T2^2 | SPHERE SUPPORT CHART



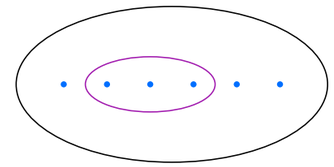
17. C13.HL2.1 | H | SPHERE SUPPORT CHART



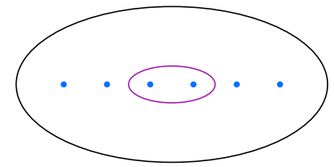


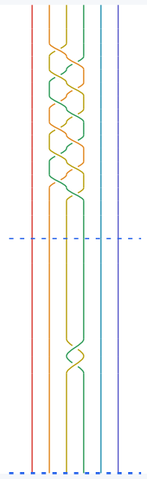
After

13. C13.recombined | T3^2 | SPHERE SUPPORT CHART



14. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART





II - step 059 - continued

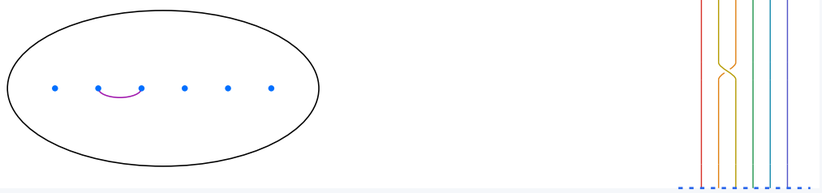
Reverse the two PC half lanterns and retain the inverse pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

18. C13.HL2.2 | H | SPHERE SUPPORT CHART



After

II - step 060

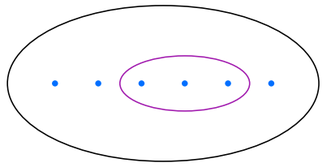
F5 down past F6

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

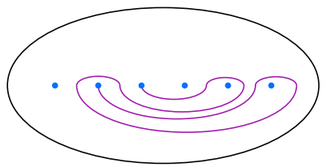
3 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

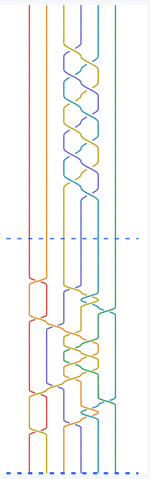
Before

4. F5 | T3^2 | SPHERE SUPPORT CHART



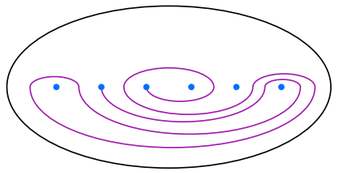
5. F6 | H | SPHERE SUPPORT CHART



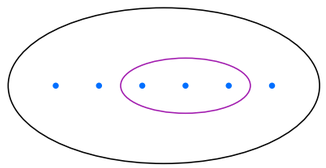


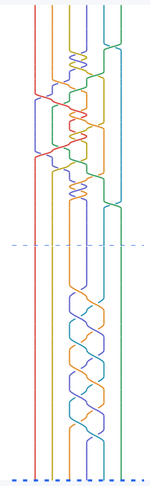
After

4. F6 | H | SPHERE SUPPORT CHART



5. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 061

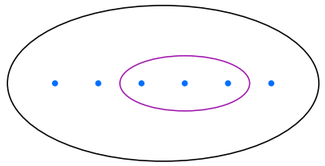
F5 down past new.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

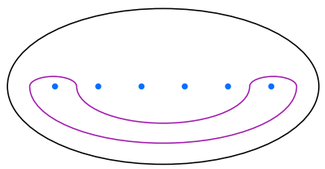
4 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

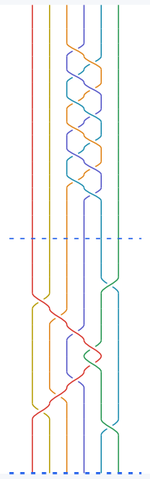
Before

5. F5 | T3^2 | SPHERE SUPPORT CHART



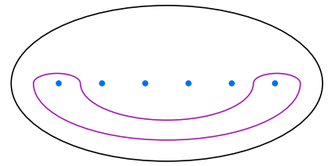
6. new.P34 | T2 | SPHERE SUPPORT CHART



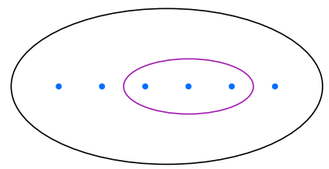


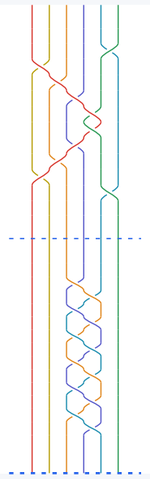
After

5. new.P34 | T2 | SPHERE SUPPORT CHART



6. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 062

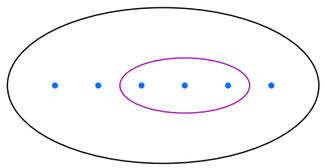
F5 down past new.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

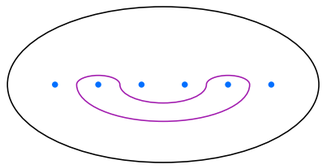
5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

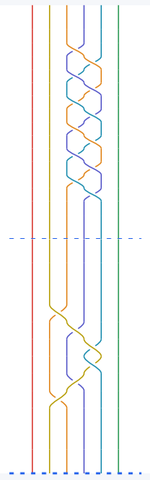
Before

6. F5 | T3^2 | SPHERE SUPPORT CHART



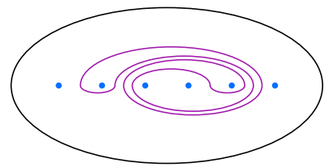
7. new.P26 | T2 | SPHERE SUPPORT CHART



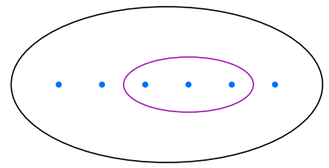


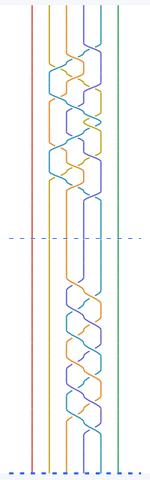
After

6. new.P26 | T2 | SPHERE SUPPORT CHART



7. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 063

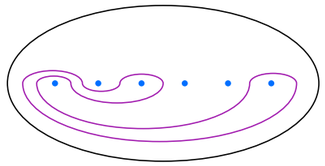
F5 down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

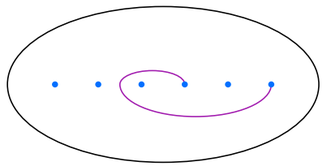
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

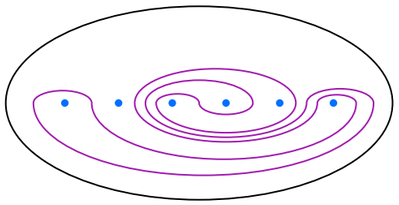
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



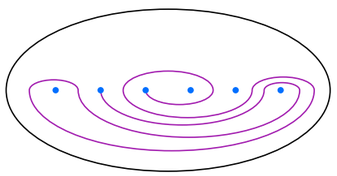
2. F3 | H



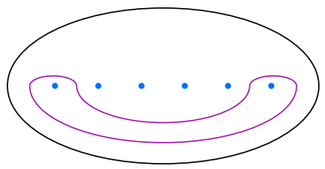
3. F4 | T3^2 | SPHERE SUPPORT CHART

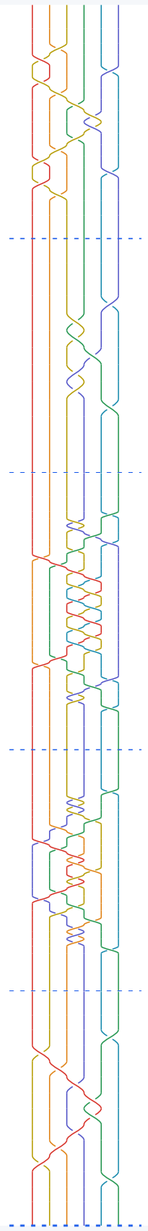


4. F6 | H | SPHERE SUPPORT CHART



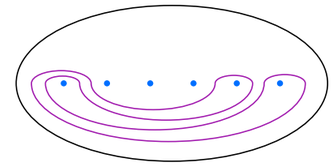
5. new.P34 | T2 | SPHERE SUPPORT CHART



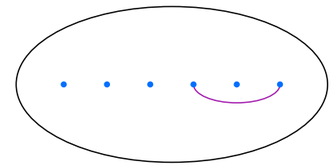


After

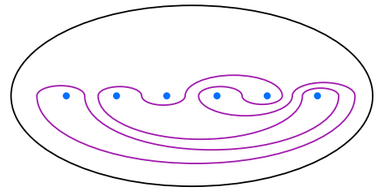
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



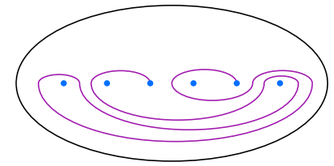
2. F3 | H



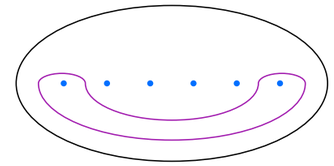
3. F4 | T3^2 | SPHERE SUPPORT CHART

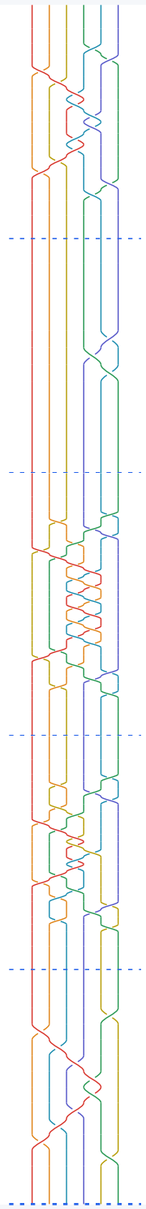


4. F6 | H | SPHERE SUPPORT CHART



5. new.P34 | T2 | SPHERE SUPPORT CHART





II - step 063 - continued

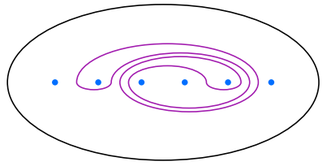
F5 down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

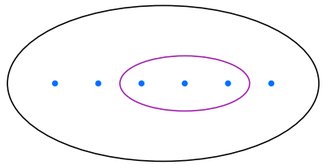
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

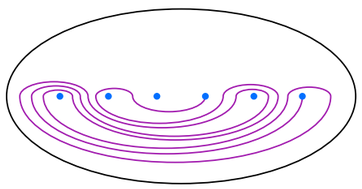
6. new.P26 | T2 | SPHERE SUPPORT CHART



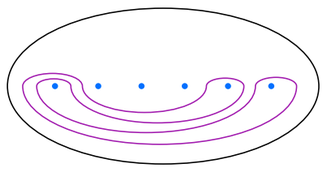
7. F5 | T3^2 | SPHERE SUPPORT CHART



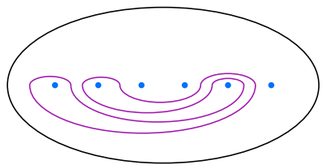
8. F9 | H | SPHERE SUPPORT CHART

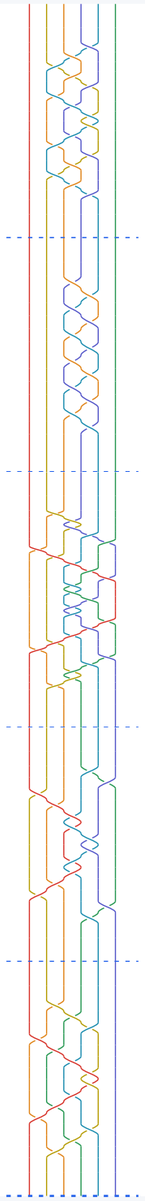


9. residual.P34 | T2 | SPHERE SUPPORT CHART



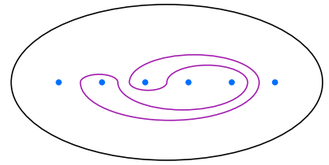
10. residual.P26 | T2 | SPHERE SUPPORT CHART



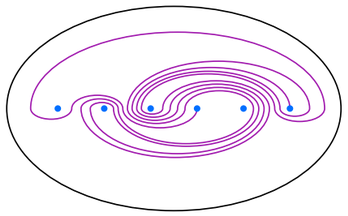


After

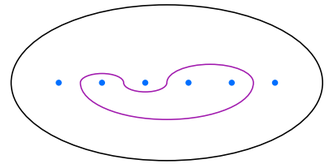
6. new.P26 | T2 | SPHERE SUPPORT CHART



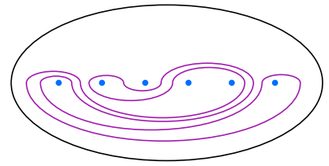
7. F9 | H | SPHERE SUPPORT CHART



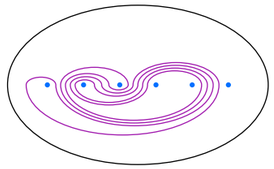
8. F5 | T3^2 | SPHERE SUPPORT CHART

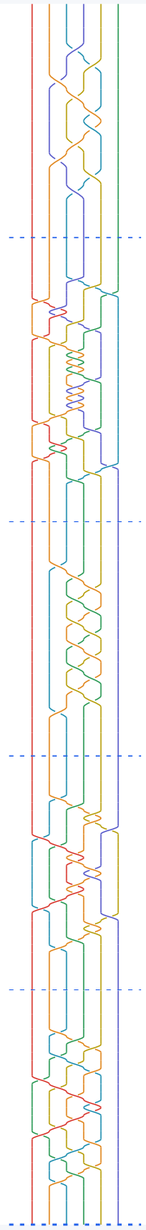


9. residual.P34 | T2 | SPHERE SUPPORT CHART



10. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 063 - continued

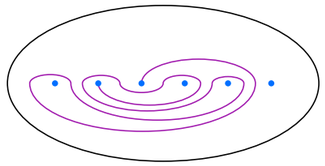
F5 down past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

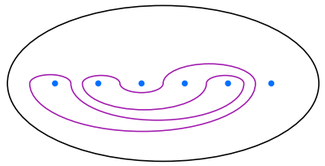
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

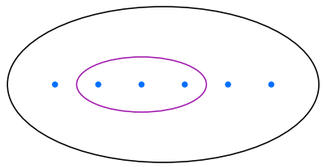
11. F11 | H | SPHERE SUPPORT CHART



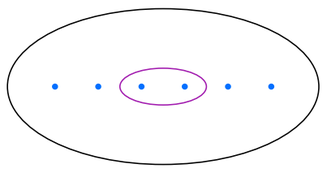
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

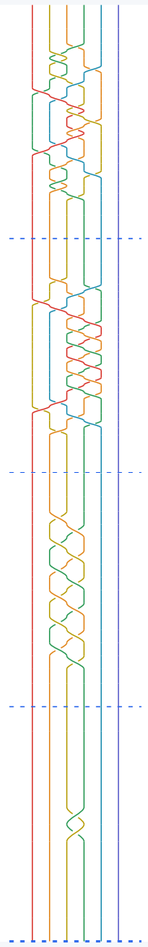


13. C13.recombined | T3^2 | SPHERE SUPPORT CHART



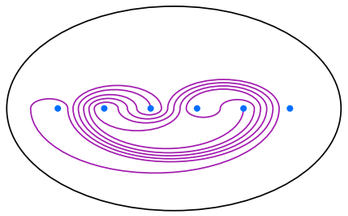
14. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART



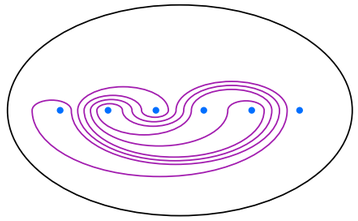


After

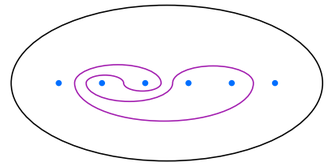
11. F11 | H | SPHERE SUPPORT CHART



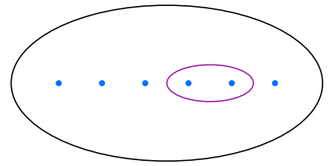
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

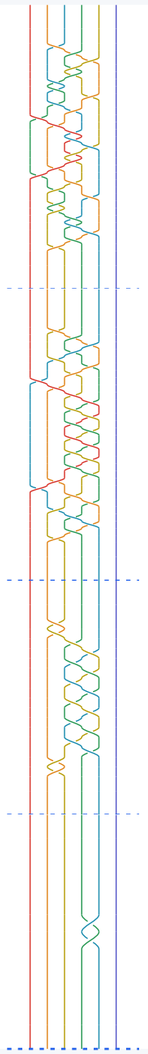


13. C13.recombined | T3^2 | SPHERE SUPPORT CHART



14. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART





II - step 064

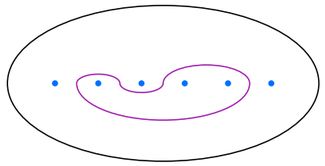
F5 down past residual.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

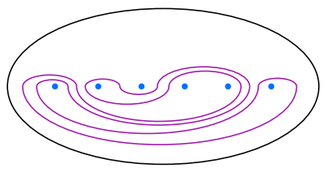
7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

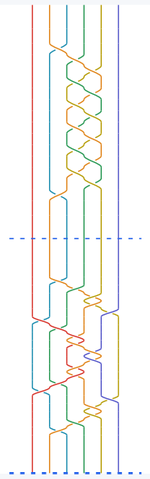
Before

8. F5 | T3^2 | SPHERE SUPPORT CHART



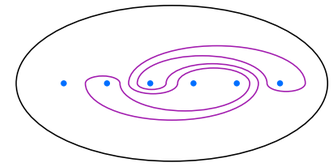
9. residual.P34 | T2 | SPHERE SUPPORT CHART



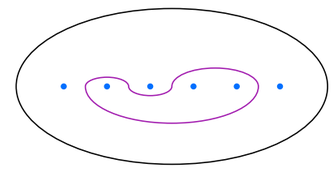


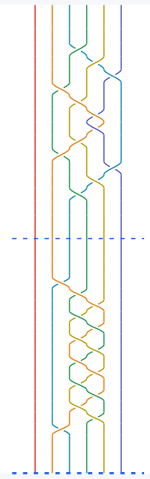
After

8. residual.P34 | T2 | SPHERE SUPPORT CHART



9. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 065

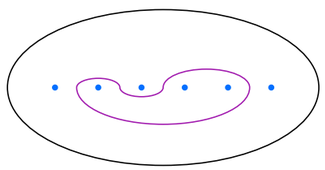
F5 down past residual.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

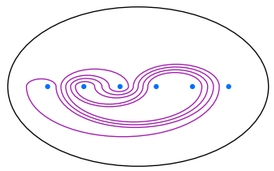
8 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

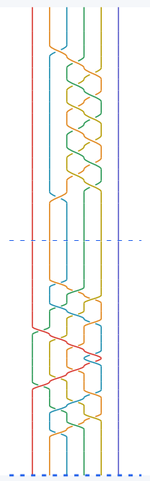
Before

9. F5 | T3^2 | SPHERE SUPPORT CHART



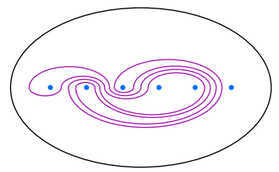
10. residual.P26 | T2 | SPHERE SUPPORT CHART



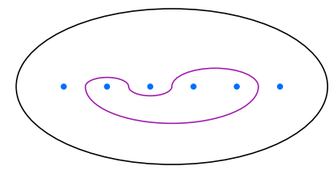


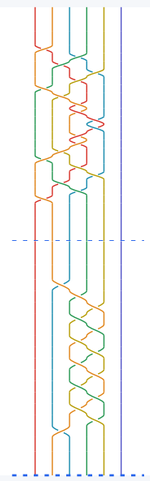
After

9. residual.P26 | T2 | SPHERE SUPPORT CHART



10. F5 | T3^2 | SPHERE SUPPORT CHART





II - step 066

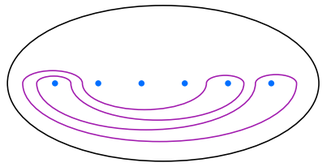
F5 down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

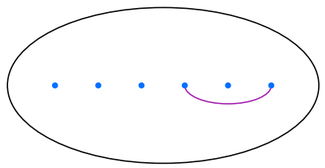
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

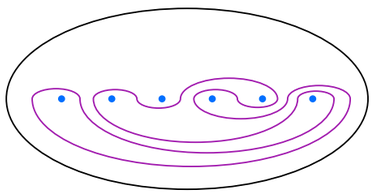
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



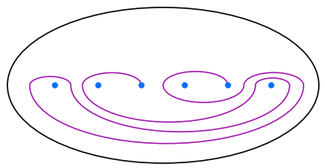
2. F3 | H



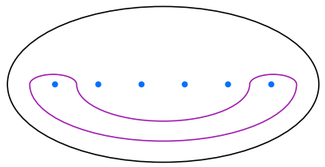
3. F4 | T3^2 | SPHERE SUPPORT CHART

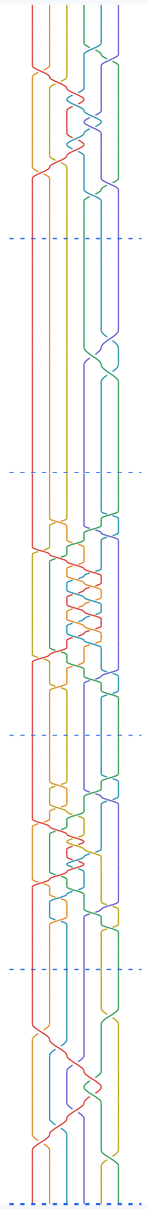


4. F6 | H | SPHERE SUPPORT CHART



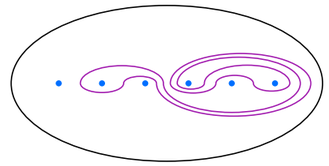
5. new.P34 | T2 | SPHERE SUPPORT CHART



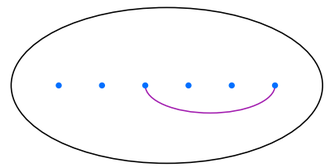


After

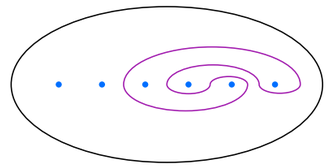
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



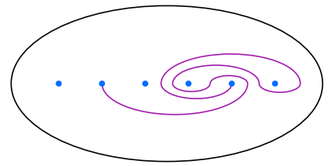
2. F3 | H



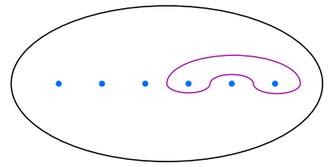
3. F4 | T3^2 | SPHERE SUPPORT CHART

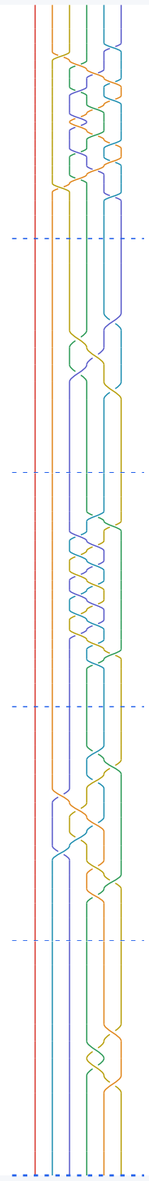


4. F6 | H | SPHERE SUPPORT CHART



5. new.P34 | T2





II - step 066 - continued

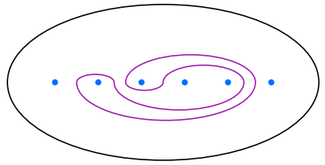
F5 down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

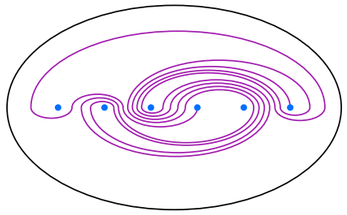
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

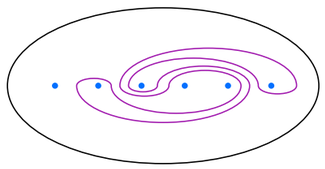
6. new.P26 | T2 | SPHERE SUPPORT CHART



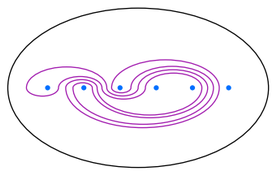
7. F9 | H | SPHERE SUPPORT CHART



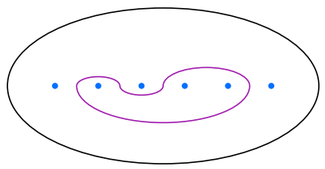
8. residual.P34 | T2 | SPHERE SUPPORT CHART

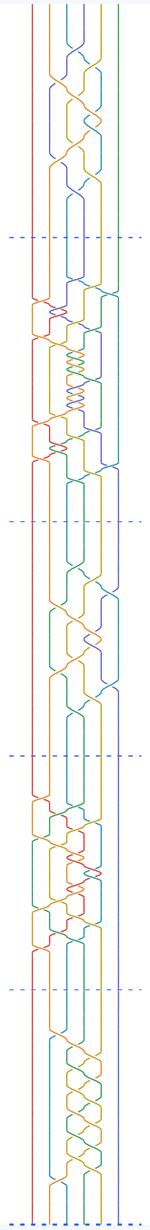


9. residual.P26 | T2 | SPHERE SUPPORT CHART



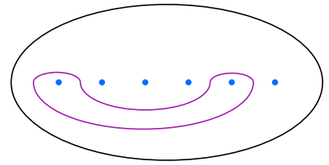
10. F5 | T3^2 | SPHERE SUPPORT CHART



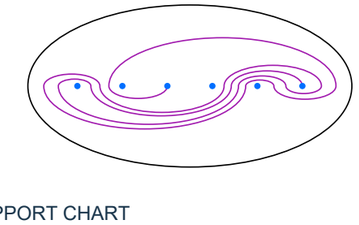


After

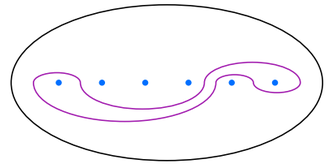
6. new.P26 | T2 | SPHERE SUPPORT CHART



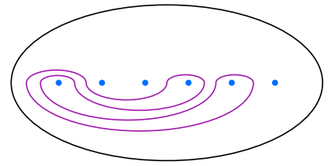
7. F9 | H | SPHERE SUPPORT CHART



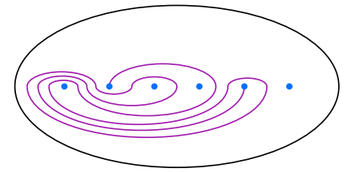
8. residual.P34 | T2 | SPHERE SUPPORT CHART

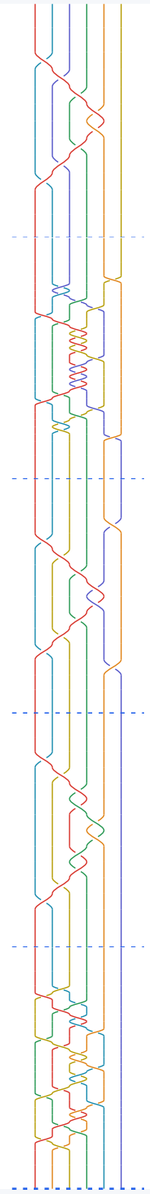


9. residual.P26 | T2 | SPHERE SUPPORT CHART



10. F11 | H | SPHERE SUPPORT CHART





II - step 066 - continued

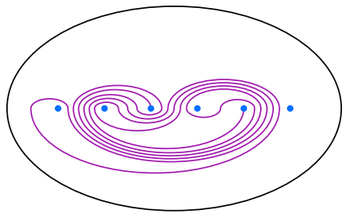
F5 down past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

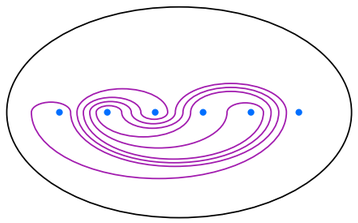
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

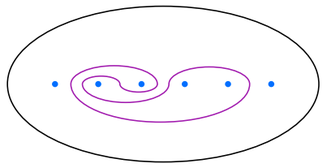
11. F11 | H | SPHERE SUPPORT CHART



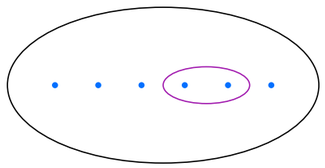
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

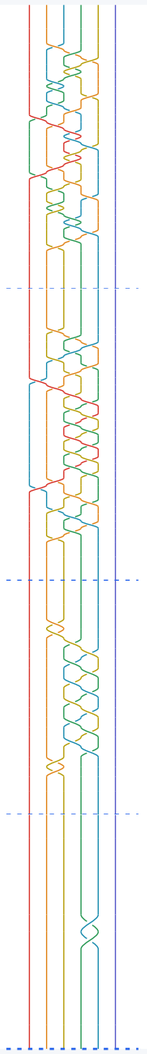


13. C13.recombined | T3^2 | SPHERE SUPPORT CHART



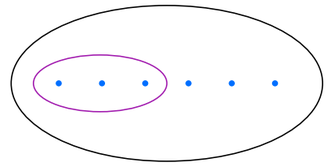
14. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART



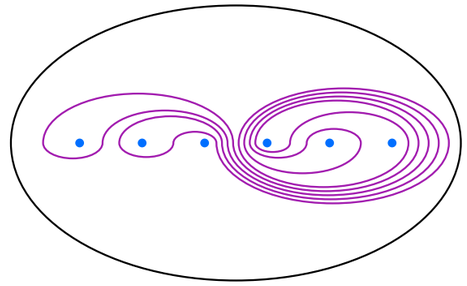


After

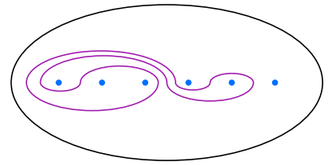
11. F5 | T3^2 | SPHERE SUPPORT CHART



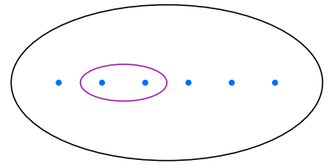
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

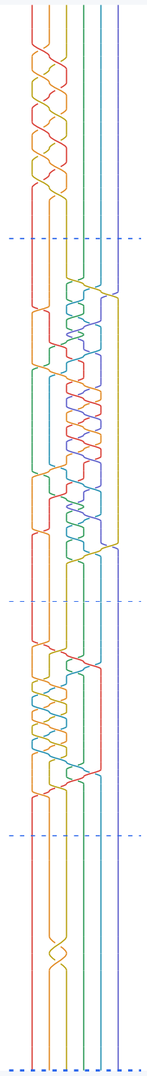


13. C13.recombined | T3^2 | SPHERE SUPPORT CHART



14. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART





II - step 067

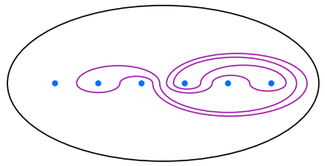
F5 down past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

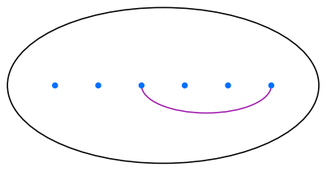
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

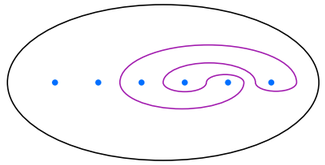
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



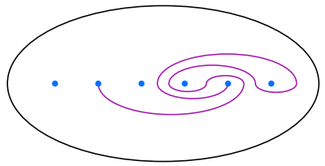
2. F3 | H



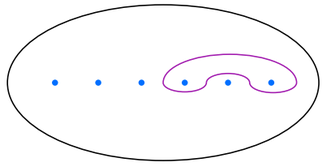
3. F4 | T3^2 | SPHERE SUPPORT CHART

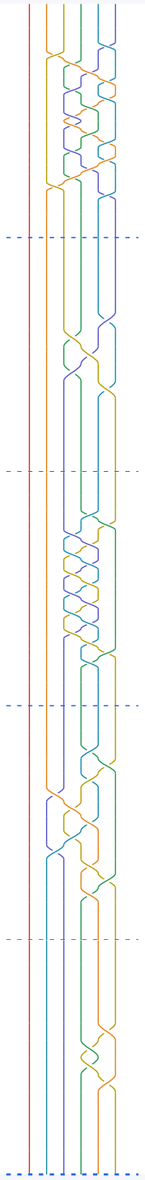


4. F6 | H | SPHERE SUPPORT CHART



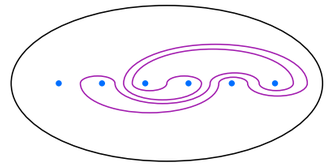
5. new.P34 | T2



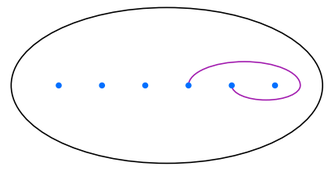


After

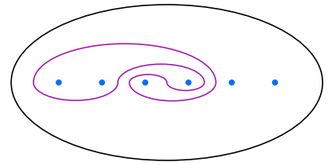
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



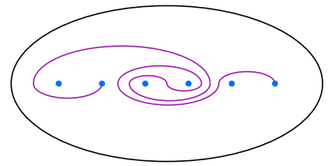
2. F3 | H



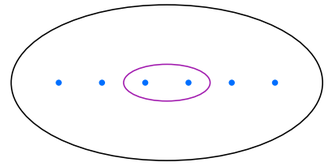
3. F4 | T3^2 | SPHERE SUPPORT CHART

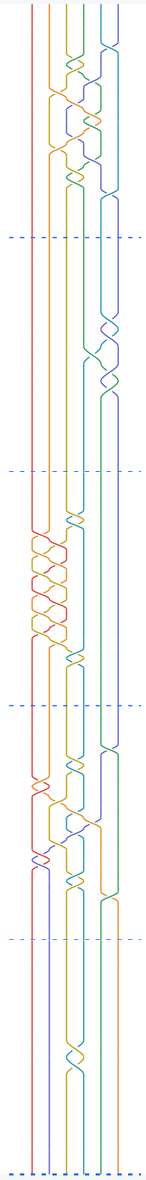


4. F6 | H



5. new.P34 | T2





II - step 067 - continued

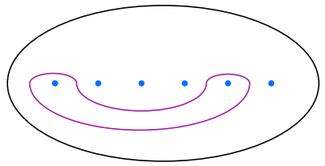
F5 down past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

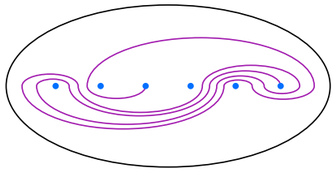
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

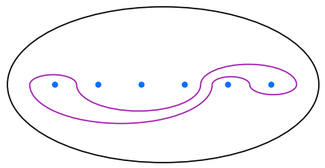
6. new.P26 | T2 | SPHERE SUPPORT CHART



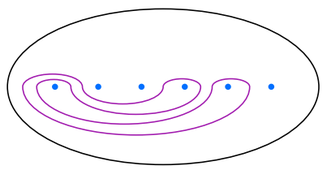
7. F9 | H | SPHERE SUPPORT CHART



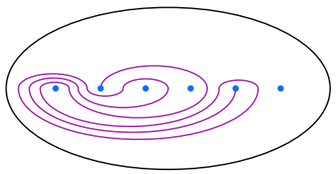
8. residual.P34 | T2 | SPHERE SUPPORT CHART

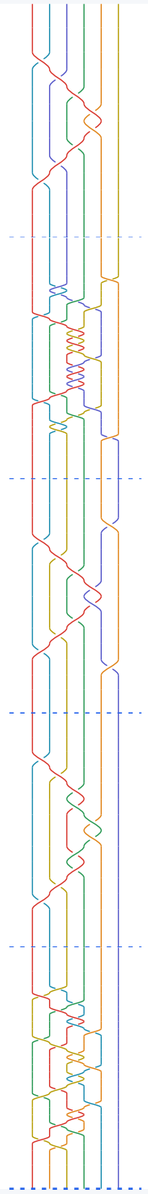


9. residual.P26 | T2 | SPHERE SUPPORT CHART



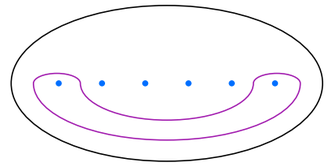
10. F11 | H | SPHERE SUPPORT CHART



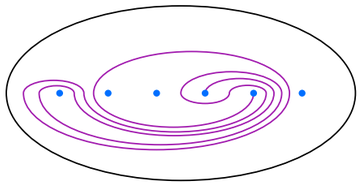


After

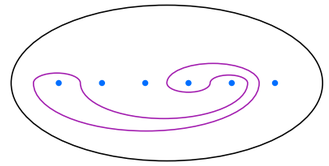
6. new.P26 | T2 | SPHERE SUPPORT CHART



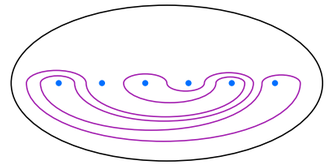
7. F9 | H | SPHERE SUPPORT CHART



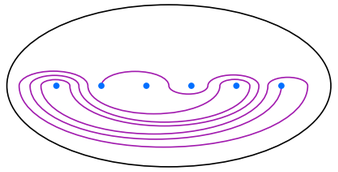
8. residual.P34 | T2 | SPHERE SUPPORT CHART

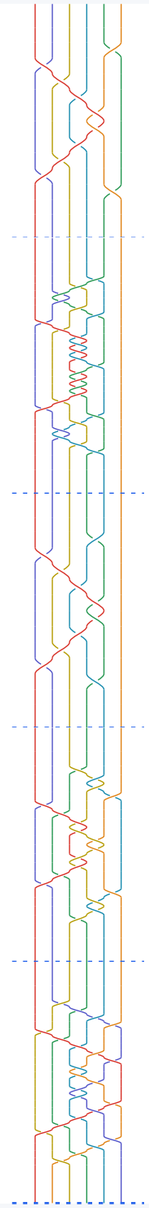


9. residual.P26 | T2 | SPHERE SUPPORT CHART



10. F11 | H | SPHERE SUPPORT CHART





II - step 067 - continued

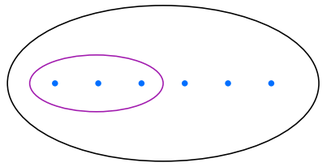
F5 down past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

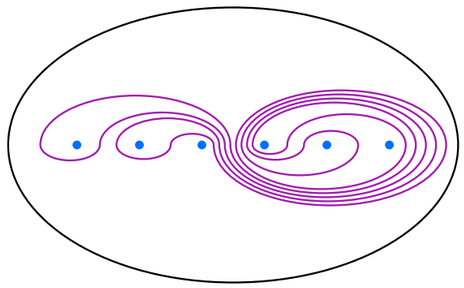
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

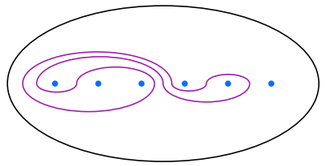
11. F5 | T3^2 | SPHERE SUPPORT CHART



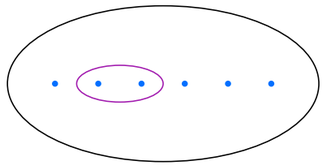
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

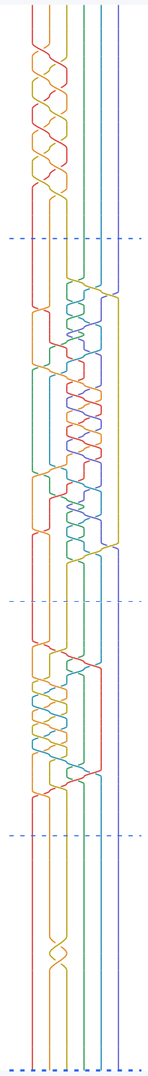


13. C13.recombined | T3^2 | SPHERE SUPPORT CHART



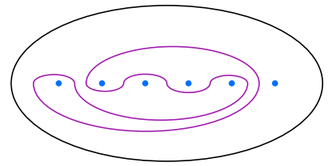
14. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART



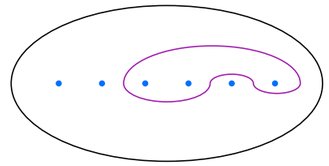


After

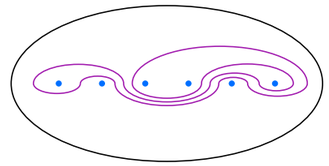
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



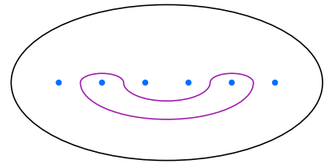
12. F5 | T3^2

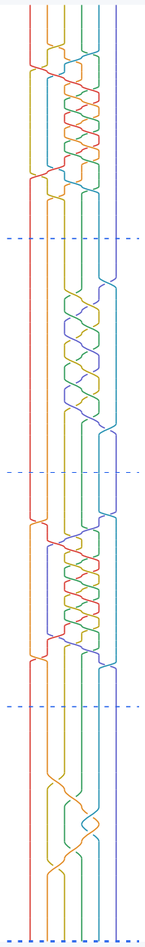


13. C13.recombined | T3^2 | SPHERE SUPPORT CHART



14. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART





II - step 068

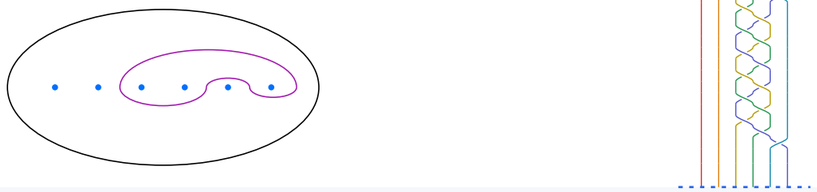
Rechart the whole C5 square along its complementary {1,3,5} triple

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

12. F5 | T3^2



After

12. C5.complement.square | T3^2 | SPHERE SUPPORT CHART



II - step 069

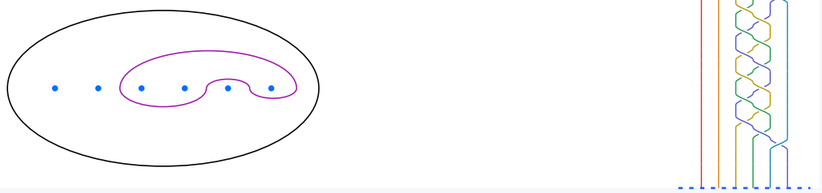
Use two cyclic PC roses, each ending in the P15 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

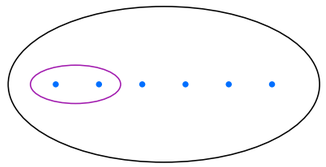
Before

12. C5.complement.square | T3^2 | SPHERE SUPPORT CHART

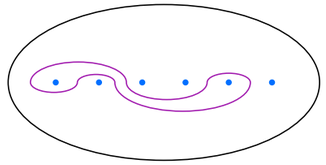


After

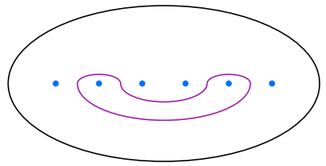
12. C5.rose.1.1 | T2



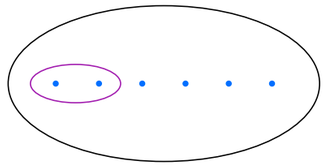
13. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



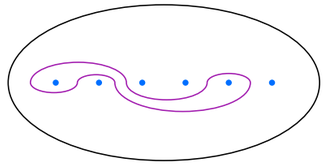
14. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART

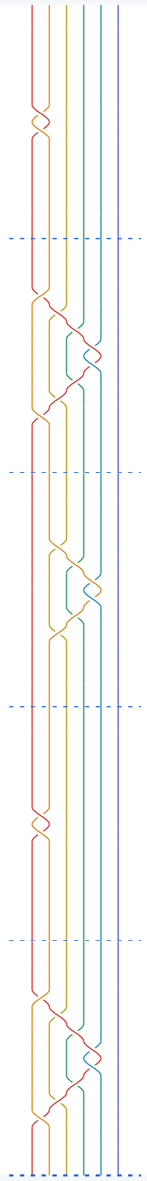


15. C5.rose.2.1 | T2



16. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART





II - step 069 - continued

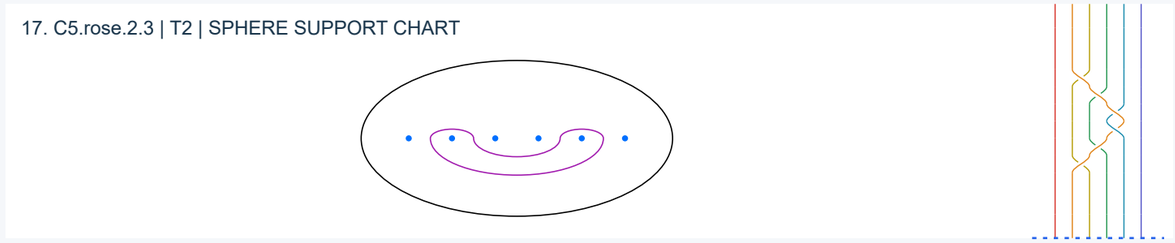
Use two cyclic PC roses, each ending in the P15 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

After



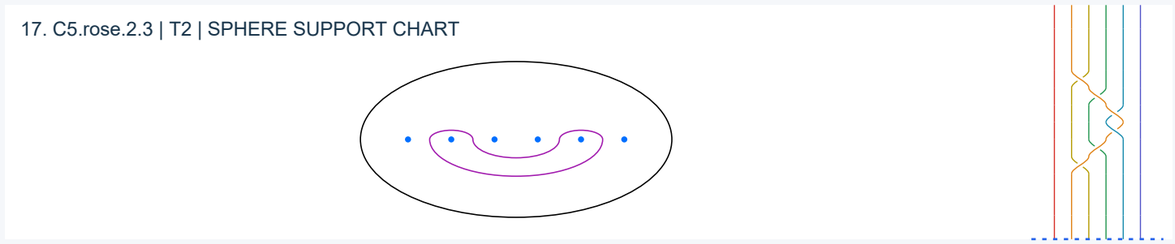
II - step 070

Rechart the last petal along the retained inverse pair

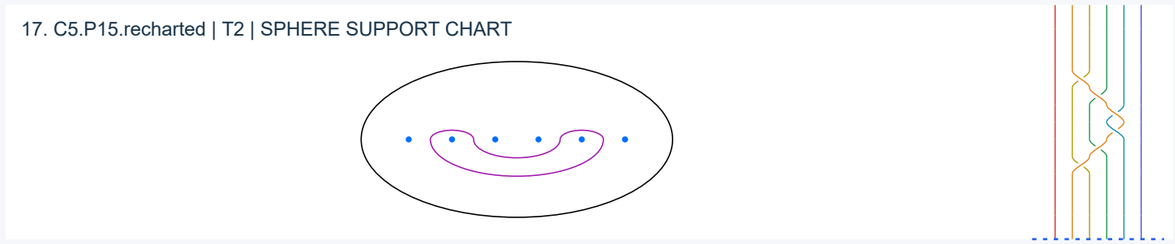
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 071

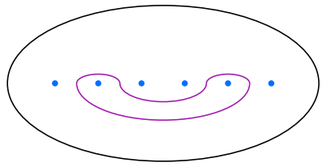
C5.P15.recharted down past C13.recombined

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

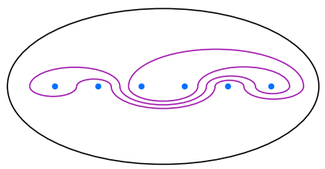
16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

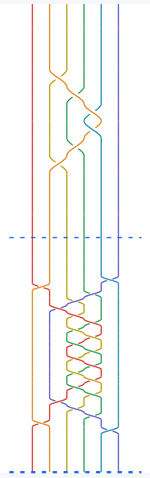
Before

17. C5.P15.recharted | T2 | SPHERE SUPPORT CHART



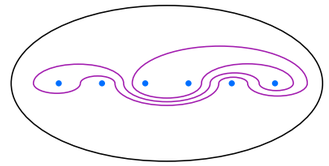
18. C13.recombined | T3^2 | SPHERE SUPPORT CHART



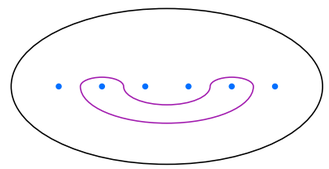


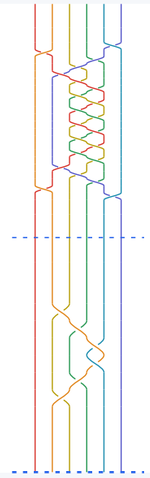
After

17. C13.recombined | T3^2 | SPHERE SUPPORT CHART



18. C5.P15.recharted | T2 | SPHERE SUPPORT CHART





II - step 072

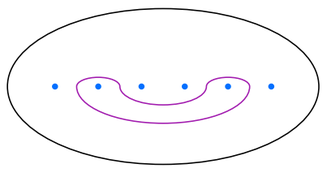
Cancel the last petal against the inverse pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

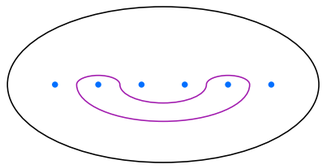
17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

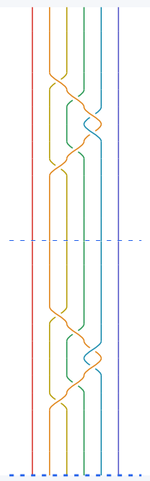
Before

18. C5.P15.recharted | T2 | SPHERE SUPPORT CHART



19. inverse.P15 | T2^-1 | SPHERE SUPPORT CHART





After

Identity / no factors in this interval.

II - step 073

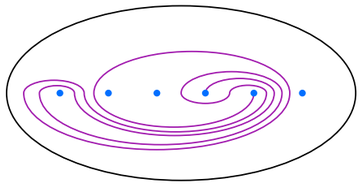
F9 down past residual.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

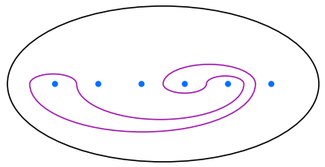
6 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

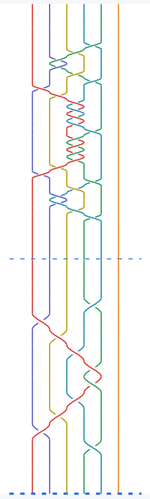
Before

7. F9 | H | SPHERE SUPPORT CHART



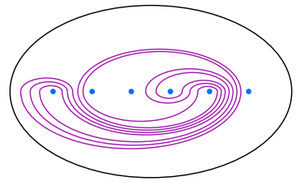
8. residual.P34 | T2 | SPHERE SUPPORT CHART



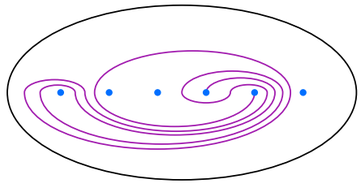


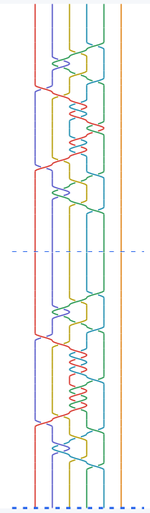
After

7. residual.P34 | T2 | SPHERE SUPPORT CHART



8. F9 | H | SPHERE SUPPORT CHART





II - step 074

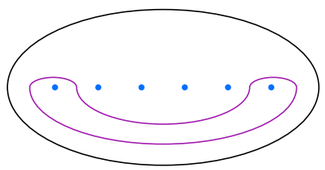
new.P26 down past residual.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

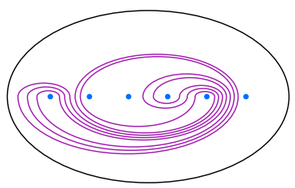
5 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

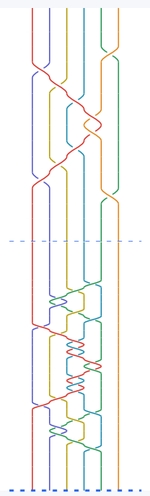
Before

6. new.P26 | T2 | SPHERE SUPPORT CHART



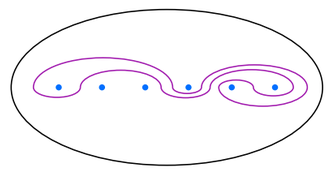
7. residual.P34 | T2 | SPHERE SUPPORT CHART



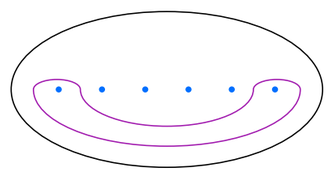


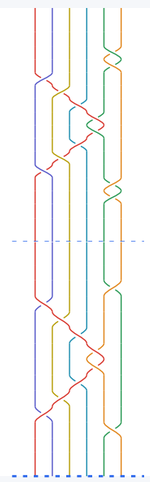
After

6. residual.P34 | T2 | SPHERE SUPPORT CHART



7. new.P26 | T2 | SPHERE SUPPORT CHART





II - step 075

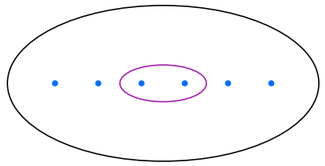
new.P34 down past residual.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

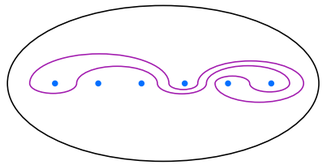
4 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

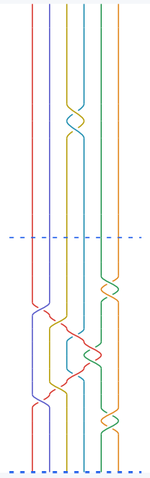
Before

5. new.P34 | T2



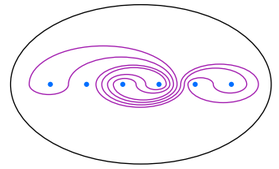
6. residual.P34 | T2 | SPHERE SUPPORT CHART



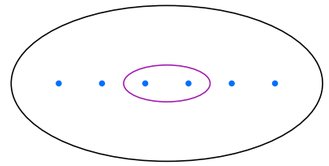


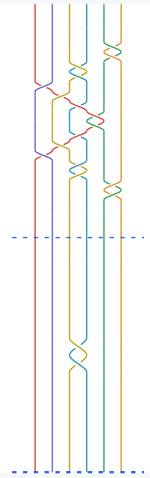
After

5. residual.P34 | T2 | SPHERE SUPPORT CHART



6. new.P34 | T2





II - step 076

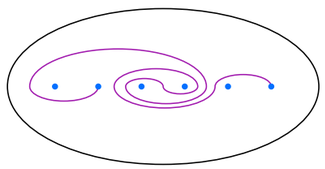
F6 down past residual.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

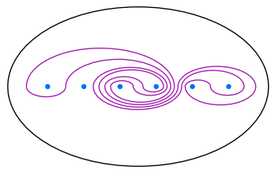
3 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

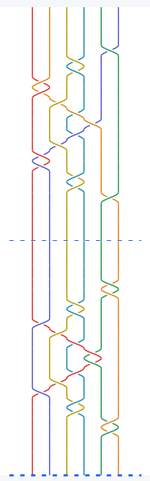
Before

4. F6 | H



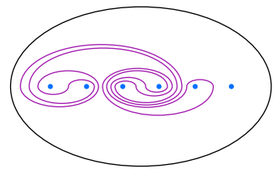
5. residual.P34 | T2 | SPHERE SUPPORT CHART



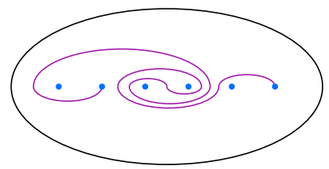


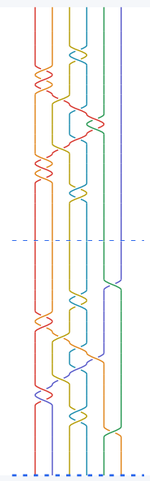
After

4. residual.P34 | T2 | SPHERE SUPPORT CHART



5. F6 | H





II - step 077

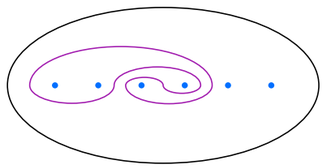
F4 down past residual.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

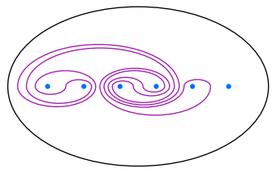
2 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

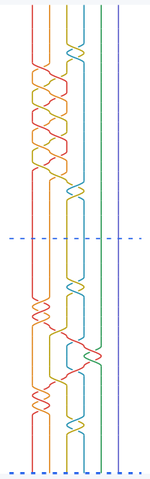
Before

3. F4 | T3^2 | SPHERE SUPPORT CHART



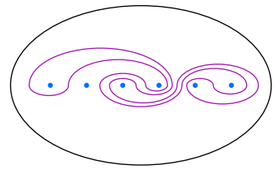
4. residual.P34 | T2 | SPHERE SUPPORT CHART



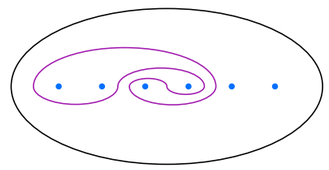


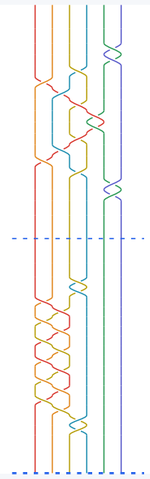
After

3. residual.P34 | T2 | SPHERE SUPPORT CHART



4. F4 | T3^2 | SPHERE SUPPORT CHART





II - step 078

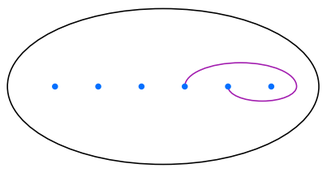
F3 down past residual.P34

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

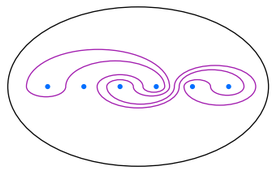
1 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

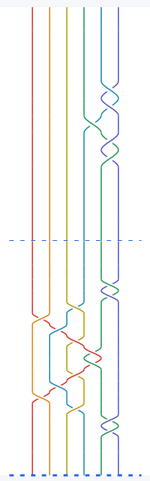
Before

2. F3 | H



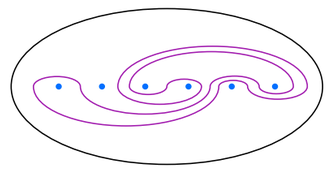
3. residual.P34 | T2 | SPHERE SUPPORT CHART



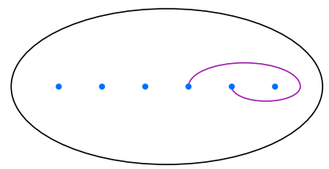


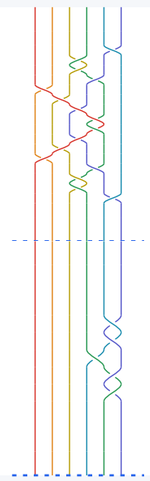
After

2. residual.P34 | T2 | SPHERE SUPPORT CHART



3. F3 | H





II - step 079

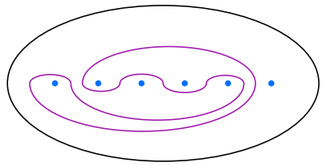
C5.rose.1.1 up past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

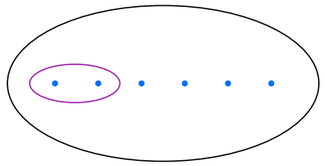
10 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

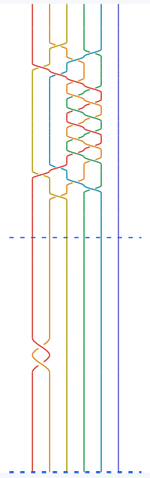
Before

11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



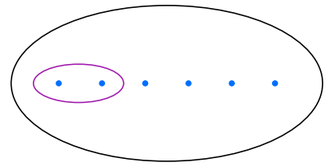
12. C5.rose.1.1 | T2



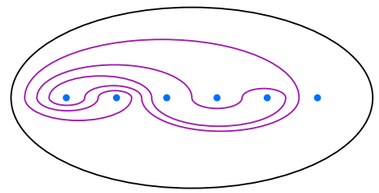


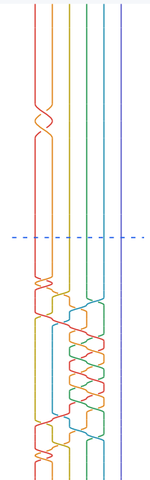
After

11. C5.rose.1.1 | T2



12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART





II - step 080

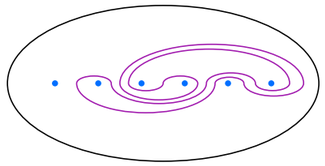
C5.rose.1.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

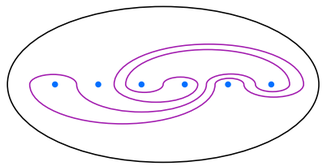
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

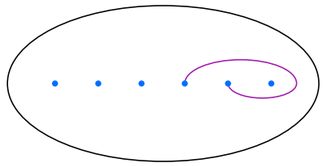
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



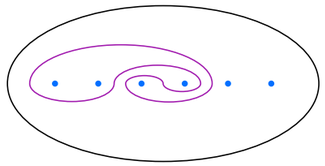
2. residual.P34 | T2 | SPHERE SUPPORT CHART



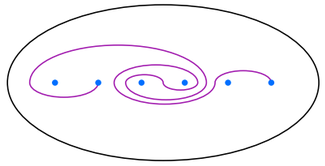
3. F3 | H

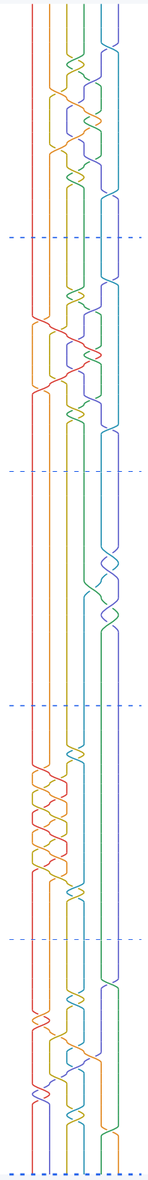


4. F4 | T3^2 | SPHERE SUPPORT CHART



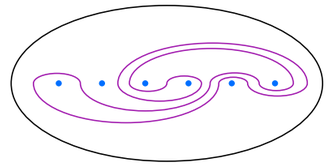
5. F6 | H



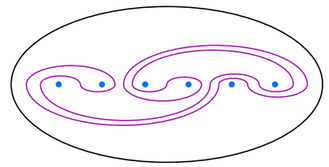


After

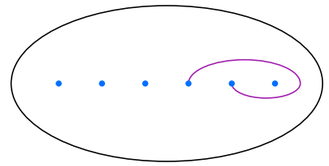
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



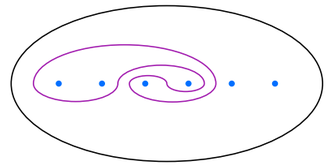
2. residual.P34 | T2 | SPHERE SUPPORT CHART



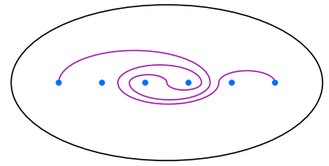
3. F3 | H

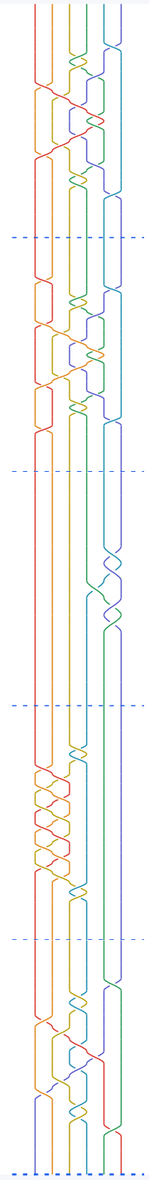


4. F4 | T3^2 | SPHERE SUPPORT CHART



5. F6 | H





II - step 080 - continued

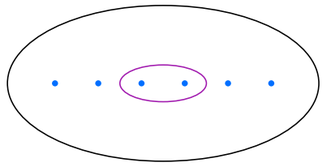
C5.rose.1.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

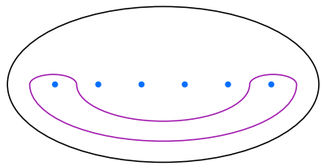
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

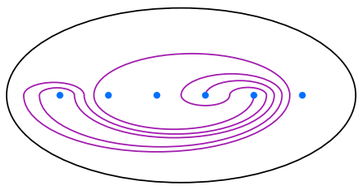
6. new.P34 | T2



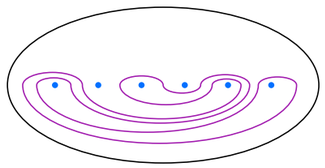
7. new.P26 | T2 | SPHERE SUPPORT CHART



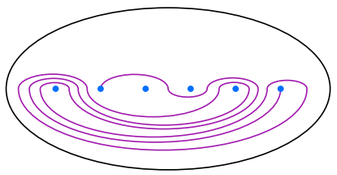
8. F9 | H | SPHERE SUPPORT CHART

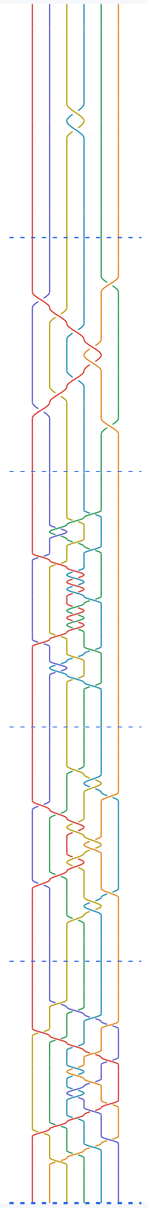


9. residual.P26 | T2 | SPHERE SUPPORT CHART



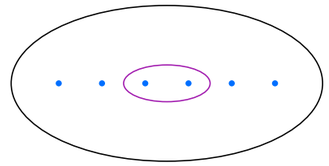
10. F11 | H | SPHERE SUPPORT CHART



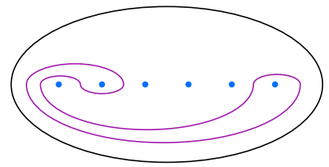


After

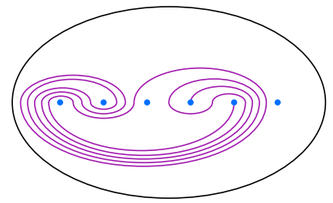
6. new.P34 | T2



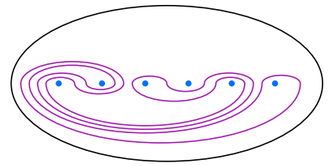
7. new.P26 | T2 | SPHERE SUPPORT CHART



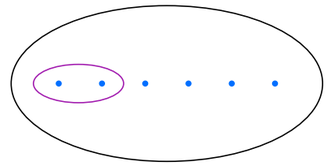
8. F9 | H | SPHERE SUPPORT CHART

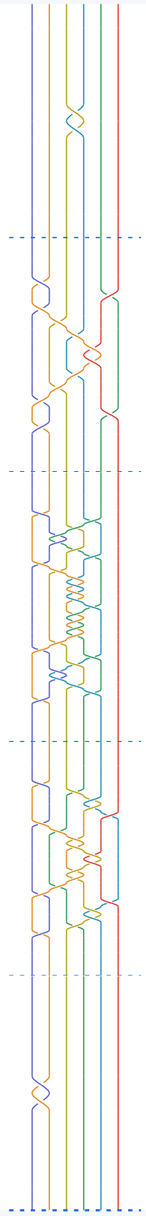


9. residual.P26 | T2 | SPHERE SUPPORT CHART



10. C5.rose.1.1 | T2





II - step 080 - continued

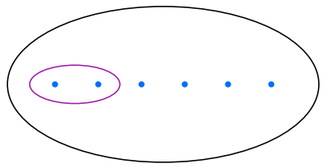
C5.rose.1.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

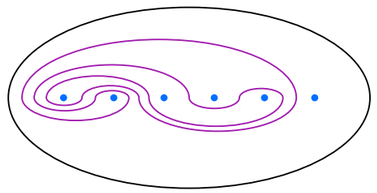
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

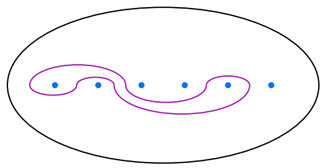
11. C5.rose.1.1 | T2



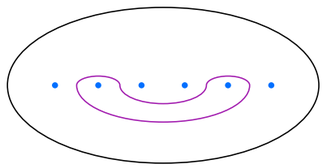
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



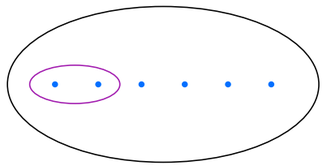
13. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

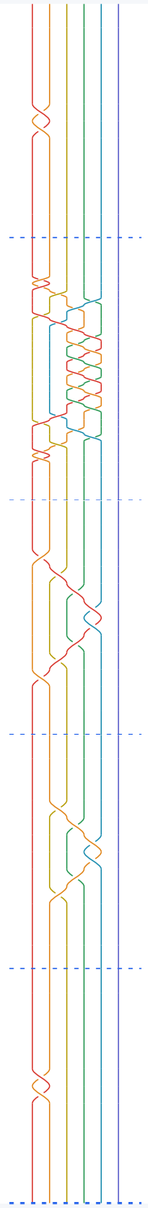


14. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



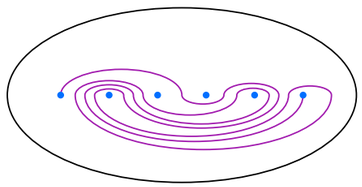
15. C5.rose.2.1 | T2



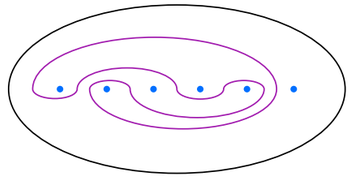


After

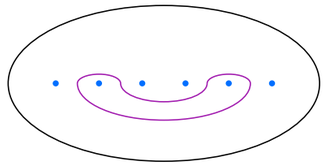
11. F11 | H | SPHERE SUPPORT CHART



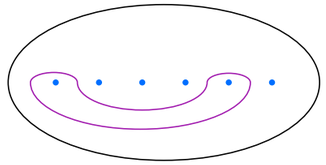
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



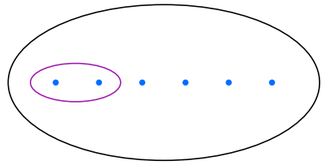
13. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

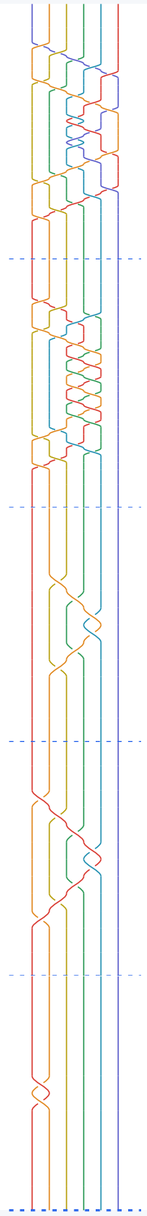


14. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



15. C5.rose.2.1 | T2





II - step 080 - continued

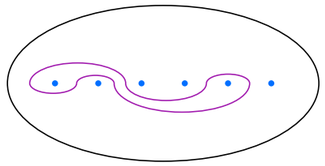
C5.rose.1.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

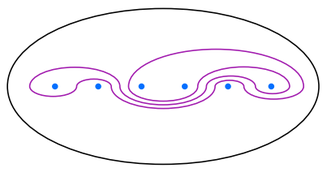
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

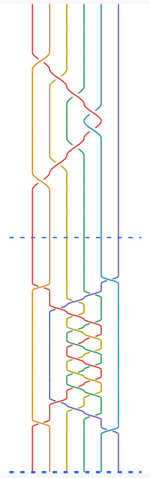
Before

16. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



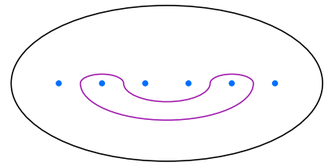
17. C13.recombined | T3^2 | SPHERE SUPPORT CHART



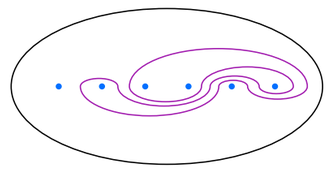


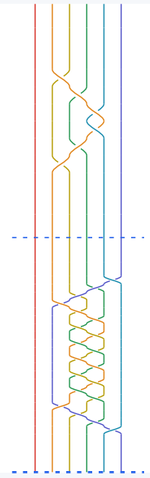
After

16. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



17. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 081

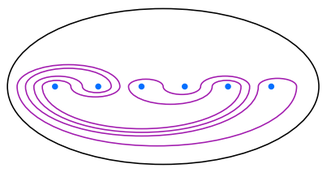
C5.rose.1.1 up past residual.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

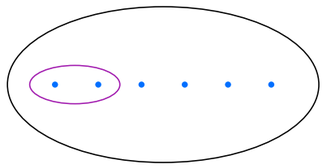
8 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

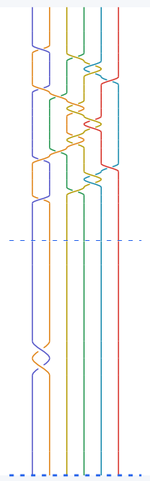
Before

9. residual.P26 | T2 | SPHERE SUPPORT CHART



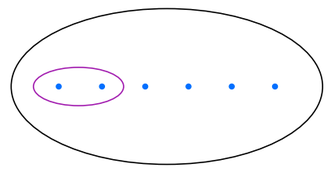
10. C5.rose.1.1 | T2



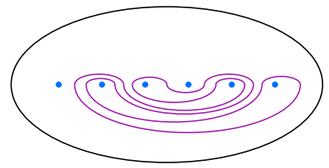


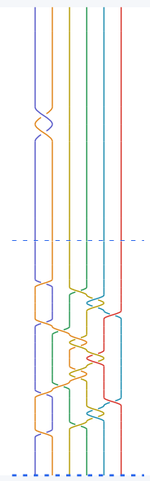
After

9. C5.rose.1.1 | T2



10. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 082

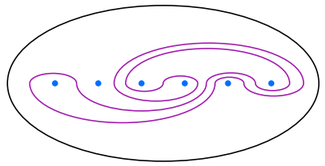
C5.rose.1.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

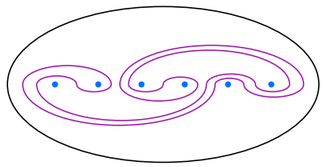
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

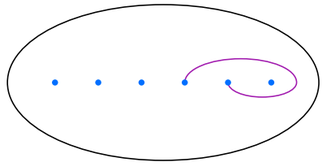
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



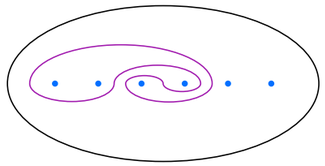
2. residual.P34 | T2 | SPHERE SUPPORT CHART



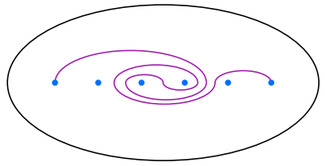
3. F3 | H

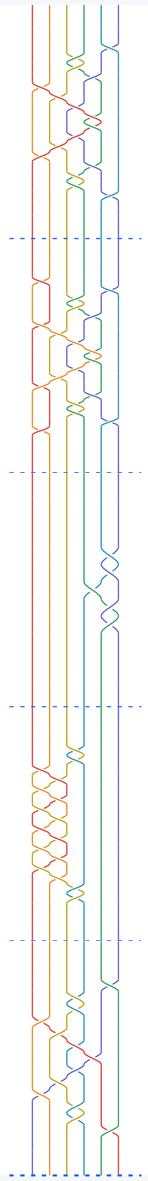


4. F4 | T3^2 | SPHERE SUPPORT CHART



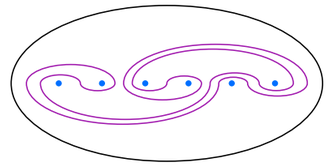
5. F6 | H



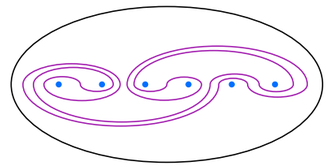


After

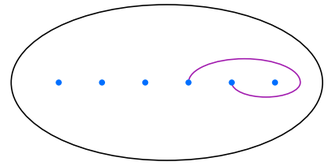
1. prefix.P45 | T2 | SPHERE SUPPORT CHART



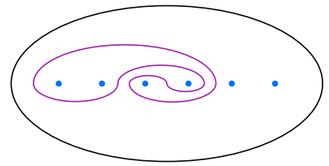
2. residual.P34 | T2 | SPHERE SUPPORT CHART



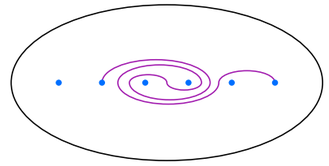
3. F3 | H

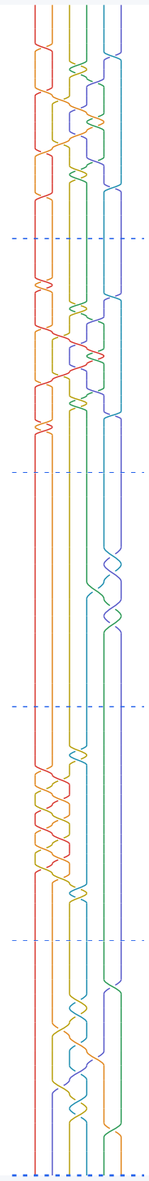


4. F4 | T3^2 | SPHERE SUPPORT CHART



5. F6 | H





II - step 082 - continued

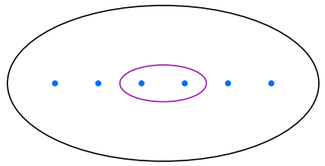
C5.rose.1.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

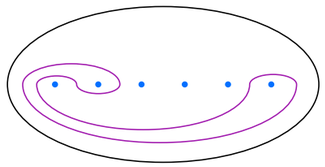
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

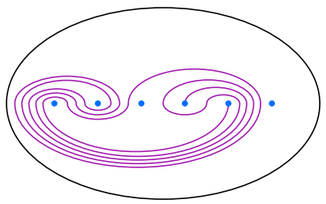
6. new.P34 | T2



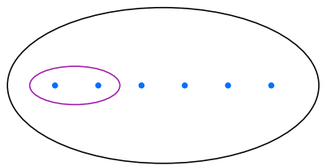
7. new.P26 | T2 | SPHERE SUPPORT CHART



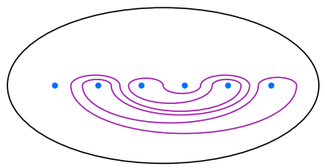
8. F9 | H | SPHERE SUPPORT CHART

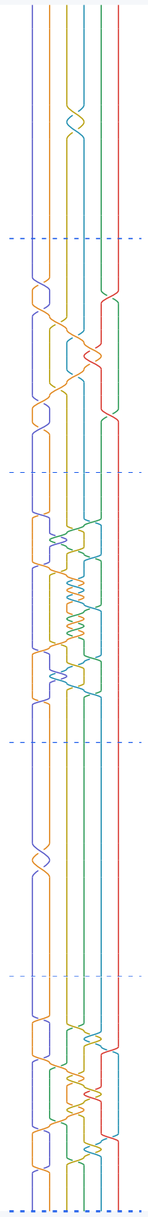


9. C5.rose.1.1 | T2



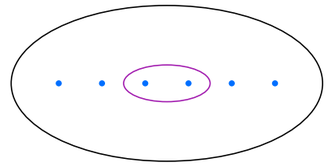
10. residual.P26 | T2 | SPHERE SUPPORT CHART



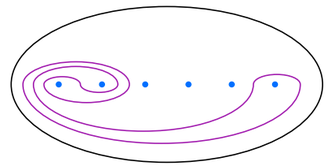


After

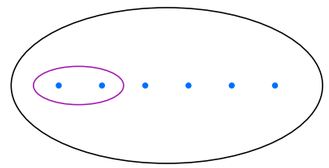
6. new.P34 | T2



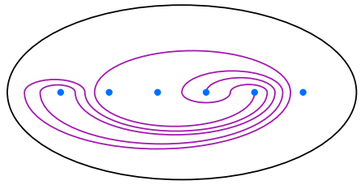
7. new.P26 | T2 | SPHERE SUPPORT CHART



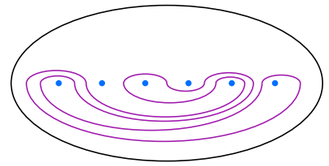
8. C5.rose.1.1 | T2

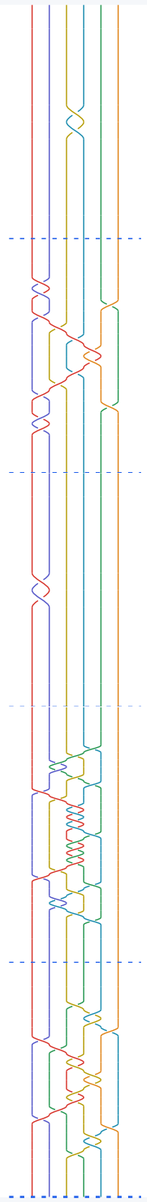


9. F9 | H | SPHERE SUPPORT CHART



10. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 082 - continued

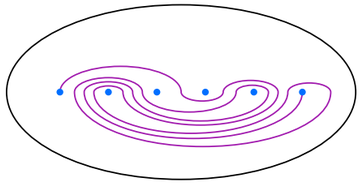
C5.rose.1.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

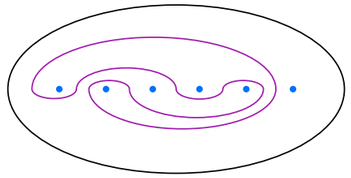
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

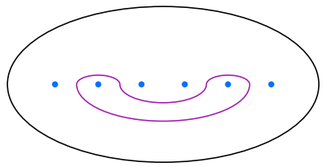
11. F11 | H | SPHERE SUPPORT CHART



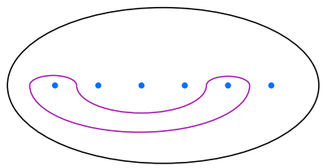
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



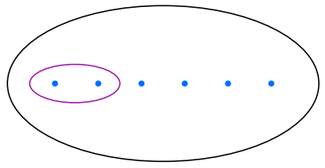
13. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

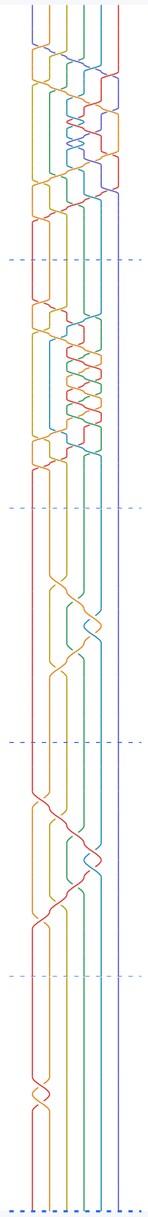


14. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



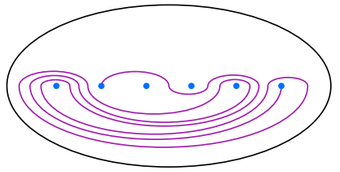
15. C5.rose.2.1 | T2



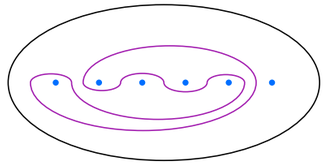


After

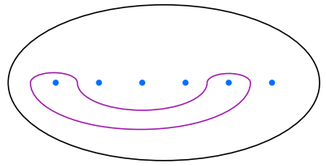
11. F11 | H | SPHERE SUPPORT CHART



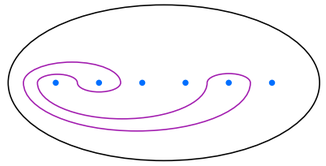
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



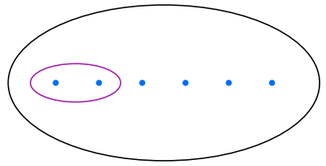
13. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

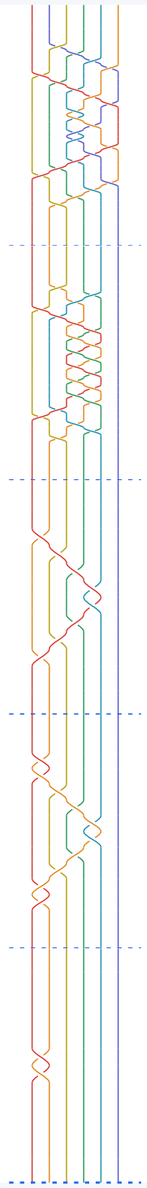


14. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



15. C5.rose.2.1 | T2





II - step 082 - continued

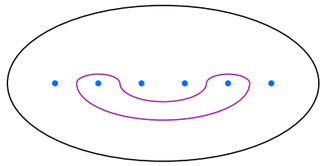
C5.rose.1.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

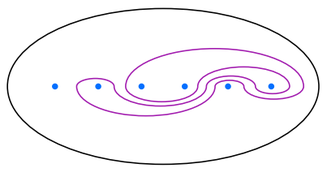
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

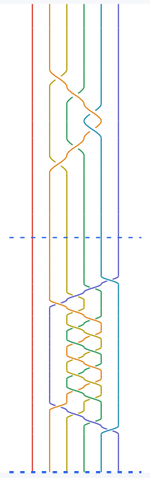
Before

16. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



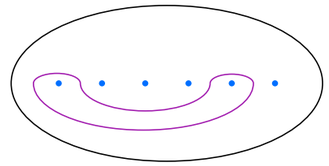
17. C13.recombined | T3^2 | SPHERE SUPPORT CHART



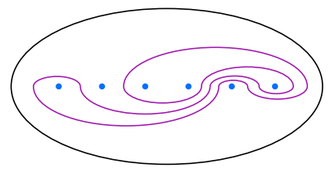


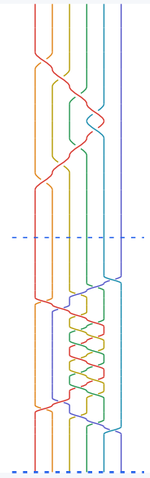
After

16. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



17. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 083

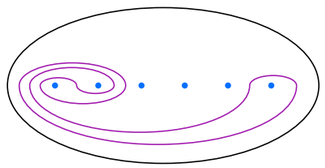
C5.rose.1.1 up past new.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

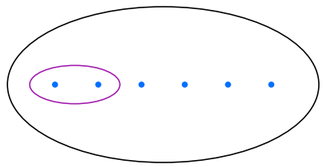
6 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

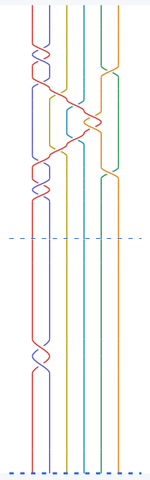
Before

7. new.P26 | T2 | SPHERE SUPPORT CHART



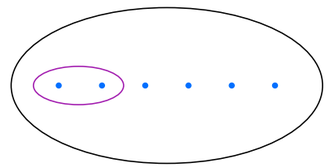
8. C5.rose.1.1 | T2



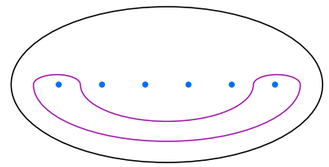


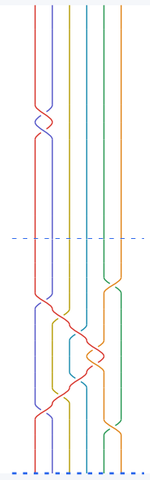
After

7. C5.rose.1.1 | T2



8. new.P26 | T2 | SPHERE SUPPORT CHART





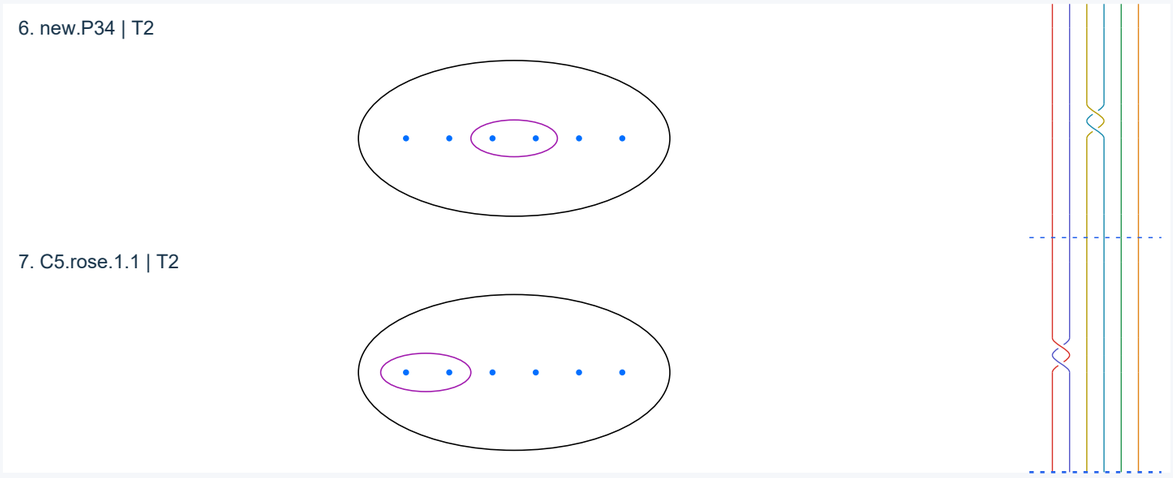
II - step 084

C5.rose.1.1 up past new.P34

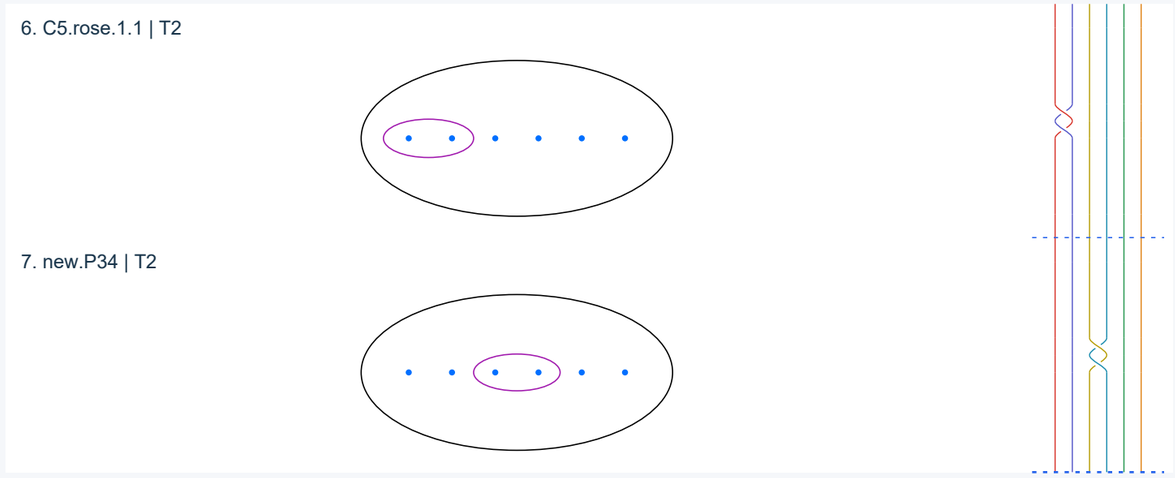
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



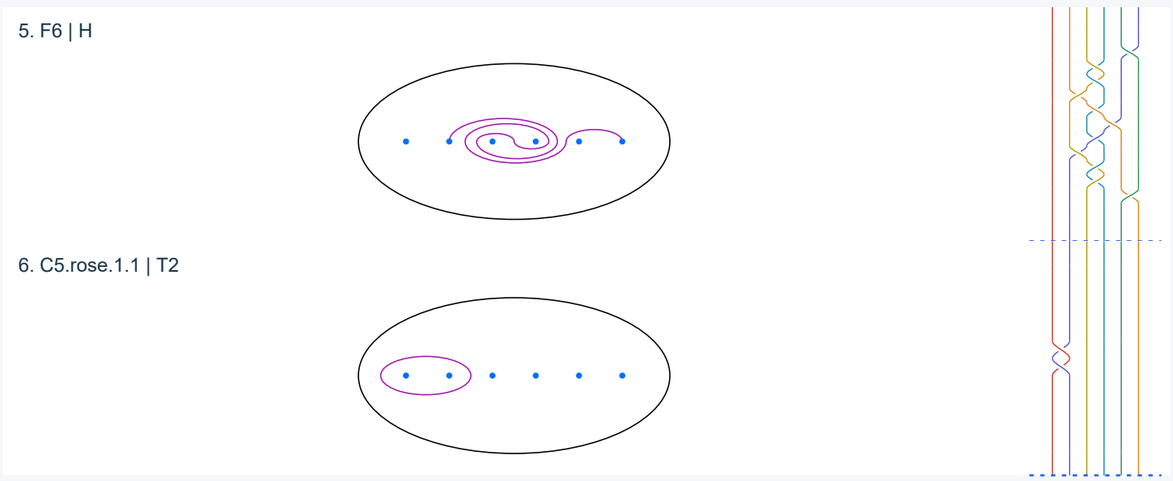
II - step 085

C5.rose.1.1 up past F6

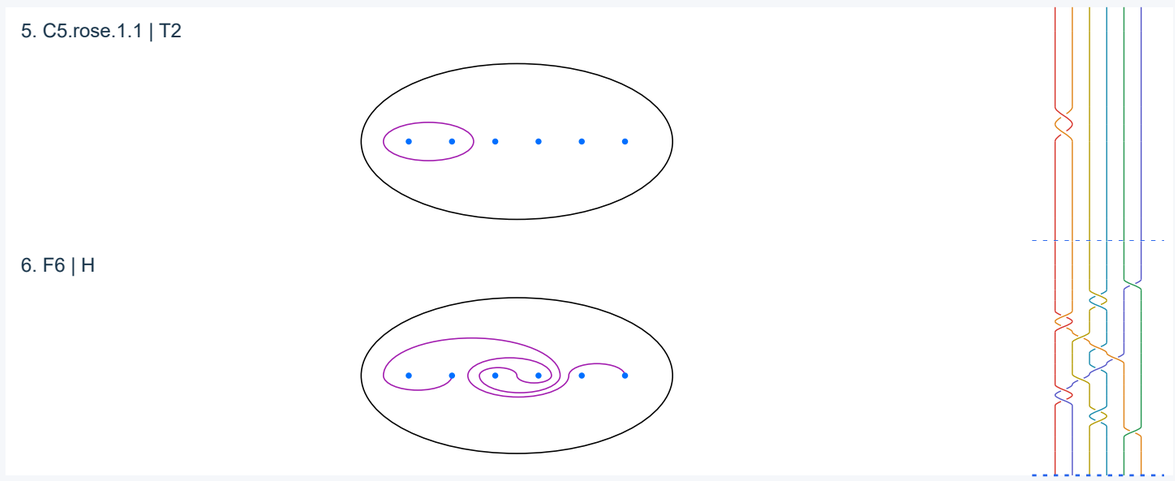
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



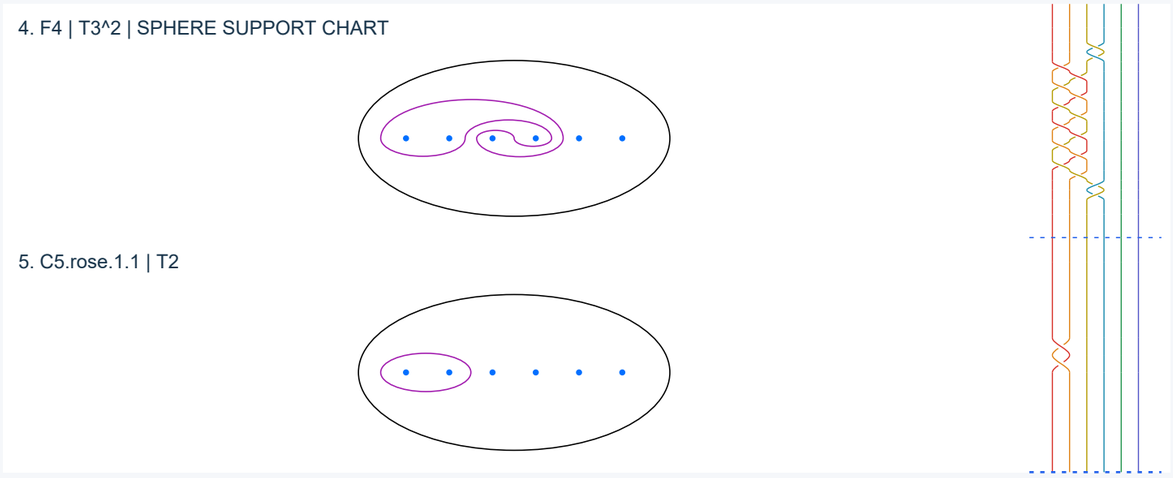
II - step 086

C5.rose.1.1 up past F4

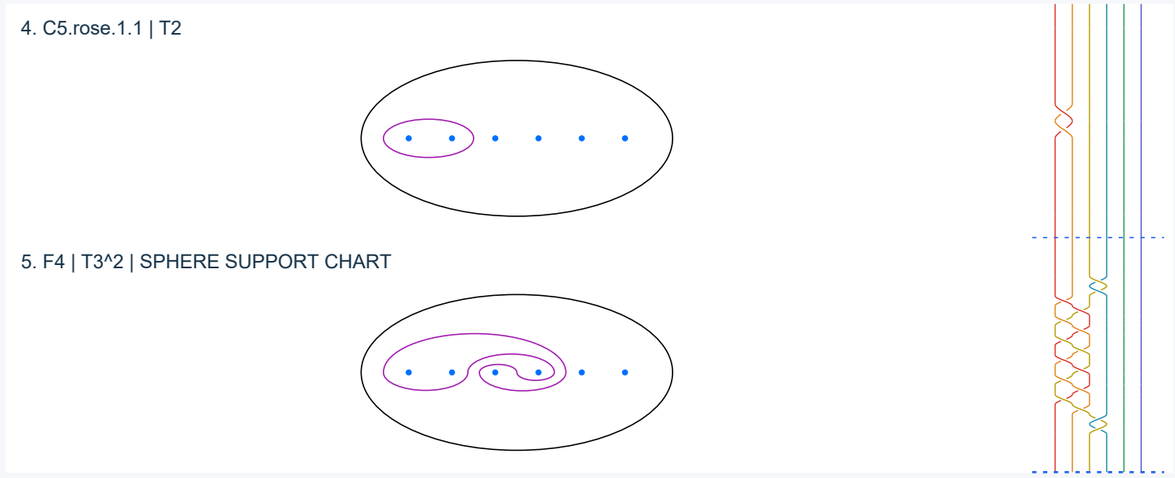
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



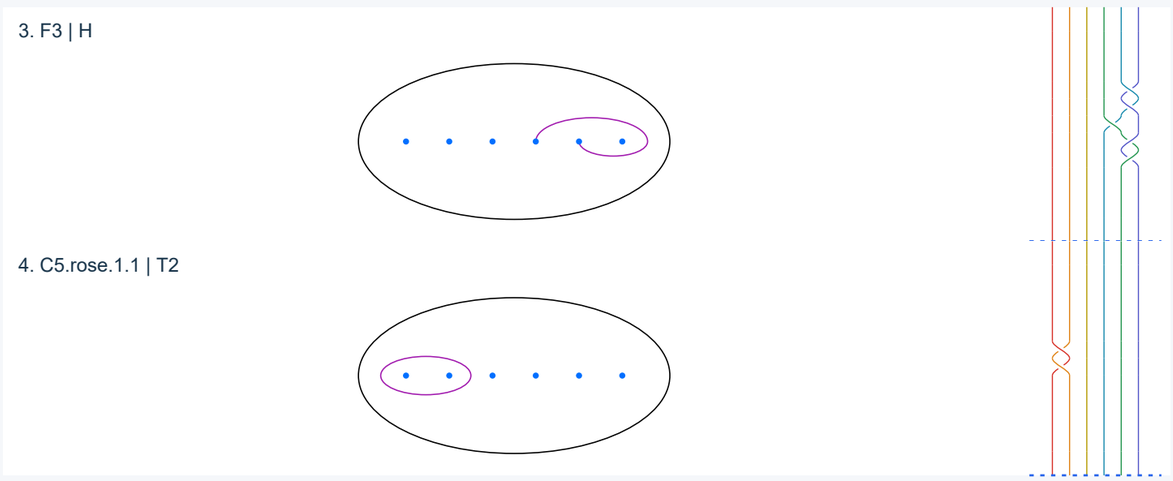
II - step 087

C5.rose.1.1 up past F3

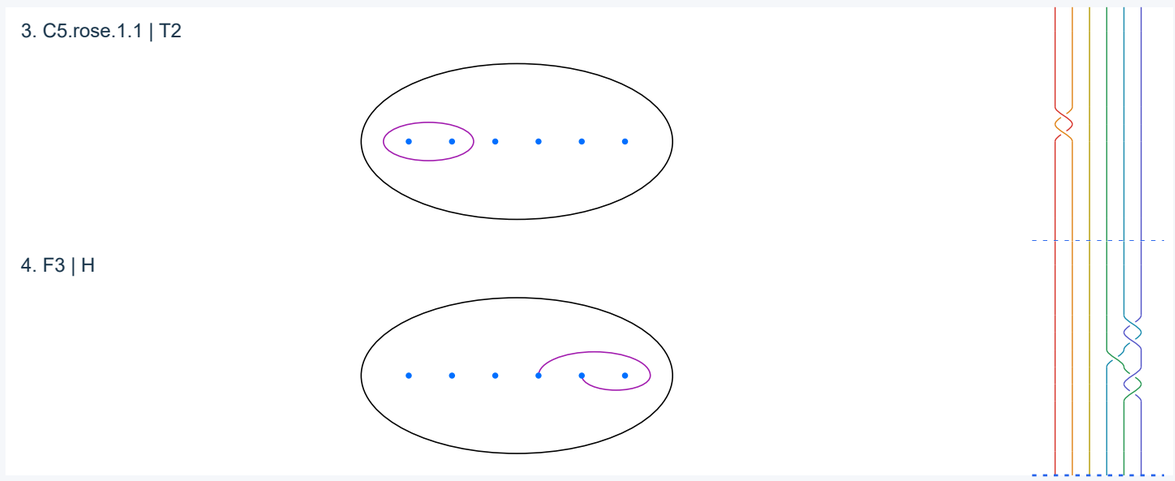
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 088

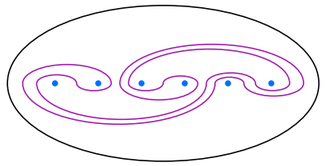
Reverse the transported PC lantern on {3,4,5}

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

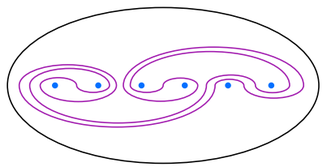
0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

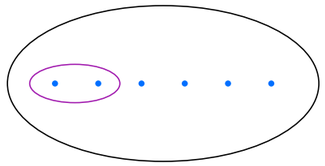
1. prefix.P45 | T2 | SPHERE SUPPORT CHART

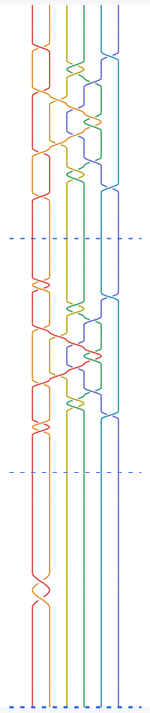


2. residual.P34 | T2 | SPHERE SUPPORT CHART



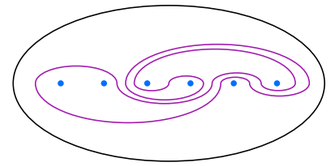
3. C5.rose.1.1 | T2

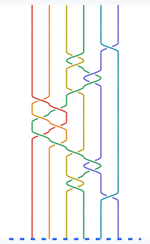




After

1. cross.rose.root | T3 | SPHERE SUPPORT CHART





II - step 089

F3 down past F4

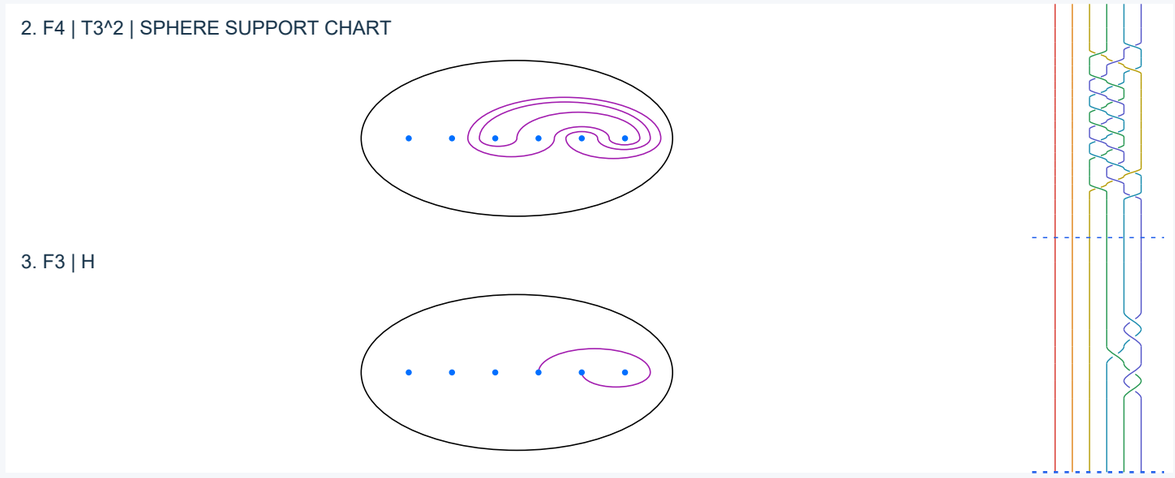
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



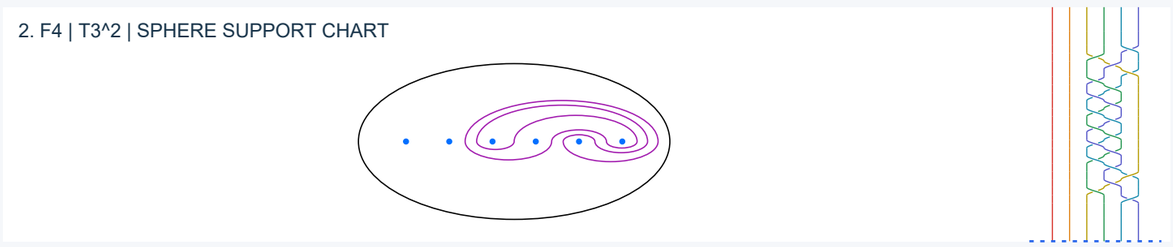
II - step 090

Put the transported C4 square in the common {35},4,6 chart

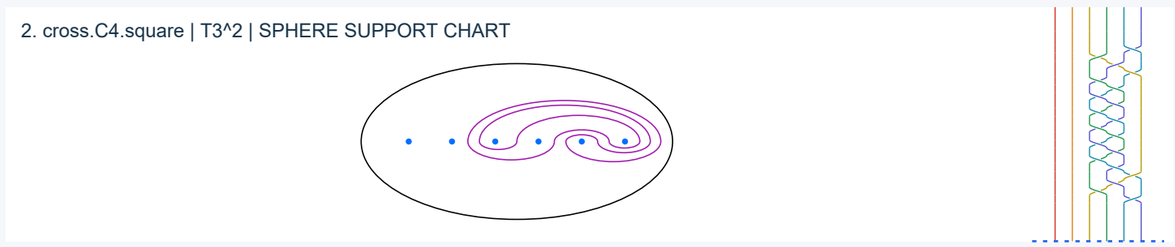
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 091

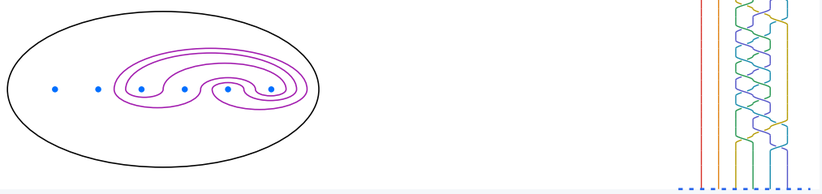
Split the common-chart C4 square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

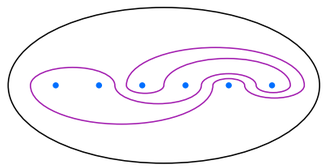
Before

2. cross.C4.square | T3^2 | SPHERE SUPPORT CHART

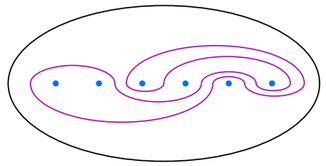


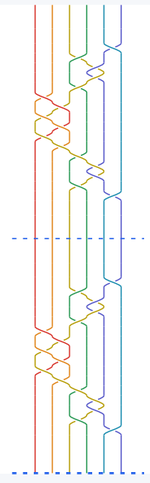
After

2. cross.C4.root | T3 | SPHERE SUPPORT CHART



3. cross.C4.unused | T3 | SPHERE SUPPORT CHART





II - step 092

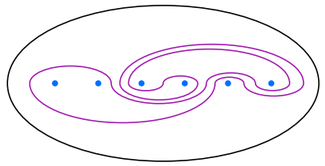
Apply the transported cluster lantern AB=O P35 Q46^-1

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

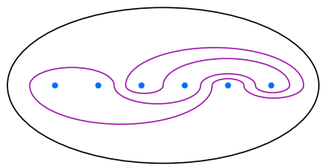
0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

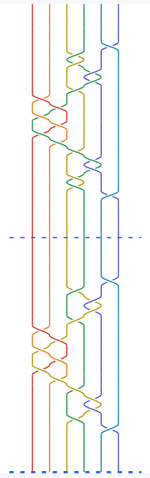
Before

1. cross.rose.root | T3 | SPHERE SUPPORT CHART



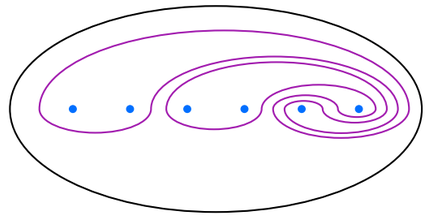
2. cross.C4.root | T3 | SPHERE SUPPORT CHART



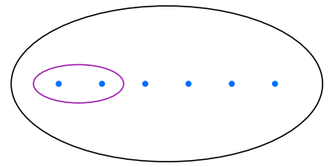


After

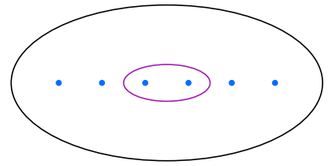
1. cross.O3456 | T4

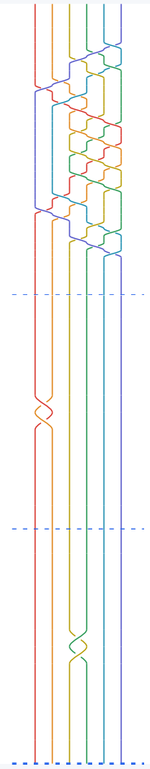


2. cross.P35 | T2



3. cross.inverse.Q46 | T2^-1





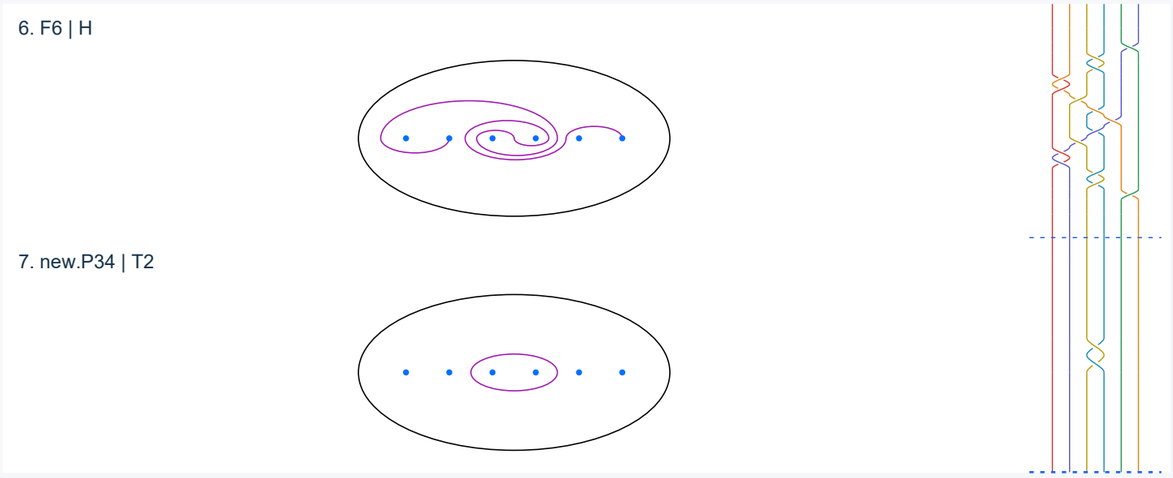
II - step 093

new.P34 up past F6

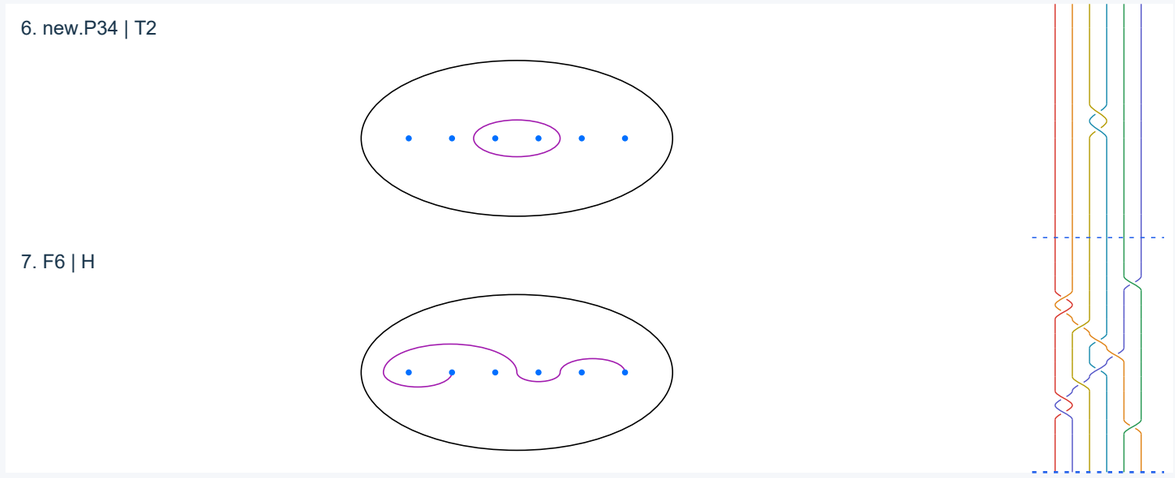
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



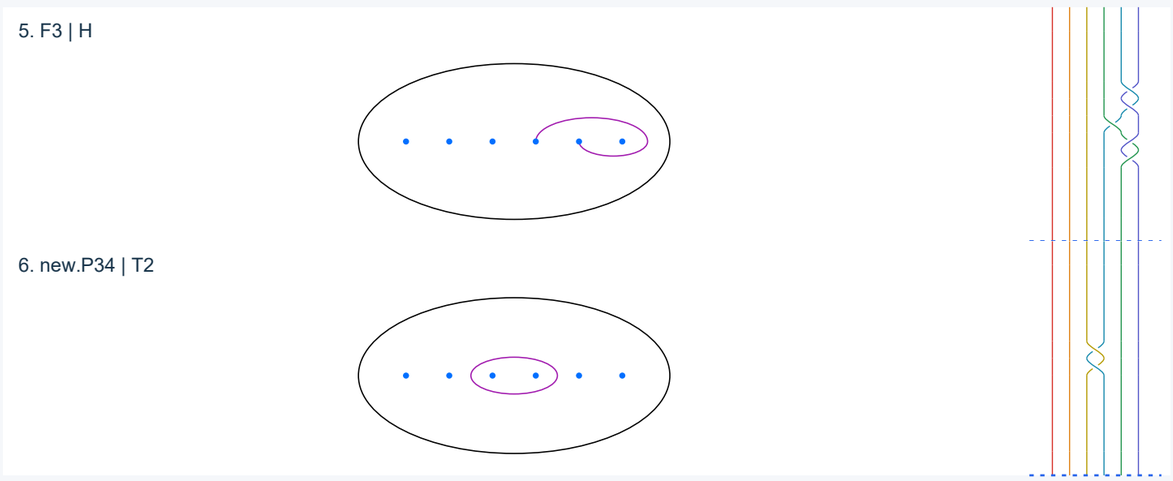
II - step 094

new.P34 up past F3

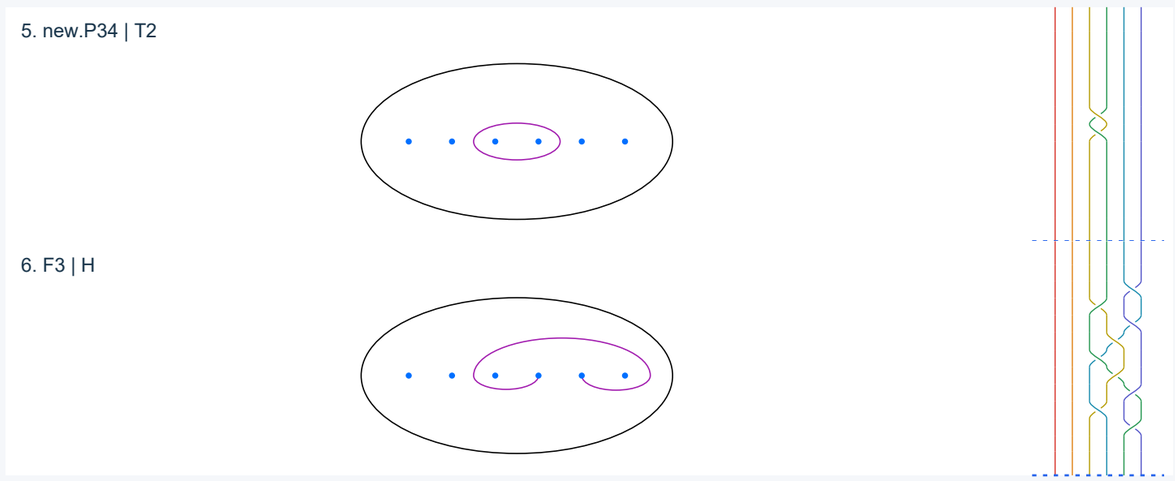
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 095

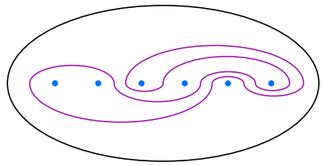
new.P34 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

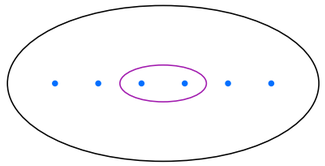
3 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

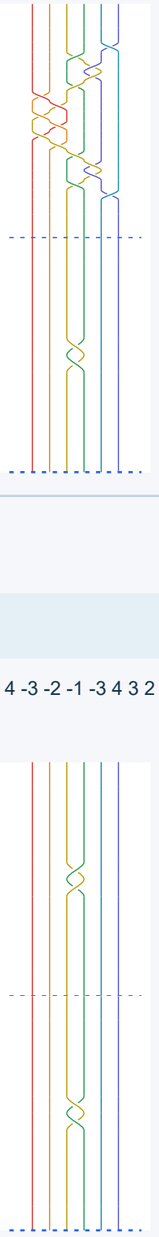
Before

4. cross.C4.unused | T3 | SPHERE SUPPORT CHART



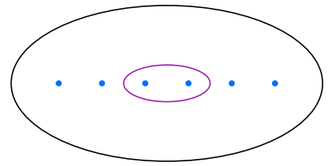
5. new.P34 | T2



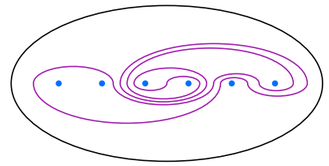


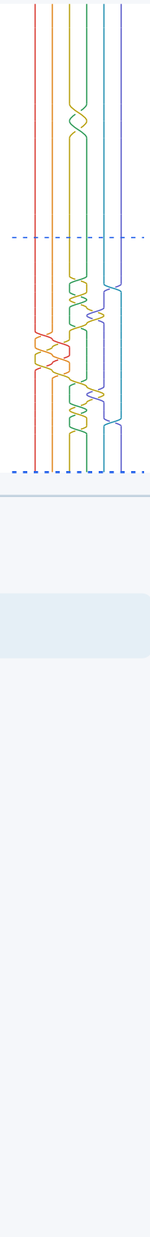
After

4. new.P34 | T2



5. cross.C4.unused | T3 | SPHERE SUPPORT CHART





II - step 096

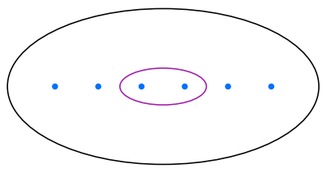
Cancel the equal inverse and positive Q46 pairs in the disk

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

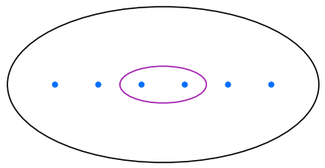
2 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

3. cross.inverse.Q46 | T2^-1



4. new.P34 | T2



After

Identity / no factors in this interval.

II - step 097

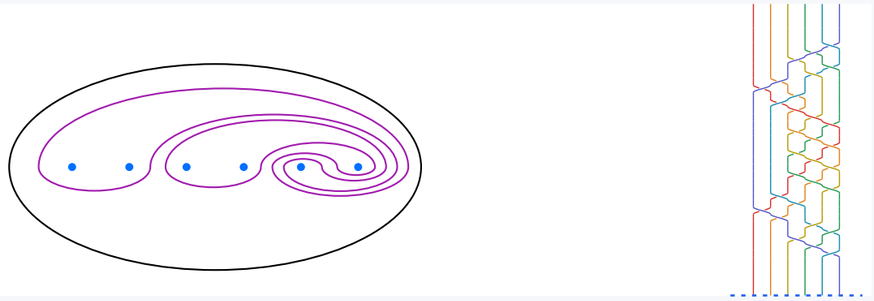
Draw O3456 along its complementary P12 sphere curve

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

1. cross.O3456 | T4



After

1. cross.P12 | T2



II - step 098

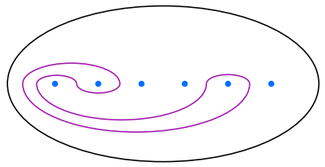
C5.rose.2.1 up past C5.rose.1.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

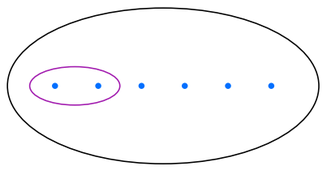
11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

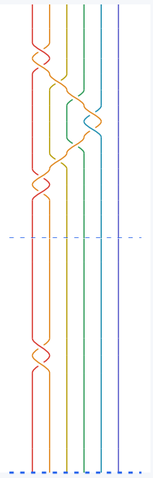
Before

12. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



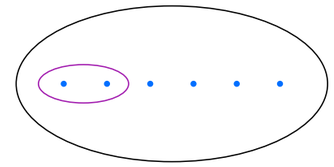
13. C5.rose.2.1 | T2



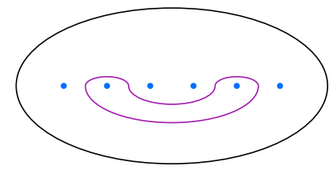


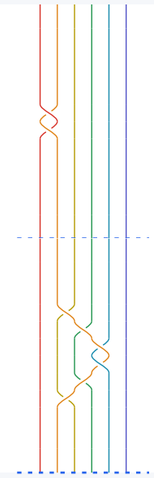
After

12. C5.rose.2.1 | T2



13. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART





II - step 099

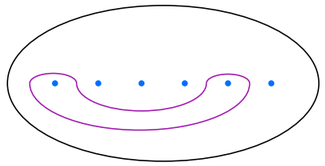
C5.rose.2.1 up past C5.rose.1.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

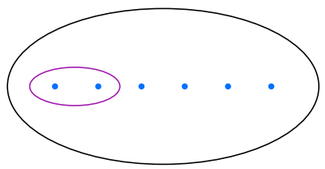
10 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

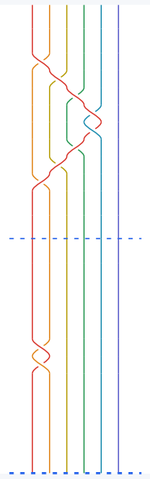
Before

11. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



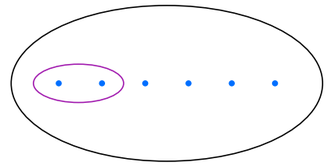
12. C5.rose.2.1 | T2



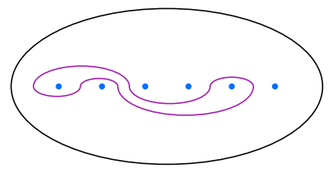


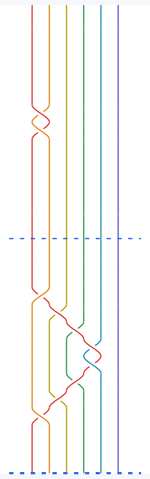
After

11. C5.rose.2.1 | T2



12. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART





II - step 100

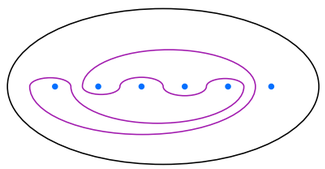
C5.rose.2.1 up past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

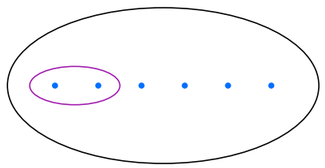
9 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

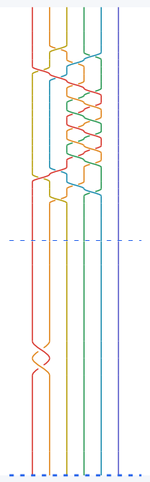
Before

10. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



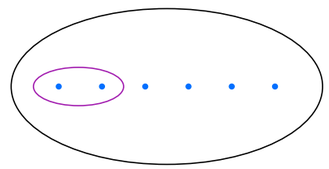
11. C5.rose.2.1 | T2



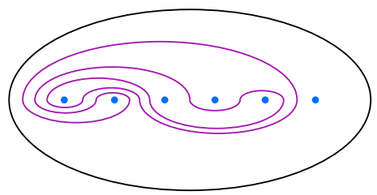


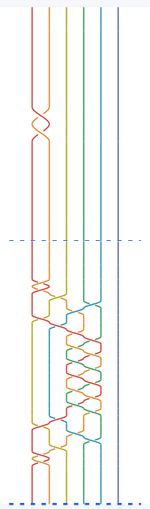
After

10. C5.rose.2.1 | T2



11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART





II - step 101

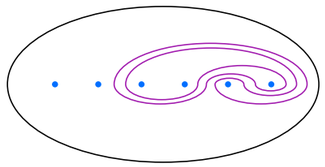
C5.rose.2.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

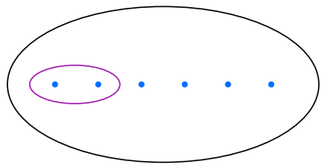
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

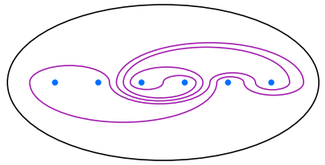
1. cross.P12 | T2



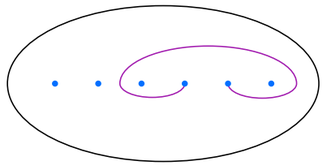
2. cross.P35 | T2



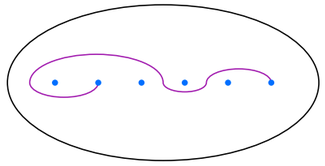
3. cross.C4.unused | T3 | SPHERE SUPPORT CHART

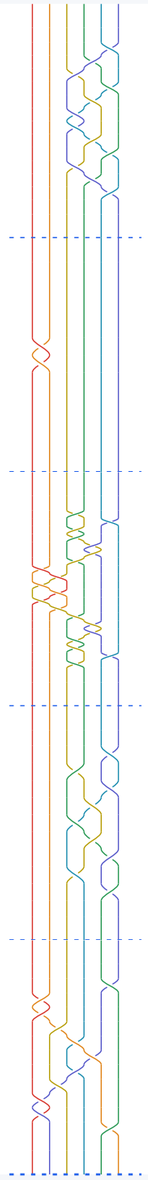


4. F3 | H



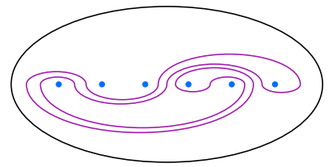
5. F6 | H



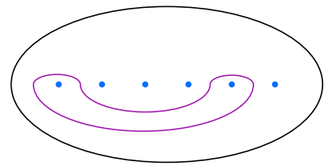


After

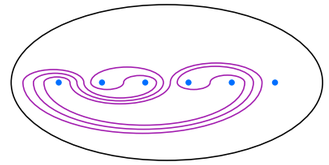
1. cross.P12 | T2 | SPHERE SUPPORT CHART



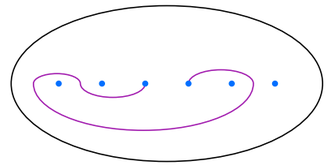
2. cross.P35 | T2



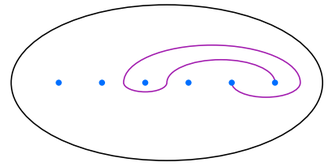
3. cross.C4.unused | T3 | SPHERE SUPPORT CHART

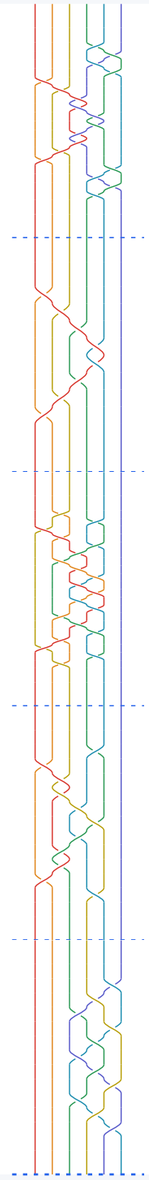


4. F3 | H | SPHERE SUPPORT CHART



5. F6 | H | SPHERE SUPPORT CHART





II - step 101 - continued

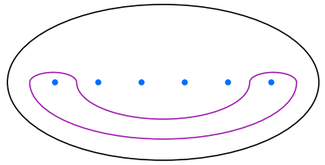
C5.rose.2.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

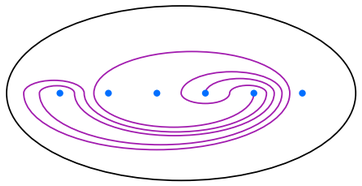
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

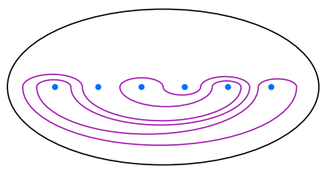
6. new.P26 | T2 | SPHERE SUPPORT CHART



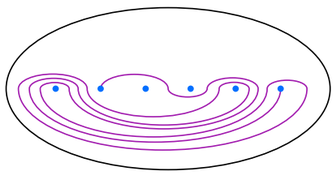
7. F9 | H | SPHERE SUPPORT CHART



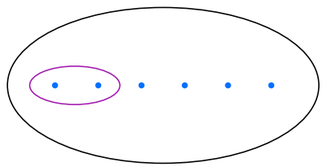
8. residual.P26 | T2 | SPHERE SUPPORT CHART

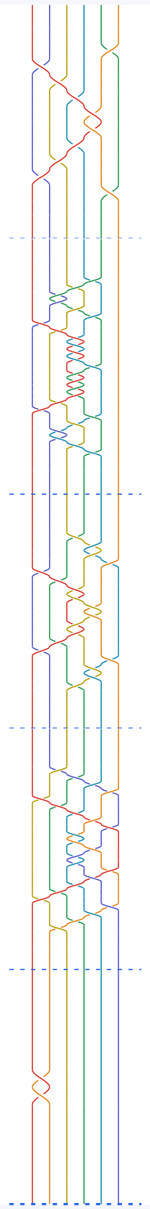


9. F11 | H | SPHERE SUPPORT CHART



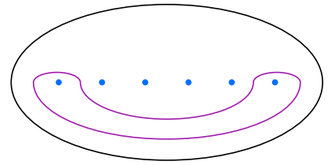
10. C5.rose.2.1 | T2



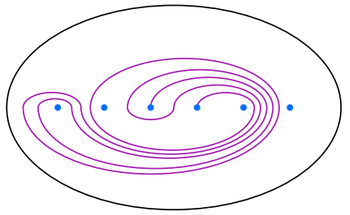


After

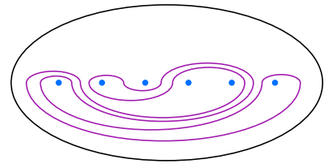
6. new.P26 | T2 | SPHERE SUPPORT CHART



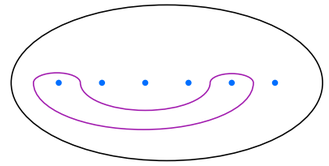
7. F9 | H | SPHERE SUPPORT CHART



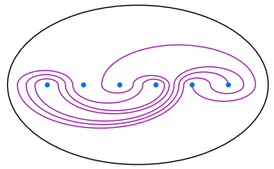
8. residual.P26 | T2 | SPHERE SUPPORT CHART

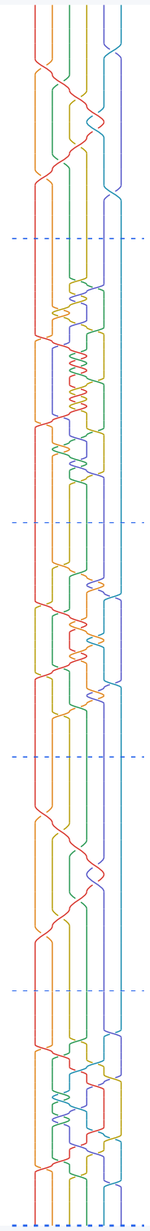


9. C5.rose.2.1 | T2



10. F11 | H | SPHERE SUPPORT CHART





II - step 101 - continued

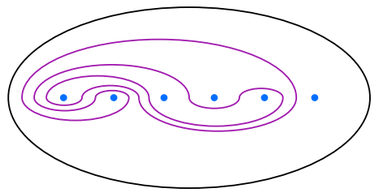
C5.rose.2.1 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

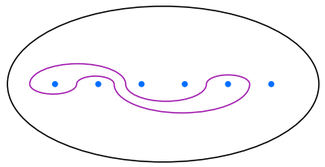
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

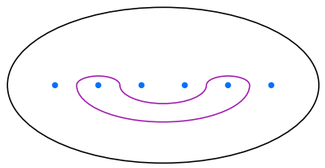
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



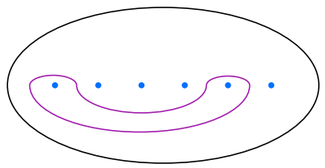
12. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



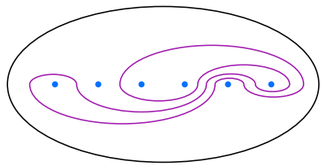
13. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART

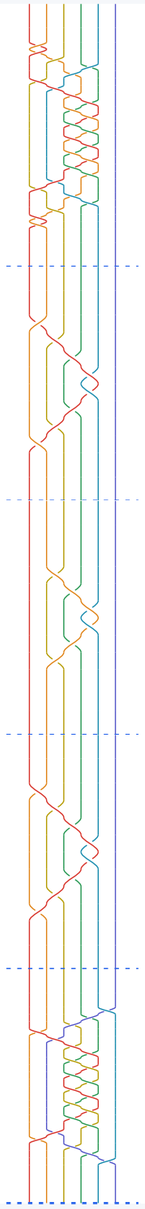


14. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



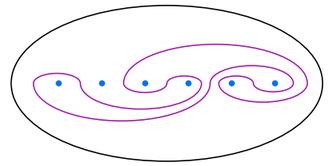
15. C13.recombined | T3^2 | SPHERE SUPPORT CHART



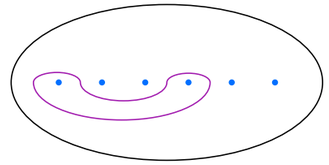


After

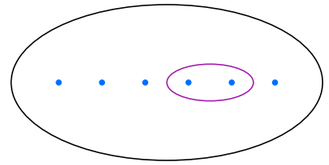
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



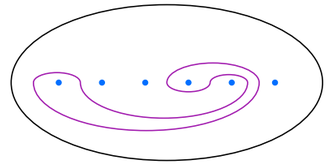
12. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



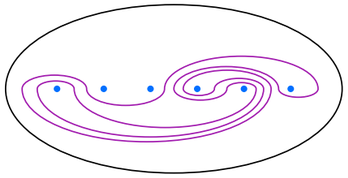
13. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART

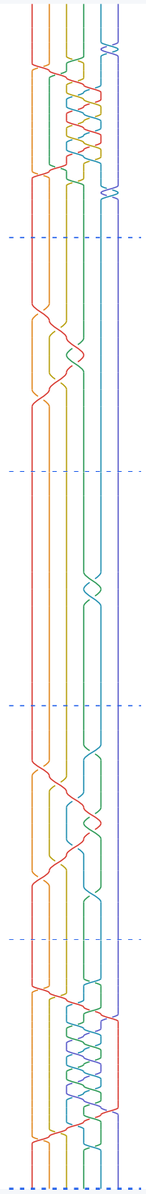


14. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



15. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 102

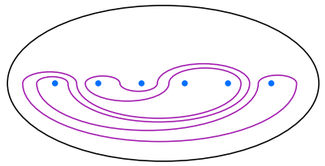
C5.rose.2.1 up past residual.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

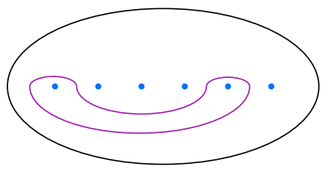
7 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

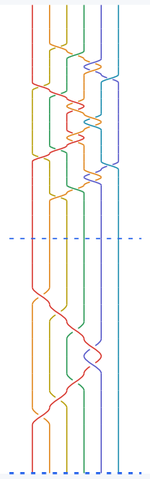
Before

8. residual.P26 | T2 | SPHERE SUPPORT CHART



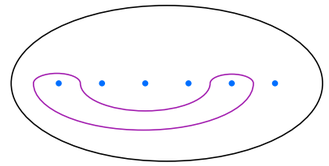
9. C5.rose.2.1 | T2



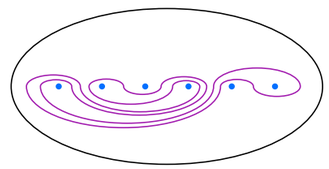


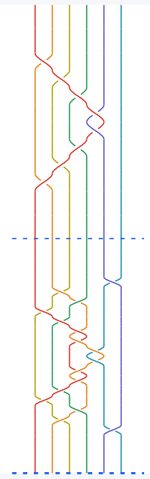
After

8. C5.rose.2.1 | T2



9. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 103

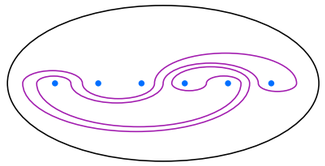
C5.rose.2.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

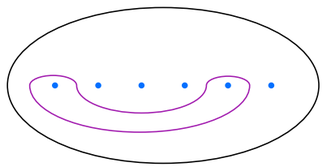
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

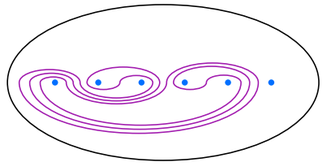
1. cross.P12 | T2 | SPHERE SUPPORT CHART



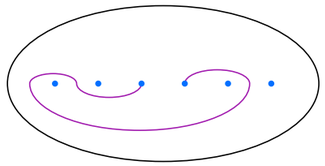
2. cross.P35 | T2



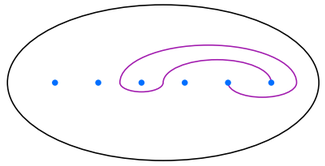
3. cross.C4.unused | T3 | SPHERE SUPPORT CHART

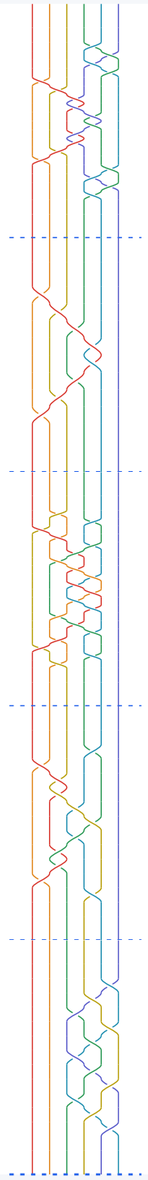


4. F3 | H | SPHERE SUPPORT CHART



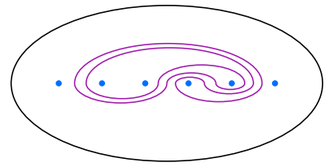
5. F6 | H | SPHERE SUPPORT CHART



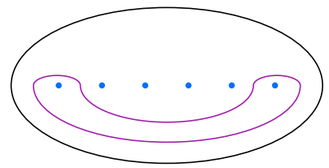


After

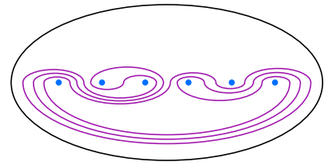
1. cross.P12 | T2



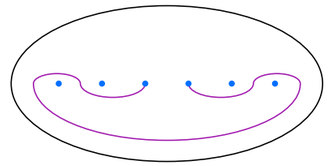
2. cross.P35 | T2



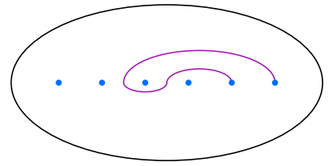
3. cross.C4.unused | T3 | SPHERE SUPPORT CHART

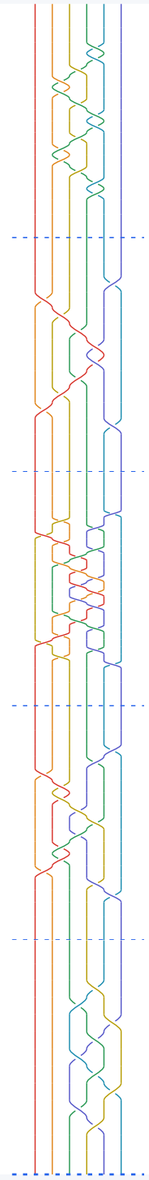


4. F3 | H | SPHERE SUPPORT CHART



5. F6 | H | SPHERE SUPPORT CHART





II - step 103 - continued

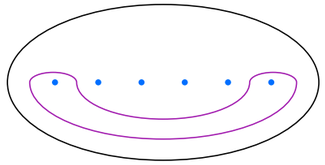
C5.rose.2.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

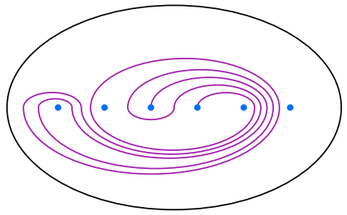
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

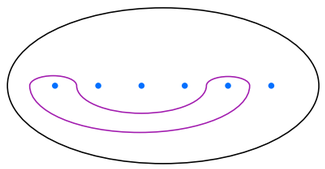
6. new.P26 | T2 | SPHERE SUPPORT CHART



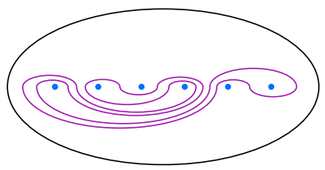
7. F9 | H | SPHERE SUPPORT CHART



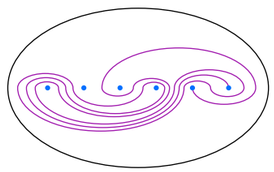
8. C5.rose.2.1 | T2

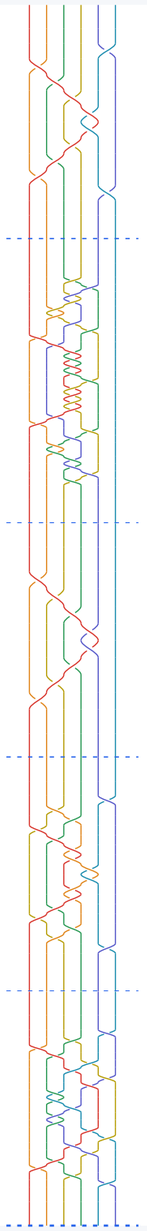


9. residual.P26 | T2 | SPHERE SUPPORT CHART



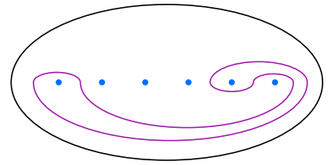
10. F11 | H | SPHERE SUPPORT CHART



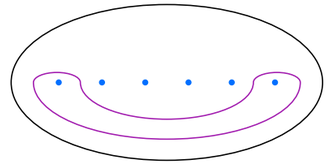


After

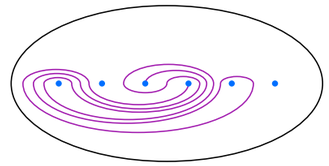
6. new.P26 | T2 | SPHERE SUPPORT CHART



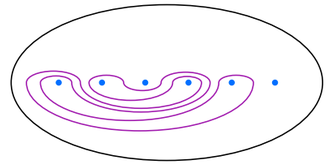
7. C5.rose.2.1 | T2



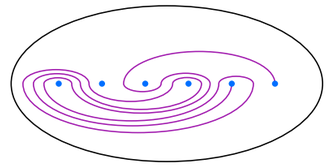
8. F9 | H | SPHERE SUPPORT CHART

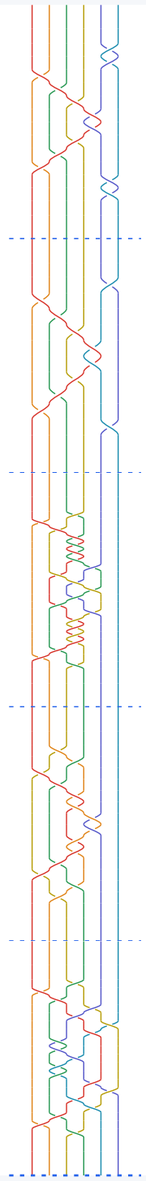


9. residual.P26 | T2 | SPHERE SUPPORT CHART



10. F11 | H | SPHERE SUPPORT CHART





II - step 103 - continued

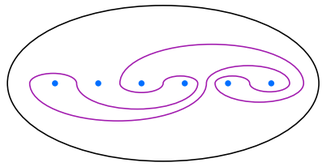
C5.rose.2.1 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

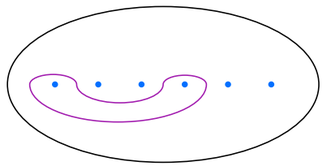
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

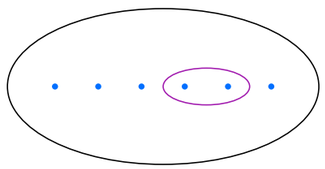
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



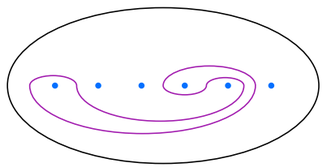
12. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



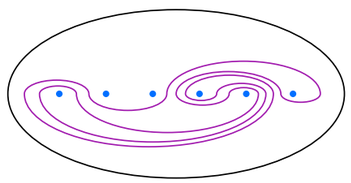
13. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART

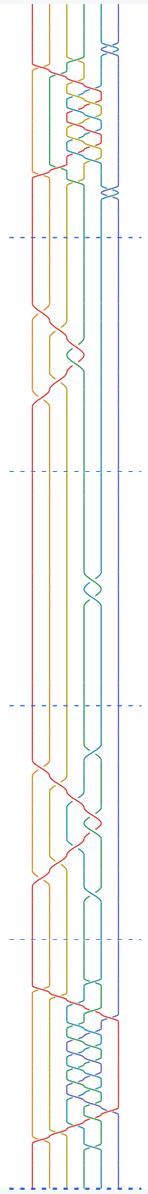


14. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



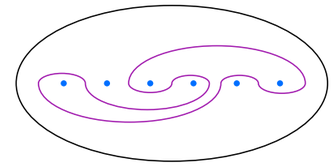
15. C13.recombined | T3^2 | SPHERE SUPPORT CHART



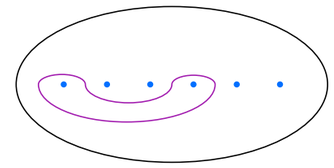


After

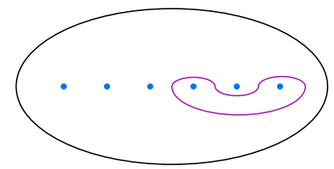
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



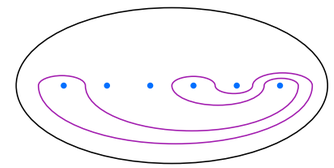
12. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



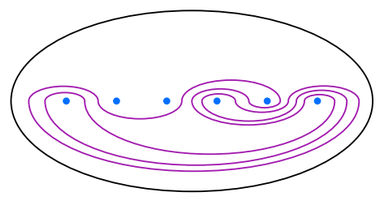
13. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART

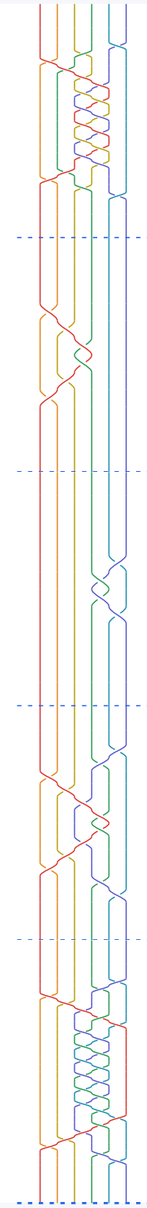


14. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



15. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 104

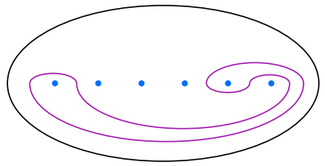
C5.rose.2.1 up past new.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

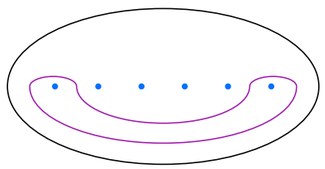
5 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

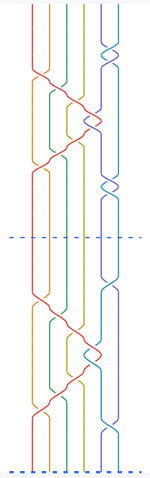
Before

6. new.P26 | T2 | SPHERE SUPPORT CHART



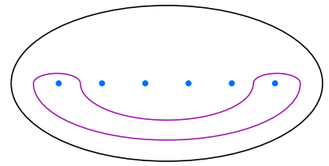
7. C5.rose.2.1 | T2



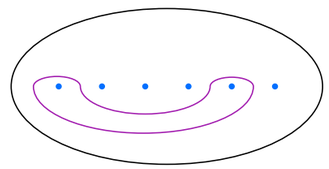


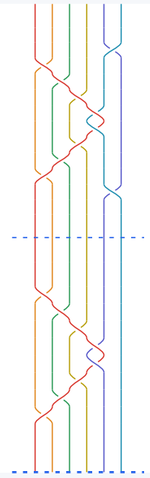
After

6. C5.rose.2.1 | T2



7. new.P26 | T2 | SPHERE SUPPORT CHART





II - step 105

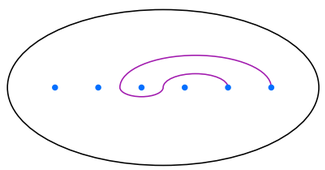
C5.rose.2.1 up past F6

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

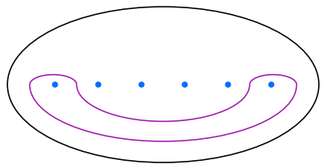
4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

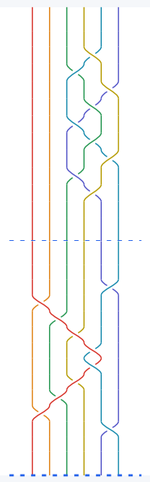
Before

5. F6 | H | SPHERE SUPPORT CHART



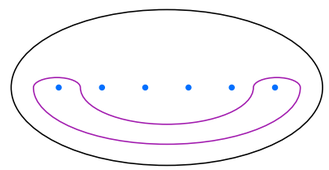
6. C5.rose.2.1 | T2



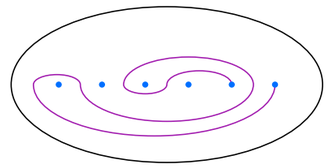


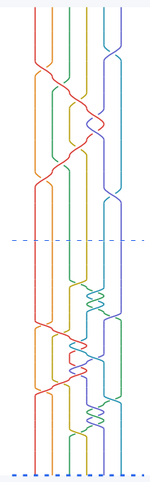
After

5. C5.rose.2.1 | T2



6. F6 | H | SPHERE SUPPORT CHART





II - step 106

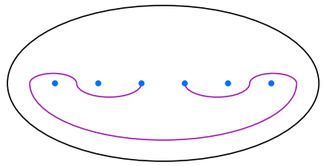
C5.rose.2.1 up past F3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

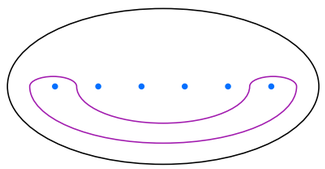
3 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

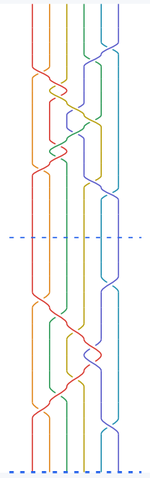
Before

4. F3 | H | SPHERE SUPPORT CHART



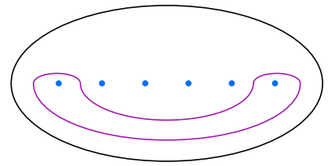
5. C5.rose.2.1 | T2



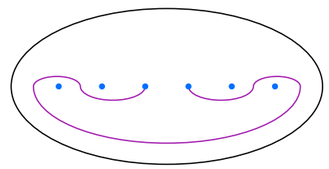


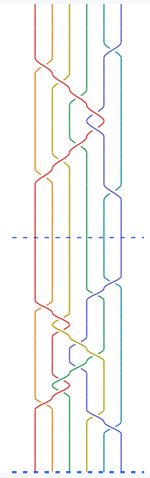
After

4. C5.rose.2.1 | T2



5. F3 | H | SPHERE SUPPORT CHART





II - step 107

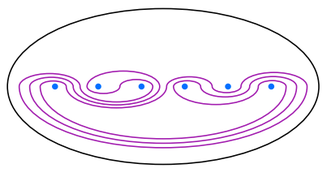
C5.rose.2.1 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

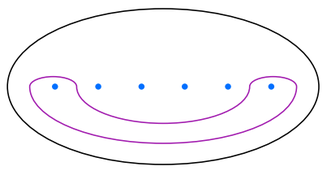
2 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

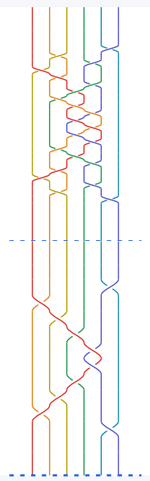
Before

3. cross.C4.unused | T3 | SPHERE SUPPORT CHART



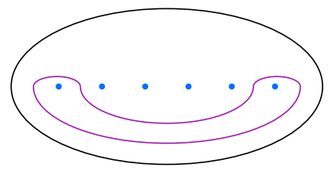
4. C5.rose.2.1 | T2



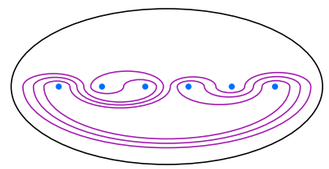


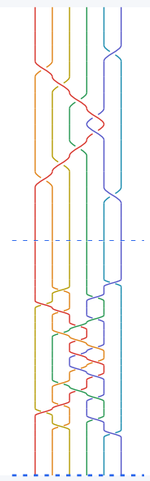
After

3. C5.rose.2.1 | T2



4. cross.C4.unused | T3 | SPHERE SUPPORT CHART





II - step 108

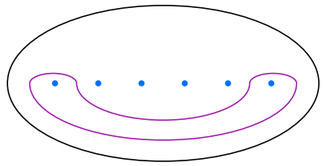
Combine the two equal P35 twists in the disk

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

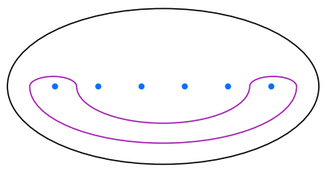
1 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

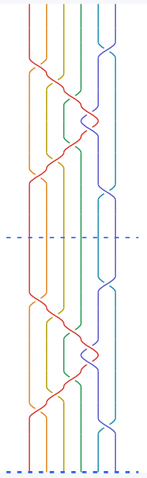
Before

2. cross.P35 | T2



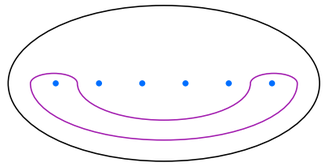
3. C5.rose.2.1 | T2

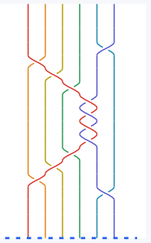




After

2. cross.P35.square | T2^2





II - step 109

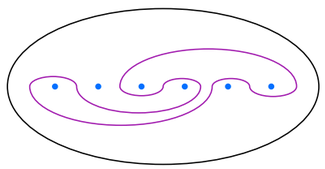
C5.rose.1.2 up past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

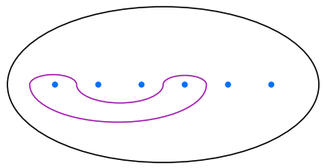
9 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

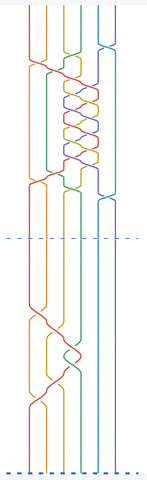
Before

10. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



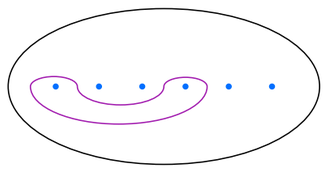
11. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



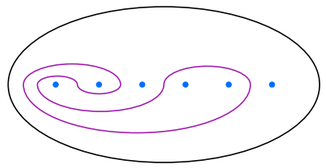


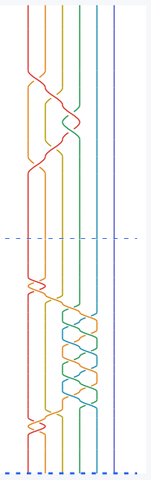
After

10. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART





II - step 110

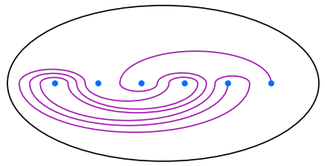
C5.rose.1.2 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

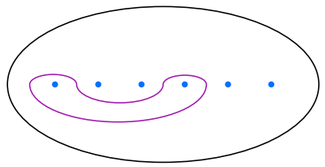
8 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

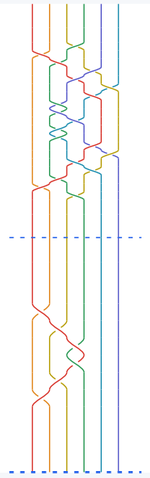
Before

9. F11 | H | SPHERE SUPPORT CHART



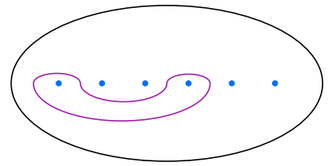
10. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



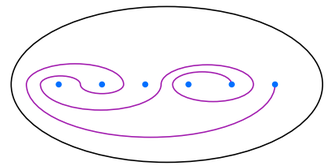


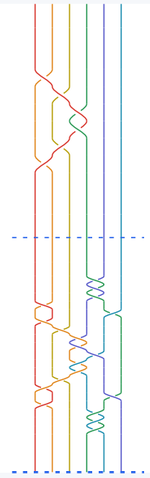
After

9. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



10. F11 | H | SPHERE SUPPORT CHART





II - step 111

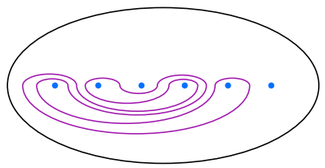
C5.rose.1.2 up past residual.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

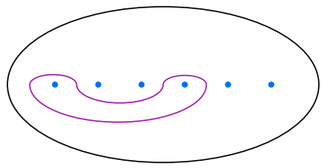
7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

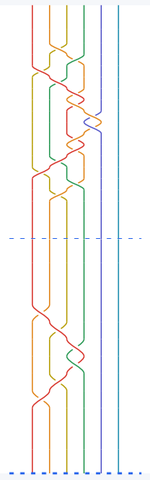
Before

8. residual.P26 | T2 | SPHERE SUPPORT CHART



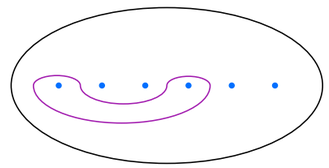
9. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



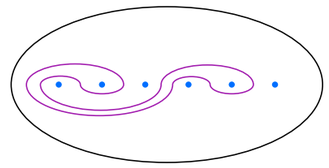


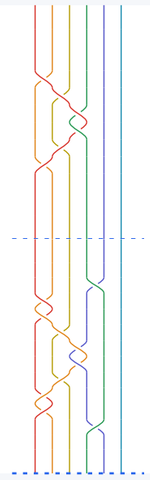
After

8. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



9. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 112

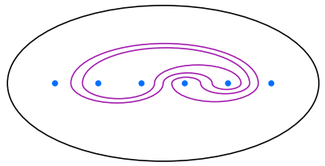
C5.rose.1.2 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

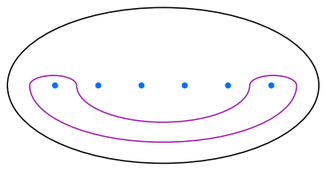
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

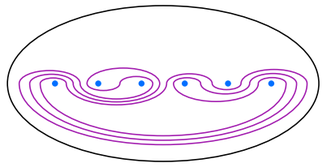
1. cross.P12 | T2



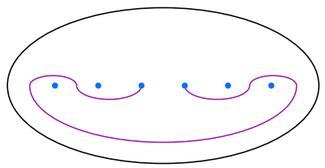
2. cross.P35.square | T2^2



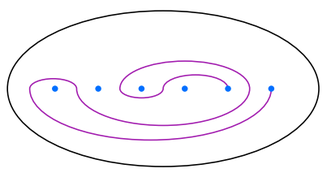
3. cross.C4.unused | T3 | SPHERE SUPPORT CHART

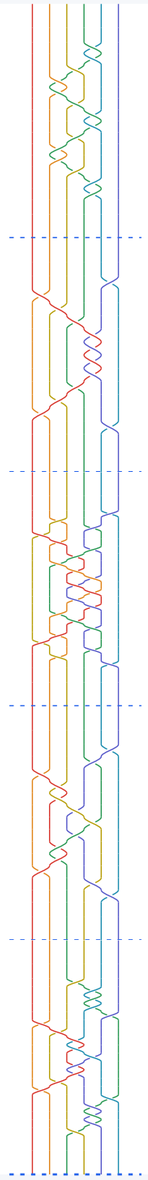


4. F3 | H | SPHERE SUPPORT CHART



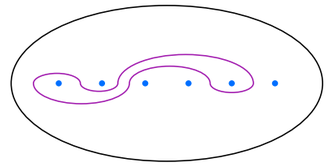
5. F6 | H | SPHERE SUPPORT CHART



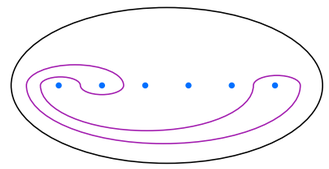


After

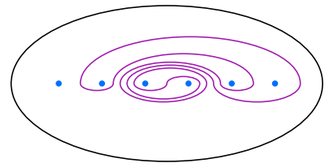
1. cross.P12 | T2



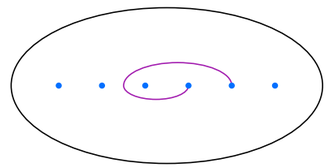
2. cross.P35.square | T2^2



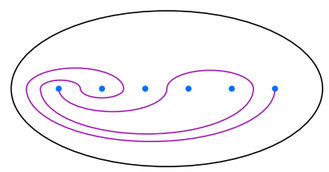
3. cross.C4.unused | T3 | SPHERE SUPPORT CHART

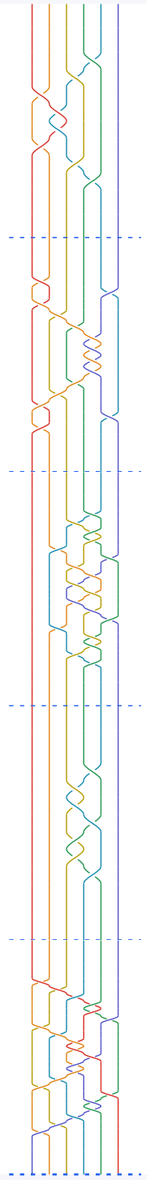


4. F3 | H



5. F6 | H | SPHERE SUPPORT CHART





II - step 112 - continued

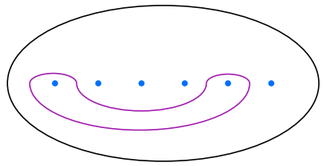
C5.rose.1.2 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

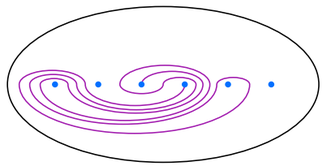
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

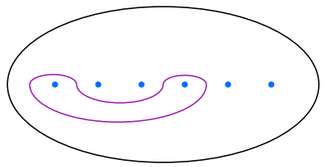
6. new.P26 | T2 | SPHERE SUPPORT CHART



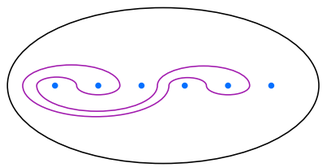
7. F9 | H | SPHERE SUPPORT CHART



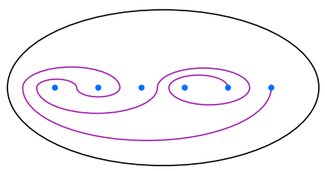
8. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

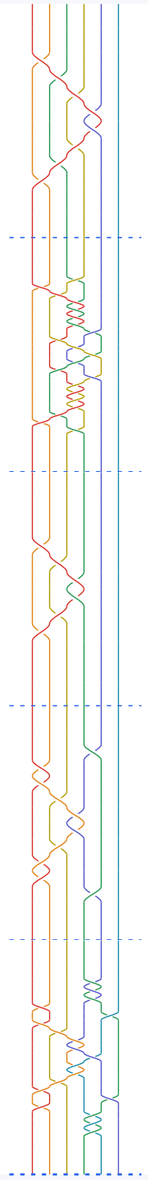


9. residual.P26 | T2 | SPHERE SUPPORT CHART



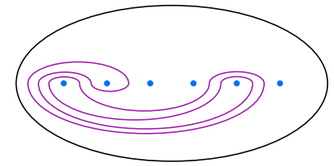
10. F11 | H | SPHERE SUPPORT CHART



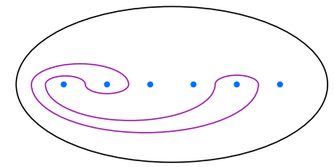


After

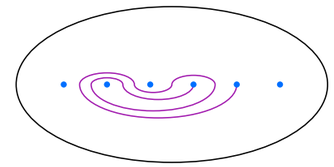
6. new.P26 | T2 | SPHERE SUPPORT CHART



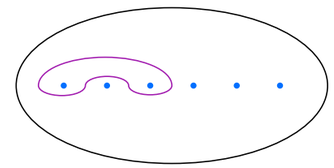
7. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



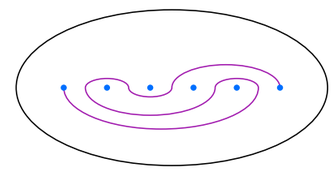
8. F9 | H | SPHERE SUPPORT CHART

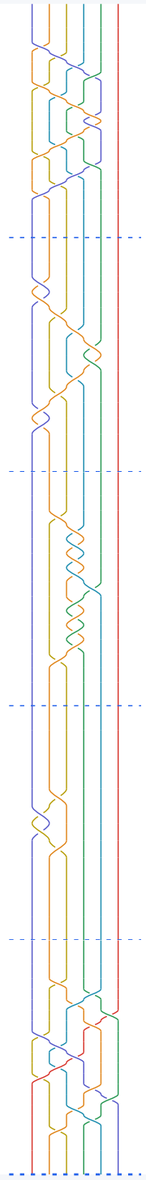


9. residual.P26 | T2 | SPHERE SUPPORT CHART



10. F11 | H | SPHERE SUPPORT CHART





II - step 112 - continued

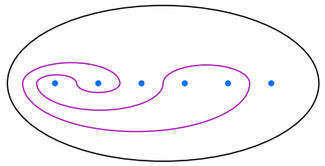
C5.rose.1.2 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

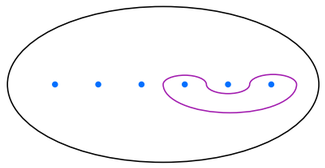
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

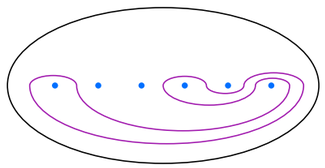
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



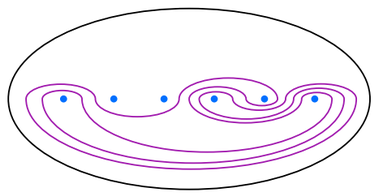
12. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART

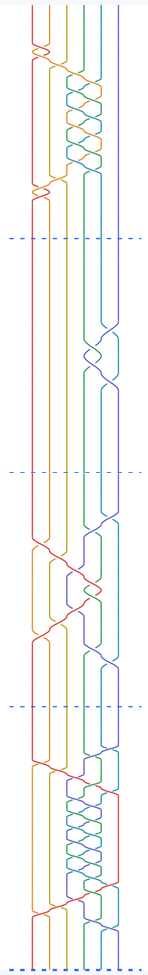


13. C5.rose.2.2 | T2 | SPHERE SUPPORT CHART



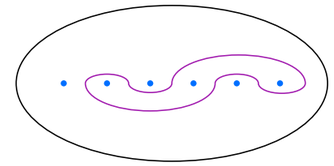
14. C13.recombined | T3^2 | SPHERE SUPPORT CHART



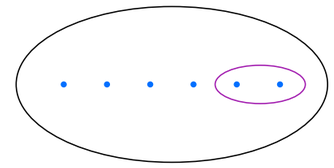


After

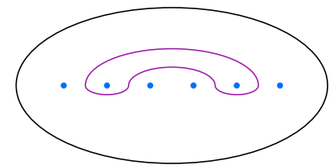
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



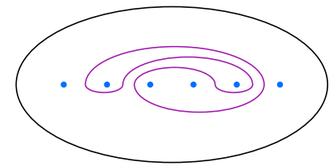
12. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART

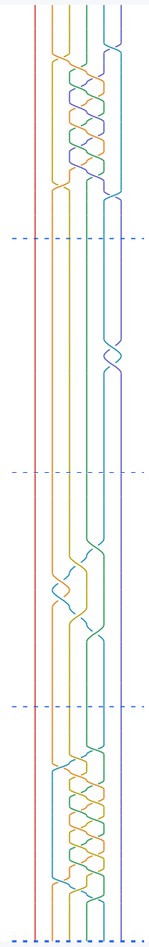


13. C5.rose.2.2 | T2



14. C13.recombined | T3^2 | SPHERE SUPPORT CHART





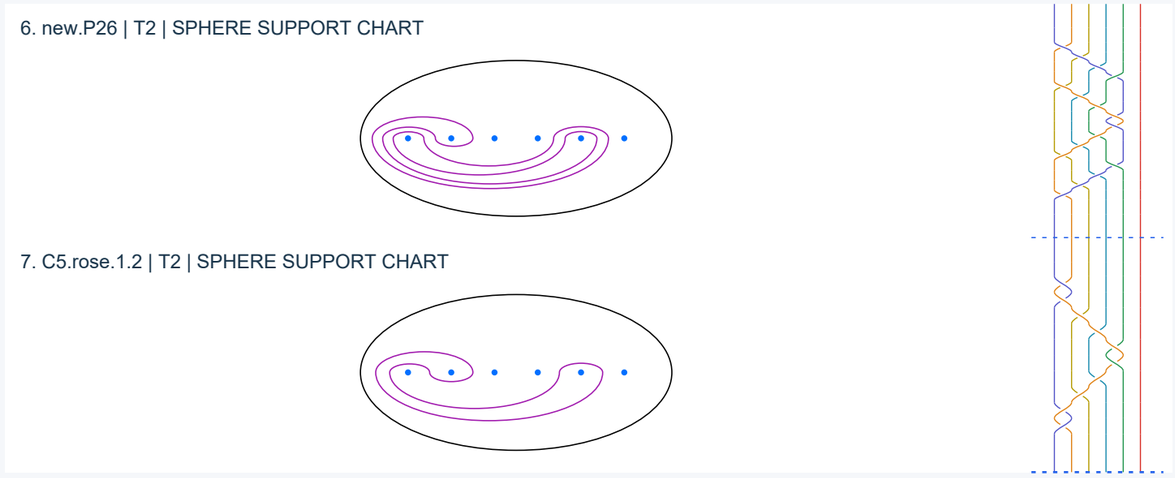
II - step 113

C5.rose.1.2 up past new.P26

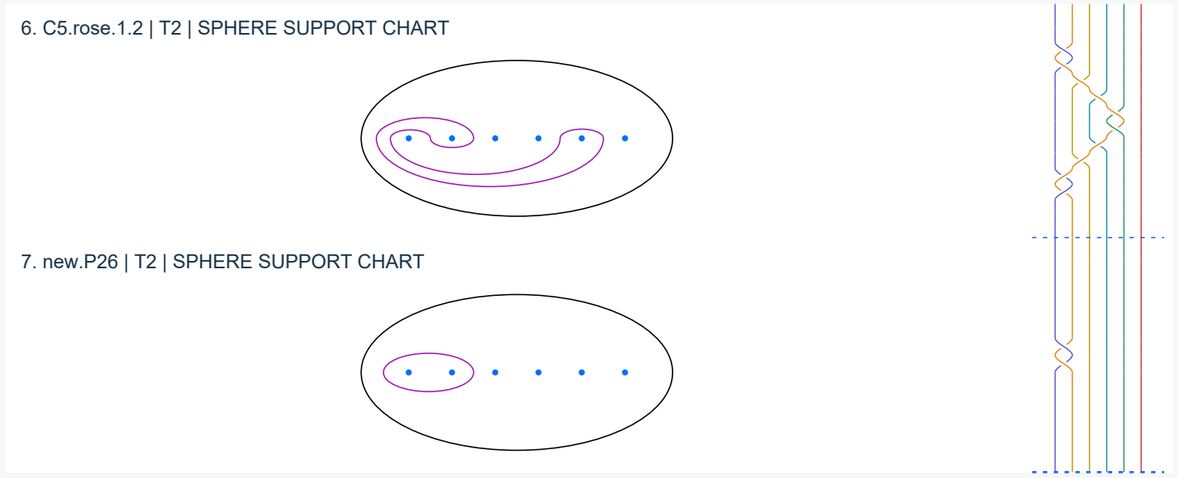
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



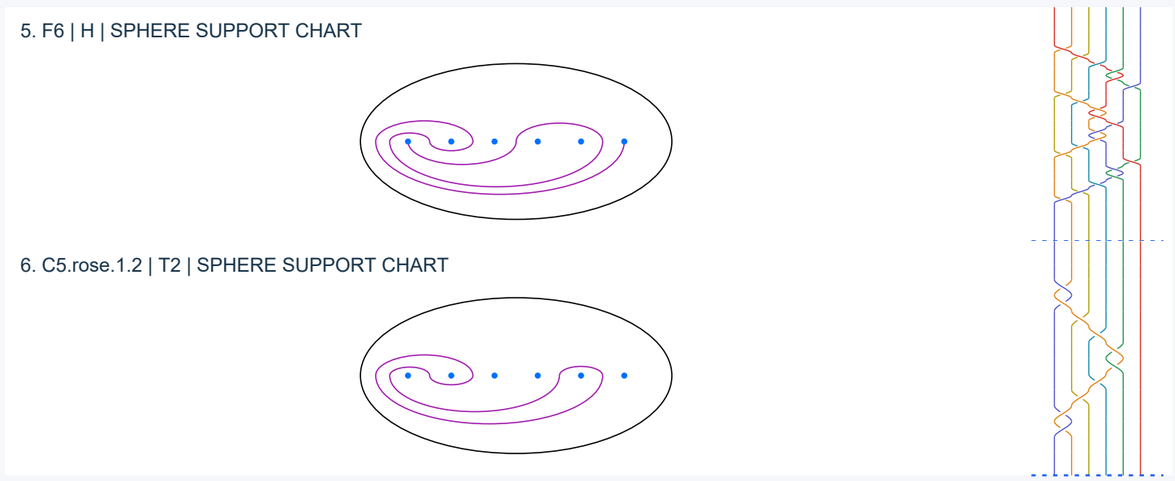
II - step 114

C5.rose.1.2 up past F6

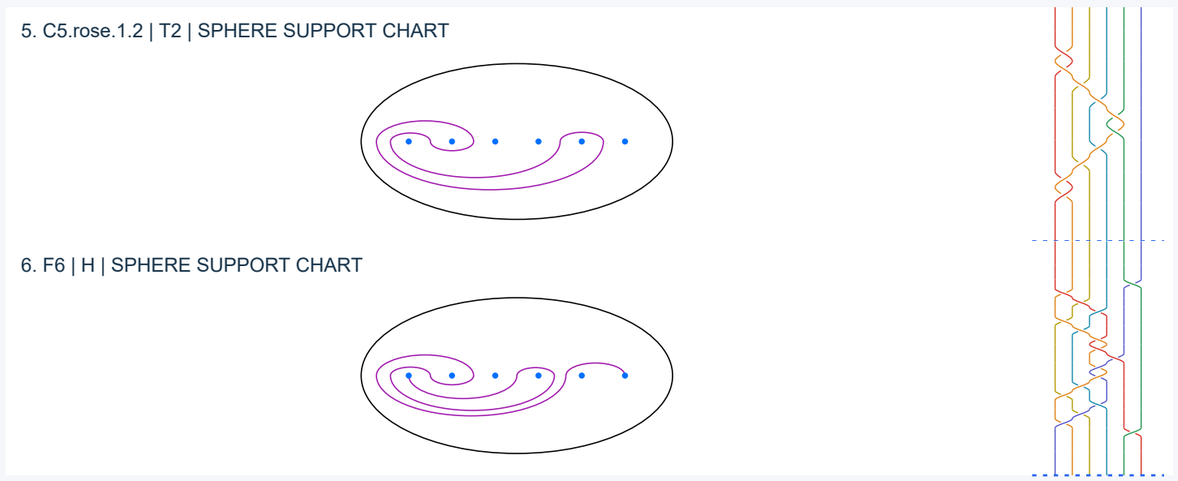
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 115

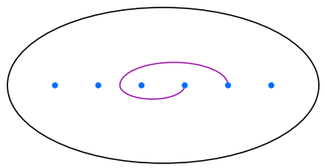
C5.rose.1.2 up past F3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

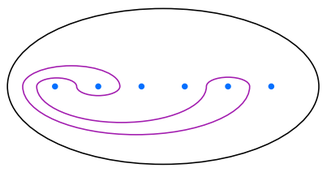
3 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

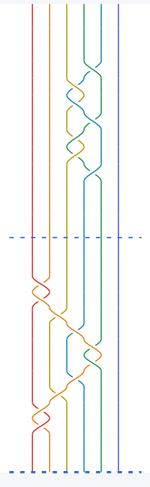
Before

4. F3 | H



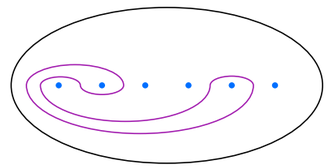
5. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



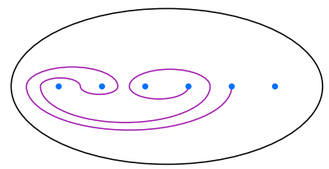


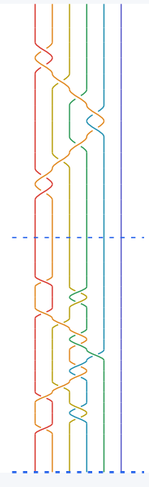
After

4. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



5. F3 | H | SPHERE SUPPORT CHART





II - step 116

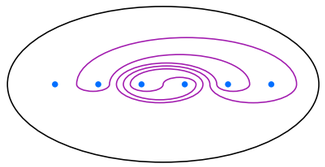
C5.rose.1.2 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

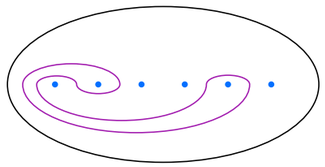
2 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

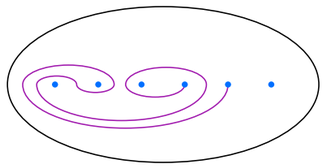
3. cross.C4.unused | T3 | SPHERE SUPPORT CHART



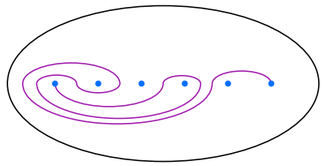
4. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



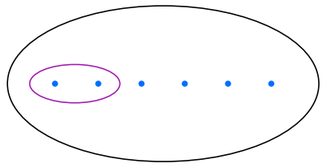
5. F3 | H | SPHERE SUPPORT CHART

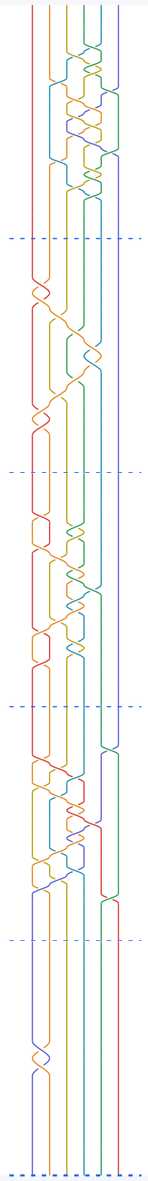


6. F6 | H | SPHERE SUPPORT CHART



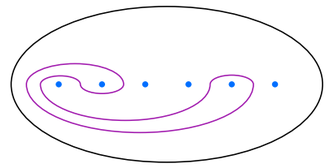
7. new.P26 | T2 | SPHERE SUPPORT CHART



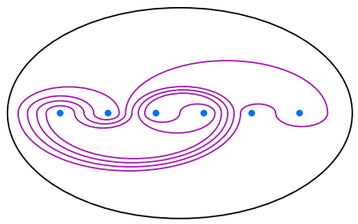


After

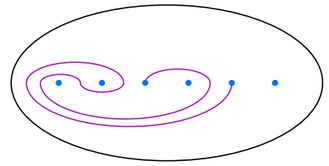
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



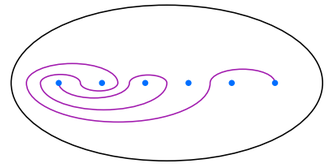
4. cross.C4.unused | T3 | SPHERE SUPPORT CHART



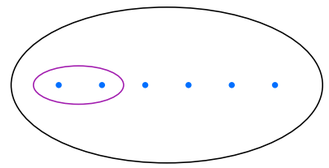
5. F3 | H | SPHERE SUPPORT CHART

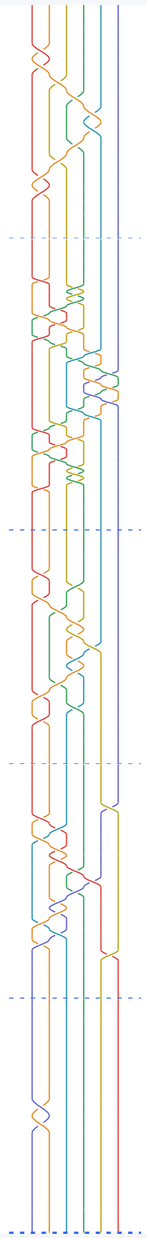


6. F6 | H | SPHERE SUPPORT CHART



7. new.P26 | T2 | SPHERE SUPPORT CHART





II - step 116 - continued

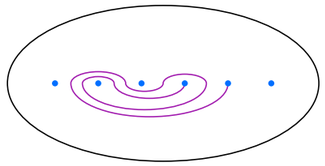
C5.rose.1.2 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

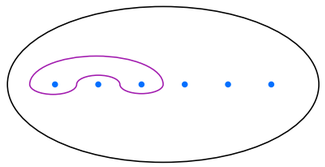
2 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

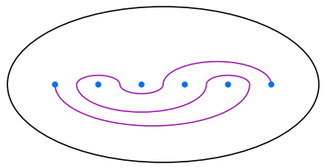
8. F9 | H | SPHERE SUPPORT CHART



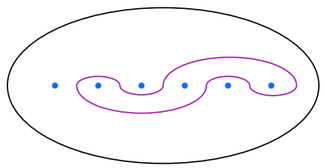
9. residual.P26 | T2 | SPHERE SUPPORT CHART

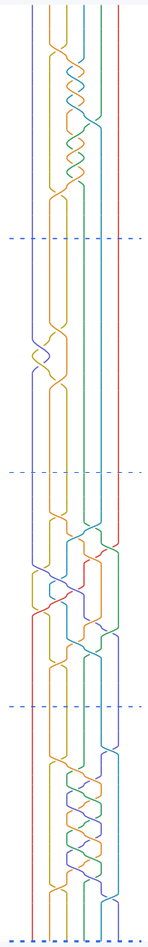


10. F11 | H | SPHERE SUPPORT CHART



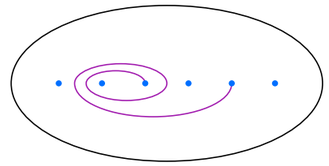
11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



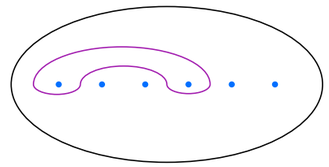


After

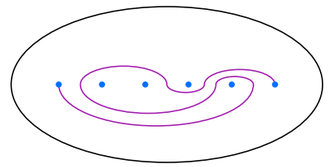
8. F9 | H | SPHERE SUPPORT CHART



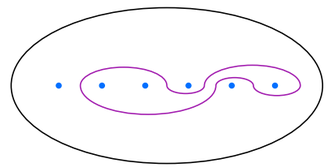
9. residual.P26 | T2 | SPHERE SUPPORT CHART

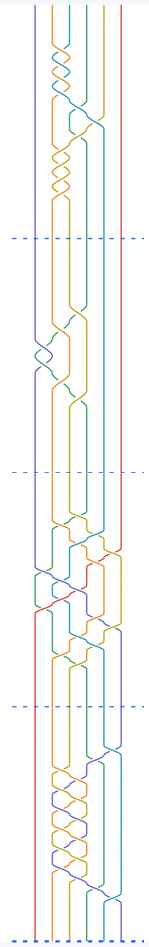


10. F11 | H | SPHERE SUPPORT CHART



11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART





II - step 117

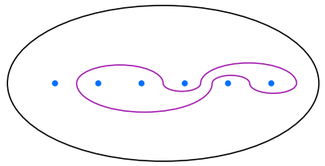
C5.rose.1.3 up past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

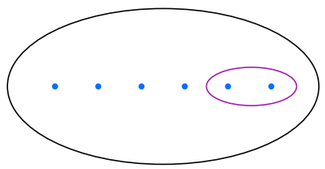
10 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 3 2 4 -2 -2 1 2 -4 -2 5.

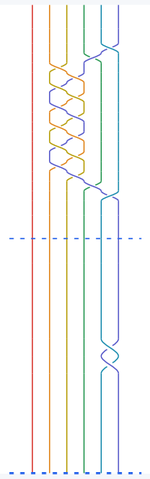
Before

11. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



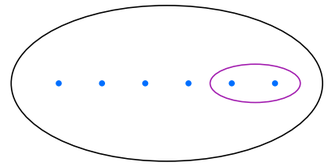
12. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



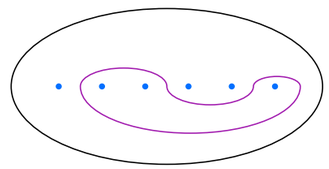


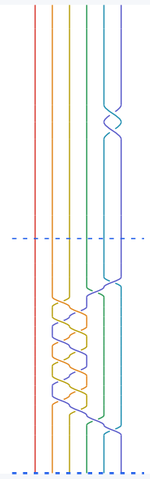
After

11. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART





II - step 118

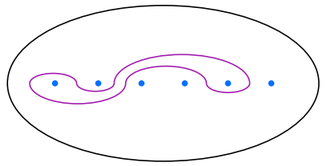
C5.rose.1.3 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

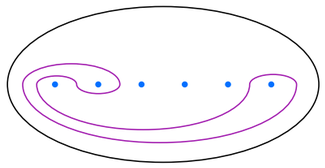
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before

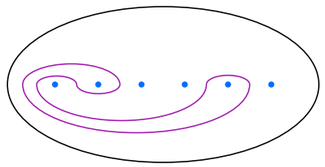
1. cross.P12 | T2



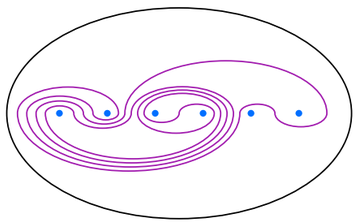
2. cross.P35.square | T2^2



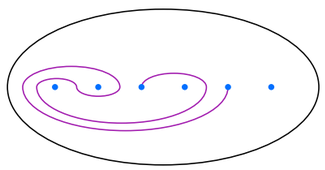
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

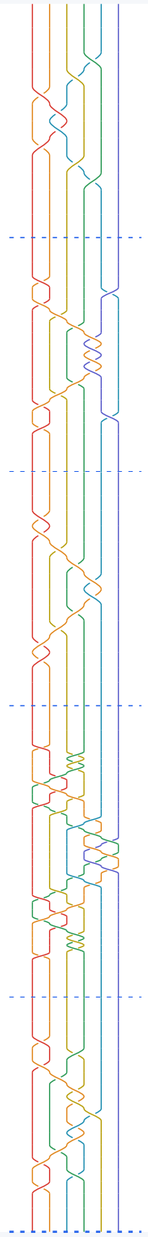


4. cross.C4.unused | T3 | SPHERE SUPPORT CHART



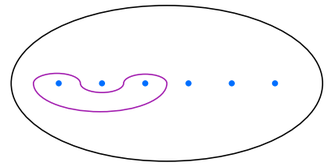
5. F3 | H | SPHERE SUPPORT CHART



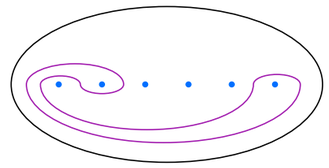


After

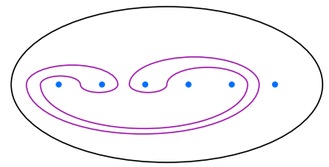
1. cross.P12 | T2



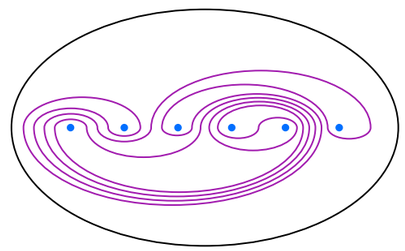
2. cross.P35.square | T2^2



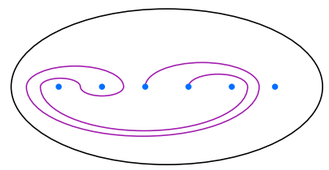
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

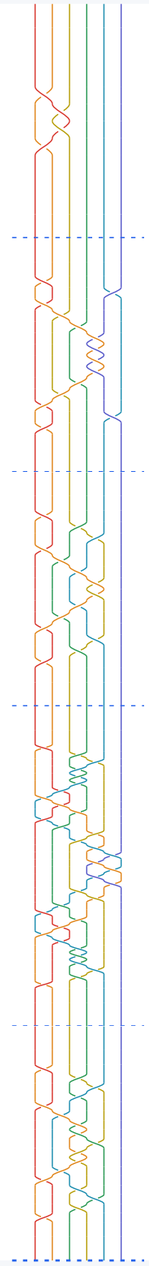


4. cross.C4.unused | T3 | SPHERE SUPPORT CHART



5. F3 | H | SPHERE SUPPORT CHART





II - step 118 - continued

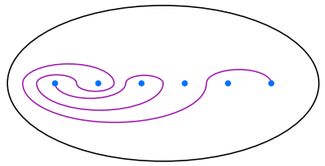
C5.rose.1.3 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

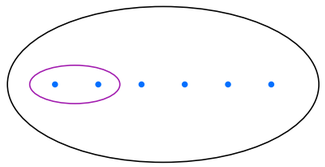
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before

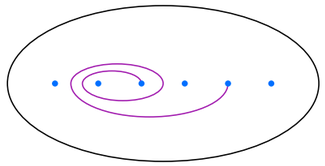
6. F6 | H | SPHERE SUPPORT CHART



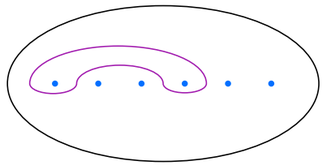
7. new.P26 | T2 | SPHERE SUPPORT CHART



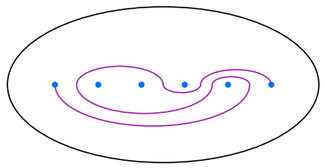
8. F9 | H | SPHERE SUPPORT CHART

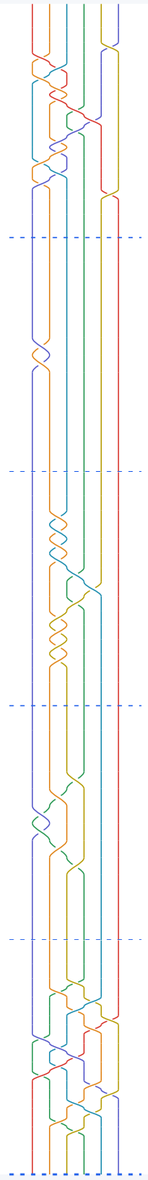


9. residual.P26 | T2 | SPHERE SUPPORT CHART



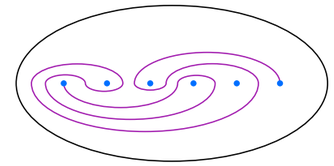
10. F11 | H | SPHERE SUPPORT CHART



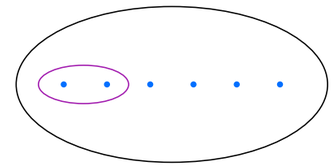


After

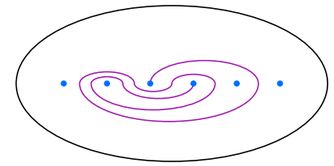
6. F6 | H | SPHERE SUPPORT CHART



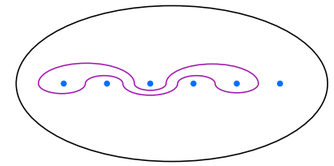
7. new.P26 | T2 | SPHERE SUPPORT CHART



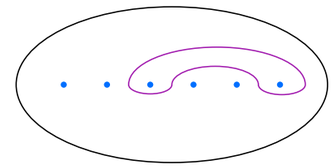
8. F9 | H | SPHERE SUPPORT CHART

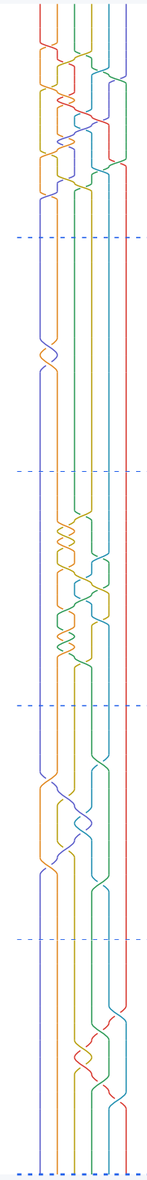


9. residual.P26 | T2 | SPHERE SUPPORT CHART



10. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART





II - step 118 - continued

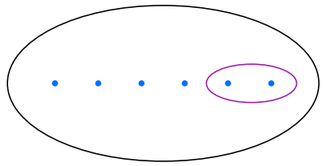
C5.rose.1.3 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

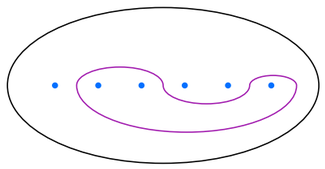
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before

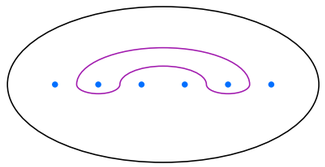
11. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



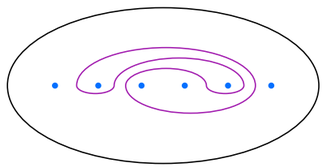
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

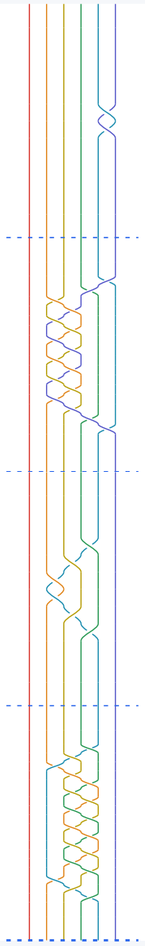


13. C5.rose.2.2 | T2



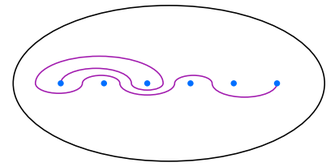
14. C13.recombined | T3^2 | SPHERE SUPPORT CHART



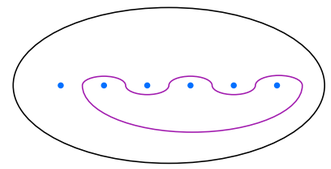


After

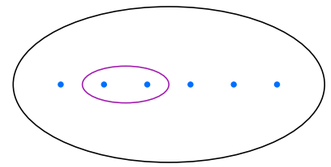
11. F11 | H | SPHERE SUPPORT CHART



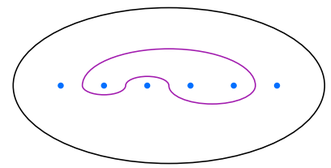
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

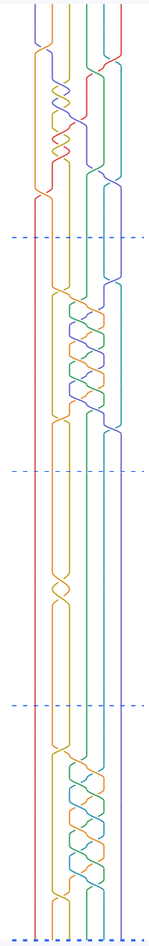


13. C5.rose.2.2 | T2



14. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 119

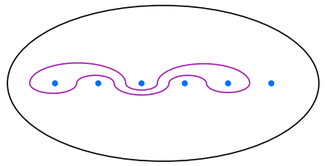
C5.rose.1.3 up past residual.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

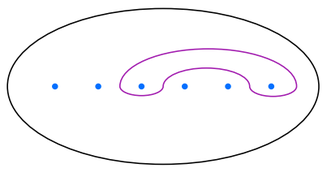
8 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

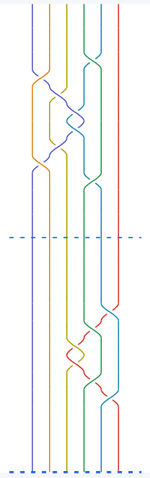
Before

9. residual.P26 | T2 | SPHERE SUPPORT CHART



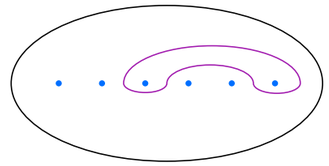
10. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



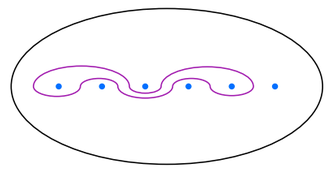


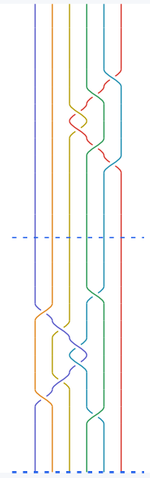
After

9. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



10. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 120

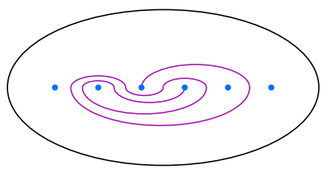
C5.rose.1.3 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

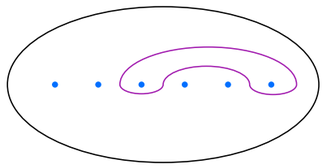
7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

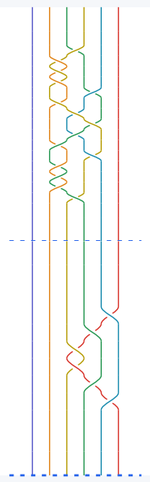
Before

8. F9 | H | SPHERE SUPPORT CHART



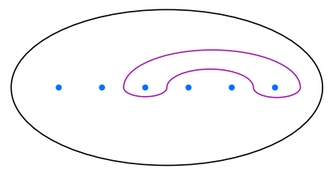
9. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



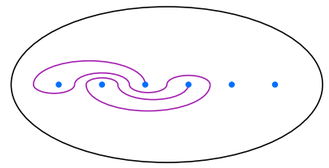


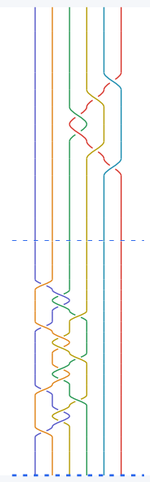
After

8. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



9. F9 | H | SPHERE SUPPORT CHART





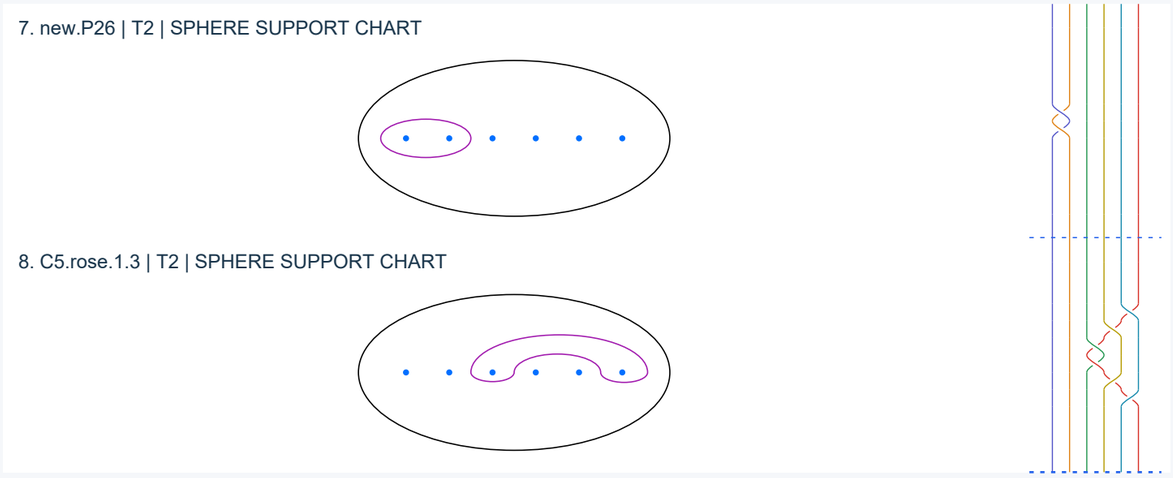
II - step 121

C5.rose.1.3 up past new.P26

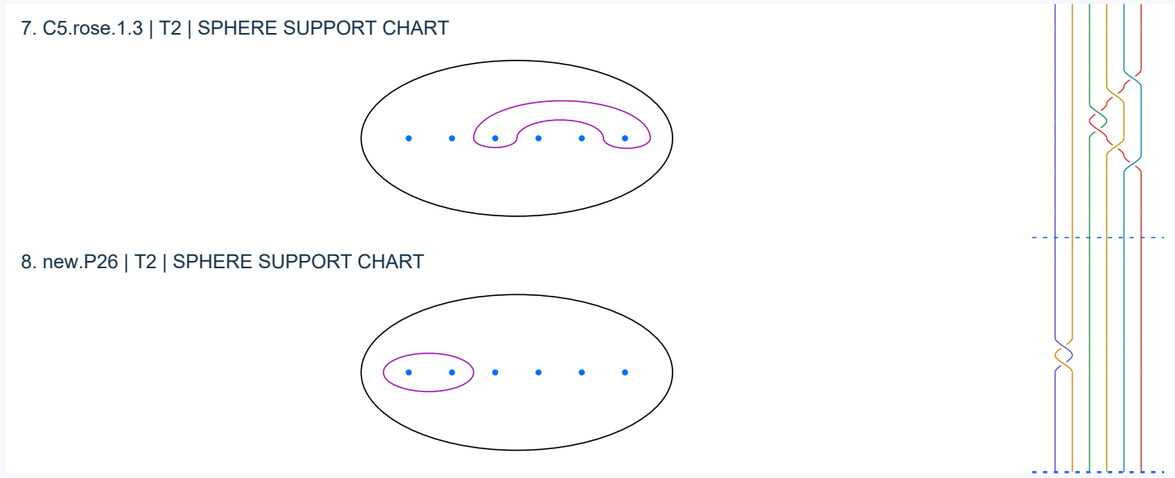
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



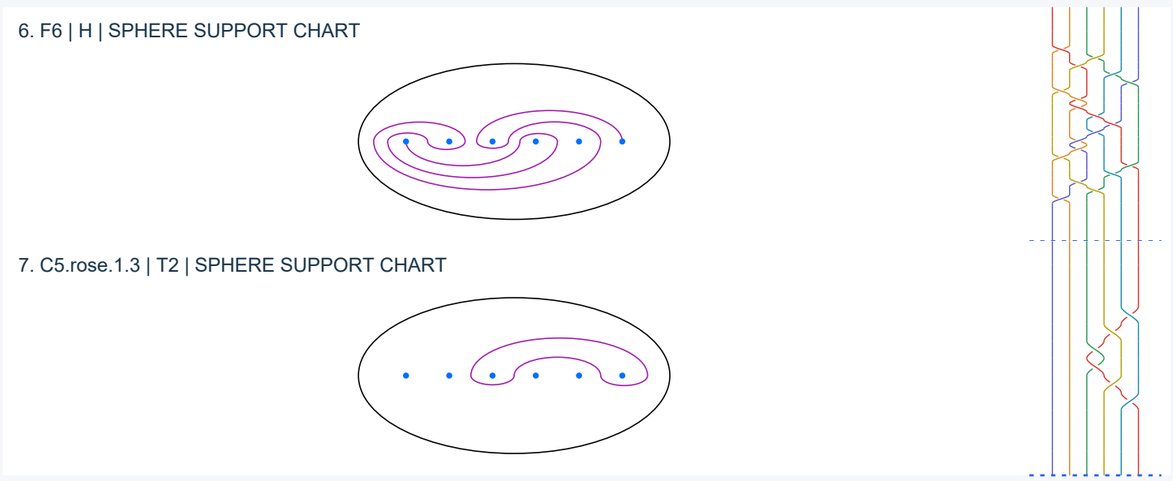
II - step 122

C5.rose.1.3 up past F6

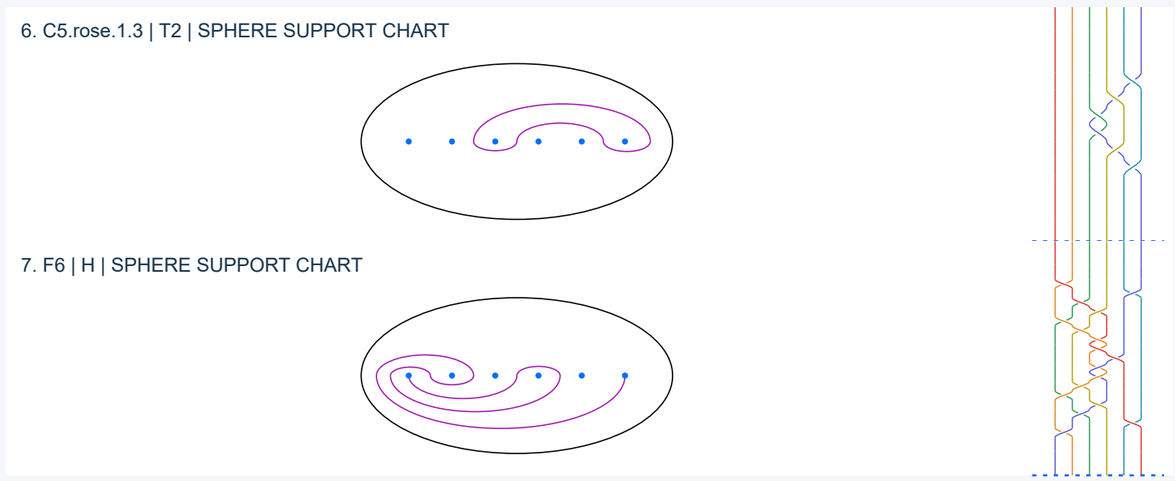
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 123

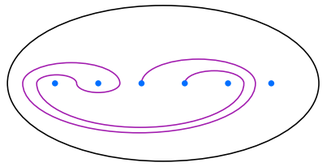
C5.rose.1.3 up past F3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

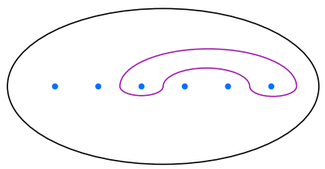
4 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

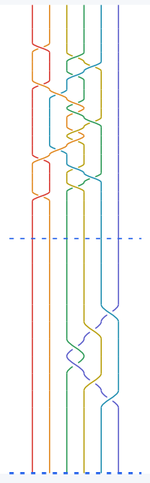
Before

5. F3 | H | SPHERE SUPPORT CHART



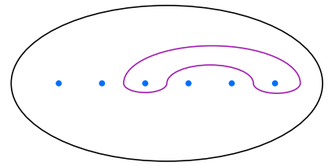
6. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



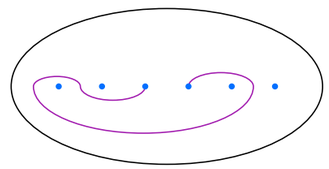


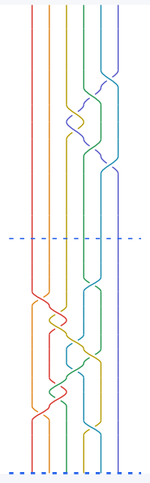
After

5. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



6. F3 | H | SPHERE SUPPORT CHART





II - step 124

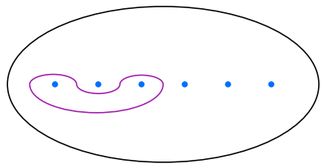
C5.rose.1.3 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

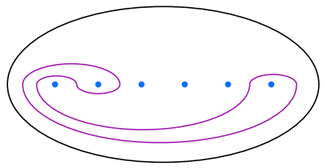
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=4 -2 -2 1 2 -4 -2 5.

Before

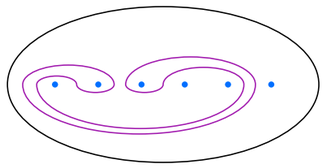
1. cross.P12 | T2



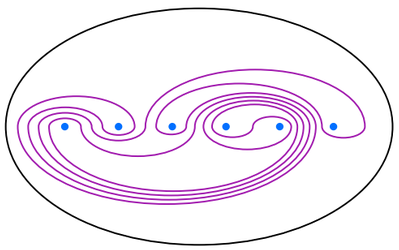
2. cross.P35.square | T2^2



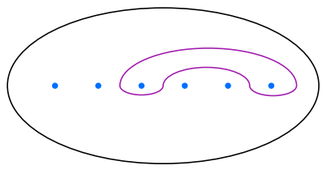
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

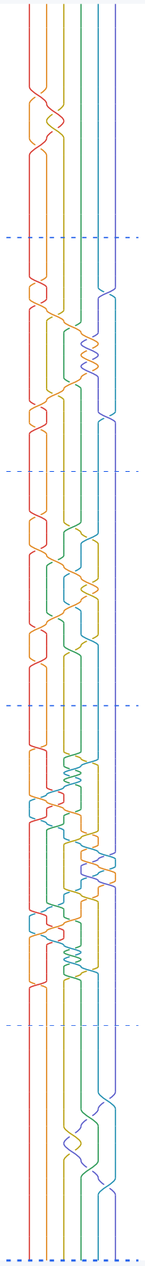


4. cross.C4.unused | T3 | SPHERE SUPPORT CHART



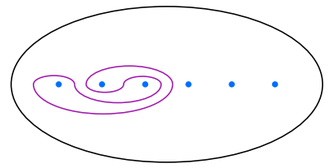
5. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



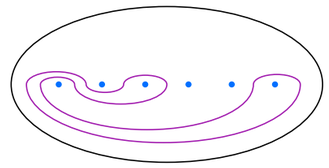


After

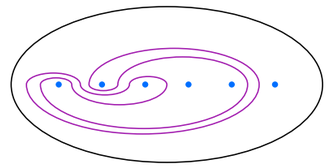
1. cross.P12 | T2



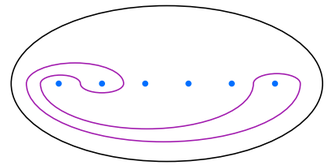
2. cross.P35.square | T2^2



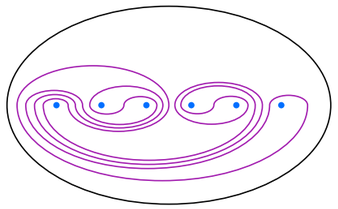
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

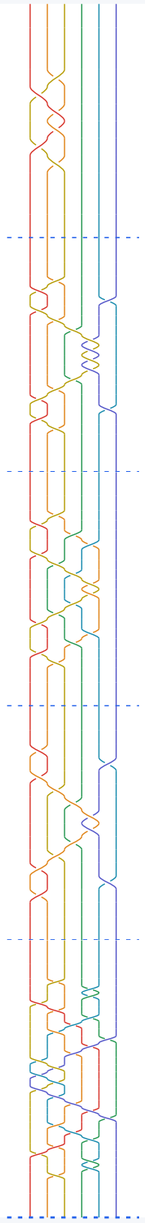


4. C5.rose.1.3 | T2



5. cross.C4.unused | T3 | SPHERE SUPPORT CHART





II - step 124 - continued

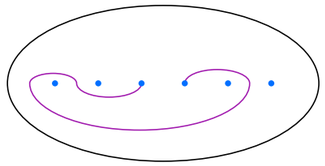
C5.rose.1.3 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

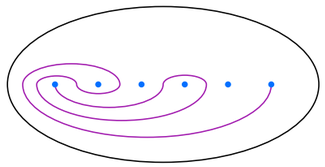
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=4 -2 -2 1 2 -4 -2 5.

Before

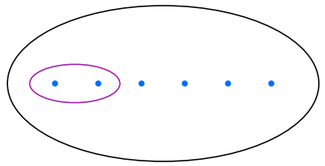
6. F3 | H | SPHERE SUPPORT CHART



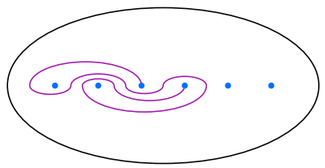
7. F6 | H | SPHERE SUPPORT CHART



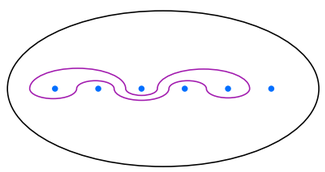
8. new.P26 | T2 | SPHERE SUPPORT CHART

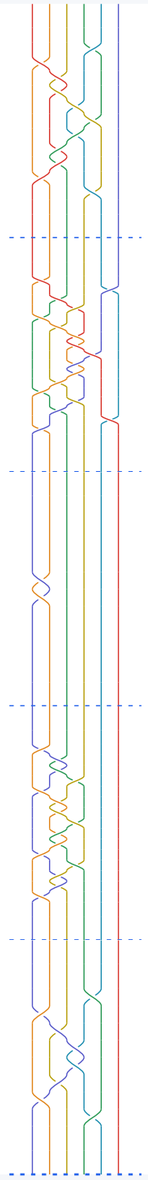


9. F9 | H | SPHERE SUPPORT CHART



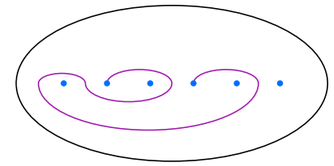
10. residual.P26 | T2 | SPHERE SUPPORT CHART



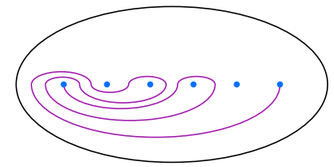


After

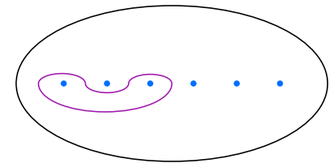
6. F3 | H | SPHERE SUPPORT CHART



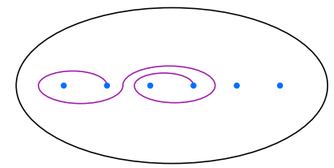
7. F6 | H | SPHERE SUPPORT CHART



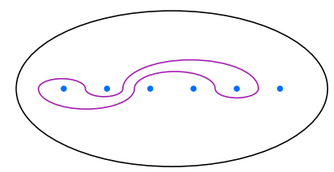
8. new.P26 | T2 | SPHERE SUPPORT CHART

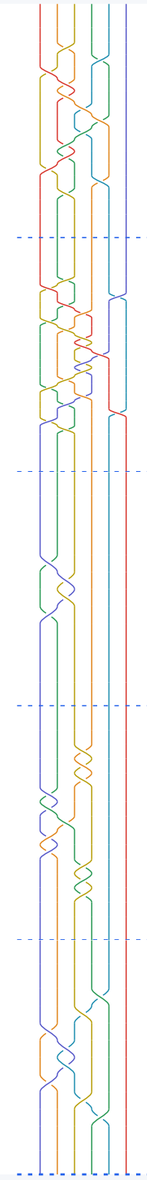


9. F9 | H | SPHERE SUPPORT CHART



10. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 124 - continued

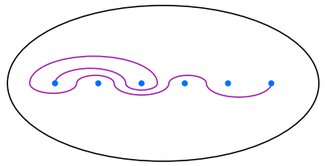
C5.rose.1.3 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

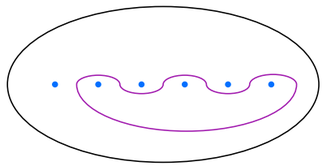
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=4 -2 -2 1 2 -4 -2 5.

Before

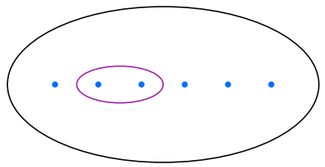
11. F11 | H | SPHERE SUPPORT CHART



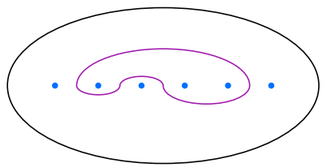
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

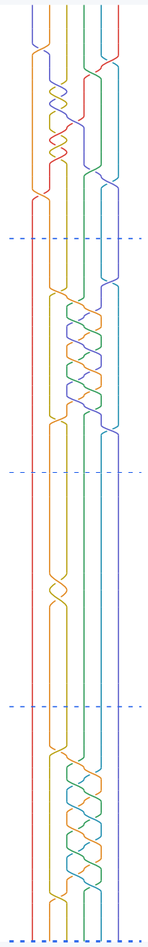


13. C5.rose.2.2 | T2



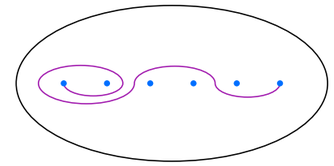
14. C13.recombined | T3^2 | SPHERE SUPPORT CHART



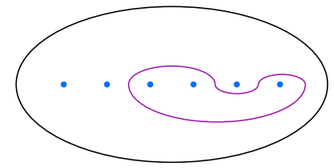


After

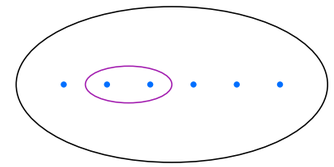
11. F11 | H | SPHERE SUPPORT CHART



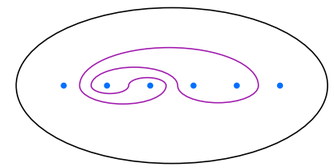
12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART

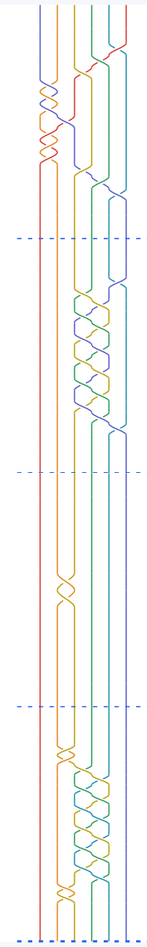


13. C5.rose.2.2 | T2



14. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 125

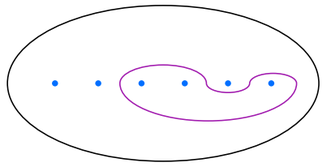
C5.rose.2.2 up past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

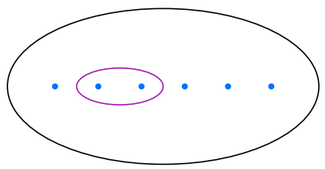
11 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=4 -2 -2 1 2 -4 -2 5; after g=4 -2 -2 1 2 -4 -2 5.

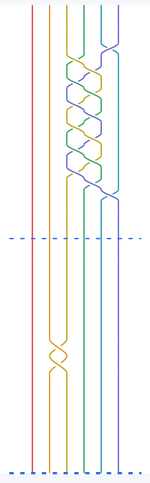
Before

12. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



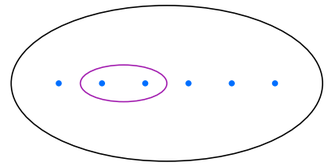
13. C5.rose.2.2 | T2



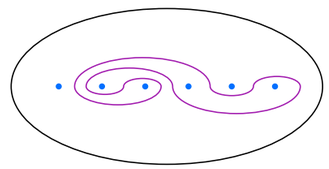


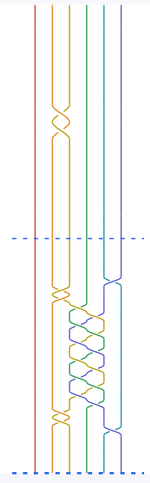
After

12. C5.rose.2.2 | T2



13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART





II - step 126

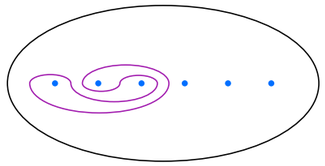
C5.rose.2.2 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

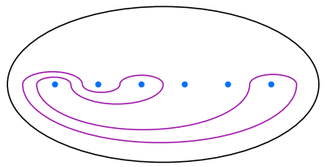
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -2 -2 1 2 -4 -2 5; after g=-2 1 2 -4 -2 5.

Before

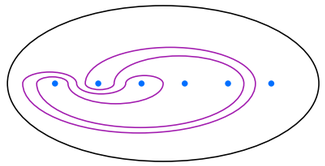
1. cross.P12 | T2



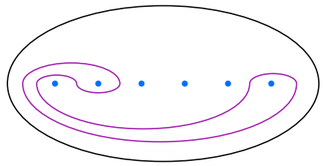
2. cross.P35.square | T2^2



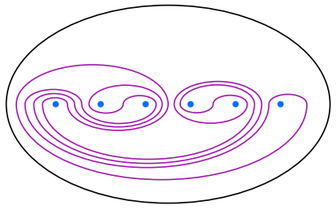
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

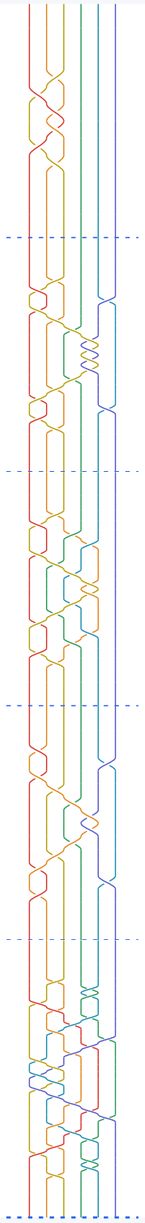


4. C5.rose.1.3 | T2



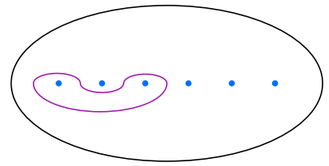
5. cross.C4.unused | T3 | SPHERE SUPPORT CHART



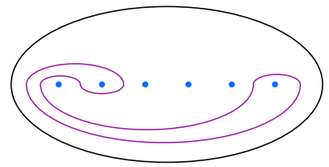


After

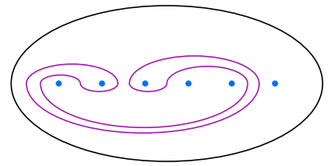
1. cross.P12 | T2



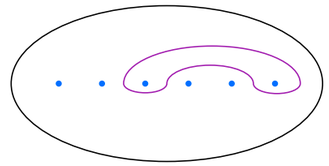
2. cross.P35.square | T2^2



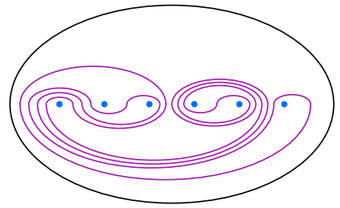
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

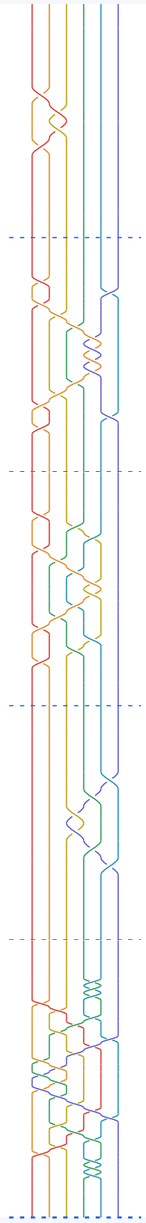


4. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



5. cross.C4.unused | T3 | SPHERE SUPPORT CHART





II - step 126 - continued

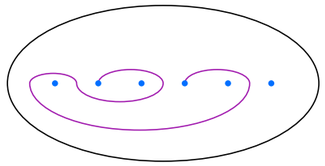
C5.rose.2.2 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

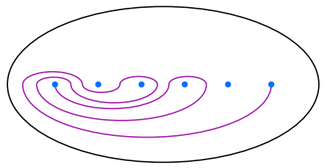
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -2 -2 1 2 -4 -2 5; after g=-2 1 2 -4 -2 5.

Before

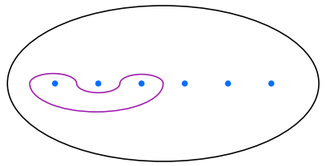
6. F3 | H | SPHERE SUPPORT CHART



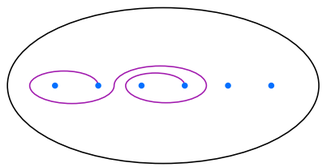
7. F6 | H | SPHERE SUPPORT CHART



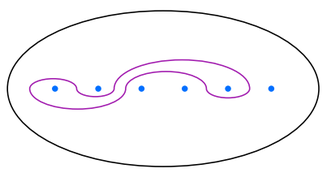
8. new.P26 | T2 | SPHERE SUPPORT CHART

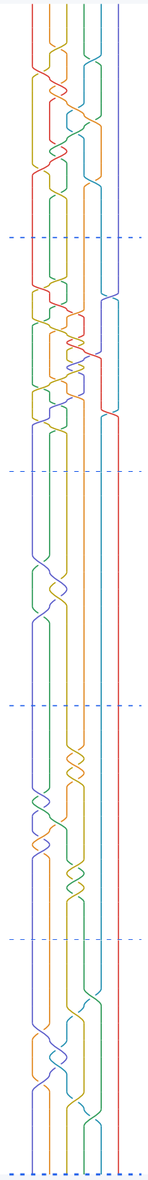


9. F9 | H | SPHERE SUPPORT CHART



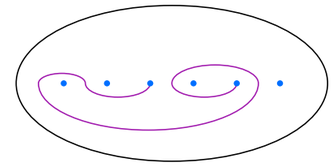
10. residual.P26 | T2 | SPHERE SUPPORT CHART



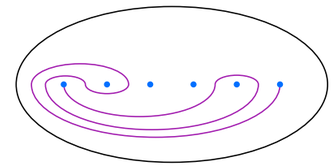


After

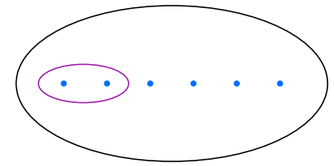
6. F3 | H | SPHERE SUPPORT CHART



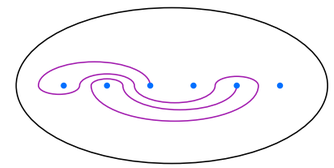
7. F6 | H | SPHERE SUPPORT CHART



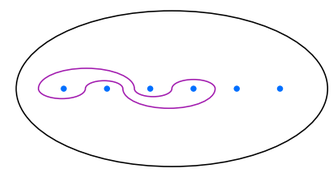
8. new.P26 | T2 | SPHERE SUPPORT CHART

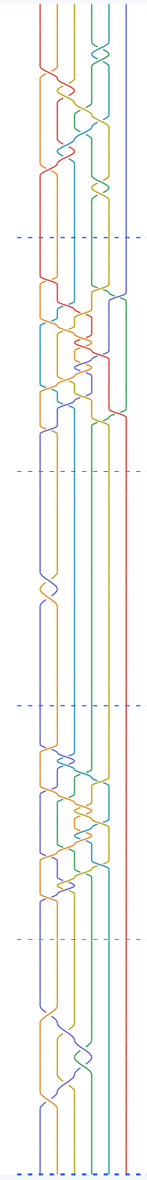


9. F9 | H | SPHERE SUPPORT CHART



10. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 126 - continued

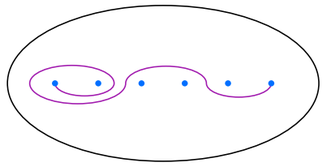
C5.rose.2.2 up past F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

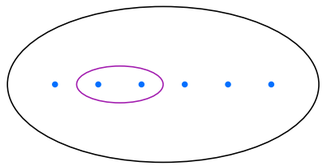
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -2 -2 1 2 -4 -2 5; after g=-2 1 2 -4 -2 5.

Before

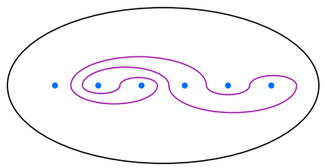
11. F11 | H | SPHERE SUPPORT CHART



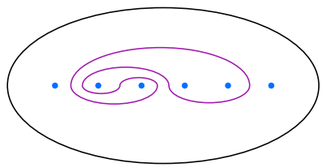
12. C5.rose.2.2 | T2

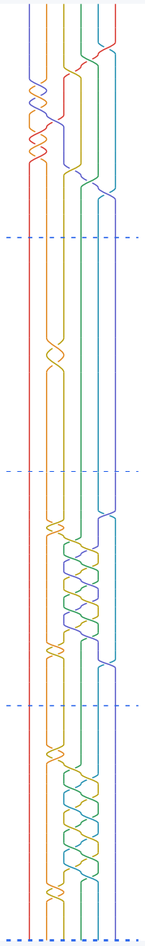


13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



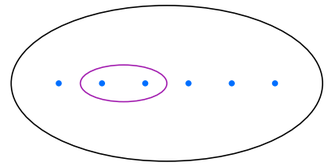
14. C13.recombined | T3^2 | SPHERE SUPPORT CHART



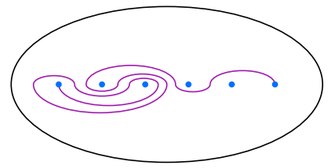


After

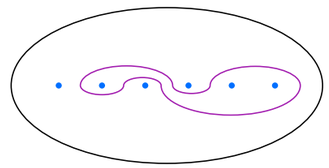
11. C5.rose.2.2 | T2



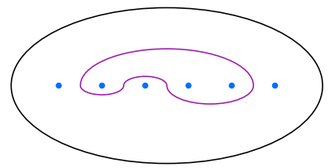
12. F11 | H | SPHERE SUPPORT CHART

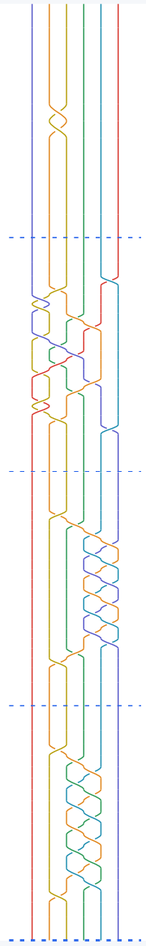


13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



14. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 127

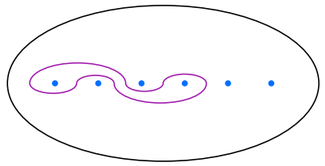
C5.rose.2.2 up past residual.P26

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

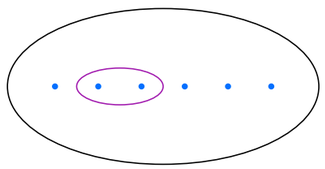
9 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-2 1 2 -4 -2 5; after g=-2 1 2 -4 -2 5.

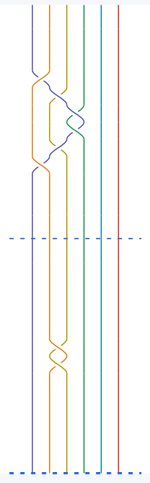
Before

10. residual.P26 | T2 | SPHERE SUPPORT CHART



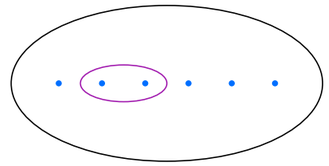
11. C5.rose.2.2 | T2



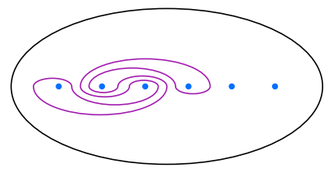


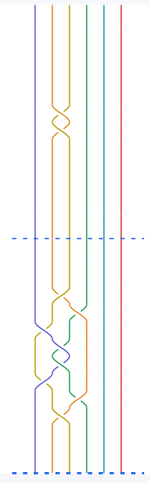
After

10. C5.rose.2.2 | T2



11. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 128

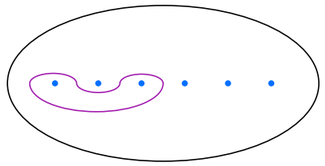
C5.rose.2.2 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

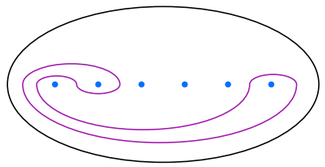
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 1 2 -4 -2 5; after g=2 -4 -2 5.

Before

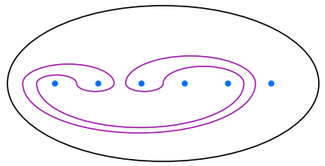
1. cross.P12 | T2



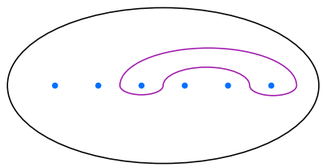
2. cross.P35.square | T2^2



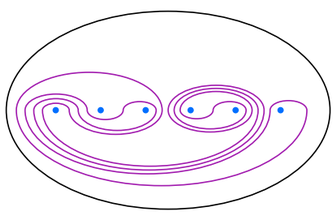
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

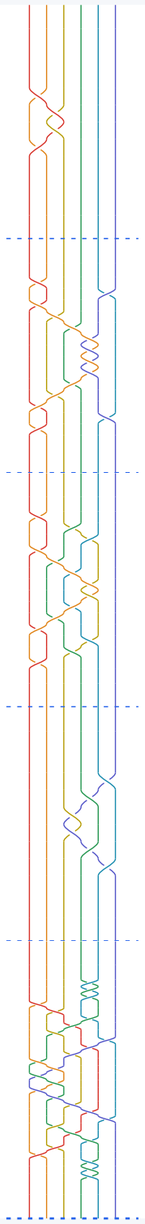


4. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



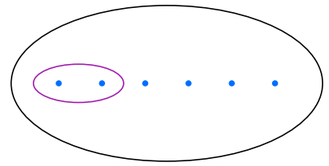
5. cross.C4.unused | T3 | SPHERE SUPPORT CHART



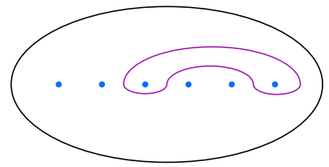


After

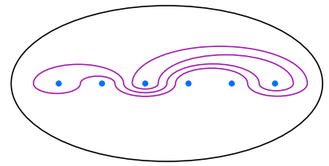
1. cross.P12 | T2



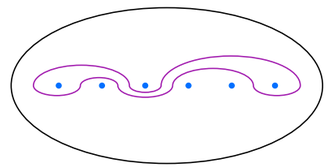
2. cross.P35.square | T2^2 | SPHERE SUPPORT CHART



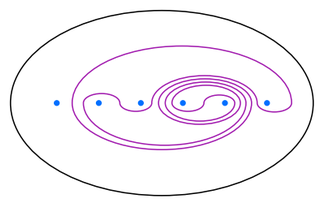
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

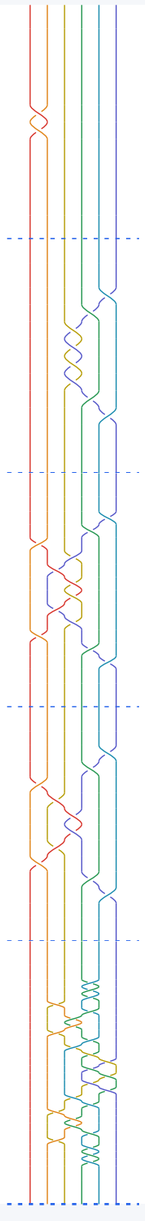


4. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



5. cross.C4.unused | T3 | SPHERE SUPPORT CHART





II - step 128 - continued

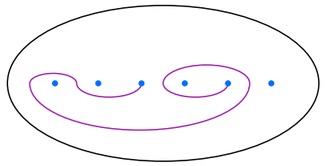
C5.rose.2.2 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

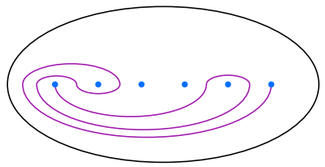
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 1 2 -4 -2 5; after g=2 -4 -2 5.

Before

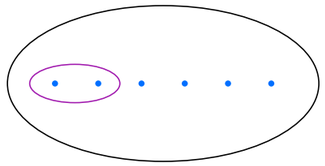
6. F3 | H | SPHERE SUPPORT CHART



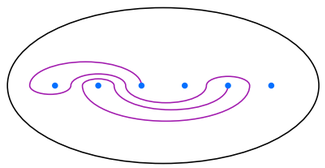
7. F6 | H | SPHERE SUPPORT CHART



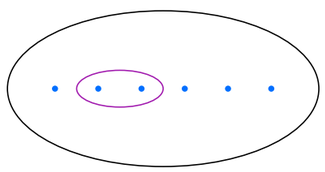
8. new.P26 | T2 | SPHERE SUPPORT CHART

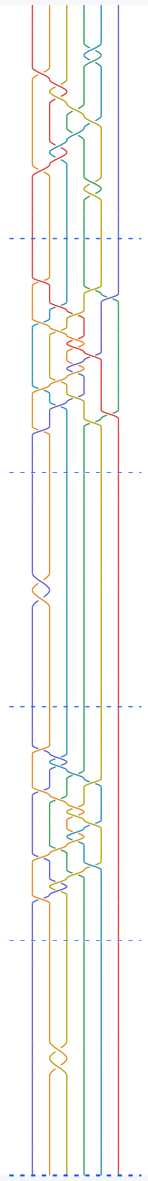


9. F9 | H | SPHERE SUPPORT CHART



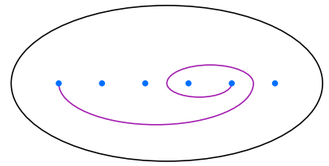
10. C5.rose.2.2 | T2



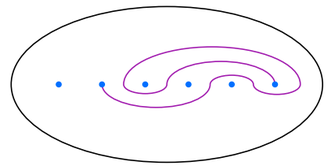


After

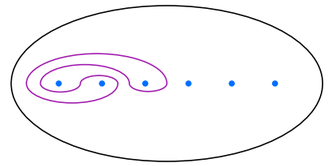
6. F3 | H | SPHERE SUPPORT CHART



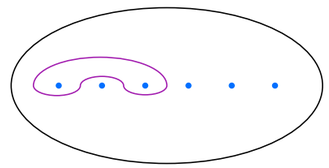
7. F6 | H | SPHERE SUPPORT CHART



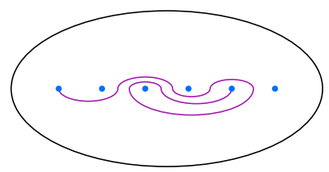
8. new.P26 | T2 | SPHERE SUPPORT CHART

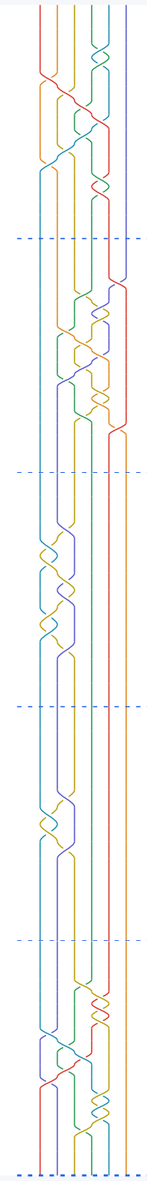


9. C5.rose.2.2 | T2



10. F9 | H | SPHERE SUPPORT CHART





II - step 128 - continued

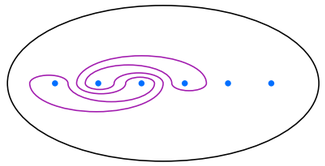
C5.rose.2.2 up past F9

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

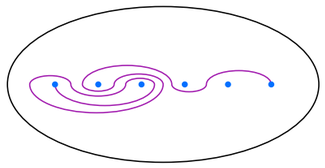
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 1 2 -4 -2 5; after g=2 -4 -2 5.

Before

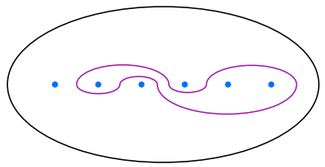
11. residual.P26 | T2 | SPHERE SUPPORT CHART



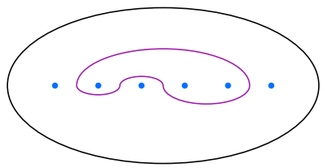
12. F11 | H | SPHERE SUPPORT CHART

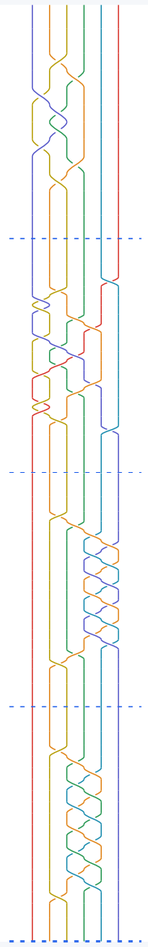


13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



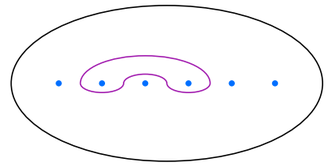
14. C13.recombined | T3^2 | SPHERE SUPPORT CHART



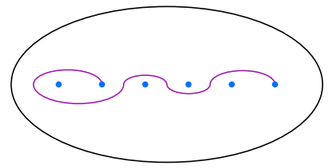


After

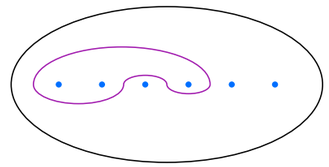
11. residual.P26 | T2 | SPHERE SUPPORT CHART



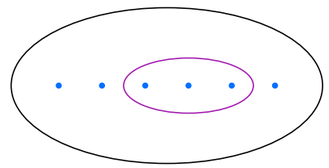
12. F11 | H | SPHERE SUPPORT CHART

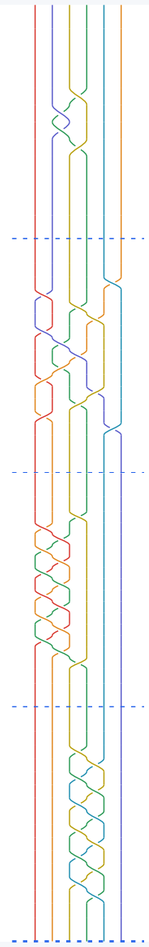


13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



14. C13.recombined | T3^2 | SPHERE SUPPORT CHART





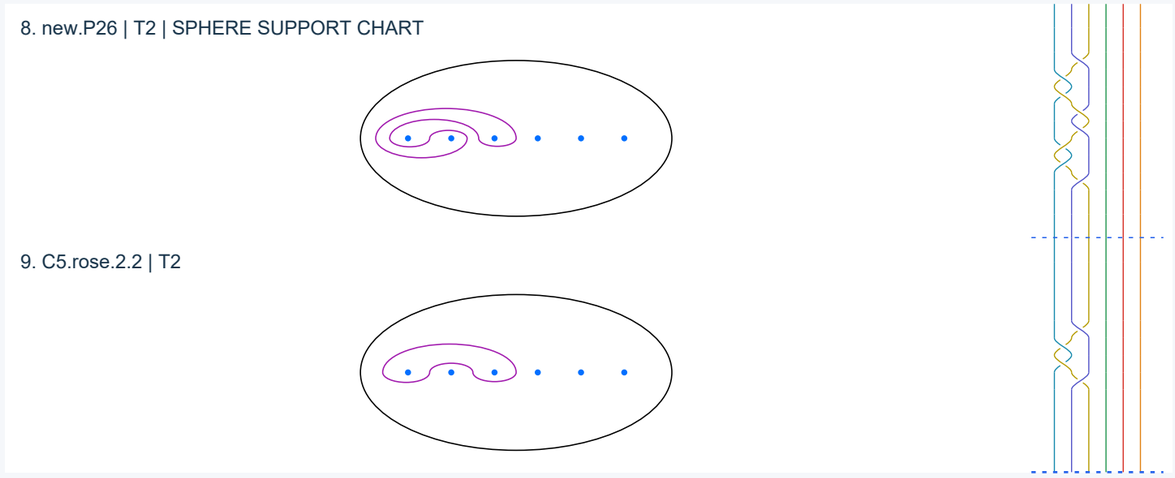
II - step 129

C5.rose.2.2 up past new.P26

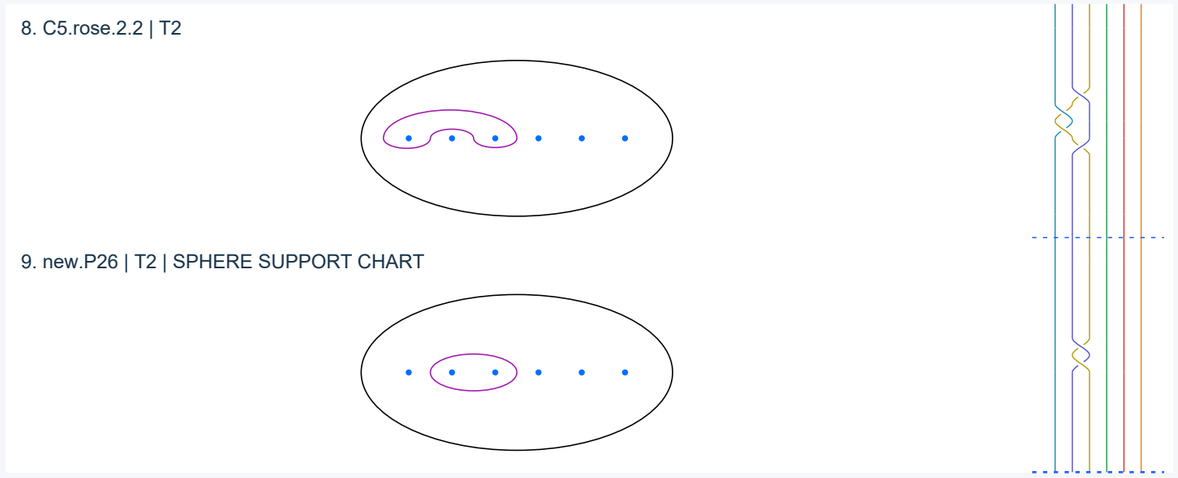
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=2 -4 -2 5; after g=2 -4 -2 5.

Before



After



II - step 130

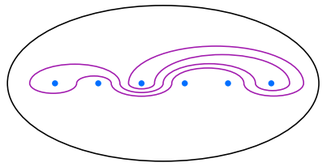
C5.rose.2.2 up past F6

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

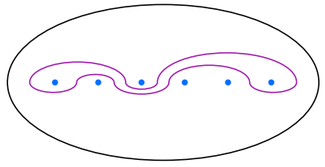
2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=2 -4 -2 5; after g=1 2 -4 -2 5.

Before

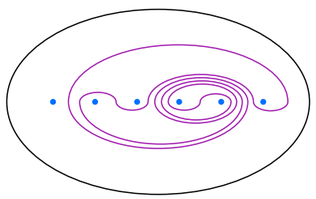
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



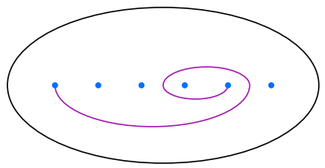
4. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



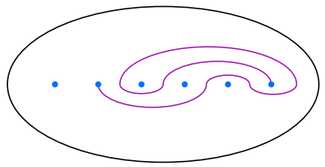
5. cross.C4.unused | T3 | SPHERE SUPPORT CHART

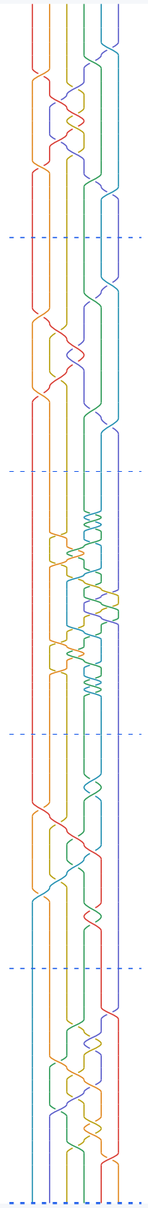


6. F3 | H | SPHERE SUPPORT CHART



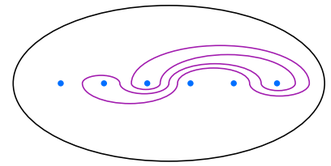
7. F6 | H | SPHERE SUPPORT CHART



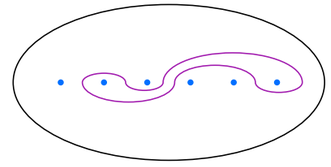


After

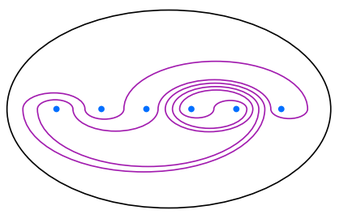
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART



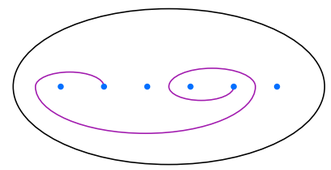
4. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



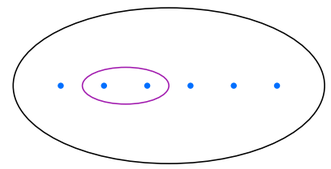
5. cross.C4.unused | T3 | SPHERE SUPPORT CHART

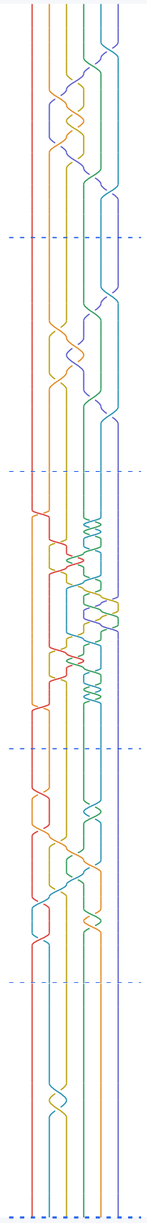


6. F3 | H | SPHERE SUPPORT CHART



7. C5.rose.2.2 | T2





II - step 130 - continued

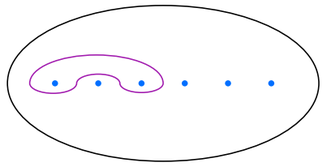
C5.rose.2.2 up past F6

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

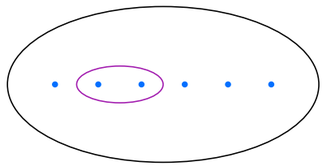
2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=2 -4 -2 5; after g=1 2 -4 -2 5.

Before

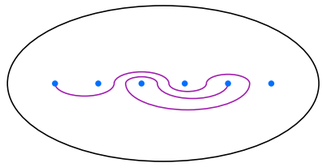
8. C5.rose.2.2 | T2



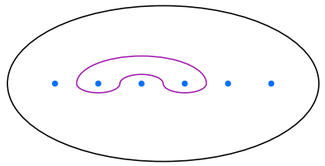
9. new.P26 | T2 | SPHERE SUPPORT CHART



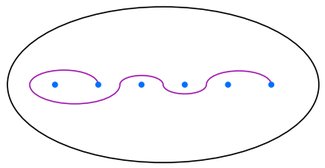
10. F9 | H | SPHERE SUPPORT CHART

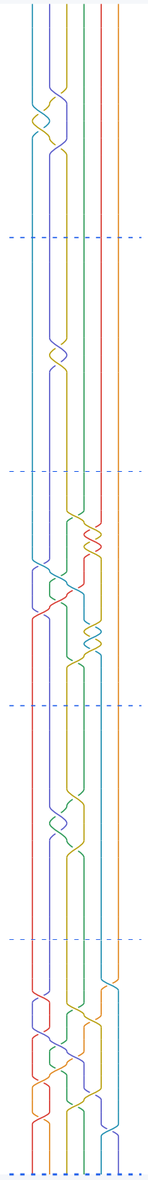


11. residual.P26 | T2 | SPHERE SUPPORT CHART



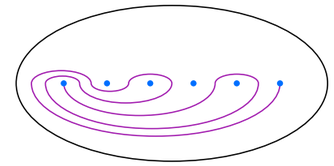
12. F11 | H | SPHERE SUPPORT CHART



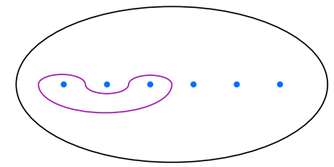


After

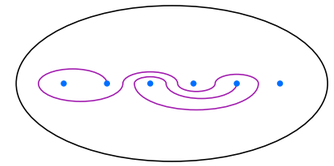
8. F6 | H | SPHERE SUPPORT CHART



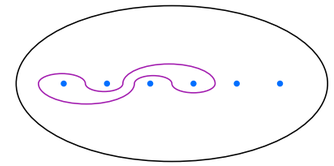
9. new.P26 | T2 | SPHERE SUPPORT CHART



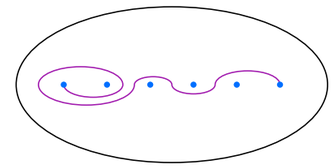
10. F9 | H | SPHERE SUPPORT CHART

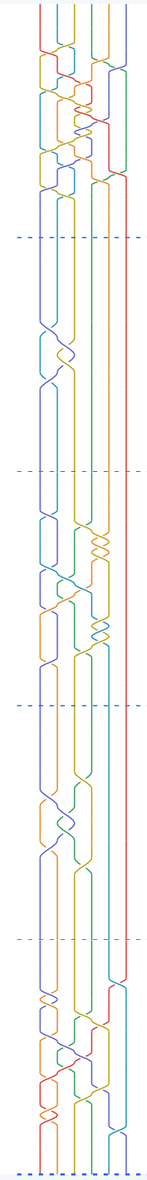


11. residual.P26 | T2 | SPHERE SUPPORT CHART



12. F11 | H | SPHERE SUPPORT CHART





II - step 131

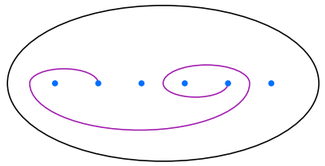
C5.rose.2.2 up past F3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

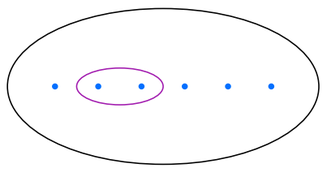
5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=1 2 -4 -2 5; after g=1 2 -4 -2 5.

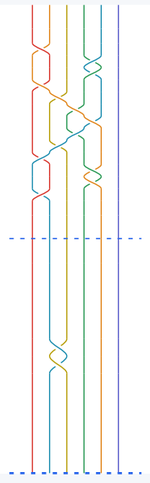
Before

6. F3 | H | SPHERE SUPPORT CHART



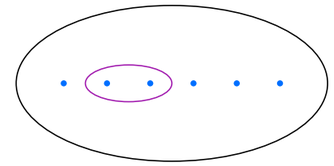
7. C5.rose.2.2 | T2



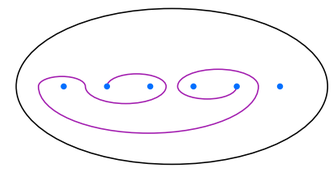


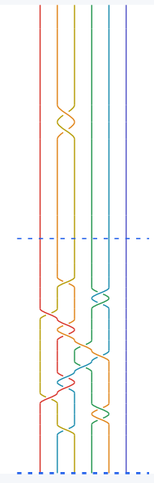
After

6. C5.rose.2.2 | T2



7. F3 | H | SPHERE SUPPORT CHART





II - step 132

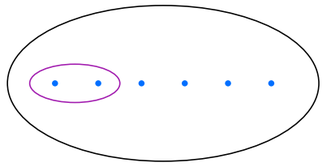
C5.rose.2.2 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

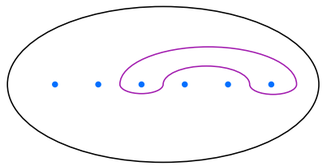
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 -2 5; after g=-4 -2 5.

Before

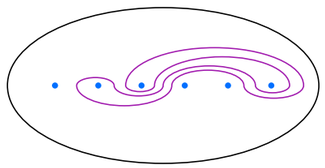
1. cross.P12 | T2



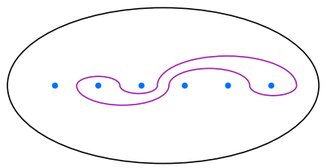
2. cross.P35.square | T2^2 | SPHERE SUPPORT CHART



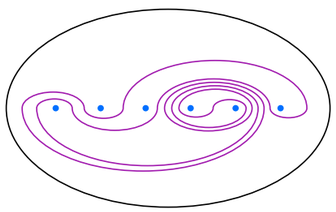
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

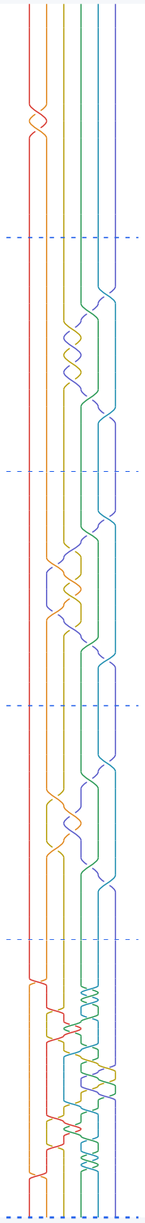


4. C5.rose.1.3 | T2 | SPHERE SUPPORT CHART



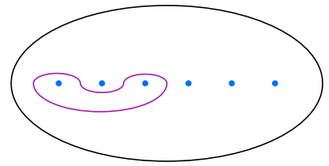
5. cross.C4.unused | T3 | SPHERE SUPPORT CHART



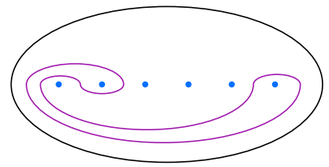


After

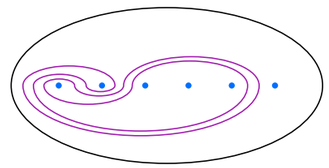
1. cross.P12 | T2



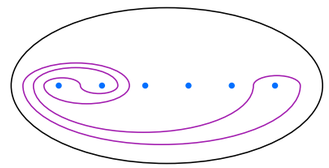
2. cross.P35.square | T2^2



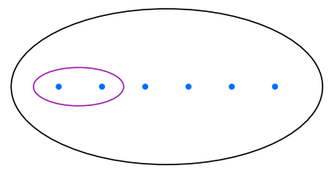
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

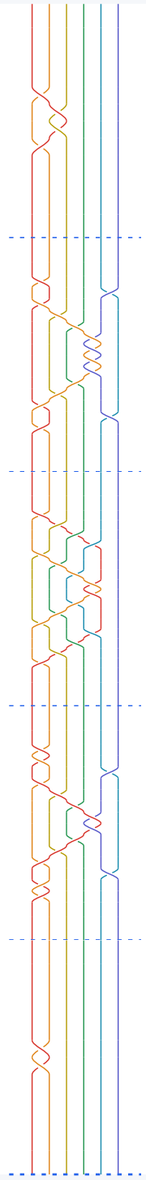


4. C5.rose.1.3 | T2



5. C5.rose.2.2 | T2





II - step 132 - continued

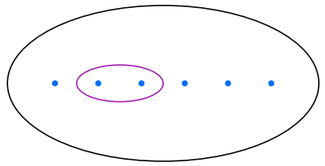
C5.rose.2.2 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

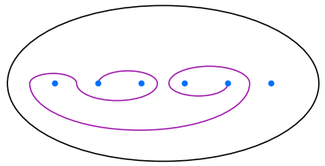
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 -2 5; after g=-4 -2 5.

Before

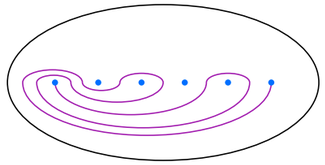
6. C5.rose.2.2 | T2



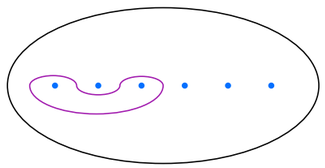
7. F3 | H | SPHERE SUPPORT CHART



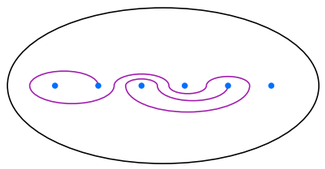
8. F6 | H | SPHERE SUPPORT CHART

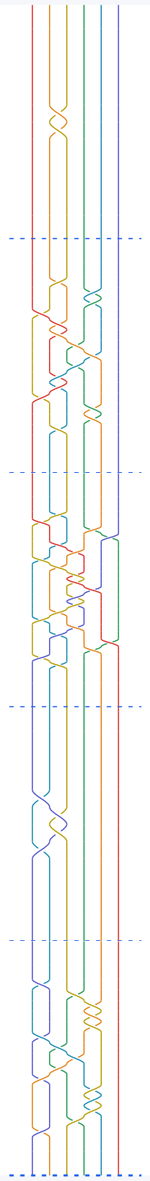


9. new.P26 | T2 | SPHERE SUPPORT CHART



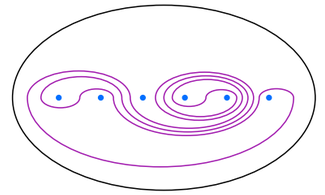
10. F9 | H | SPHERE SUPPORT CHART



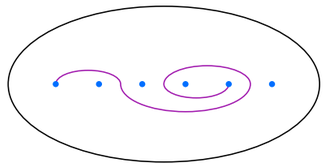


After

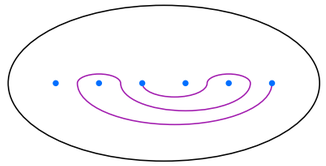
6. cross.C4.unused | T3 | SPHERE SUPPORT CHART



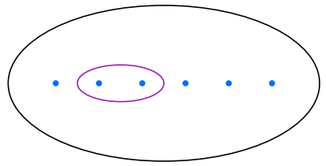
7. F3 | H | SPHERE SUPPORT CHART



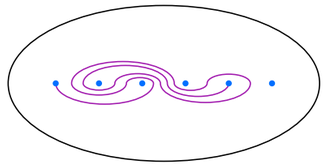
8. F6 | H | SPHERE SUPPORT CHART

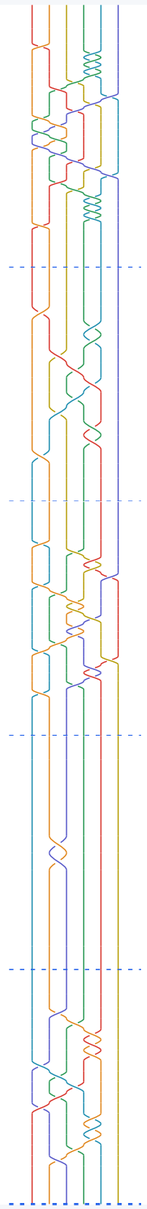


9. new.P26 | T2 | SPHERE SUPPORT CHART



10. F9 | H | SPHERE SUPPORT CHART





II - step 132 - continued

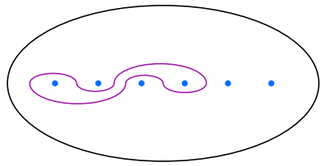
C5.rose.2.2 up past cross.C4.unused

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

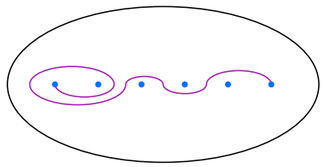
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 -2 5; after g=-4 -2 5.

Before

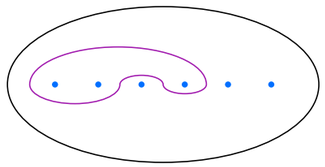
11. residual.P26 | T2 | SPHERE SUPPORT CHART



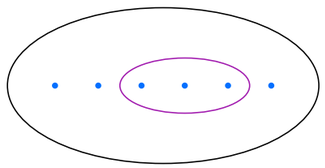
12. F11 | H | SPHERE SUPPORT CHART

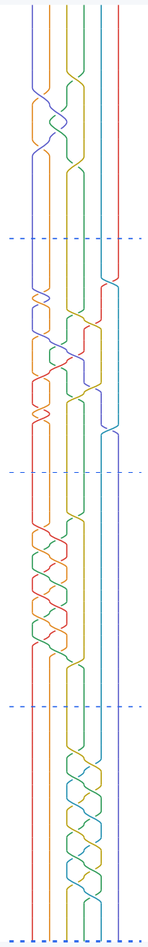


13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



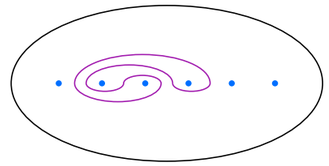
14. C13.recombined | T3^2 | SPHERE SUPPORT CHART



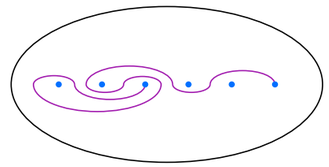


After

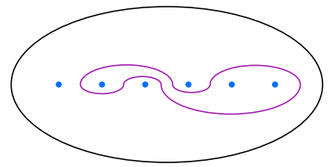
11. residual.P26 | T2 | SPHERE SUPPORT CHART



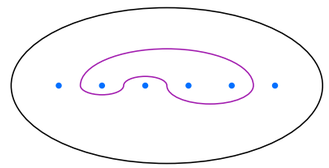
12. F11 | H | SPHERE SUPPORT CHART

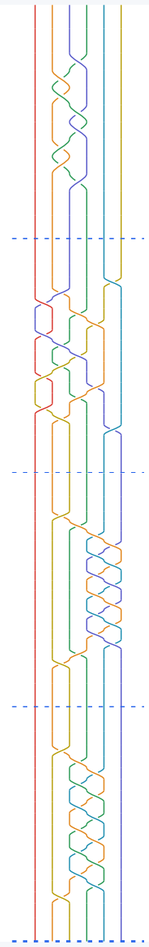


13. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



14. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 133

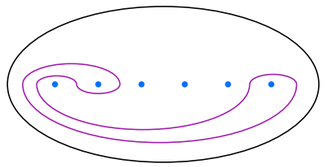
Compress the five 135 petals and retain inverse P15

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

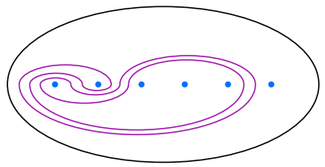
1 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=-4 -2 5.

Before

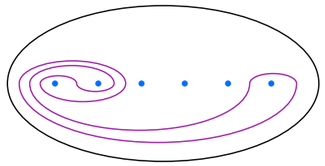
2. cross.P35.square | T2^2



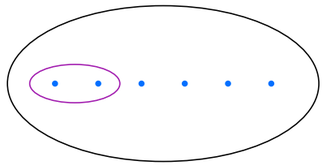
3. C5.rose.1.2 | T2 | SPHERE SUPPORT CHART

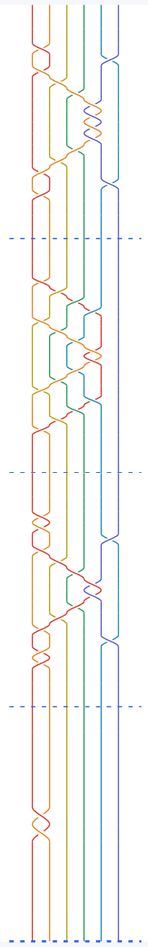


4. C5.rose.1.3 | T2



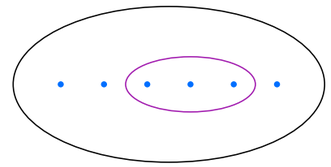
5. C5.rose.2.2 | T2



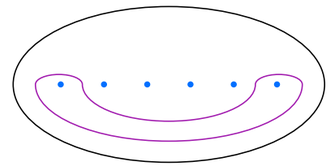


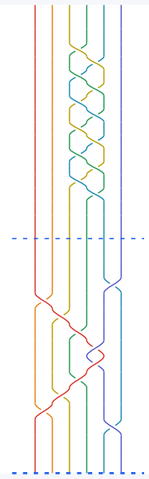
After

2. alternate.C5.square | T3^2 | SPHERE SUPPORT CHART



3. alternate.inverse.P15 | T2^-1





II - step 134

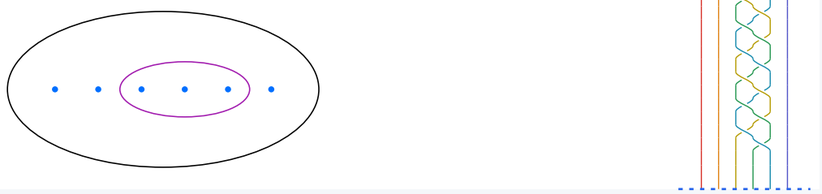
Use the half-rotated PC half lantern twice

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=-4 -2 5.

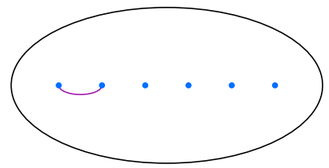
Before

2. alternate.C5.square | T3^2 | SPHERE SUPPORT CHART

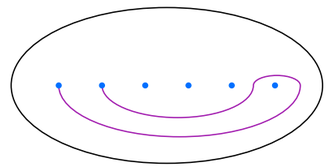


After

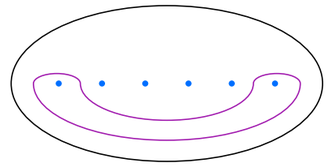
2. alternate.HL1.1 | H



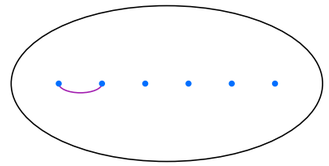
3. alternate.HL1.2 | H



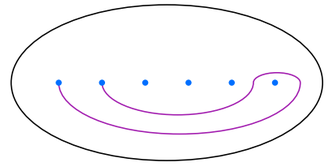
4. alternate.HL1.3 | T2^2

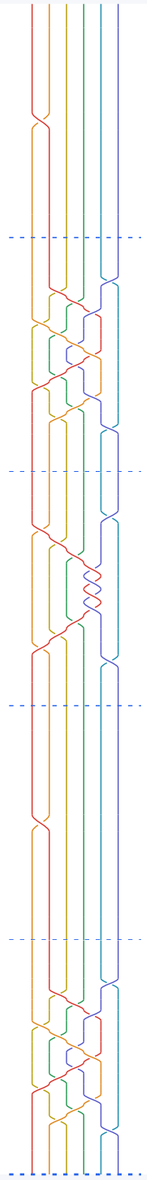


5. alternate.HL2.1 | H



6. alternate.HL2.2 | H





II - step 134 - continued

Use the half-rotated PC half lantern twice

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=-4 -2 5.

Before

After

7. alternate.HL2.3 | T2^2



II - step 135

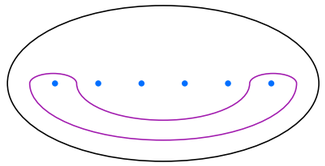
Cancel inverse P15 against one unit of the last squared pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

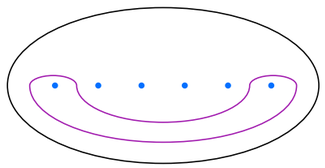
6 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=-4 -2 5.

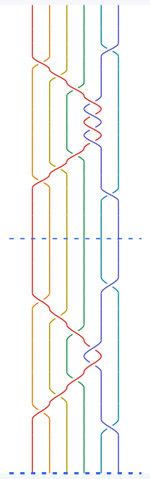
Before

7. alternate.HL2.3 | T2^2



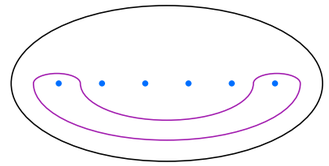
8. alternate.inverse.P15 | T2^-1

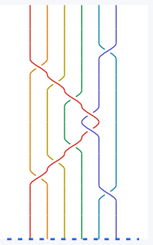




After

7. alternate.P15.remaining | T2





II - step 136

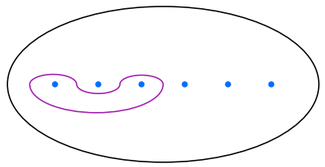
Expand the last ordinary C4 root by a PC rose

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

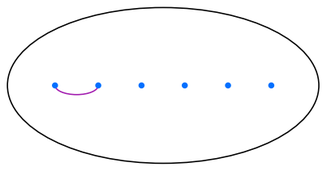
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=1 -4 -2 5.

Before

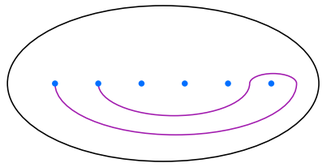
1. cross.P12 | T2



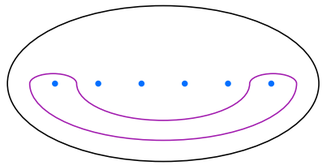
2. alternate.HL1.1 | H



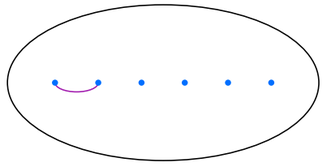
3. alternate.HL1.2 | H

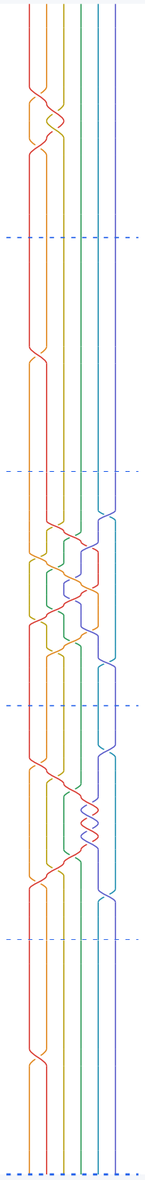


4. alternate.HL1.3 | T2^2



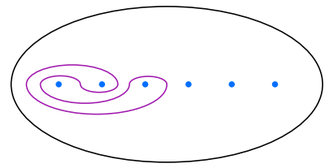
5. alternate.HL2.1 | H



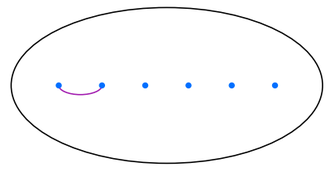


After

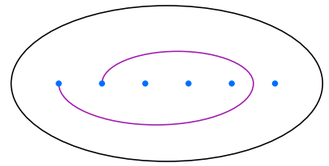
1. cross.P12 | T2



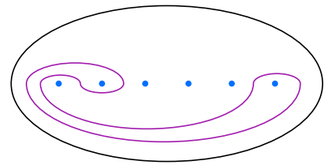
2. alternate.HL1.1 | H



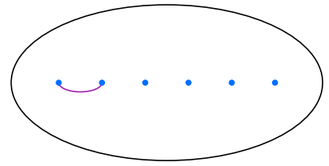
3. alternate.HL1.2 | H | SPHERE SUPPORT CHART

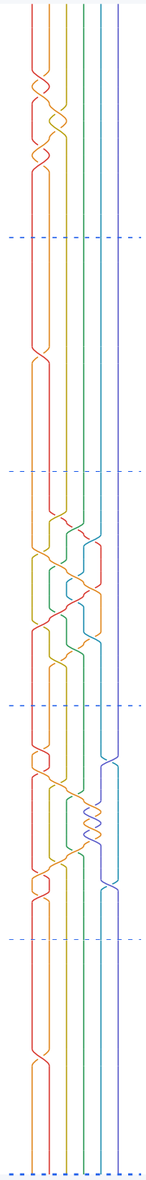


4. alternate.HL1.3 | T2^2



5. alternate.HL2.1 | H





II - step 136 - continued

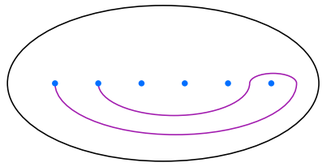
Expand the last ordinary C4 root by a PC rose

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

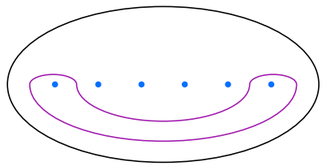
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=1 -4 -2 5.

Before

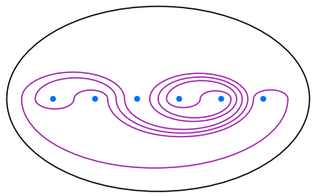
6. alternate.HL2.2 | H



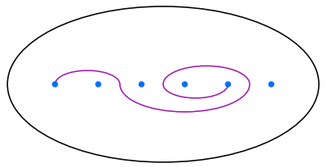
7. alternate.P15.remaining | T2



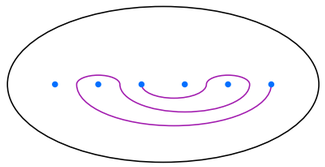
8. cross.C4.unused | T3 | SPHERE SUPPORT CHART

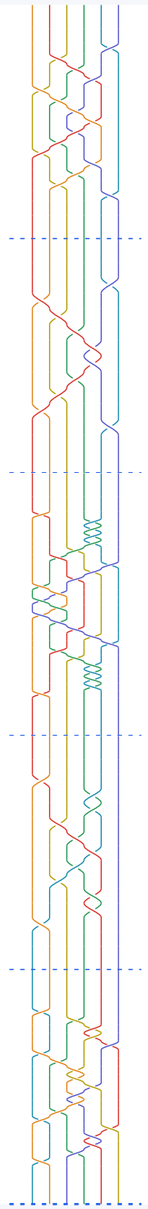


9. F3 | H | SPHERE SUPPORT CHART



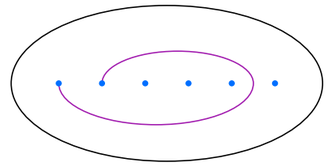
10. F6 | H | SPHERE SUPPORT CHART



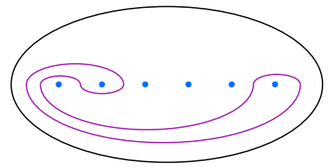


After

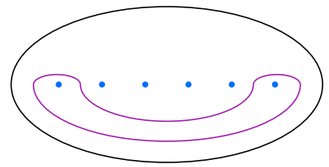
6. alternate.HL2.2 | H | SPHERE SUPPORT CHART



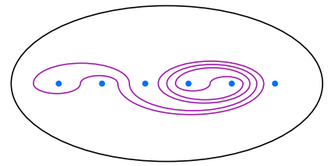
7. alternate.P15.remaining | T2



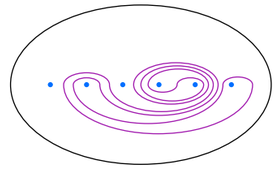
8. alternate.C4.rose.1 | T2

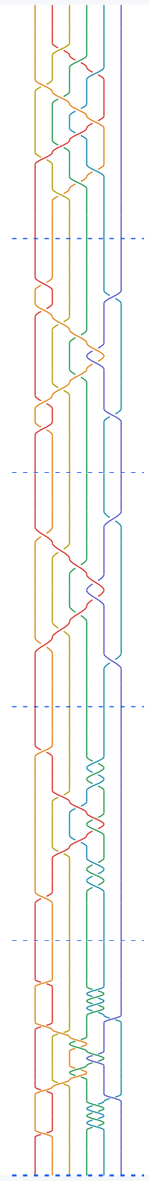


9. alternate.C4.rose.2 | T2 | SPHERE SUPPORT CHART



10. alternate.C4.rose.3 | T2 | SPHERE SUPPORT CHART





II - step 136 - continued

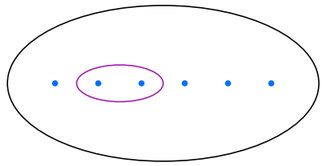
Expand the last ordinary C4 root by a PC rose

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

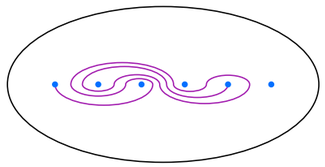
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=1 -4 -2 5.

Before

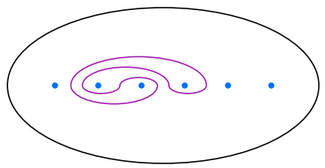
11. new.P26 | T2 | SPHERE SUPPORT CHART



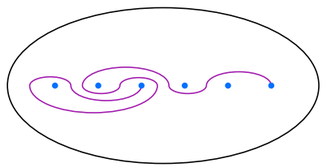
12. F9 | H | SPHERE SUPPORT CHART



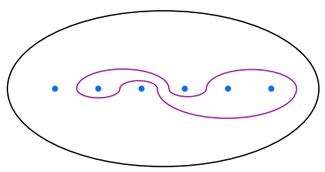
13. residual.P26 | T2 | SPHERE SUPPORT CHART

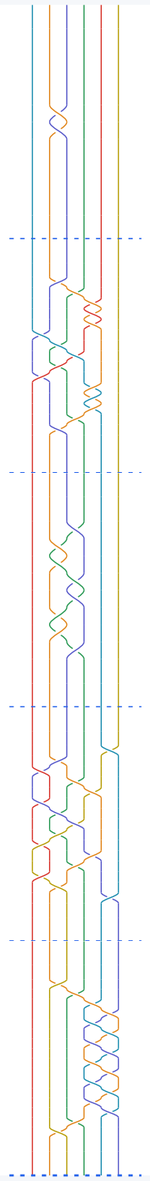


14. F11 | H | SPHERE SUPPORT CHART



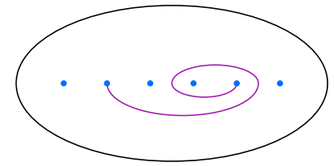
15. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



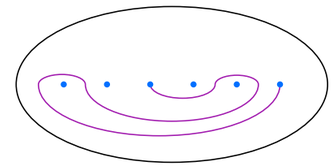


After

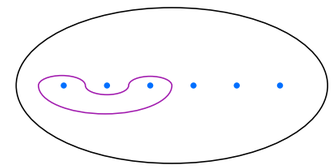
11. F3 | H | SPHERE SUPPORT CHART



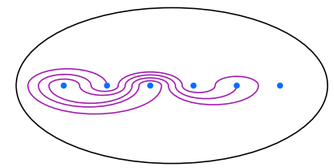
12. F6 | H | SPHERE SUPPORT CHART



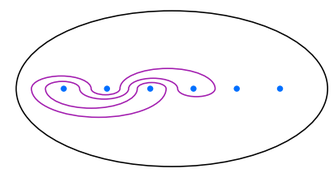
13. new.P26 | T2 | SPHERE SUPPORT CHART

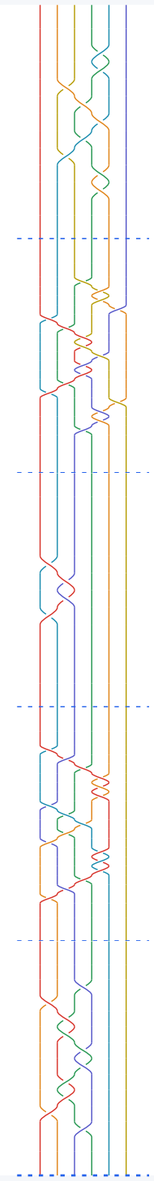


14. F9 | H | SPHERE SUPPORT CHART



15. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 136 - continued

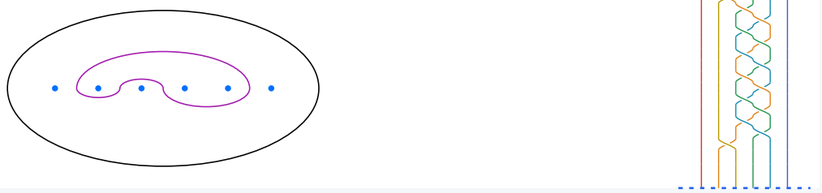
Expand the last ordinary C4 root by a PC rose

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=1 -4 -2 5.

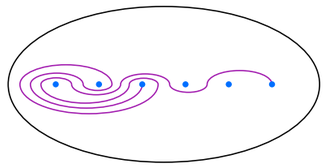
Before

16. C13.recombined | T3^2 | SPHERE SUPPORT CHART

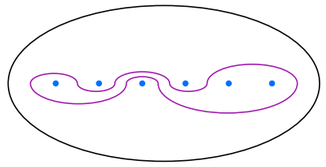


After

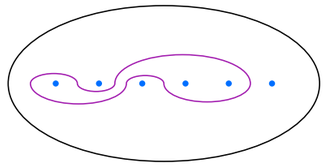
16. F11 | H | SPHERE SUPPORT CHART

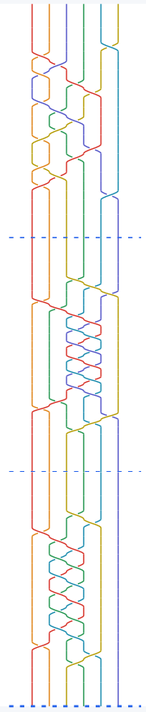


17. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



18. C13.recombined | T3^2 | SPHERE SUPPORT CHART





II - step 137

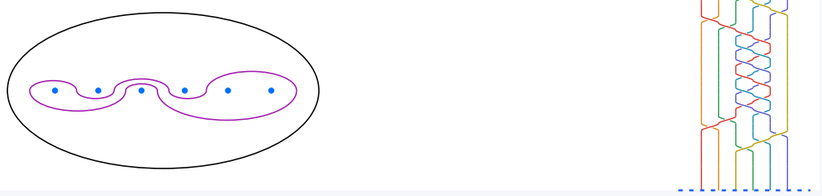
Draw the whole C12 square as H56 inverse applied to C13

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

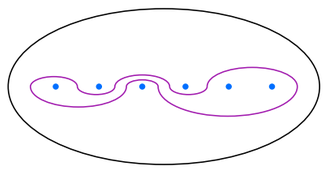
Before

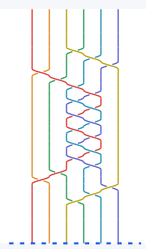
17. F12.root+F12.2+F12.root+F12.3 | T3^2 | SPHERE SUPPORT CHART



After

17. last.simple.C12 | T3^2 | SPHERE SUPPORT CHART





II - step 138

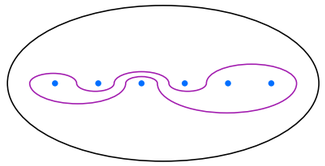
Apply the ordered two-lantern consequence; retain both inverses

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

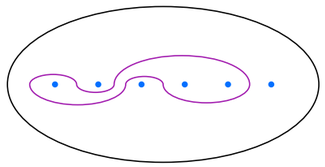
16 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

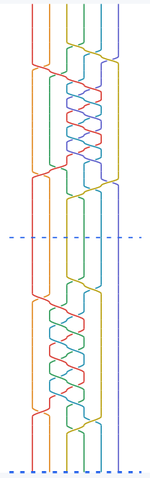
Before

17. last.simple.C12 | T3^2 | SPHERE SUPPORT CHART



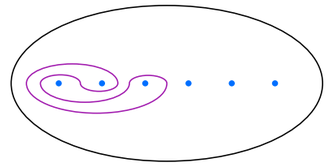
18. C13.recombined | T3^2 | SPHERE SUPPORT CHART



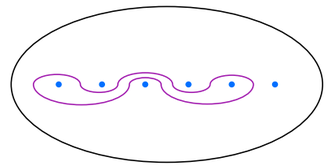


After

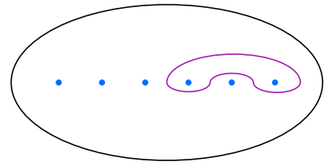
17. last.P12.square | T2^2



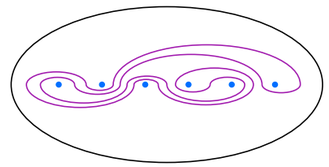
18. last.P34.square | T2^2

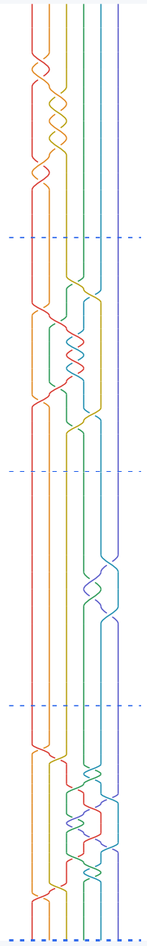


19. last.inverse.Q56 | T2^-1



20. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART





II - step 139

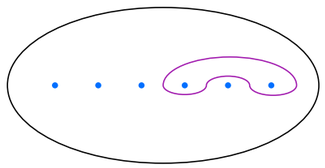
Transport the second inverse up across last.inverse.Q56

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

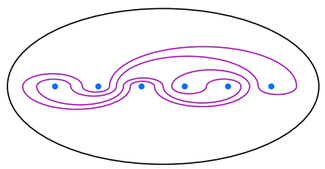
18 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

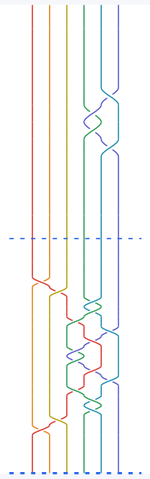
Before

19. last.inverse.Q56 | T2^-1



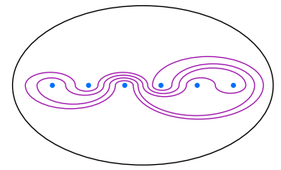
20. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



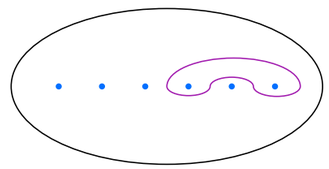


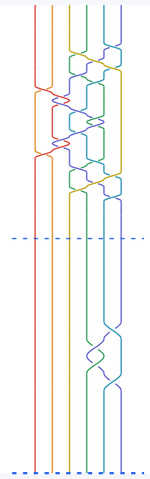
After

19. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



20. last.inverse.Q56 | T2^-1





II - step 140

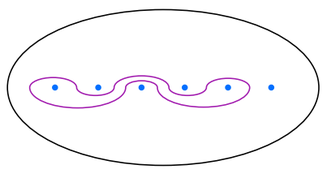
Transport the second inverse up across last.P34.square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

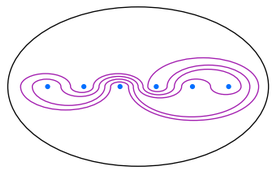
17 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

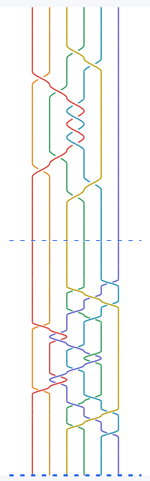
Before

18. last.P34.square | T2^2



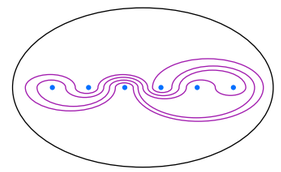
19. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



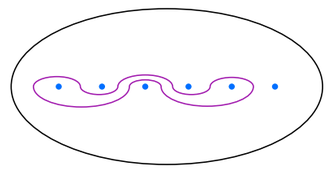


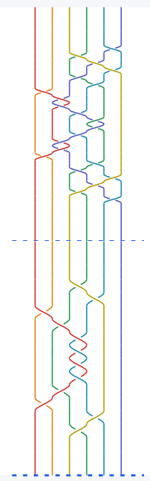
After

18. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



19. last.P34.square | T2^2





II - step 141

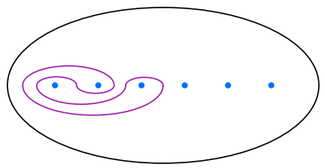
Transport the second inverse up across last.P12.square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

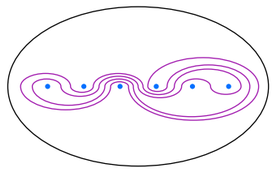
16 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

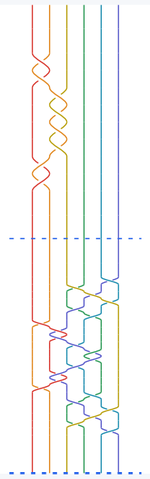
Before

17. last.P12.square | T2^2



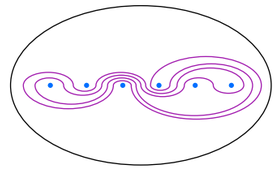
18. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



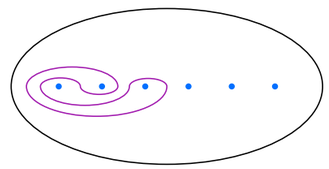


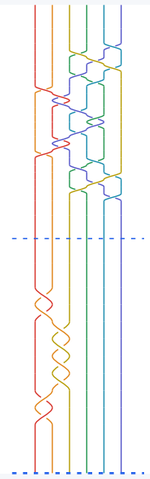
After

17. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



18. last.P12.square | T2^2





II - step 142

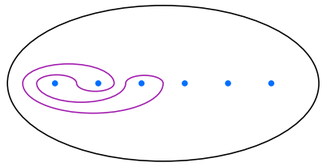
Transport the second inverse up across F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

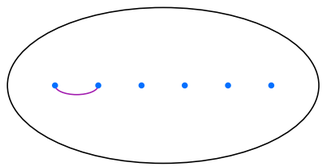
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=-2 5.

Before

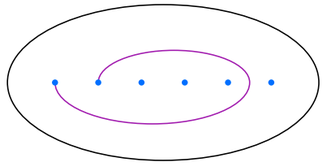
1. cross.P12 | T2



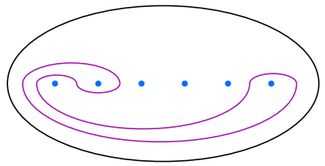
2. alternate.HL1.1 | H



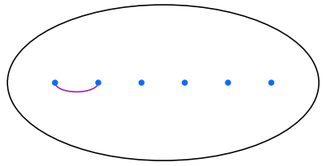
3. alternate.HL1.2 | H | SPHERE SUPPORT CHART

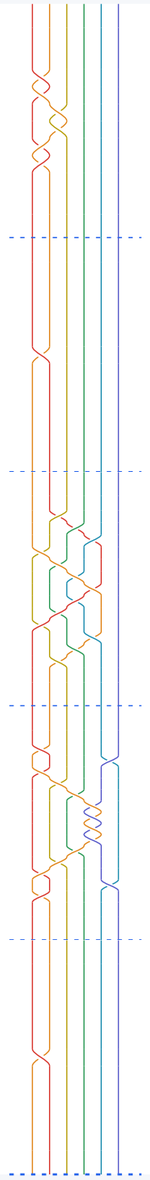


4. alternate.HL1.3 | T2^2



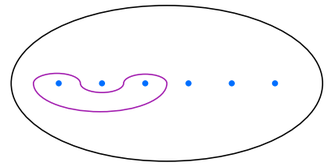
5. alternate.HL2.1 | H



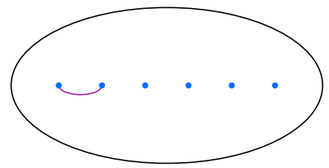


After

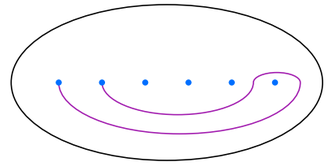
1. cross.P12 | T2



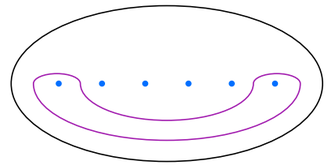
2. alternate.HL1.1 | H



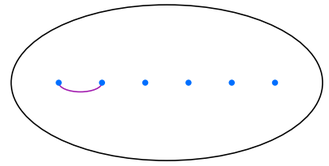
3. alternate.HL1.2 | H

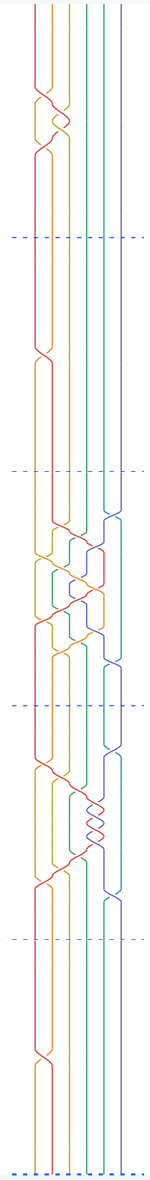


4. alternate.HL1.3 | T2^2



5. alternate.HL2.1 | H





II - step 142 - continued

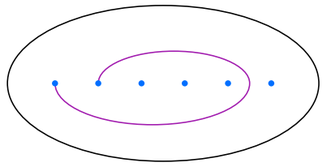
Transport the second inverse up across F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

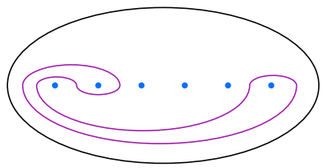
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=-2 5.

Before

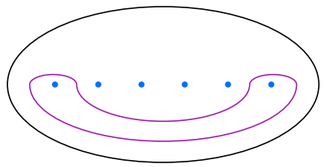
6. alternate.HL2.2 | H | SPHERE SUPPORT CHART



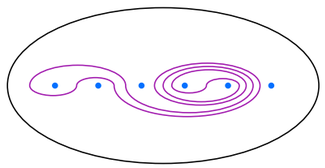
7. alternate.P15.remaining | T2



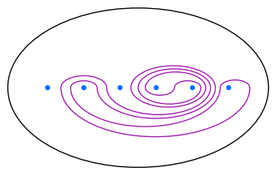
8. alternate.C4.rose.1 | T2

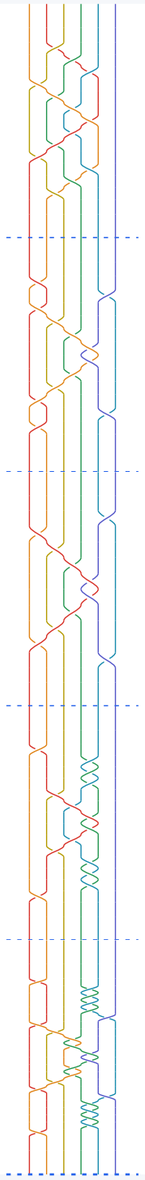


9. alternate.C4.rose.2 | T2 | SPHERE SUPPORT CHART



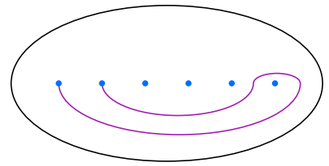
10. alternate.C4.rose.3 | T2 | SPHERE SUPPORT CHART



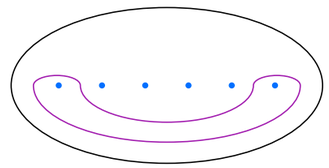


After

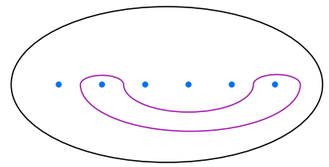
6. alternate.HL2.2 | H



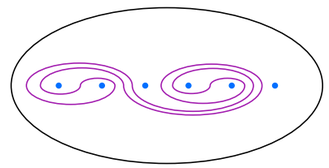
7. alternate.P15.remaining | T2



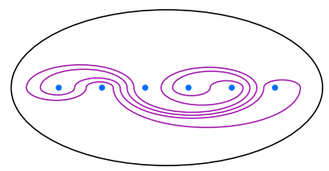
8. alternate.C4.rose.1 | T2

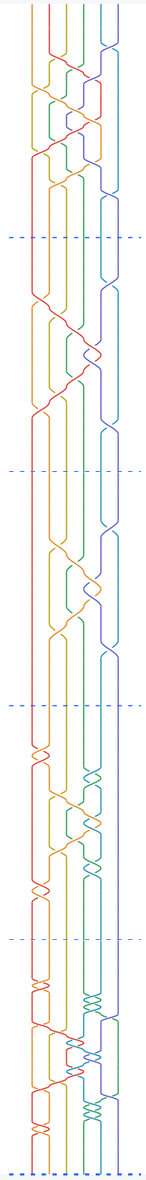


9. alternate.C4.rose.2 | T2 | SPHERE SUPPORT CHART



10. alternate.C4.rose.3 | T2 | SPHERE SUPPORT CHART





II - step 142 - continued

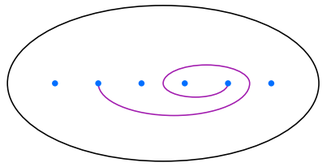
Transport the second inverse up across F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

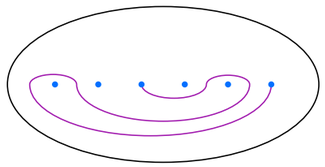
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=-2 5.

Before

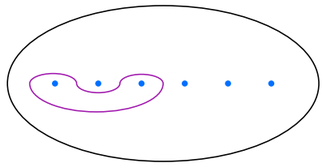
11. F3 | H | SPHERE SUPPORT CHART



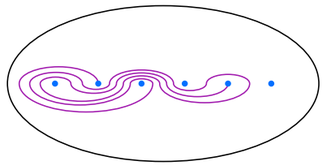
12. F6 | H | SPHERE SUPPORT CHART



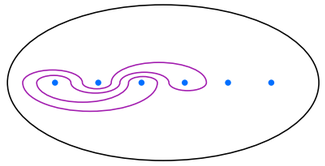
13. new.P26 | T2 | SPHERE SUPPORT CHART

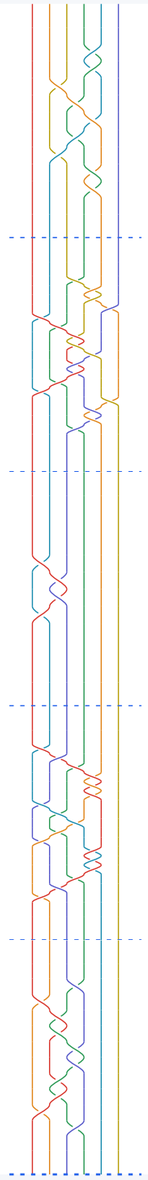


14. F9 | H | SPHERE SUPPORT CHART



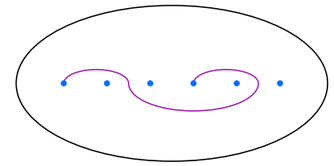
15. residual.P26 | T2 | SPHERE SUPPORT CHART



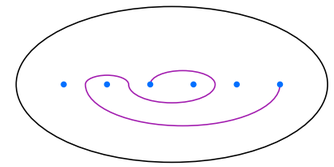


After

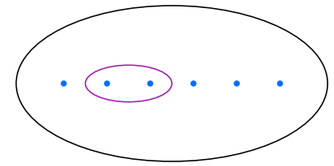
11. F3 | H | SPHERE SUPPORT CHART



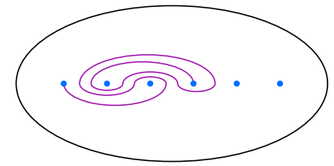
12. F6 | H | SPHERE SUPPORT CHART



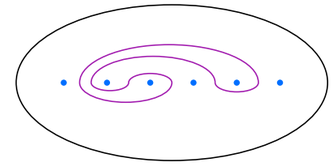
13. new.P26 | T2 | SPHERE SUPPORT CHART

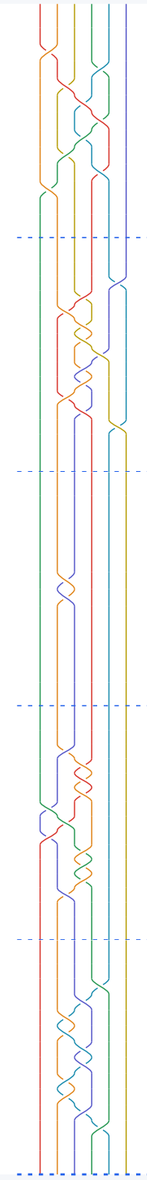


14. F9 | H | SPHERE SUPPORT CHART



15. residual.P26 | T2 | SPHERE SUPPORT CHART





II - step 142 - continued

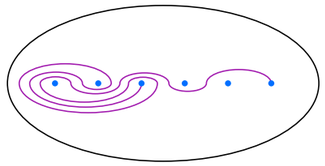
Transport the second inverse up across F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

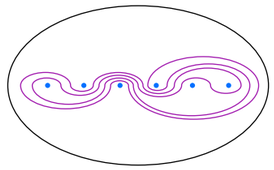
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=-2 5.

Before

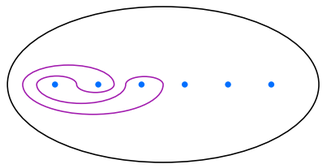
16. F11 | H | SPHERE SUPPORT CHART



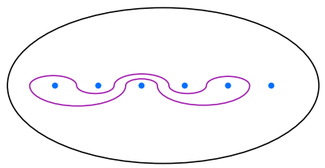
17. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



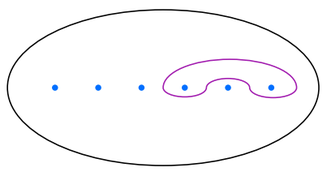
18. last.P12.square | T2^2

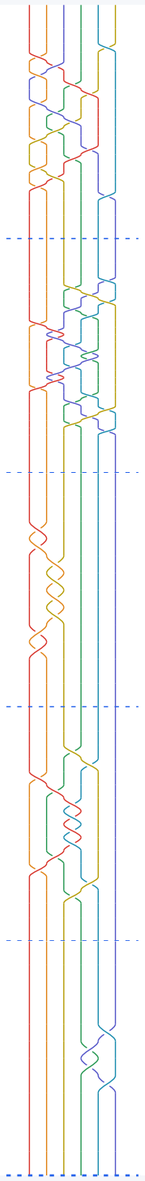


19. last.P34.square | T2^2



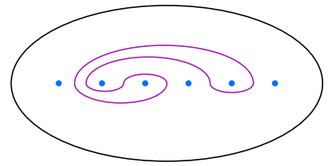
20. last.inverse.Q56 | T2^-1



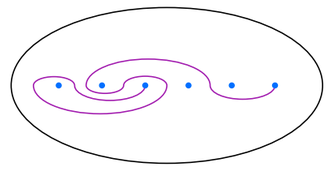


After

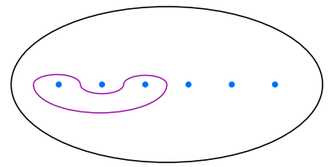
16. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART



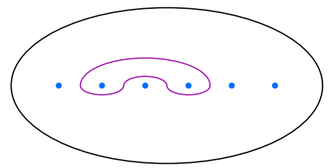
17. F11 | H | SPHERE SUPPORT CHART



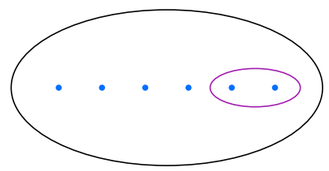
18. last.P12.square | T2^2

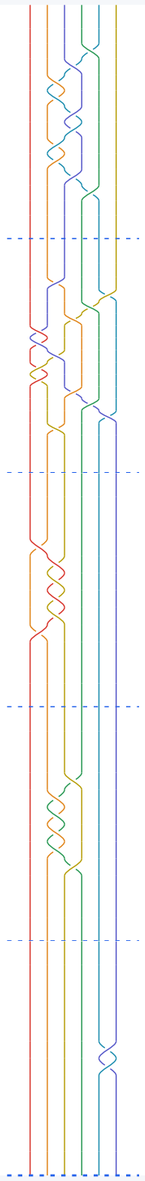


19. last.P34.square | T2^2



20. last.inverse.Q56 | T2^-1





II - step 143

Use the exact sphere match with residual.P26

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

20 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

No displayed row changes. The exact certificate records the change of source presentations.

II - step 144

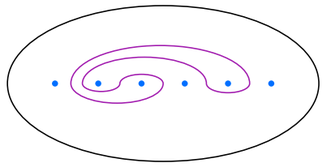
Cancel the matched positive and inverse pair exactly in the disk

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

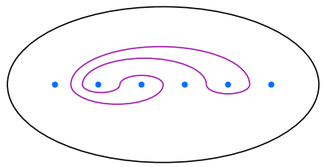
14 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

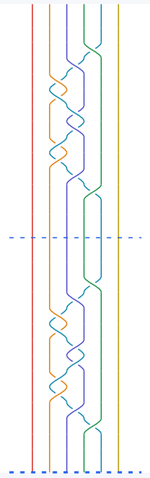
Before

15. residual.P26 | T2 | SPHERE SUPPORT CHART



16. last.inverse.BQ56 | T2^-1 | SPHERE SUPPORT CHART





After

Identity / no factors in this interval.

II - step 145

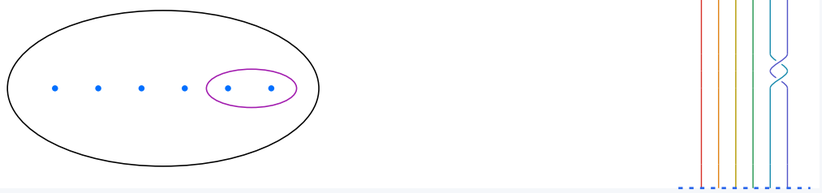
Replace inverse P56 using the squared sphere relation

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

Before

18. last.inverse.Q56 | T2^-1



After

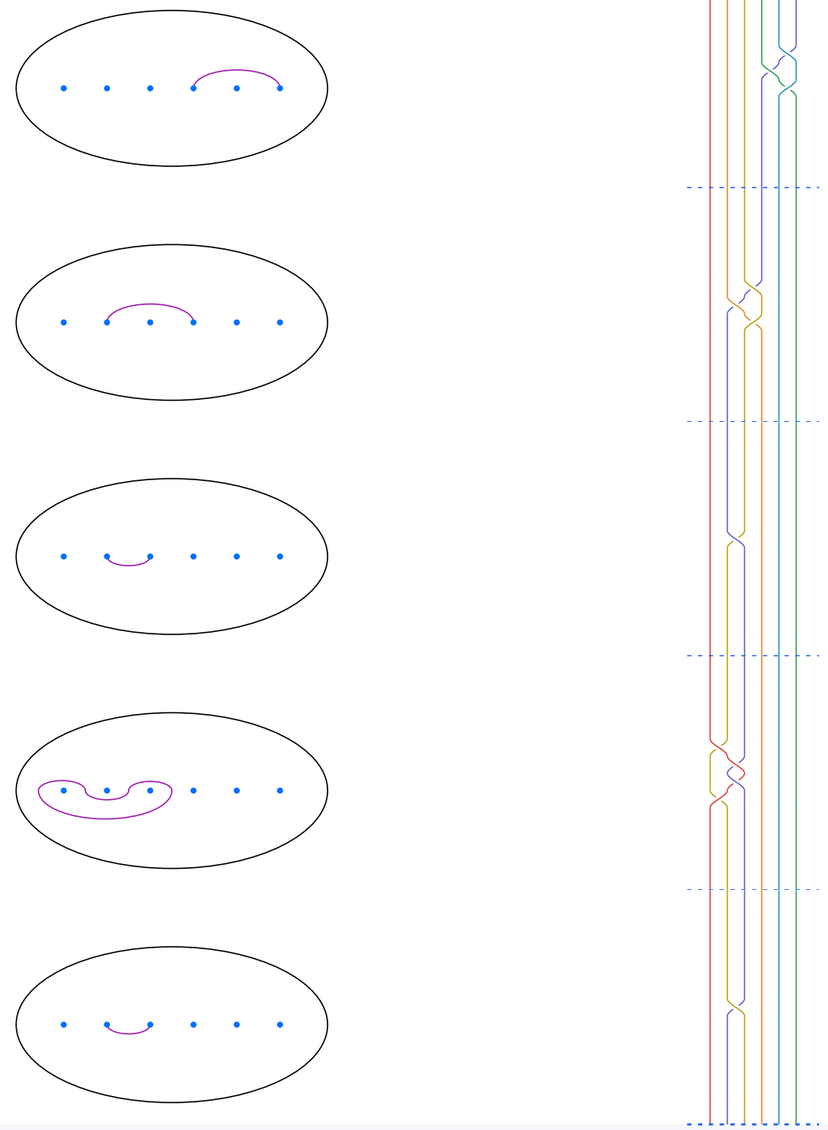
18. sphere.A1.1 | H

19. sphere.A1.2 | H

20. sphere.A1.3 | H

21. sphere.A1.4 | T2

22. sphere.A1.5 | H



II - step 145 - continued

Replace inverse P56 using the squared sphere relation

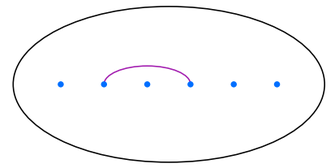
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

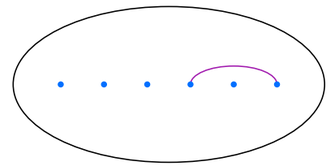
Before

After

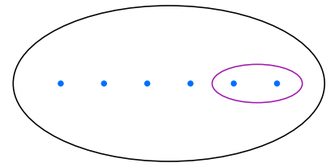
23. sphere.A1.6 | H



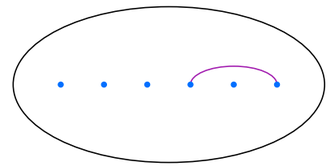
24. sphere.A1.7 | H



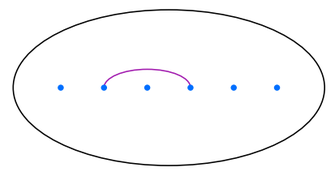
25. sphere.positive.P56 | T2

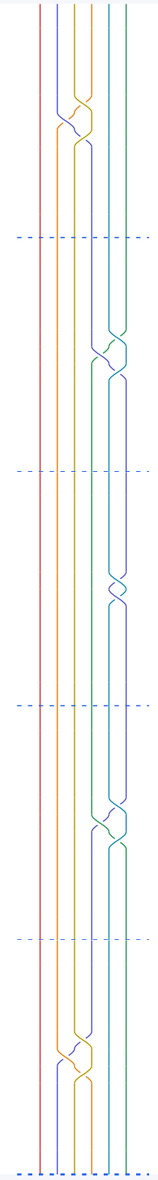


26. sphere.A2.1 | H



27. sphere.A2.2 | H





II - step 145 - continued

Replace inverse P56 using the squared sphere relation

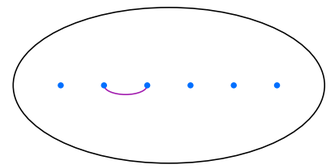
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

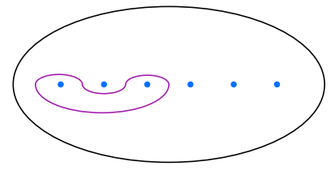
Before

After

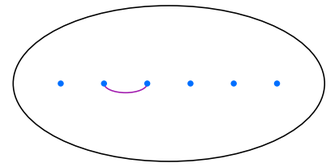
28. sphere.A2.3 | H



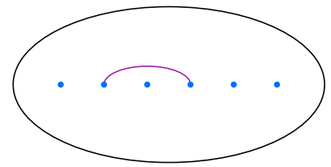
29. sphere.A2.4 | T2



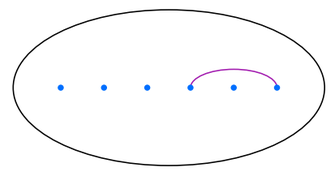
30. sphere.A2.5 | H

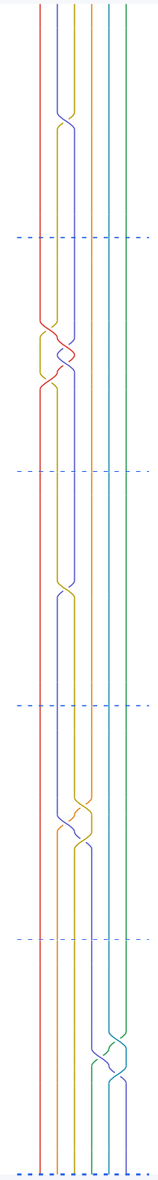


31. sphere.A2.6 | H



32. sphere.A2.7 | H

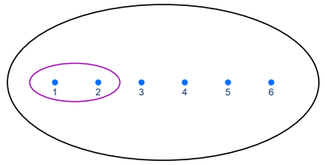




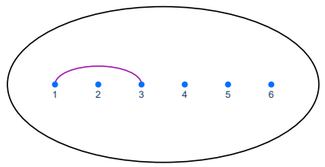
The actual positive fifty: complete factorization

FULL ORDERED FACTORIZATION. Read down the left column, then down the right. Braids continue with strand colors retained.

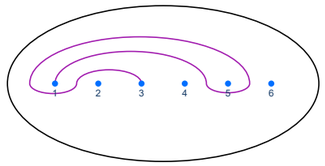
1. cross.P12 | T2



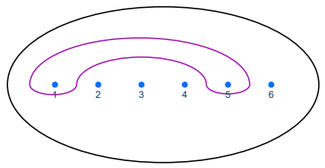
2. alternate.HL1.1 | H



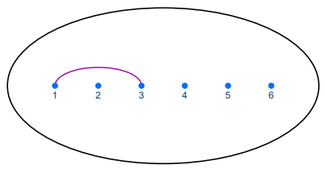
3. alternate.HL1.2 | H | SPHERE SUPPORT CHART



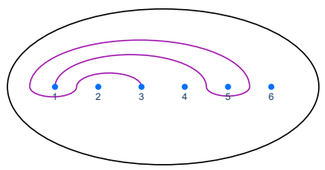
4. alternate.HL1.3 | T2^2 | SPHERE SUPPORT CHART

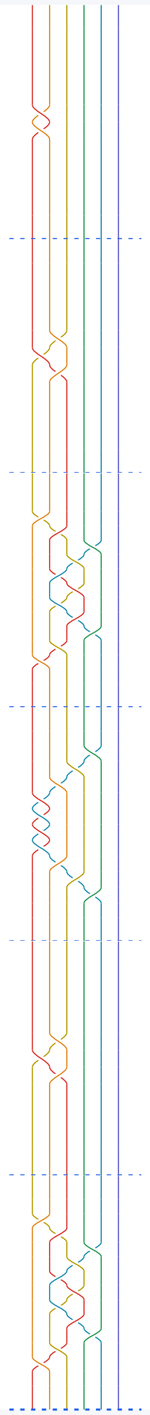


5. alternate.HL2.1 | H

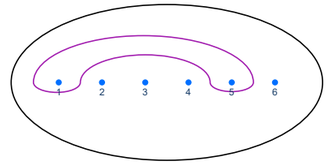


6. alternate.HL2.2 | H | SPHERE SUPPORT CHART

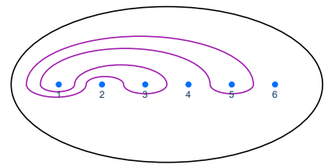




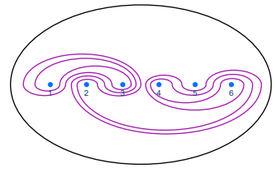
7. alternate.P15.remaining | T2 | SPHERE SUPPORT CHART



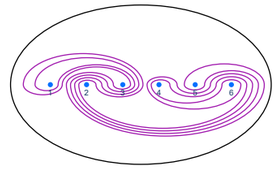
8. alternate.C4.rose.1 | T2 | SPHERE SUPPORT CHART



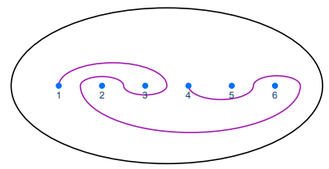
9. alternate.C4.rose.2 | T2 | SPHERE SUPPORT CHART



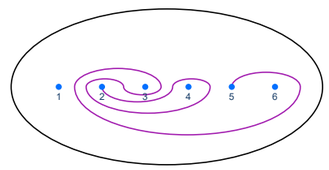
10. alternate.C4.rose.3 | T2 | SPHERE SUPPORT CHART

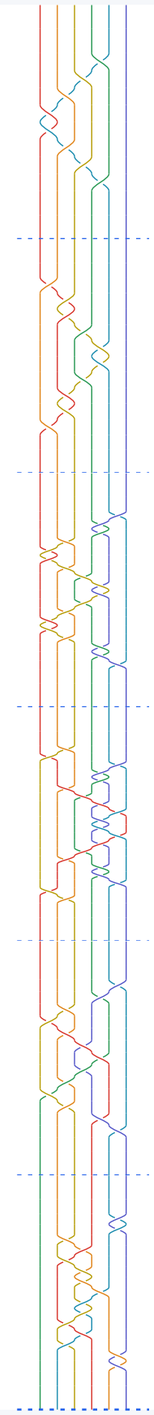


11. F3 | H | SPHERE SUPPORT CHART



12. F6 | H | SPHERE SUPPORT CHART

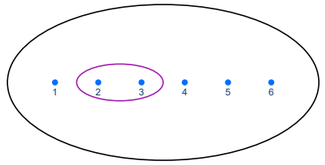




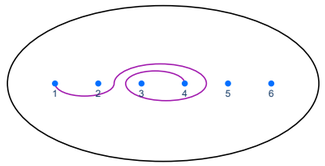
The actual positive fifty: complete factorization - continued

FULL ORDERED FACTORIZATION. Read down the left column, then down the right. Braids continue with strand colors retained.

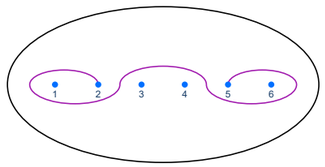
13. new.P26 | T2 | SPHERE SUPPORT CHART



14. F9 | H | SPHERE SUPPORT CHART



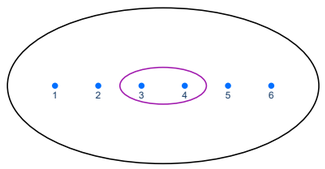
15. F11 | H | SPHERE SUPPORT CHART



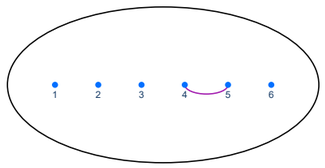
16. last.P12.square | T2^2

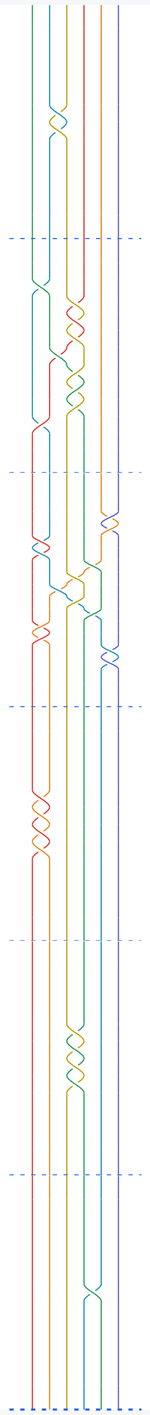


17. last.P34.square | T2^2

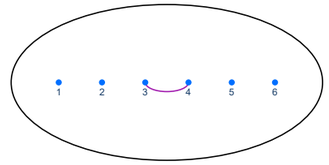


18. sphere.A1.1 | H

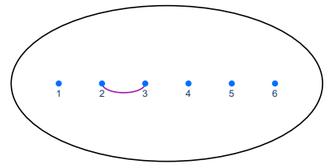




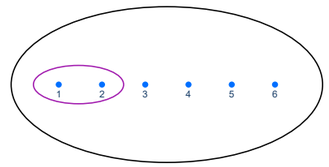
19. sphere.A1.2 | H



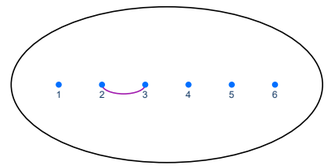
20. sphere.A1.3 | H



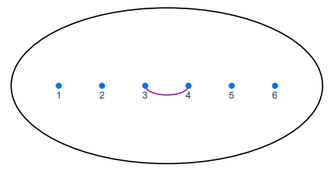
21. sphere.A1.4 | T2



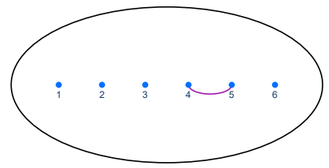
22. sphere.A1.5 | H

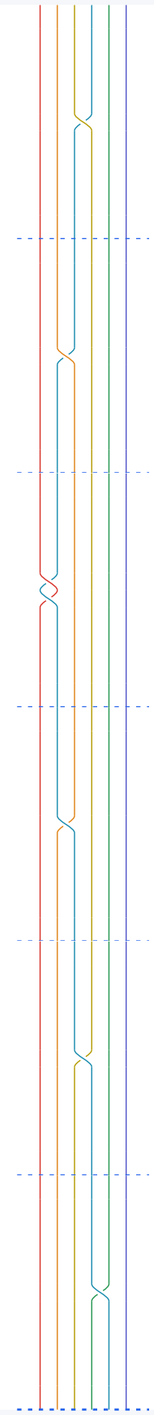


23. sphere.A1.6 | H



24. sphere.A1.7 | H





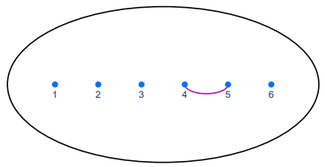
The actual positive fifty: complete factorization - continued

FULL ORDERED FACTORIZATION. Read down the left column, then down the right. Braids continue with strand colors retained.

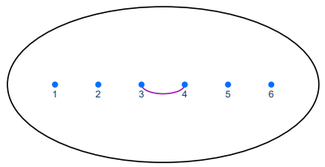
25. sphere.positive.P56 | T2



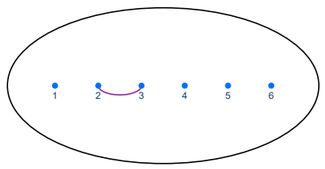
26. sphere.A2.1 | H



27. sphere.A2.2 | H



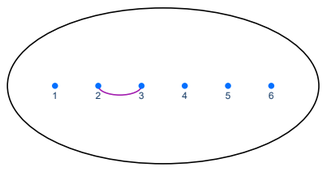
28. sphere.A2.3 | H



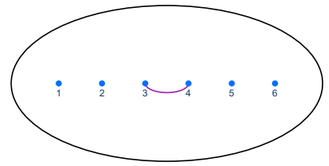
29. sphere.A2.4 | T2



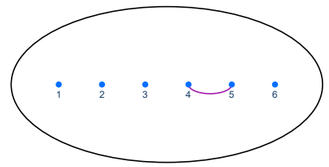
30. sphere.A2.5 | H

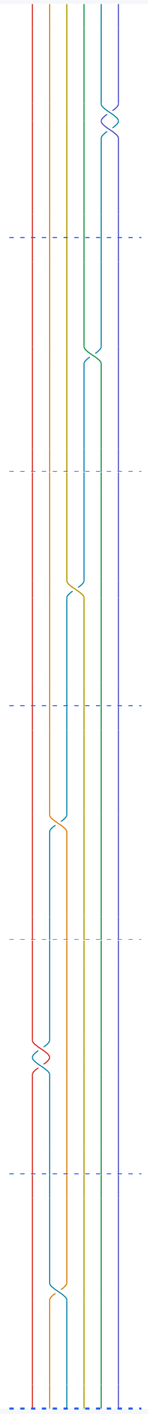


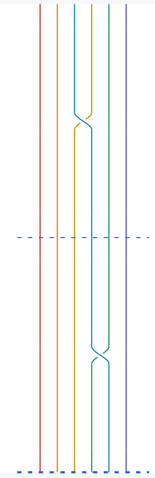
31. sphere.A2.6 | H



32. sphere.A2.7 | H







III. Isolate an actual standard twenty

Seven disk Hurwitz moves and one explicit sphere rechart isolate the twenty at the bottom. It is a disk conjugate of W0. The remaining thirty is not assumed to be W1.

III - step 001

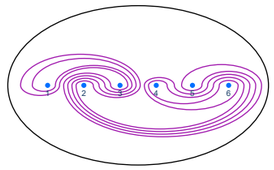
Move F3 up: transport the adjacent {5,6} pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

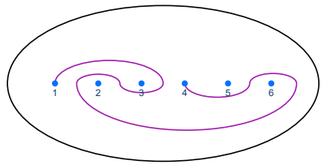
9 unchanged displayed factors above; 21 below. Only the changed interval is shown.

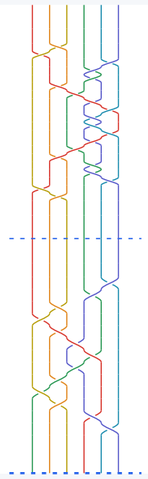
Before

10. alternate.C4.rose.3 | T2 | SPHERE SUPPORT CHART



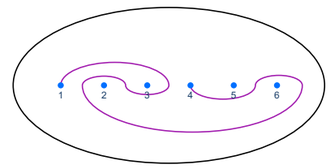
11. F3 | H | SPHERE SUPPORT CHART



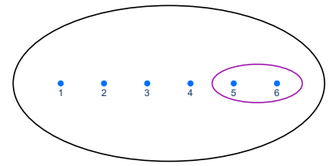


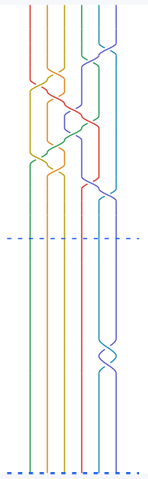
After

10. F3 | H | SPHERE SUPPORT CHART



11. alternate.C4.rose.3 | T2 | SPHERE SUPPORT CHART





III - step 002

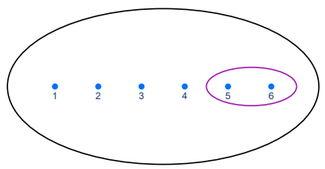
SPHERE MOVE: rechart the transported pair as the round T56

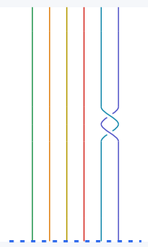
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

10 unchanged displayed factors above; 21 below. Only the changed interval is shown.

Before

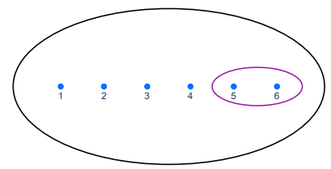
11. alternate.C4.rose.3 | T2 | SPHERE SUPPORT CHART

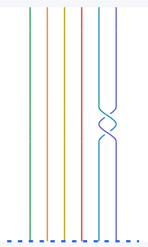




After

11. alternate.C4.rose.3 | T2





III - step 003

Move the round T56 down, conjugating the crossed factor

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

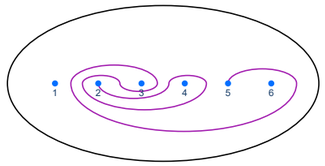
10 unchanged displayed factors above; 20 below. Only the changed interval is shown.

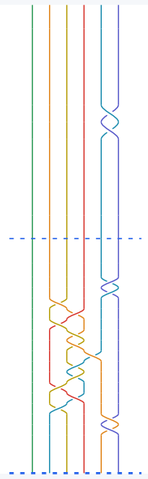
Before

11. alternate.C4.rose.3 | T2



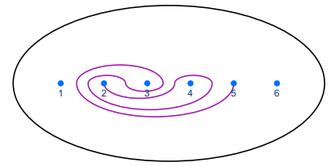
12. F6 | H | SPHERE SUPPORT CHART



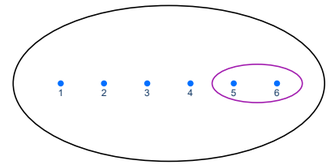


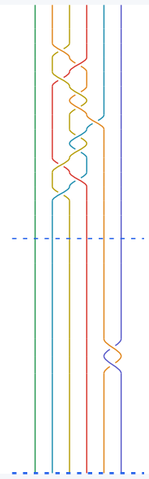
After

11. F6 | H | SPHERE SUPPORT CHART



12. alternate.C4.rose.3 | T2





III - step 004

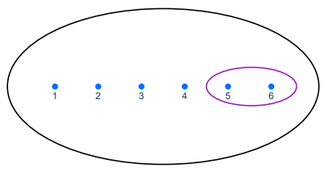
Move the round T56 down, conjugating the crossed factor

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

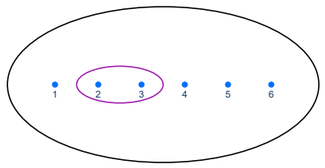
11 unchanged displayed factors above; 19 below. Only the changed interval is shown.

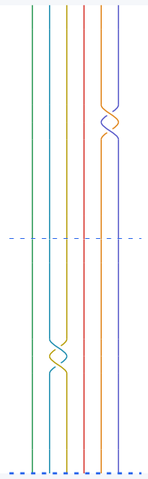
Before

12. alternate.C4.rose.3 | T2



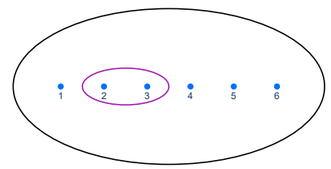
13. new.P26 | T2 | SPHERE SUPPORT CHART



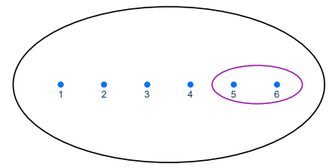


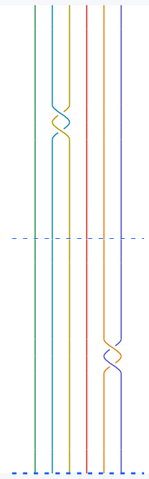
After

12. new.P26 | T2 | SPHERE SUPPORT CHART



13. alternate.C4.rose.3 | T2





III - step 005

Move the round T56 down, conjugating the crossed factor

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

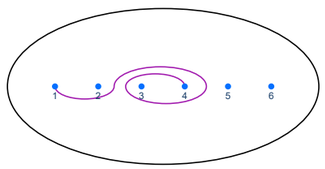
12 unchanged displayed factors above; 18 below. Only the changed interval is shown.

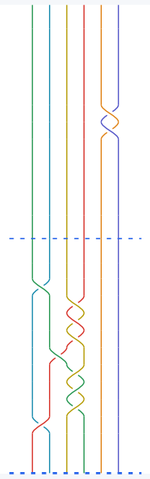
Before

13. alternate.C4.rose.3 | T2



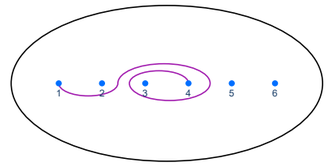
14. F9 | H | SPHERE SUPPORT CHART



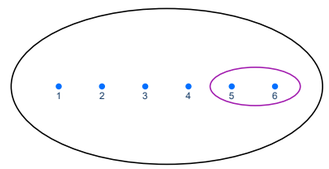


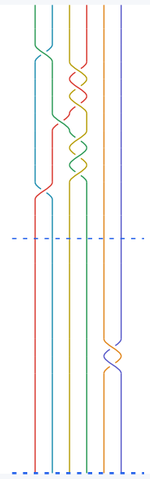
After

13. F9 | H | SPHERE SUPPORT CHART



14. alternate.C4.rose.3 | T2





III - step 006

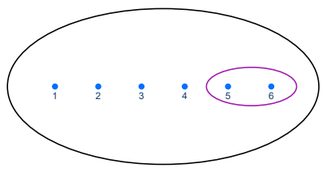
Move the round T56 down, conjugating the crossed factor

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

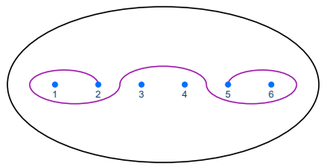
13 unchanged displayed factors above; 17 below. Only the changed interval is shown.

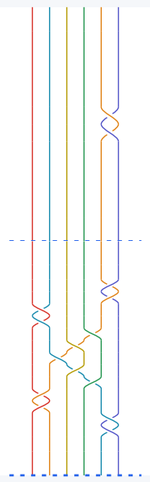
Before

14. alternate.C4.rose.3 | T2



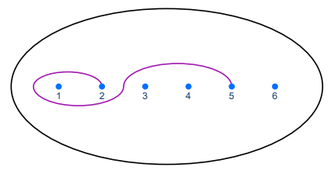
15. F11 | H | SPHERE SUPPORT CHART



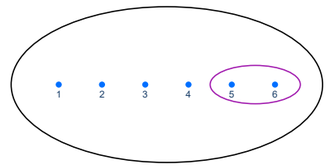


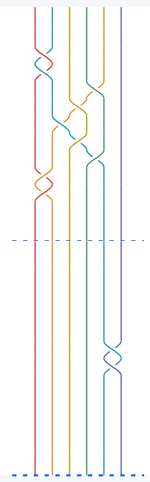
After

14. F11 | H | SPHERE SUPPORT CHART



15. alternate.C4.rose.3 | T2





III - step 007

Move the round T56 down, conjugating the crossed factor

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 16 below. Only the changed interval is shown.

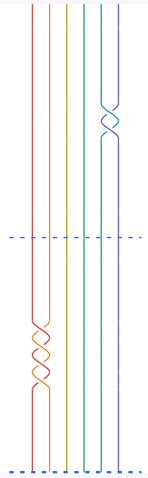
Before

15. alternate.C4.rose.3 | T2



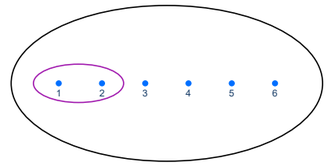
16. last.P12.square | T2^2



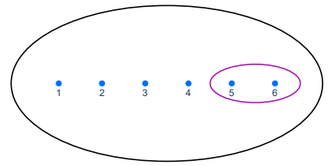


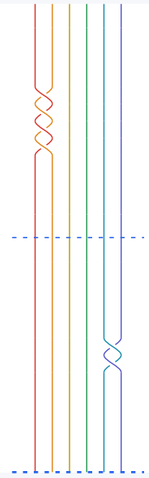
After

15. last.P12.square | T2^2



16. alternate.C4.rose.3 | T2





III - step 008

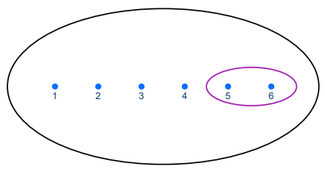
Move the round T56 down, conjugating the crossed factor

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

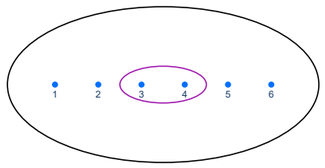
15 unchanged displayed factors above; 15 below. Only the changed interval is shown.

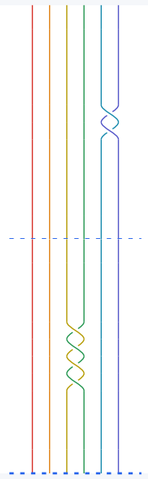
Before

16. alternate.C4.rose.3 | T2



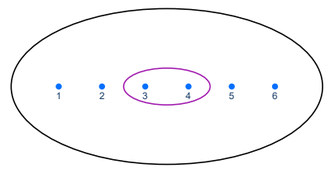
17. last.P34.square | T2^2



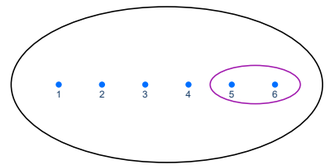


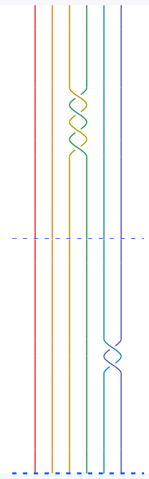
After

16. last.P34.square | T2^2



17. alternate.C4.rose.3 | T2





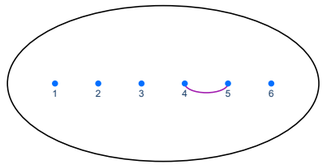
The isolated standard twenty

FULL ORDERED FACTORIZATION. Read down the left column, then down the right. Braids continue with strand colors retained.

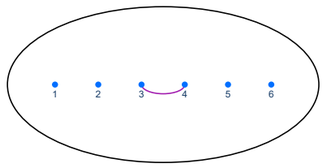
1. alternate.C4.rose.3 | T2



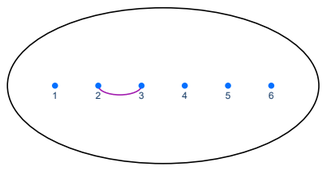
2. sphere.A1.1 | H



3. sphere.A1.2 | H



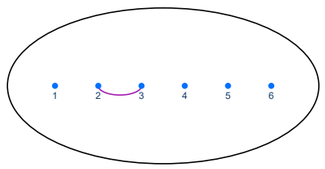
4. sphere.A1.3 | H



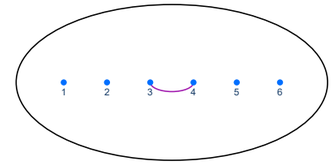
5. sphere.A1.4 | T2



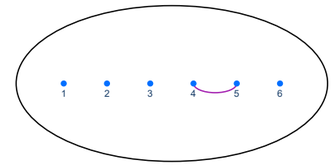
6. sphere.A1.5 | H



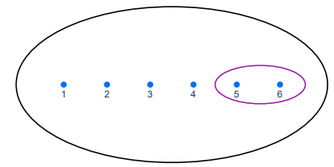
7. sphere.A1.6 | H



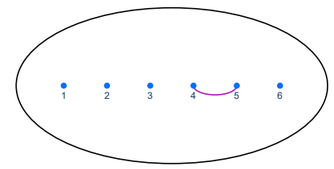
8. sphere.A1.7 | H



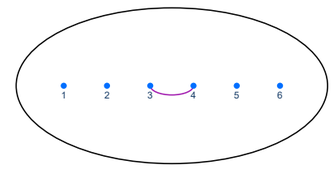
9. sphere.positive.P56 | T2



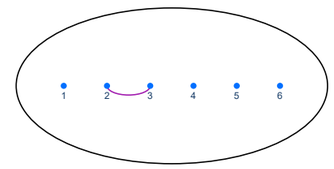
10. sphere.A2.1 | H



11. sphere.A2.2 | H



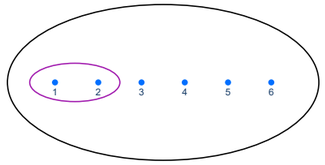
12. sphere.A2.3 | H



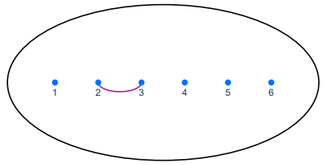
The isolated standard twenty - continued

FULL ORDERED FACTORIZATION. Read down the left column, then down the right. Braids continue with strand colors retained.

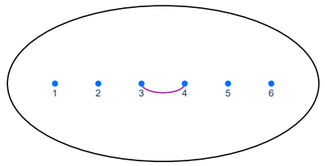
13. sphere.A2.4 | T2



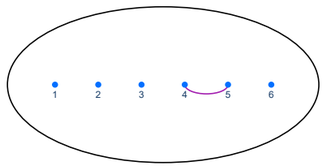
14. sphere.A2.5 | H

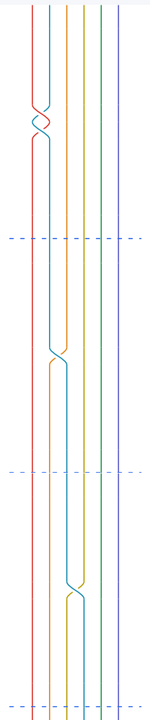


15. sphere.A2.6 | H



16. sphere.A2.7 | H





IV. The thirty and its exact disk-boundary ending

First remove the independently verified standard twenty in an explicit sphere move. The remaining proof does not assume the thirty product is identity. Every chart change is checked locally, the ordered disk product is checked on all five based meridians, and the final double cap is explicit.

IV - step 001

SPHERE MOVE: close the isolated standard twenty (W0)

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

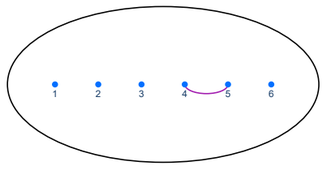
16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

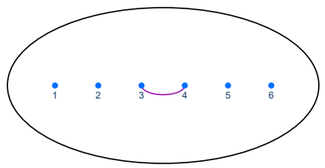
17. alternate.C4.rose.3 | T2



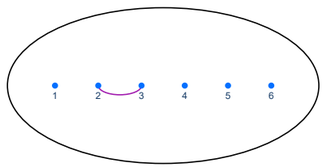
18. sphere.A1.1 | H



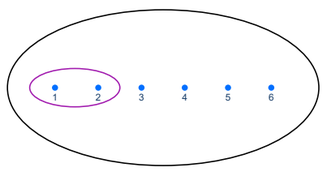
19. sphere.A1.2 | H

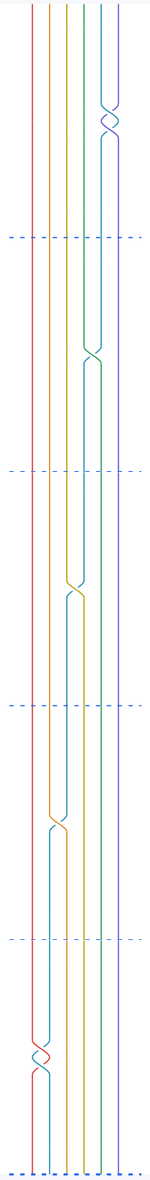


20. sphere.A1.3 | H



21. sphere.A1.4 | T2





After

Identity / no factors in this interval.

IV - step 001 - continued

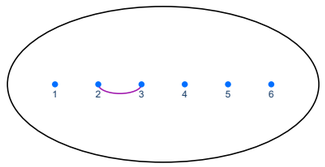
SPHERE MOVE: close the isolated standard twenty (W0)

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

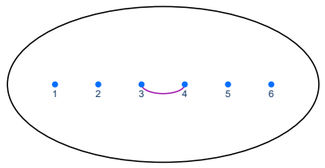
16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

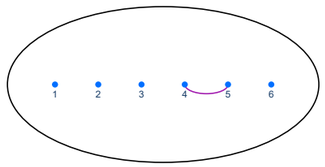
22. sphere.A1.5 | H



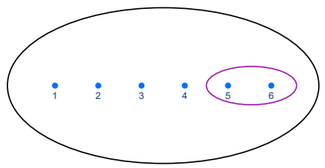
23. sphere.A1.6 | H



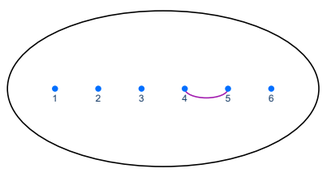
24. sphere.A1.7 | H

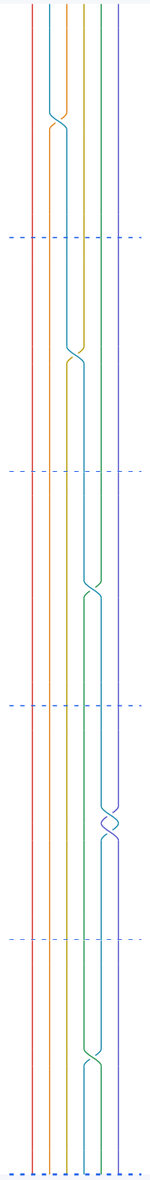


25. sphere.positive.P56 | T2



26. sphere.A2.1 | H





After

Identity / no factors in this interval.

IV - step 001 - continued

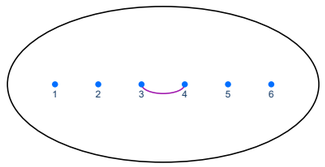
SPHERE MOVE: close the isolated standard twenty (W0)

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

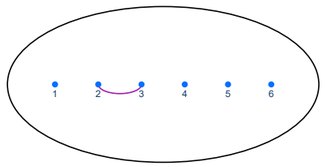
16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

27. sphere.A2.2 | H



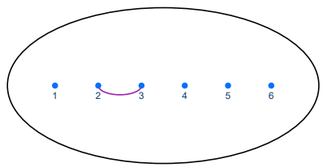
28. sphere.A2.3 | H



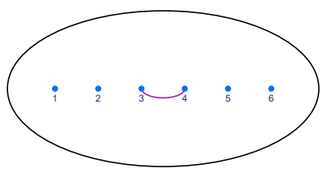
29. sphere.A2.4 | T2

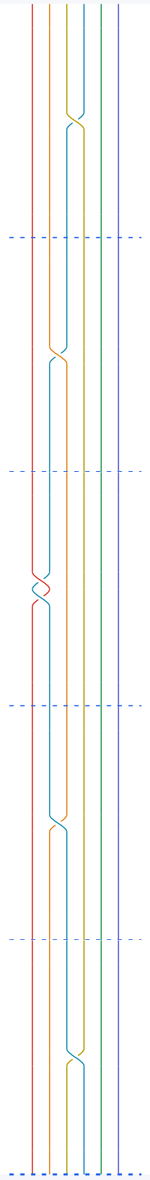


30. sphere.A2.5 | H



31. sphere.A2.6 | H





After

Identity / no factors in this interval.

IV - step 001 - continued

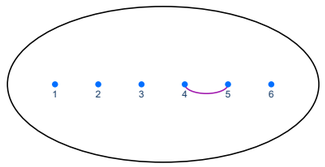
SPHERE MOVE: close the isolated standard twenty (W0)

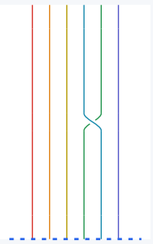
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

32. sphere.A2.7 | H





After

Identity / no factors in this interval.

IV - step 002

Use a disk-equal five-point representative of cross.P12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 003

Use a disk-equal five-point representative of alternate.HL1.1

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 004

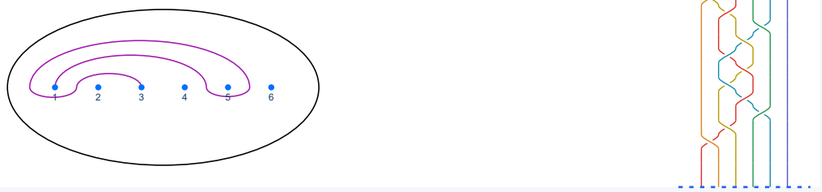
SPHERE MOVE: put point 6 at infinity for alternate.HL1.2

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

2 unchanged displayed factors above; 13 below. Only the changed interval is shown.

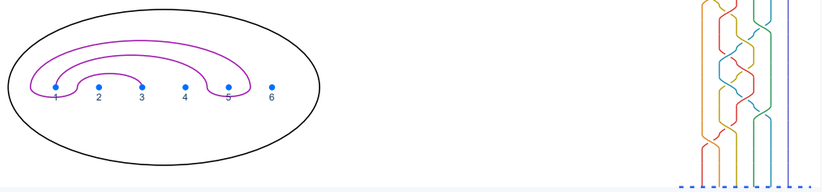
Before

3. alternate.HL1.2 | H | SPHERE SUPPORT CHART



After

3. alternate.HL1.2 | H



IV - step 005

SPHERE MOVE: put point 6 at infinity for alternate.HL1.3

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

3 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before

4. alternate.HL1.3 | T2^2 | SPHERE SUPPORT CHART



After

4. alternate.HL1.3 | T2^2



IV - step 006

Use a disk-equal five-point representative of alternate.HL2.1

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 007

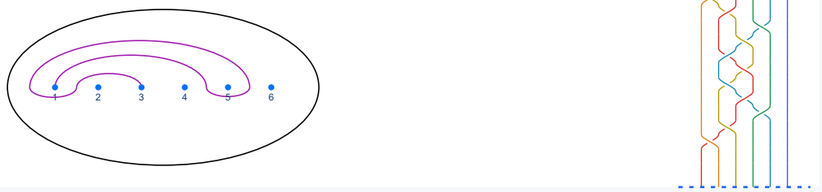
SPHERE MOVE: put point 6 at infinity for alternate.HL2.2

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

5 unchanged displayed factors above; 10 below. Only the changed interval is shown.

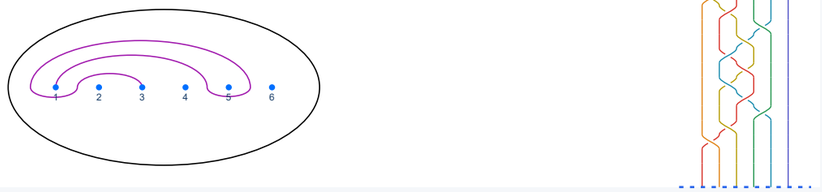
Before

6. alternate.HL2.2 | H | SPHERE SUPPORT CHART



After

6. alternate.HL2.2 | H



IV - step 008

SPHERE MOVE: put point 6 at infinity for alternate.P15.remaining

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

6 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before

7. alternate.P15.remaining | T2 | SPHERE SUPPORT CHART



After

7. alternate.P15.remaining | T2



IV - step 009

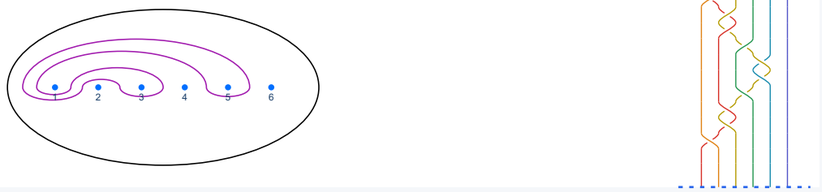
SPHERE MOVE: put point 6 at infinity for alternate.C4.rose.1

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

7 unchanged displayed factors above; 8 below. Only the changed interval is shown.

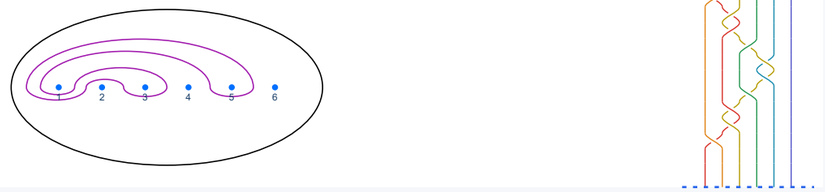
Before

8. alternate.C4.rose.1 | T2 | SPHERE SUPPORT CHART



After

8. alternate.C4.rose.1 | T2



IV - step 010

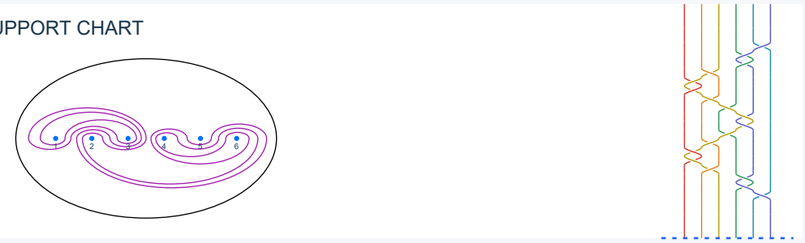
SPHERE MOVE: put point 6 at infinity for alternate.C4.rose.2

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

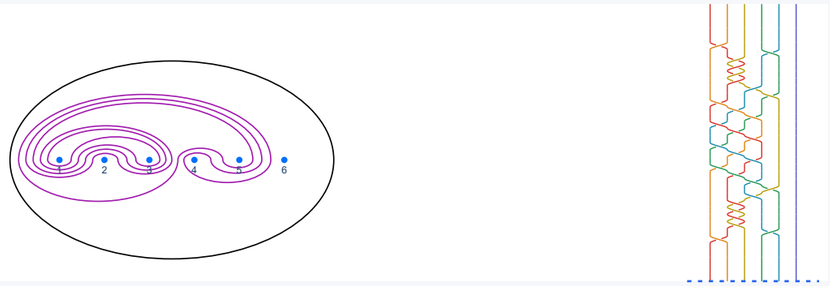
Before

9. alternate.C4.rose.2 | T2 | SPHERE SUPPORT CHART



After

9. alternate.C4.rose.2 | T4



IV - step 011

SPHERE MOVE: put point 6 at infinity for F3

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 012

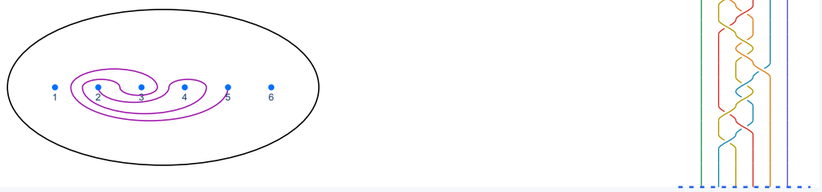
SPHERE MOVE: put point 6 at infinity for F6

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

10 unchanged displayed factors above; 5 below. Only the changed interval is shown.

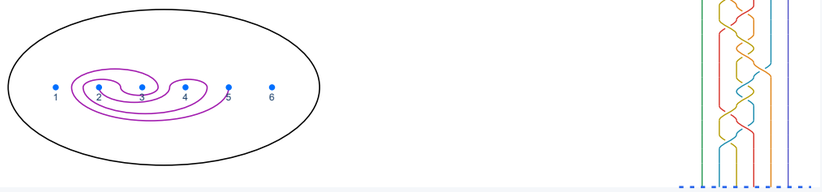
Before

11. F6 | H | SPHERE SUPPORT CHART



After

11. F6 | H



IV - step 013

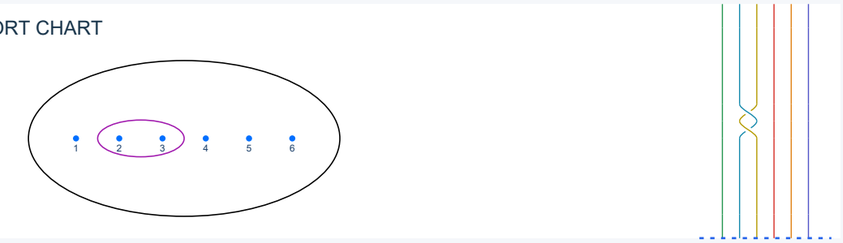
SPHERE MOVE: put point 6 at infinity for new.P26

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

11 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before

12. new.P26 | T2 | SPHERE SUPPORT CHART



After

12. new.P26 | T2



IV - step 014

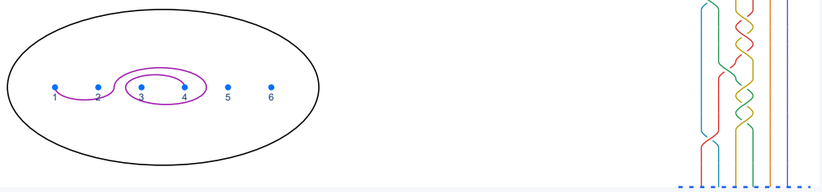
SPHERE MOVE: put point 6 at infinity for F9

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

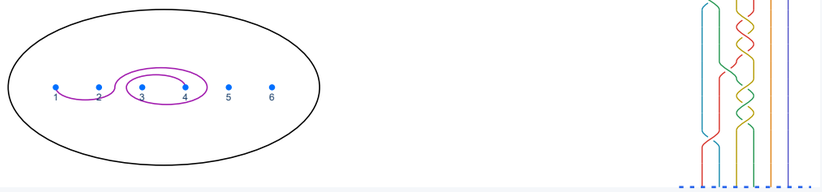
Before

13. F9 | H | SPHERE SUPPORT CHART



After

13. F9 | H



IV - step 015

SPHERE MOVE: put point 6 at infinity for F11

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

13 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before

14. F11 | H | SPHERE SUPPORT CHART



After

14. F11 | H



IV - step 016

Use a disk-equal five-point representative of last.P12.square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 017

Use a disk-equal five-point representative of last.P34.square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 018

Exact five-point disk relation: double outer-boundary twist

EXACT DISK BOUNDARY RELATION: independently calculated based Artin actions agree.

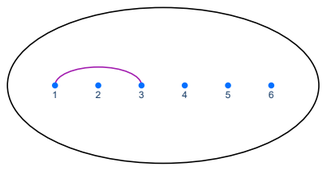
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

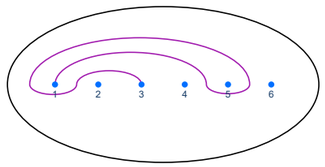
1. cross.P12 | T2



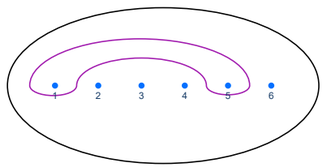
2. alternate.HL1.1 | H



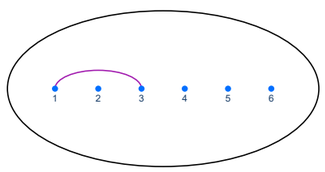
3. alternate.HL1.2 | H

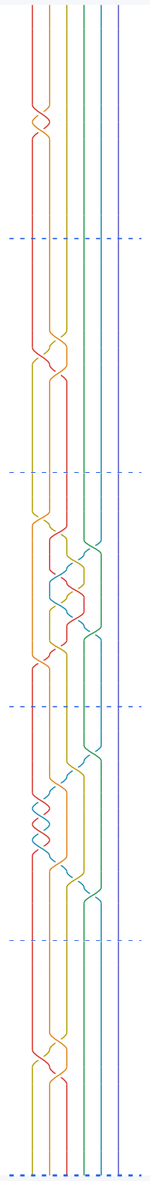


4. alternate.HL1.3 | T2^2



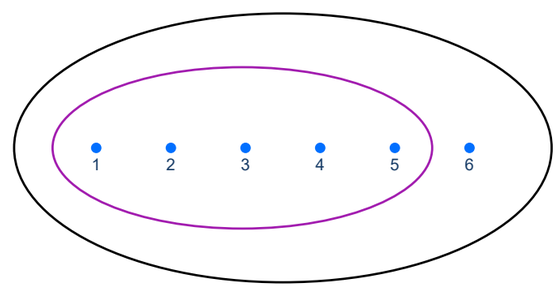
5. alternate.HL2.1 | H

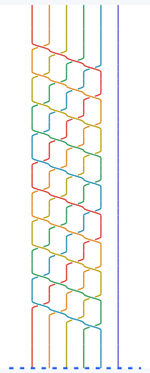




After

1. boundary.five.double | T5^2





IV - step 018 - continued

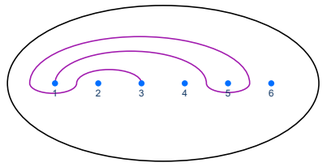
Exact five-point disk relation: double outer-boundary twist

EXACT DISK BOUNDARY RELATION: independently calculated based Artin actions agree.

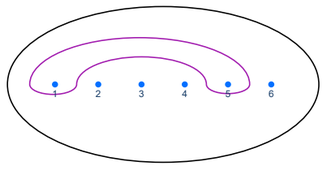
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

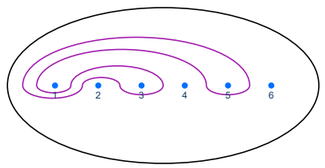
6. alternate.HL2.2 | H



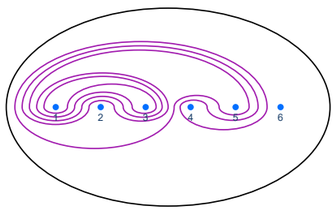
7. alternate.P15.remaining | T2



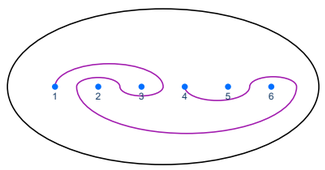
8. alternate.C4.rose.1 | T2



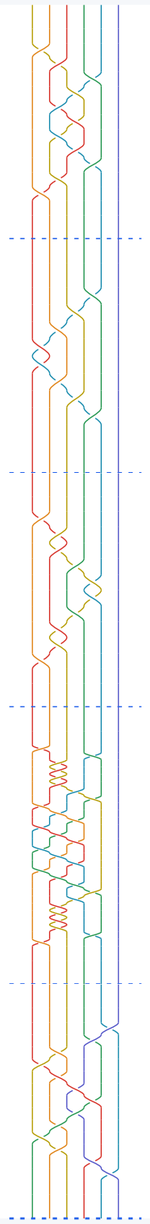
9. alternate.C4.rose.2 | T4



10. F3 | H | SPHERE SUPPORT CHART



After



IV - step 018 - continued

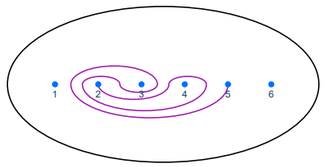
Exact five-point disk relation: double outer-boundary twist

EXACT DISK BOUNDARY RELATION: independently calculated based Artin actions agree.

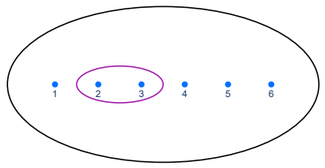
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

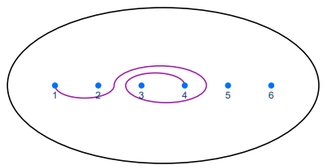
11. F6 | H



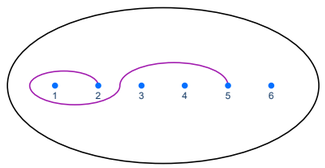
12. new.P26 | T2



13. F9 | H



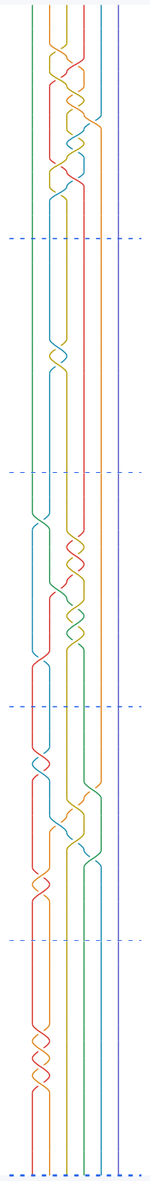
14. F11 | H



15. last.P12.square | T2^2



After



IV - step 018 - continued

Exact five-point disk relation: double outer-boundary twist

EXACT DISK BOUNDARY RELATION: independently calculated based Artin actions agree.

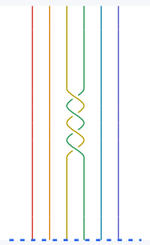
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

16. last.P34.square | T2^2



After



IV - step 019

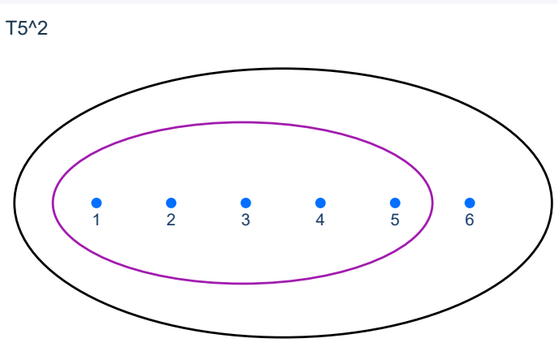
SPHERE MOVE: cap the double boundary around point 6

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

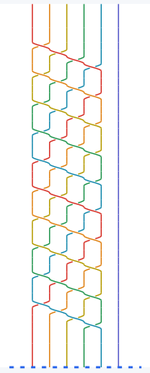
0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

1. boundary.five.double | T5^2



After

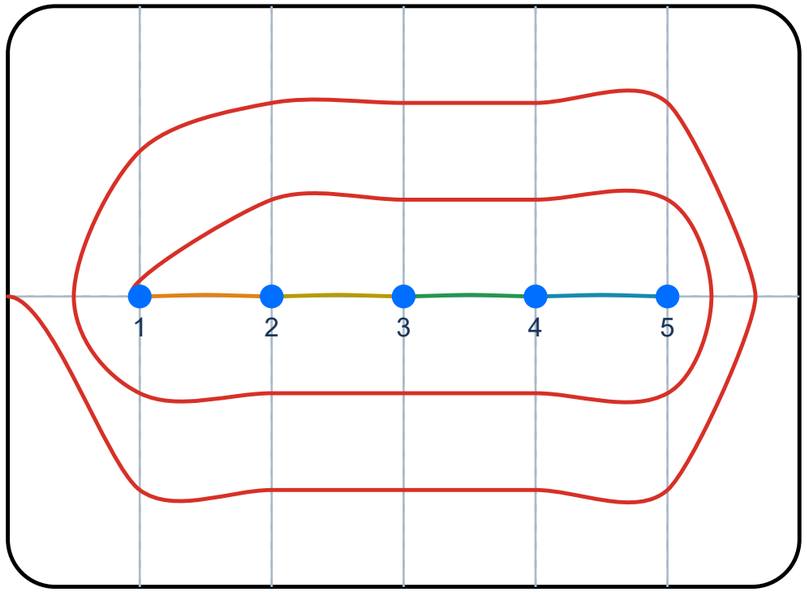


Identity / no factors in this interval.

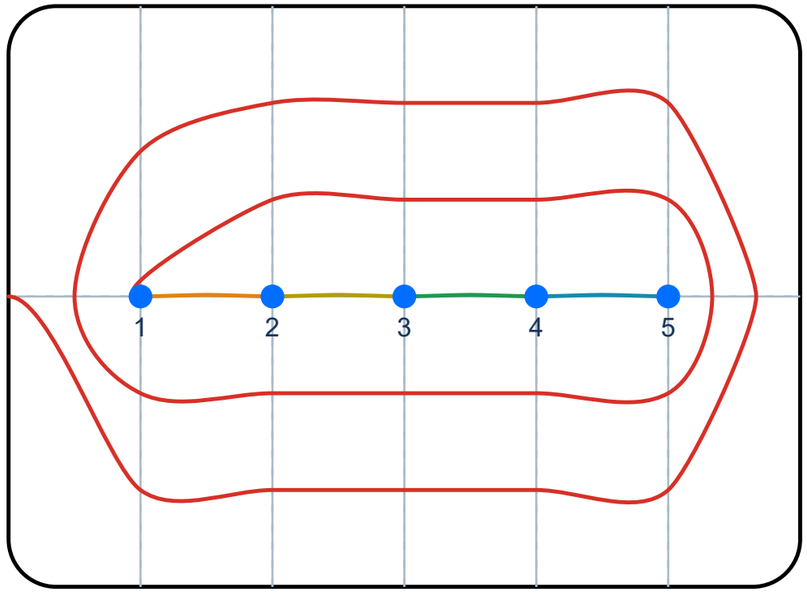
V. The decisive joint cut-system certificate

The ordered five-point model and the double outer-boundary twist have independently calculated, identical based actions. The reconstructed disk chain has five paths and a joint normal-placement certificate. This is different from a six-point sphere reference chain.

Calculated ordered five-point product



Double outer-boundary twist



The boundary-to-first-point arc has itinerary cuts (5,0,5); the four inner arcs are straight. The normal engine verifies all five paths jointly. The exact actual/reference meridian images are listed next.

x1 -> 1 2 3 4 5 1 2 3 4 5 1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x2 -> 1 2 3 4 5 1 2 3 4 5 2 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x3 -> 1 2 3 4 5 1 2 3 4 5 3 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x4 -> 1 2 3 4 5 1 2 3 4 5 4 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x5 -> 1 2 3 4 5 1 2 3 4 5 -4 -3 -2 -1 -5 -4 -3 -2 -1

VI. Sources, limitations and downloadable certificates

The new ending is an independent exact boundary-action proof. The classification theorem is not used to infer identity. A shorter geometric derivation and an elementary full W0 W1 Hurwitz route remain separate improvements.

Graphical working seed and saved substitution route: https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-compact/index.html

New twenty extraction, thirty boundary proof and certificates: https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html

Auroux, Fiber sums of genus 2 Lefschetz fibrations, https://arxiv.org/pdf/math/0204285. Lemmas 6-7 concern standard-block invariance; Theorem 9 recognizes transitive positive identity factorizations. Its identity hypothesis must not be reversed.

Scope: sphere mapping-class action and the separately checked closed genus-two lift. No hidden disk equality, cumulative joint support drawing, or unperformed elementary Hurwitz normal form is claimed.

Source SHA256: 05dbc99bc8137ce2a716bef51080ef25abfd13fa29f08c95f9bde68ffbe8c417

All 177 atomic transitions in the linked chain are retained: 5 working-chart joins, 145 saved substitutions, 8 twenty-extraction moves, 1 explicit twenty closure, and 18 thirty-closure steps.